Critical sets in Latin squares

Full research proofs and explicit examples

Research review prepared for Richard Bean 17 October 2026 This companion volume

collects the proof developments from the present investigation, gives the exact completion and partial square for every archived matrix pair included in its inventory, and studies inverse transversal prolongation. It preserves the distinction between published inputs, new deductions, numerical certificates and unresolved experiments. "New" or "unpublished" means developed in this investigation; it is not an assertion of priority following a comprehensive literature search.

The strongest verified examples currently used are $\mathrm{lcs}(9) \geq 47$, $\mathrm{lcs}(15) \geq 150$, $\mathrm{lcs}(27) \geq 447$ and $\mathrm{lcs}(33) \geq 844$, together with the other orders in the record table. In particular,

$$844 > 834 = \lceil 33^2 - 33^{\log_2 3} \rceil.$$

The rounded BCC upper proposal is therefore false. The former sharp value 838 at order33 is also false as an equality. Neither failure disproves the separate equality proposal at powers of two.

The collection includes the complete $n^2 - \lceil 7n/2 - 4 \rceil$ upper proof for $n \geq 14$, its one-hole-symbol case, the earlier $n \geq 8$ argument, safe binary and ternary lifts, arbitrary mixed lifts, the Fu–Fu–Liao one-clue improvement, the dyadic and thesis prolongations, the $4m + 3$ refinement, the product converse and the strengthened counting argument. Published seed theorems are identified explicitly. In particular, $(5m^2 - 3m)/2$ at order $2m$ is Donovan's 1998 theorem, not an original result of Bean's thesis.

Independent audits here are separate AI-agent reconstructions and, where specified, independent programs or certificate replays. They are not human refereeing or formal proof-assistant verification. Solver-dependent uniqueness and solver-free certificate replay are distinguished.

Inverse prolongation. The requested operation reduces the order by one. Some older sources use "stripping a transversal" for the forward operation; throughout this volume *stripping* means contraction, as requested. Its precise criterion and transfer proof are included. A large defining superset is never counted as a critical set before minimality is proved or protected clues are counted.

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1 Definitions, evidence and the record atlas

A partial Latin square is *defining* if it has exactly one Latin completion. It is *critical* if removing any clue destroys uniqueness. The notation $\mathrm{lcs}(n)$ (also $\ell(n)$ in the earlier notes) denotes the largest critical-set size over all Latin squares of order n. A private trade for a clue is a Latin trade whose intersection with the partial square is exactly that clue. Switching to its mate proves that clue necessary. An intercalate is the four-entry trade on a 2 by 2 Latin subsquare. Counting clues with such witnesses is different from counting the witnesses themselves.

Lemma 1.1 (Protected reduction)**.** *If U defines L and each clue in $F \subseteq U$ has a private trade in U, every defining subset of U contains F. In particular an inclusion-minimal defining subset exists and has at least $|F|$ clues.*

Proof. If a defining subset omitted a clue $f \in F$, its private trade would be disjoint from that subset, and switching the trade would give a second completion. Starting from finite U, repeatedly remove a clue whenever unique completion survives. Termination gives a critical subset and the first assertion proves its size bound. The witnesses remain private through every deletion that retains their designated clues. □

The atlas first gives one current best available example at each order 9–33. It then includes the other distinct archived matrix pairs, including smaller construction experiments and older records. It is a reproducible snapshot of saved artifacts, not an enumeration of every transient search state. All displayed grids have been checked afresh for Latinness, agreement with their partials, and the stated number of clues. These checks alone do not certify uniqueness or minimality.

Status I means a previously independently verified example; T means an explicit theorem construction; P means producer-reported critical, with no independent verification asserted here; U means unresolved and contributes no lower bound. Exact verification notes accompany each pair. The machine readable companion `lcs_full_review/examples.json` contains the same arrays, source paths, statuses and full hashes. Symbol labels are normalized to $0, \ldots, n-1$ and every hole is encoded by -1.

The inventory contains 152 distinct pairs: 25 current representatives and 127 additional archived instances.

n	Record	Holes	U_p	Atlas
9	47	34	55	E001
10	59	41	71	E002
11	74	47	88	E003
12	90	54	108	E004
13	107	62	129	E005
14	126	70	151	E006
15	150	75	176	E007
16	175	81	204	E008
17	200	89	233	E009
18	224	100	265	E010
19	245	116	298	E011
20	281	119	334	E012
21	304	137	371	E013
22	340	144	411	E014
23	367	162	452	E015
24	405	171	496	E016
25	449	176	541	E017
26	482	194	589	E018
27	447	282	638	E019
28	536	248	690	E020
29	624	217	743	E021
30	654	246	799	E022
31	721	240	856	E023
32	781	243	916	E024
33	844	245	977	E025

2 Complete proof of the constant-minus-four upper bound

This chapter integrates the proved refinement with all its elementary, frame, and local-forcing dependencies. It spells out the exceptional one-hole-symbol case. The two audit reports later in this document distinguish independent

agent review from human refereeing and disclose the reviewer's contributions.

Theorem 2.1. *For every integer $n \geq 14$,*

$$\boxed{\operatorname{lcs}(n) \leq n^2 - \left\lceil \frac{7n}{2} - 4 \right\rceil.}$$

2.1 Witnesses, covering, and exceptional lines

Let P be a critical set in its unique Latin-square completion M of order n. Put $H = n^2 - |P|$. A line is a row, a column, or the occurrences of a symbol; its degree is its number of holes. Write $RS(i), CS(j)$ for the missing-symbol sets of row i and column j. Permuting the coordinates of the triples $(\text{row}, \text{column}, \text{symbol})$ is called conjugation; it preserves criticality and permutes the three sorts of lines.

For each clue e, criticality gives a Latin square $W_e \neq M$ completing $P \setminus \{e\}$. It changes e, since otherwise it would complete P. It retains every other clue. We call it a deletion witness.

Lemma 2.2 (Intersection and covering)**.** *Every $RS(i)$ intersects every $CS(j)$. If J is the set of hole-columns of row i, then $\bigcup_{j \in J} CS(j) = [n]$, and consequently $\sum_{j \in J} d(j) \geq n$. The same assertions hold in all conjugates.*

Proof. At a hole (i, j) the value $M(i, j)$ belongs to both missing sets. At a clue use its changed value in its deletion witness. For covering, a symbol missing from row i has its M-position in a column in J. A symbol present at a clue e of row i must move in W_e to one of that row's holes. That destination column already missed it in P, because e is outside J. □

No line has degree zero: in a full row, deleting one clue cannot change the sole changeable position while retaining the other $n - 1$ entries. Conjugate for the other line types. A one-hole row has its hole in an empty column by covering, and its missing symbol is absent from P by intersection. In particular a one-hole symbol has its sole hole in an empty row and an empty column. Their union contains two holes of every other symbol, so at most one symbol has degree one; conjugation proves the analogous statements for rows and columns. Also at most one row is empty, at most one column is empty, and at most one symbol is absent, since swapping two such lines would give a second completion.

2.2 The frame case, including the sharper constant

Let the three minimum degrees be ρ, γ, σ. Call a coordinate pair exceptional when one minimum is one and the other is at most two. If every pair is exceptional, at least two minima equal one; conjugate so that a row i_* and a column j_* each have one hole. The hole of i_* lies in an empty column c and carries an absent symbol w. The hole of j_* lies in an empty row r and carries the same absent symbol, by uniqueness of the absent symbol. We have $r \neq i_*$ and $M(r, c) \neq w$.

The union F of row r, column c, and all positions of w consists of $3n - 3$ holes: the three pairwise intersections are distinct, with no triple intersection. Let E be the other holes, R_E, C_E their row and column supports, and $\pi(i)$ the column containing w in row i. Then $\pi(i_*) = c$, $\pi(r) = j_*$. Neither r, i_* meets E; neither c, j_* meets E. Define

$$X = [n] \setminus (R_E \cup \{r, i_*\}), \quad Y = [n] \setminus (C_E \cup \{c, j_*\}), \quad Z = Y \setminus \pi(R_E).$$

Thus $|X| = n - |R_E| - 2$ and $|Y| = n - |C_E| - 2$.

If $j \in Z$, put $k = \pi^{-1}(j) \in X$. For $i \in X \setminus \{k\}$ the cell (i, j) is a clue. Its only row holes are in $c, \pi(i)$ and its only column holes are in r, k. Put $u = M(i, c)$ and $v = M(r, j)$. If $u \neq v$, its witness must change the clue to w, the sole common member of $\{u, w\}$ and $\{v, w\}$. The w of row k must leave (k, j). It cannot move to (k, c): row i_* has only one changeable position, (i_*, c), which must retain w. There is no other changeable cell in row k, since $k \notin R_E$. This is impossible. Hence $u = v$ for every $i \in X \setminus \{k\}$, which column distinctness permits for at most one i. Therefore $|X| \leq 2$ if Z is nonempty.

If $|X| \geq 3$, we obtain $Y \subseteq \pi(R_E)$ and $|R_E| + |C_E| \geq n - 2$, whence $2|E| \geq n - 2$. If $|X| \leq 2$, then $|E| \geq |R_E| \geq n - 4 \geq (n - 2)/2$ for $n \geq 6$. In either case

$$H \geq 3n - 3 + \left\lceil \frac{n-2}{2} \right\rceil = \left\lceil \frac{7n}{2} - 4 \right\rceil. \tag{2.1}$$

2.3 Reduction to two stars

Assume henceforth, for a contradiction, that

$$H < 7n/2 - 4. \tag{2.2}$$

Write $k = \lfloor n/2 \rfloor \geq 7$. The integer caps are $H \leq 7k - 5$ for $n = 2k$ and $H \leq 7k - 1$ for $n = 2k + 1$. The frame case is excluded by (2.1), so orient a nonexceptional pair as rows and columns. A minimum degree at least four gives $H \geq 4n$. If one minimum is three and the other at most three, covering a line with at most three holes gives $H \geq n + 3(n - 3) = 4n - 9 \geq 7n/2 - 4$. Therefore both minima are two.

Let R, C be *all* degree-two rows and columns; set $m = |R|, q = |C|$. The two hole-rows of any degree-two column have total degree at least n; the other rows have degree at least three except for at most m degree-two rows. Thus

$$H \geq 4n - m - 6, \qquad H \geq 4n - q - 6. \tag{2.3}$$

The parity caps imply $m, q \geq k - 1 \geq 6$.

Two disjoint hole-column pairs of rows in R give $H \geq 2n + 2(n - 4) = 4n - 8$. Otherwise the family of two-element pairs is intersecting. An intersecting family either has a common member or is a triangle: given $\{u, v\}$ and $\{u, w\}$, a pair avoiding u must be $\{v, w\}$, and a pair meeting all three stays within $\{u, v, w\}$. In a triangle, covering says the union of each pair of its three missing-symbol sets is $[n]$. Each symbol belongs to at least two sets; the three column degrees sum to at least $2n$. Hence $H \geq 4n - 6$. Both alternatives contradict (2.2).

Every row in R consequently has a hole in a shared column j_0. Transposition supplies a shared hole-row i_0 for C. Each center has degree at least six and lies outside its degree-two family. There is a symbol a missing from every row in R and every column in C. To prove this, the distinct M-entries in j_0 and i_0 show that both families of missing pairs have unions of size at least six. They intersect across families by Lemma 2.2. If the column family has two disjoint pairs, the row union has size at most four. If it has a triangle, the row union has size at most three. Otherwise it is a star with center a and at least three distinct leaves, forcing every row pair to contain a. This proves the assertion.

2.4 A strengthened duplicate lemma, with the degree-one repair

Lemma 2.3. *Suppose $m \geq 5$ degree-two rows share a hole-column j_0 and a missing symbol a. If two have the same missing pair or the same hole-column pair, then*

$$H \geq 3n + m^2 - 6m + 4.$$

The estimate allows an exceptional degree-one column and holds in all the conjugates needed for the four row/column duplication types.

Proof. First suppose rows u, v both miss $\{a, b\}$. Relabel so that $M(u, j_0) = a, M(v, j_0) = b$. Their other hole-columns d_u, d_v carry b, a, respectively. Every other selected row t misses $\{a, b_t\}$, with $M(t, j_0) = b_t$ and $M(t, d_t) = a$. The b_t are distinct and avoid a, b. The $m - 2$ columns d_t, column d_v, and j_0 are pairwise distinct.

For $s \neq t$ among these other rows, column d_t misses b_s. Otherwise the b_s in row t is a clue. Deleting it forces b_s into (t, j_0), since d_t contains a retained b_s. Then (s, j_0) must become a. But the two unchanged-clue rows u, v still put the distinct values a, b in their two j_0 positions, so one of these positions is also a. This contradicts column distinctness. Thus d_t misses a and the $m - 3$ symbols b_s for $s \neq t$, giving degree at least $m - 2$. We do not claim that d_t misses its own b_t.

Covering row v gives $d(j_0) + d(d_v) \geq n$. The remaining $n - m$ columns have degree at least two, except possibly one of degree one. The disjoint count is $H \geq n + (m - 2)^2 + 2(n - m) - 1 = 3n + m^2 - 6m + 3$. If there is no one-hole column, the last constant is already four.

If a one-hole column λ exists, its hole-row r is empty and its missing symbol is globally absent. Intersecting the selected missing pairs identifies that symbol as a. None of their second-hole columns is λ. Also $d_u \neq d_v$: equality would leave an intercalate entirely in holes, and switching it would violate uniqueness. If $d_u = d_t$ for some t, this column additionally misses b, giving one more hole than the $m - 2$ counted there. Otherwise d_u is outside all distinguished columns. It has three distinct holes: its b in row u, its absent-a position, and its empty-row position in r. The last two differ because row r's a is in λ. This recovers the lost unit. Interchanging columns and symbols proves the repeated-hole-pair case; allowing a degree-one column makes that conjugation legitimate. Transposition supplies the two column versions. □

The polynomial increases for $m \geq 5$. At $m = k - 1$, its excess over $7n/2 - 4$ is $(k - 7)(k - 2) + 1$ for even order and $(k - 7)(k - 2) + \frac{1}{2}$ for odd order. Thus all four duplicate types are excluded. Write the row missing pairs as $\{a, b_i\}$ and their hole-columns as $\{j_0, d_i\}$. The b_i and the d_i are distinct; transpose likewise.

2.5 The common symbol is missing from both centers

Suppose a were retained as a clue in j_0. For each $p \in R$, its missing a must occupy (p, d_p). For $i \neq p$ in R, cell (i, d_p) is a clue. If d_p did not miss b_i, deleting this clue would change it to a, the only possible common member of its row and column missing sets. The a of row p would have to move from d_p to its only other changeable position, j_0, blocked by the retained a-clue there. Therefore every d_p misses a and all $m-1$ other b_i, and has degree at least m. Counting these m columns, j_0, and all other columns gives

$$H \geq m^2 + m + 2(n - m - 1) = 2n + m^2 - m - 2.$$

For $m \geq k-1, k \geq 7$ this exceeds (2.2). Transpose to obtain

$$a \in CS(j_0) \cap RS(i_0). \tag{2.4}$$

2.6 Positive and zero overlaps

Let x count holes in $R \times C$. They form a matching: each selected row and column has only one hole apart from its center. Distinct second-hole columns outside C give

$$m + q \leq n + x - 1. \tag{2.5}$$

If $x \geq 1$, covering a paired row and using (2.4) gives $d(j_0) \geq n-1$: the complement of $CS(j_0)$ is contained in its paired column's missing pair with a removed. Similarly $d(i_0) \geq n-1$. Counting the center, all degree-two columns, and the others gives $H \geq 4n - q - 4$ and its transpose. Therefore $m, q > n/2$; incidence gives at least two paired holes (at least three in even order).

Two paired columns have distinct missing pairs intersecting only in a. The complement of $CS(j_0)$ lies in this intersection but excludes a, by (2.4); it is empty. Both centers are consequently empty. Now $H \geq 4n - q - 3$ and its transpose force $m, q \geq k+2$. Incidence gives $x \geq 5$ in even order and $x \geq 4$ in odd order.

It remains to exclude $x = 0$. Put $h = d(j_0)$ and $B = [n] \setminus CS(j_0)$. The m distinct second-hole columns lie outside $C \cup \{j_0\}$; each has degree at least three and misses B by covering. Also $a \notin B$. At most one selected row places its a in j_0, so at least $m-1$ of the second-hole columns miss a as well. Hence

$$\begin{aligned} H \geq h + 3n - q - 3 + (m-1)\max(0, n-h-2) \\ + \max(0, n-h-3). \end{aligned} \tag{2.6}$$

Its unique minimum as a function of integer h is $4n - q - 5$ at $h = n-2$. Indeed, with $e = n - h$, the variable part decreases until $e = 2$, then rises by $m-2$ at $e = 3$ and by $m-1$ per subsequent step. The contrary caps now force $m, q \geq k$. For even n this contradicts (2.5). For odd $n = 2k+1$, it forces $m = q = k$, and equality in (2.6) gives $H = 7k - 1$ and $h = n - 2$.

Now $B = \{b, c\}$ avoids a. Each of b, c is missing from all k second-hole columns, giving $d_b, d_c \geq k$. The a-holes in the selected rows and columns are disjoint since $x = 0$, so $d_a \geq 2k$. Charging all other symbols two holes except for at most one degree-one exception yields

$$H \geq 2k + k + k + 2(n-3) - 1 = 4n - 9 = 8k - 5 > 7k - 1,$$

a contradiction. Thus the structural reduction is established for every coordinate pair with minima two: stars without duplicates, $m, q > n/2$, and at least four paired holes.

2.7 All local forcing arguments

The following arguments use the stars, nonduplication, and $x \geq 4$ only. They impose no minimum degree on unrelated lines. Index the paired holes (i_v, j_v), with missing pairs $\{a, b_v\}$ in rows and $\{a, c_v\}$ in columns; the b_v are distinct, as are the c_v.

Pair agreement. If $b_v \neq c_v$, choose distinct $p, q \neq v$ with $b_v \neq c_p, c_q$; $x \geq 4$ ensures this. Delete the clue (i_v, j_p), whose original value is s. The missing b_v cannot occupy j_v or j_p, which have retained b_v entries, so it moves to j_0. The changed clue becomes a, and the remaining hole (i_v, j_v) must receive $s = c_v$. Repeating with j_q gives $M(i_v, j_q) = c_v$ too, contradicting row distinctness. Hence $b_v = c_v$ for every v.

No leaf values in the paired submatrix. If a clue (i_1, j_2) contains b_3, the indices are distinct because row i_1 misses b_1 and column j_2 misses b_2. Its witness forces

$$\begin{aligned} &W(i_1, j_0) = b_3, \quad W(i_3, j_0) = a, \quad W(i_3, j_3) = b_3, \\ &W(i_0, j_3) = a, \quad W(i_0, j_1) = b_1, \quad W(i_1, j_1) = a, \\ &W(i_1, j_2) = b_1. \end{aligned}$$

At each step the relevant degree-two row or column has only its two holes available, and any value already placed is forbidden at the next position. The last value is impossible: column j_2 did not miss b_1, and the deleted clue contained b_3.

Forcing a missing leaf in any other row. Suppose a clue (i, j_1) contains b_2. Its row is neither a paired row, by the preceding paragraph, nor i_0, whose position there is a hole. The deletion witness forces

$$\begin{aligned} &W(i_0, j_1) = b_2, \quad W(i_0, j_2) = a, \quad W(i_2, j_2) = b_2, \\ &W(i_2, j_0) = a, \quad W(i_1, j_0) = b_1, \quad W(i_1, j_1) = a, \\ &W(i, j_1) = b_1. \end{aligned}$$

Thus row i misses b_1. If $i \in R$, its missing pair would be $\{a, b_1\}$, duplicating i_1. Consequently no row in R contains any b_v in any paired column; transpose for columns.

Fix s. In column j_s, all b_v for $v \neq s$ occur as clues, since the missing pair is $\{a, b_s\}$. They occupy distinct rows outside $R \cup \{i_0\}$, and the preceding forcing rule makes b_s missing in each such row. Transposition gives at least $x - 1$ holes of b_s in distinct columns outside $C \cup \{j_0\}$. These disjoint locations are essential to the counts below.

2.8 The third coordinate without a one-hole symbol

Let T be all degree-two symbols, $t = |T|$. Let y count holes incident with a row of R and a symbol of T, and z those incident with a column of C and a symbol of T. If no symbol has degree one, its minimum cannot be at least three: covering after conjugation on a degree-two row gives at least $n + 3(n - 2) \geq 4n - 9$, contradicting (2.2). Its minimum is therefore two. Apply the strengthened reduction to row–symbol and column–symbol coordinates. It gives $t > n/2$ and $y, z \geq 4$.

The shared symbol-line of the degree-two rows must be the original a, the unique common member of their distinct missing pairs. Similarly the common missing column-label of those rows is j_0. Applying both conjugate reductions identifies the degree-two symbols' shared row and column as i_0, j_0: the column–symbol reduction supplies their common missing row-label i_0, and the row–symbol reduction their common missing column-label j_0. Thus the three configurations are compatible and have the incidence inequalities

$$m + q \leq n + x - 1, \quad m + t \leq n + y - 1, \quad q + t \leq n + z - 1. \tag{2.7}$$

Covering a paired row in row–symbol coordinates gives $d_a \geq n - 2$.

2.9 The exceptional one-hole-symbol case in full

Suppose a symbol w has degree one. Its hole forces an empty row and column. These must be i_0, j_0: a different empty column, together with j_0, would give every row in R the same two hole-columns; transpose for rows. Therefore $w = M(i_0, j_0)$ and $d(i_0) = d(j_0) = n$.

Each degree-two symbol has exactly one hole in i_0 and one in j_0. These holes are distinct, since the intersection contains w, which has degree one. Their other hole-rows are distinct by column distinctness in j_0, and their other hole-columns are distinct by row distinctness in i_0. This directly gives two further stars without duplicated pairs:

- In row–symbol coordinates, the degree-two rows share symbol-line a, the degree-two symbols share row i_0, and both families miss the common column-label j_0.
- In column–symbol coordinates, the degree-two columns share a, the degree-two symbols share column j_0, and both families miss the common row-label i_0.

Here $a \notin T$ since $d_a \geq m \geq 6$. The elementary incidence counts $m + t \leq n + y - 1$ and $q + t \leq n + z - 1$ hold even when t, y, z are small. For example the m distinct second missing symbols of the rows consist of y members of T and $m - y$ outside $T \cup \{a\}$, so $m - y \leq n - t - 1$. No minimum-two reduction has been applied to a coordinate pair of minimum one.

For each original paired hole, let $\epsilon_v = 1$ if it carries b_v, and zero if it carries a. The a-holes from the selected rows and columns and the disjoint leaf holes from Section 2.7 give

$$d_a \geq m + q - x + \sum_v \epsilon_v, \qquad d_{b_v} \geq x + 1 - \epsilon_v.$$

For the second inequality, the selected row and column supply $2 - \epsilon_v$ leaf holes, in $C \cup \{j_0\}$; the local forcing lemma supplies $x - 1$ more outside that set. Hence

$$d_a + \sum_v d_{b_v} \geq m + q + x^2. \tag{2.8}$$

All these $x + 1$ symbols have degree at least four. They are distinct from w and the t degree-two symbols. Every remaining symbol has degree at least three, so

$$H \geq 3n + m + q - t + x^2 - 3x - 5. \tag{2.9}$$

If $y \leq 3$, substitute $t \leq n + y - 1 - m$ to obtain

$$\begin{aligned} H &\geq 2n + 2m + q + x^2 - 3x - y - 4 \\ &\geq \frac{7n}{2} - \frac{3}{2}. \end{aligned}$$

The last step uses $m, q \geq n/2 + 1/2$, $x \geq 4$, and $y \leq 3$. This contradicts (2.2); transpose to exclude $z \leq 3$. Thus $y, z \geq 4$. The local forcing proofs of Section 2.7 apply to both directly constructed stars, since they impose no degree condition on unrelated lines. Covering in a paired row–symbol row gives $d_a + 2 \geq n$. All three common lines therefore have degree at least $n - 2$, also in this exceptional case.

2.10 The final three-coordinate count

In either case we have all three configurations and (2.7), with $x, y, z \geq 4$. For either case, define $\epsilon_v = 1$ if the hole at the intersection of paired row i_v and column j_v contains b_v, and $\epsilon_v = 0$ otherwise. The disjoint-location count used above gives $d_{b_v} \geq x + 1 - \epsilon_v$ independently of whether a one-hole symbol exists: the selected row and column supply $2 - \epsilon_v$ holes in $C \cup \{j_0\}$, and local forcing supplies $x - 1$ outside that set. In particular these leaf symbols have degree at least four and do not belong to the set T of degree-two symbols. In the original configuration at most one ϵ_v is one: otherwise two rows would put a in j_0. Therefore

$$\sum_v d_{b_v} \geq x(x + 1) - 1.$$

Count symbols in four disjoint classes: a; the x leaf symbols b_v; the t degree-two symbols; and the rest. Charge the rest three holes, allowing a loss of two for at most one degree-one exception. This gives

$$\begin{aligned} H &\geq (n - 2) + x(x + 1) - 1 + 2t + 3(n - x - t - 1) - 2 \\ &= 4n - t + x^2 - 2x - 8. \end{aligned}$$

The other two conjugates give

$$H \geq 4n - q + y^2 - 2y - 8, \qquad H \geq 4n - m + z^2 - 2z - 8.$$

Put $S = m + q + t$ and $V = x + y + z$. The three incidence inequalities give $2S \leq 3n + V - 3$. Adding the hole counts and substituting yields

$$H \geq \frac{7n}{2} - \frac{15}{2} + \frac{(2x^2 - 5x) + (2y^2 - 5y) + (2z^2 - 5z)}{6} \geq \frac{7n}{2} - \frac{3}{2},$$

since $2v^2 - 5v \geq 12$ for every integer $v \geq 4$. This contradicts (2.2). Integrality proves Theorem 2.1. □

Range and remaining gap. The duplicate estimate at $m \geq \lfloor n/2 \rfloor - 1$ requires $n \geq 14$. The proof does not establish the minus-four constant at orders 8–13. The sharper frame estimate reaches the current threshold exactly; a further global improvement requires more than strengthening the final count alone.

3 Safe binary lifts and thesis profiles

3.1 A safe lift using the number of private intercalates

Let Q be a critical set of size s in a Latin square L of order m. Let t count its clues which have a **private intercalate**: an intercalate meeting Q in exactly that clue. This number counts clues, not intercalates.

Then

$$\boxed{\ell(2m) \geq m^2 + 2s + t.} \tag{D1}$$

This refines the protected-subset argument of Stinson–van Rees, reproduced as Gower's Lemmas 1.19–1.21. No strong-completion or all-clues-2-critical assumption on Q is needed.

Proof. In $C_2 \times L$, fill the top-left block completely and put a copy of Q in each other block. Call this uniquely completable partial square P^*. Its uniqueness follows because the full block fixes the symbol sets in its adjacent blocks, their critical sets complete those blocks, and these in turn fix and complete the fourth block.

The following clues are essential in P^*:

1. All $3s$ clues in the three copies of Q, witnessed by trades confined to their individual subsquares.
2. The $m^2 - s$ full-block cells whose corresponding base position is outside Q, witnessed by the four copies of that base cell.
3. The t full-block cells corresponding to private-intercalate clues. If a base intercalate has rows r, r' and columns c, c', with (r, c) its sole clue, use the product intercalate with cells $(r, c), (r, m+c'), (m+r', c), (m+r', m+c')$.

Every other cell of these witness intercalates lies outside P^*. Delete redundant full-block clues until a critical subset $Q' \subseteq P^*$ remains. The listed $m^2 + 2s + t$ essential clues cannot be deleted. This proves (D1).

Moreover, at least

$$t' \geq m^2 - s + 4t \tag{D2}$$

clues of Q' have private intercalates: the last two classes above contribute $m^2 - s + t$, and the three partial blocks contribute $3t$. All these witnesses survive clue deletion.

Applying (D1) again and using (D2) gives

$$\boxed{\ell(4m) \geq 7m^2 + 3s + 6t.} \tag{D3}$$

The full conjectural bounds would be $m^2 + 3s$ and $7m^2 + 9s$. The rigorous losses here are respectively $s - t$ and $6(s - t)$. They vanish for a 2-critical seed.

3.1.1 Iterating the safe lift

Put $D = m^2 - s$ and $u = s - t$. In one lift let δ be the number of unprotected full-block clues deleted. Then

$$0 \leq \delta \leq u, \quad D' = 3D + \delta, \quad u' \leq 4u - \delta. \tag{D4}$$

Here u' is the number of clues without a private intercalate in the resulting critical set. If a lower bound for t is used instead of its exact value, the same inequalities hold with u as an upper bound.

For $k \geq 1$, define

$$b_1 = 1, \quad b_2 = 6, \quad b_k = 27 \cdot 4^{k-3} \ (k \geq 3).$$

Then a rigorous iterated family is

$$\boxed{\ell(2^k m) \geq 4^k m^2 - 3^k D - b_k u.} \tag{D5}$$

To check it, maximize a nonnegative functional $aD' + bu'$ over (D4). It is at most $3aD + [4b + \max(0, a - b)]u$. Starting with $(a, b) = (1, 0)$ and applying this recursion k times gives $a = 3^k$ and the stated b_k. This proof permits all possible deletion outcomes; it does not assume the full block remains critical.

3.2 An additional family from the thesis

In the thesis's order-$4m$ set, seven m-blocks are full and nine are triangular. The proof in § 5.3 makes every full-block clue essential using a private intercalate in an appropriate $2m$-subsquare. Consequently it supplies a seed with

$$s = (23m^2 - 9m)/2, \qquad t \geq 7m^2.$$

Insert these into (D3), with base order $4m$, to obtain

$$\boxed{\ell(16m) \geq (377m^2 - 27m)/2 \quad (m \geq 1).} \tag{F16}$$

This improves the direct application of the thesis's $4m$ formula at the same order by

$$\frac{9m(m+1)}{2}.$$

For instance it gives **4645** at order 80, compared with the thesis formula's **4510**. The deduction is rigorous from the cited construction and (D1)–(D3); its prior publication status has not been determined. It can lose to other families, including the powers-of-two construction.

The program also uses a modest refinement when the block parameter $m = 2h$ is even. Each triangular critical set has $h(h+1)/2 = m(m+2)/8$ private-intercalate clues. To see this for $H(A) = \{(i,j) : 1 \leq j \leq i \leq m-1\}$, partition the cyclic square into intercalates with rows $r, r+h$ and columns $c, c+h$, $0 \leq r, c < h$. Exactly one of their cells lies in $H(A)$ when $c = 0$, or when $1 \leq c < h$ and $r < c$. There are $h + \sum_{c=1}^{h-1} c = h(h+1)/2$ such intercalates. The other triangular blocks are isotopic or conjugate to this one.

Thus the thesis's $2m$ and $4m$ seeds respectively have at least

$$m^2 + 3\tau(m), \qquad 7m^2 + 9\tau(m)$$

private clues, where $\tau(m) = m(m+2)/8$ for even m, and $\tau(m) = 0$ for odd m. These lower bounds, not an unproved doubling conjecture, are fed into (D5).

4 Safe ternary lifts and iteration

4.1 A useful published product inequality

Gower's Theorem 2.20, together with his Theorem 2.5, implies

$$\boxed{\ell(3m) \geq 2m^2 + 5\ell(m).} \tag{G3}$$

Take the strongly critical set $A = \{(0,0;0),(1,1;2)\}$ in the cyclic square of order three and any critical set Q of order m, size s. The completable product has two full m-blocks and seven copies of Q. Its unique completion follows from the strong outer set. Explicitly, its block positions can be completed in the order $(0,1),(0,2),(1,0),(1,2),(2,0),(2,1),(2,2)$, because each step has a forced symbol class and contains a uniquely completable copy of Q.

All $7s$ clues in the partial blocks are essential: a deletion witness stays within that subsquare. In each full block, all $m^2 - s$ positions outside Q are essential: the nine copies of that base position form an order-three subsquare meeting the product in exactly the two clues of A. Deleting either clue permits a different completion in that subsquare. Reduce the product to a critical subset; these protected clues survive. Their number is $7s + 2(m^2 - s) = 2m^2 + 5s$. The coefficient of s is positive, so any lower bound for $\ell(m)$ can be substituted, and (G3) can be applied recursively.

This is an application of a published theorem omitted from the earlier numerical envelope. It requires a strong outer factor; it is not an assertion about unrestricted completable products.

Source: R. A. H. Gower, *Critical sets in products of latin squares*, Ars Combinatoria 55 (2000), 293–317, especially printed pp. 303–304, 308 and 315–316. The relevant scan pages were inspected visually.

4.2 Retaining clues with order-three witnesses

Call a clue **3-essential** when it lies in an order-three Latin subsquare in which deleting that clue from the induced partial square destroys unique completion. This is Gower's definition. Let t_3 count such clues of Q, counting clues rather than subsquares.

The local argument in Gower's Theorem 2.13 protects the two full-block copies of each of these t_3 clues, even if other seed clues have no order-three witness. Consequently

$$\boxed{\ell(3m) \geq 2m^2 + 5s + 2t_3.} \tag{T1}$$

Here is why the local assertion applies. Let S be a witnessing order-three subsquare for a seed clue e. The induced partial square $Q \cap S$ is uniquely completable, since any different completion within S would extend to a different completion of the whole seed. Gower's proof on pp. 311–312 uses only this local partial square and the necessity of e. It constructs an order-three subsquare through each full-block copy of e in which that clue is still necessary. The argument does not use 3-essentiality of any other clue of Q.

For completeness, the finite local fact was separately checked over all 512 subsets of an order-three Latin square. Exactly 466 are uniquely completable. For every necessary clue of each such subset, both corresponding full-block clues were found to have explicit order-three witnesses in its product with A. The full-block positions outside the subset were checked too: **4428 protected-clue assertions passed**. This finite check supports the cited local proof and is reproducible in `check_lcs_ternary.py`.

After reducing the product to a critical subset, its number of 3-essential clues is at least

$$t_3' \geq 2(m^2 - s) + 9t_3. \tag{T2}$$

The first term counts full-block positions outside Q; the second counts seven internal copies and two full-block copies of each original 3-essential clue. The classes are disjoint. Their witnesses survive any deletion of redundant clues: the surviving partial square is still uniquely completable, while the old local alternative after deleting the designated clue remains valid.

4.3 Iteration and combination with binary lifts

Write $D = m^2 - s$ and $u = s - t_3$. The initial product has $2m^2 + 7s$ clues, and only its $2u$ unprotected full-block clues can be removed. If δ of them are removed, then

$$0 \leq \delta \leq 2u, \qquad D' = 7D + \delta, \qquad u' \leq 9u - \delta. \tag{T3}$$

For k ternary lifts, put $a_0 = 1, b_0 = 0$ and

$$a_{k+1} = 7a_k, \qquad b_{k+1} = 9b_k + 2\max(0, a_k - b_k).$$

Maximizing the nonnegative functional $aD' + bu'$ over (T3) proves

$$\boxed{\ell(3^k m) \geq 9^k m^2 - 7^k(m^2 - s) - b_k(s - t_3).} \tag{T4}$$

The first coefficients are $b_1 = 2, b_2 = 28, b_3 = 294, b_4 = 2744$, with $b_k = 2744\,9^{k-4}$ for $k \geq 4$. In particular $t_3 = s$ recovers the usual 3-critical product recurrence. Formula (T4) uses the actual seed size; when its coefficient becomes negative, one must not substitute a lower bound for that size indiscriminately.

If the original seed also has t_2 private-intercalate clues, after k ternary lifts at least $7^k t_2$ such clues survive in the seven partial blocks at each step. Let $N = 3^k m$, and let B be the lower bound in (T4). The previously proved binary lifts therefore give

$$\boxed{\ell(2N) \geq N^2 + 2B + 7^k t_2,} \qquad \boxed{\ell(4N) \geq 7N^2 + 3B + 6 \cdot 7^k t_2.} \tag{T5}$$

These two expressions have positive coefficients of the intermediate size, so the lower bound B suffices. No explicit choice of the intervening critical subset is needed.

5 Full finite proof of the ternary local protection lemma

This section supplies the finite local step used in the safe ternary and mixed-product bounds. It gives a complete, solver-free exhaustive proof, rather than treating the number of passed tests as an unexplained premise. The enumeration below was independently reconstructed using affine lines in $\mathbb{F}_3^2$, without importing the earlier subsquare or private-witness enumerators. It is a computer-assisted finite proof; the accompanying certificate records an explicit Latin trade for every required assertion.

Local statement. Let $S(i, j) = i + j \pmod 3$, and let Q be any subset of its nine entries that uniquely completes to S. A clue $e \in Q$ is necessary if $Q \setminus \{e\}$ has a second Latin completion. On row and column pairs $(R, i), (C, j)$, form

$$G((R, i), (C, j)) = (R + C,\, i + j) \quad \text{in } \mathbb{F}_3^2.$$

Prescribe both blocks $(R, C) = (0, 0), (1, 1)$ completely, and copies of Q in the other seven blocks; call the resulting partial square $P(Q)$. For either full block, every position whose inner cell is a hole of Q or a necessary clue of Q has a private Latin trade contained in an order-three subsquare of G. That is, the changed cells of the trade meet $P(Q)$ in exactly the designated full-block clue.

Why there are only finitely many cases, with none omitted. A labelled Latin square of order three is a sequence of three permutations of $\{0, 1, 2\}$ satisfying the three column conditions. Enumerating all $6^3 = 216$ sequences and retaining exactly those satisfying these conditions gives the following complete list, with slashes separating rows:

012/120/201	012/201/120	021/102/210
021/210/102	102/021/210	102/210/021
120/012/201	120/201/012	201/012/120
201/120/012	210/021/102	210/102/021

Each of the $2^9 = 512$ subsets of S is represented by a nine-bit mask M. It is uniquely completable precisely when only S in this list agrees with all its clues. Its necessary clues are exactly those $e \in M$ for which another listed square agrees with S at every position of $M \setminus \{e\}$ and differs at e. This classifies all local inputs without a general completion solver. Exactly 466 masks are uniquely completable.

Every Latin square of order three is isotopic to S: normalizing its first row and column to $0, 1, 2$ forces the unique reduced order-three square. Consequently this classification covers every witnessing order-three subsquare of an arbitrary seed, after consistent row, column and symbol relabelling in all nine product blocks.

A finite catalogue of valid trades. The four one-dimensional subspaces of $\mathbb{F}_3^2$ have direction vectors

$$(0,1), \quad (1,0), \quad (1,1), \quad (1,2).$$

For each direction v, let the row indices be any coset of $\langle v \rangle$, and the column indices any coset of the same subgroup. There are three choices of each coset, giving $4 \cdot 3 \cdot 3 = 36$ Latin subsquares of order three in G. Their symbol sets are the corresponding sum cosets. In each subsquare, replace its entries by each of the twelve labelled Latin squares on that same three-symbol set. The identity filling is discarded. Every remaining replacement preserves the symbol set of each affected row and column, so it is a Latin trade in the full order-nine square. The catalogue has $36 \cdot 11 = 396$ nonidentity mates, with 360 distinct changed-cell supports.

It is unnecessary to prove that this catalogue includes every possible trade of G: it suffices to find a valid witness for each required clue. For each of the 466 admissible masks, the following exhaustive procedure finds such a witness in this explicit catalogue.

```
L3 := all triples of permutations of {0,1,2}
      whose columns also are permutations
S[i,j] := (i+j) mod 3
G[(R,i),(C,j)] := ((R+C) mod 3, (i+j) mod 3)
W := empty list
for v in {(0,1),(1,0),(1,1),(1,2)}:
    for each row coset I and column coset J of <v>:
        for A in L3, relabelled by the symbol set of I+J:
            replace G[I,J] by A to obtain G_A
            if G_A != G:
                append (I,J,A,{x:G_A[x]!=G[x]}) to W
for M in all 512 masks of S:
    if exactly one A in L3 agrees with S on M:
        E := {e in M : some A in L3 differs at e
                       and agrees on M minus {e}}
        P := full blocks 00 and 11, plus M in other blocks
        for full block B in {00,11}:
            for inner cell e outside M or in E:
                target := copy of e in B
                find W_a in W with support(W_a) intersect P
                                  = {target}
                assert such W_a exists; save its index
```

Every loop is finite and its range is explicitly specified. The saved certificate contains all 466 masks and all 4,428 required target–witness pairs. Its replay reconstructs each chosen mate, checks all nine rows and columns of the altered order-nine square, and checks that its changed-cell support intersects the complete product clue set in exactly the target. It also recomputes the set of admissible masks and the full required target list, so missing cases cannot be hidden by the certificate. All 4,428 assertions passed. No SAT, integer-programming, or uniqueness-oracle verdict is used by this local proof.

Transfer to an arbitrary seed. Let Q_0 be a critical set of size s in an order-m Latin square L. Let t_3 count distinct clues that are necessary within the restriction of Q_0 to at least one order-three Latin subsquare. The restriction to such a subsquare is uniquely completable: otherwise its alternative completion could replace that subsquare in L while retaining Q_0. The local statement therefore applies to it. In $C_3 \times L$, the nine copies of those three rows and columns form the required order-nine Latin subsquare. A local trade found above extends to the whole product by leaving all other cells unchanged. Its intersection with the full product's clues is still exactly its target.

Prescribe two complete order-m blocks at outer positions $00, 11$ and seven copies of Q_0. This template is defining: the symbol classes, and then the uniquely completable copies, fill partial blocks in the order

$$01,\ 02,\ 10,\ 12,\ 20,\ 21,\ 22.$$

All $7s$ partial-block clues are necessary, using their original deletion trades inside their respective order-m subsquares. Each full block has $m^2 - s$ protected positions above seed holes; their outer order-three subsquares give witnesses directly. The finite lemma additionally protects the t_3 positions above locally necessary seed clues in each full block. These classes are disjoint. Reducing the defining template to any inclusion-minimal defining subset therefore gives

$$\boxed{\mathrm{lcs}(3m) \geq 7s + 2(m^2 - s + t_3) = 2m^2 + 5s + 2t_3.}$$

The same critical subset has at least

$$\boxed{t_3' \geq 2(m^2 - s) + 9t_3}$$

locally necessary clues: the two full-block classes just described and seven internal copies of the seed's t_3 clues. Their local alternative completions survive removal of other clues. Their induced partial subsquares remain uniquely completable because the whole retained set remains defining. Hence these are valid simultaneous local witnesses for the same extracted critical set.

If δ is the number of removed full-block clues, only the $2(s - t_3)$ unprotected positions can be removed. Thus, with $N = m^2$, $D = N - s$ and $u_3 = s - t_3$,

$$0 \leq \delta \leq 2u_3, \qquad N' = 9N, \quad D' = 7D + \delta, \quad u_3' \leq 9u_3 - \delta.$$

These are the precise hypotheses used by the pure ternary and mixed-word recurrences. If the seed also has t_2 private-intercalate clues, their seven internal copies remain protected, giving $t_2' \geq 7t_2$ for the same critical subset.

Reproduction and audit scope. Run

```
python3 outputs/lcs_full_review/ternary_local_finite_proof.py
```

from the project directory. The standard-library-only script writes and replays `ternary_local_certificate.json`; counts and hashes are in `ternary_local_finite_proof_results.json`. Its independent affine-line enumeration reproduces the earlier 4,428 local assertions. The all-order transfer argument above then proves the stated bounds for arbitrary m; it is not an extrapolation from checking a finite range of seed orders. This independent agent review is not human refereeing or a priority claim.

6 Arbitrary mixed binary and ternary lifts

6.1 State and protected clues

For a critical seed of order m and known size s, let t_2 count clues with private intercalates and t_3 count clues essential in order-three subsquares. Put

$$N = m^2, \quad D = N - s, \quad u_2 = s - t_2, \quad u_3 = s - t_3.$$

Certified lower bounds on t_2, t_3 can be used, making u_2, u_3 upper bounds. The seed size s must be known: substituting only a lower bound is not generally valid.

After a binary lift, delete redundant clues until a critical subset remains. If δ clues are deleted, the protected-clue argument gives

$$0 \le \delta \le u_2, \quad N' = 4N, \quad D' = 3D + \delta, \quad u_2' \le 4u_2 - \delta, \quad u_3' \le N + 3u_3 - \delta. \tag{B}$$

The first private-clue inequality is the earlier binary result. The second follows because the three partial blocks each retain all their seed clues and all their internal order-three witnesses: $t_3' \ge 3t_3$, whereas $s' = N + 3s - \delta$.

After a ternary lift, the corresponding relations are

$$0 \le \delta \le 2u_3, \quad N' = 9N, \quad D' = 7D + \delta, \quad u_2' \le 2N + 7u_2 - \delta, \quad u_3' \le 9u_3 - \delta. \tag{T}$$

Here the seven partial blocks preserve the original private intercalates, giving $t_2' \ge 7t_2$. The stronger order-three protection is the earlier ternary lemma. An essential clue cannot be deleted during reduction, and its local alternative completion remains a witness after deleting other clues. Thus both kinds of protection survive together through any sequence of lifts.

6.2 Backward coefficient recurrence

Consider a nonnegative functional $aN + bD + cu_2 + du_3$ at the output of one lift. Substitute (B) or (T) and maximize over the allowed δ. Its coefficient is $b - c - d$. This gives the backward transforms

$$\boxed{B(a,b,c,d) = (4a + d,\ 3b,\ 4c + \max(0, b - c - d),\ 3d),}$$

$$\boxed{T(a,b,c,d) = (9a + 2c,\ 7b,\ 7c,\ 9d + 2\max(0, b - c - d)).} \tag{M}$$

All coefficients remain nonnegative, which permits upper bounds on the two witness deficits at every substitution.

Let $w = (w_1, \ldots, w_r)$, with each $w_i \in \{2, 3\}$, list the lifts in forward order. Start with $(0, 1, 0, 0)$, the final deficit, and apply the transforms in reverse word order. If the resulting tuple is (a, b, c, d) and $n = m \prod_i w_i$, then

$$\boxed{\ell(n) \ge n^2 - am^2 - b(m^2 - s) - c(s - t_2) - d(s - t_3).} \tag{L}$$

This is an existence bound: it need not identify the redundant clues removed at intermediate steps. It does not assert that every relaxed deletion count occurs or that this is an optimal product construction.

6.3 Searching all orders of the two lifts

For fixed numbers i, j of binary and ternary steps, $b = 3^i 7^j$. The implementation `lcs_mixed_lifts.py` retains coefficient tuples not dominated coordinatewise by another tuple, and records a forward word for each. A new backward transform prepends its factor to that word.

Early pruning needs care: the displayed transforms are not coordinatewise monotone. The correct justification uses pointwise order of functionals on

$$K = \{(N, D, u_2, u_3) : N, D \ge 0,\ 0 \le u_2, u_3 \le N\}.$$

Every relaxed output state in (B), using its displayed upper bounds as equalities and $0 \le \delta \le u_2$, is in K. The same is true in (T), with $0 \le \delta \le 2u_3$. For example, $N + 3u_3 - \delta$ lies between zero and $4N$, and $2N + 7u_2 - \delta$ between zero and $9N$. Thus taking the maximum over these states preserves pointwise order. Coordinatewise domination initially implies pointwise domination on K, and any preceding sequence preserves that order. Discarding a dominated tuple cannot remove the best bound obtainable from this recurrence. This does not mean that coordinatewise domination itself persists after each transform.

The table evaluates the recurrence on all 14 previously certified seeds, including both order-12 and both order-15 witness profiles, and two elementary seeds of orders two and three, and takes its maximum with the previously documented constructions. It makes no claim to cover all known critical sets or all product templates.

The elementary seeds have profiles (order, size, t_2, t_3) equal to (2, 1, 1, 0) and (3, 3, 0, 3). Their explicit squares and all four deletion alternatives are in the small-seed certificate; exhaustive enumeration verifies their unique completion and minimality. The order-three seed is itself its order-three witness subsquare.

6.4 Definitions and the invariant witness argument

Let Q be a critical set of size s in a Latin square of order m. Put

$$N = m^2, \qquad D = N - s, \qquad u_2 = s - t_2, \qquad u_3 = s - t_3.$$

Here t_2 counts clues having a private intercalate, and t_3 counts clues essential in an order-three subsquare. In both cases these are counts of distinct clues, not counts of witnesses. Known lower bounds for t_j can replace their exact counts; then u_j are upper bounds. We assume $0 \le t_j \le s$, as is true for certified counts.

If a local alternative completion disagrees with a uniquely completable partial Latin square in precisely one clue, that clue is essential. The same alternative survives every subsequent deletion of other clues. If the resulting whole partial square remains uniquely completable, its restriction to the witness subsquare also remains uniquely completable: any other local completion could be substituted into the whole Latin square. Thus the designated clue remains 2-essential or 3-essential, respectively. This proves that all protected local witnesses used below survive reduction to a critical subset.

6.5 Reconstructing the binary transition

In $C_2 \times L$, fill one $m \times m$ block and put Q in each of the other three blocks. The product is uniquely completable. The full block fixes the symbol classes in its adjacent blocks; their uniquely completable copies of Q fill them, after which the fourth block is fixed and fills.

All $3s$ partial-block clues are essential, using an alternative completion inside their individual $m \times m$ subsquare. The full block has at least $N - s + t_2$ protected clues: those outside the base set Q have the four-copy intercalate, and each base clue with a private intercalate gives the mixed-block intercalate described in the binary proof. The coordinate construction indeed uses two rows, two columns, and two symbols; its other three cells are outside the partial set.

Consequently only at most u_2 full-block clues can be removed. If precisely δ are removed when obtaining a critical subset, then

$$0 \le \delta \le u_2, \qquad N' = 4N, \qquad s' = N + 3s - \delta, \qquad D' = 3D + \delta.$$

The protected full-block clues and three internal copies give

$$t_2' \ge N - s + 4t_2.$$

Independently, every original order-three witness stays wholly within each partial block, so

$$\boxed{t_3' \ge 3t_3.}$$

These are simultaneous guarantees for the same critical subset. Subtracting from its exact size gives

$$\boxed{u_2' \le 4u_2 - \delta, \qquad u_3' \le N + 3u_3 - \delta.}$$

No 3-essentiality of a full-block clue is required for the second bound.

6.6 Reconstructing the ternary transition

Use the cyclic order-three outer square and full blocks at $(0,0)$ and $(1,1)$, with Q in its other seven blocks. The two full blocks determine the symbol class at $(0,1)$; completing that block permits the remaining partial blocks to be completed successively. Hence the product is uniquely completable. All $7s$ partial-block clues are essential, using their internal deletion witnesses.

The full blocks protect $2(N - s)$ positions outside Q, using the outer order-three subsquare at each base position. They also protect both full-block copies of every 3-essential seed clue by the local order-three product lemma in the

ternary proof. I independently checked that finite lemma as described below. Together with the seven internal copies, these guarantees give

$$t_3' \geq 2(N - s) + 9t_3.$$

Only at most $2u_3$ full-block clues can be removed. Thus

$$0 \leq \delta \leq 2u_3, \qquad N' = 9N, \qquad s' = 2N + 7s - \delta, \qquad D' = 7D + \delta.$$

Every private intercalate of a seed clue also remains within each of the seven partial blocks. Therefore, for the same resulting critical subset,

$$\boxed{t_2' \geq 7t_2, \qquad u_2' \leq 2N + 7u_2 - \delta, \qquad u_3' \leq 9u_3 - \delta.}$$

The independent finite check enumerated all 12 Latin squares of order three, all 512 subsets of the cyclic order-three square, and all 36 order-three subsquares of $C_3 \times C_3$. Those subsquares yield 360 distinct nonempty local trade supports. For each uniquely completable seed subset, intersect each support with the full product clue set; an intersection consisting of one clue certifies that clue's protection. Exactly 466 seed subsets were uniquely completable, and **all 4428 required full-block protection assertions passed**. This independently reproduces the local result without importing the existing subsquare or witness enumerator. Isotopy transfers the local result to every order-three seed subsquare.

I also directly checked **702 inherited binary order-three witness assertions** across those local seed subsets, and **28 inherited ternary intercalate witness assertions** using the four single-clue order-two seeds. These checks support the general internal-copy argument; their finite sizes are not a restriction on that argument.

6.7 Backward functional calculation

For nonnegative coefficients define

$$F(N, D, u_2, u_3) = aN + bD + cu_2 + du_3.$$

Substitute the binary transition and maximize over $0 \leq \delta \leq u_2$:

$$F' \leq (4a + d)N + 3bD + 4cu_2 + 3du_3 + (b - c - d)\delta,$$

$$\boxed{B(a, b, c, d)} = \boxed{(4a + d,\ 3b,\ 4c + \max(0, b - c - d),\ 3d)}.$$

The ternary transition similarly gives

$$\boxed{T(a, b, c, d) = (9a + 2c,\ 7b,\ 7c,\ 9d + 2\max(0, b - c - d)).}$$

All output coefficients remain nonnegative. Therefore replacing actual u_j by certified upper bounds is legitimate at every step.

For a forward lift word $w \in \{2, 3\}^*$, start with the final deficit functional $(0, 1, 0, 0)$ and apply these operators in reverse word order. If the resulting coefficients are (a, b, c, d), then

$$\boxed{\ell\Big(m \prod w\Big) \geq \Big(m \prod w\Big)^2 - am^2 - b(m^2 - s) - c(s - t_2) - d(s - t_3).}$$

The actual seed size s, or an independently justified bound on this entire expression, is needed. Its coefficient after expansion need not be positive; indiscriminately substituting a lower bound for s would not be valid.

For fixed binary and ternary counts i, j, one has $b = 3^i 7^j$. Appending an operator to an already computed backward coefficient tuple **prepends** the corresponding factor to the forward construction word.

As an independent arithmetic check, exhaustive enumeration of every relaxed deletion choice agrees exactly with the backward coefficient evaluation in **525 cases**, covering every word of length at most three and every integer state $1 \leq N \leq 2$, $0 \leq D, u_2, u_3 \leq N$. This checks the maximizing sign, factor order, and interactions between the two witness deficiencies.

6.8 Why intermediate Pareto pruning is valid

At a fixed pair of factor counts, a coefficient tuple with no larger a, c, d and the same b yields no larger deficit bound for every nonnegative state. This directly justifies pruning at the final stage. There is a subtlety at intermediate

stages: B and T are **not** componentwise monotone as maps of coefficient tuples. For example, increasing d may decrease the new binary c. A componentwise monotonicity assertion would be false.

The needed stronger argument uses the invariant state cone

$$\mathcal{K} = \{(N, D, u_2, u_3) : N, D \geq 0,\ 0 \leq u_2, u_3 \leq N\}.$$

Every legitimate input lies in this cone. Every binary relaxed successor

$$(4N, 3D + \delta, 4u_2 - \delta, N + 3u_3 - \delta), \qquad 0 \leq \delta \leq u_2,$$

also lies in it: in particular its last component is nonnegative because $u_2 \leq N$, and both deficiency components are at most $4N$. Every ternary relaxed successor

$$(9N, 7D + \delta, 2N + 7u_2 - \delta, 9u_3 - \delta), \qquad 0 \leq \delta \leq 2u_3,$$

likewise lies in it, using $u_3 \leq N$.

Viewed as maximization operators over these successor states, B and T preserve pointwise functional domination on $\mathcal{K}$. Namely, $F \leq G$ on $\mathcal{K}$ implies

$$\max_{\delta} F(S_\delta x) \leq \max_{\delta} G(S_\delta x) \qquad (x \in \mathcal{K}).$$

Thus a coefficientwise dominated tuple may be discarded before arbitrary further backward operators: some retained tuple remains at least as good on every feasible seed, even if their subsequently transformed coefficients cease to be componentwise comparable. Repeating this argument proves that the proposed dynamic program retains the best **evaluated bound from these relaxed recurrences** over all words with the specified counts.

This does not prove that the bound is best among actual constructions. Actual deletion choices, extra witnesses, and relations between the state variables may improve it.

6.9 Audit scope and verdict

The two cross-preservation inequalities, the simultaneous deficiency transitions, both backward coefficient maps, and the corrected justification of intermediate Pareto pruning are valid. Their composition gives an unconditional lower bound for every finite mixture of the established binary and ternary constructions, using certified seed data.

I subsequently audited the saved exposition and implementation. The implementation applies the operators in the correct direction, preserves the recorded forward words, and compares coefficient tuples correctly. Its lexicographic ordering makes the one-pass dominance filter valid: a later tuple cannot componentwise dominate an earlier one.

Independent implementation checks enumerated **3431 unpruned words**, covering every pair of factor counts $0 \leq i, j \leq 6$, and compared the full and pruned minima in **2009 evaluations** on the original 14 certified profiles and a grid of small states in the invariant cone. They agreed throughout. The pure binary and pure ternary coefficient formulas also agreed through exponent nine. All eight numerical construction bounds displayed in the initial audited exposition were independently recalculated from their recorded seed profiles and words; the later elementary-seed extension is checked below. The earlier-envelope comparison column is historical bookkeeping; any subsequent strengthening of another construction can only increase the combined envelope.

In particular, word 3222 has coefficient tuple $(54, 189, 189, 0)$. At order $24m$, subtracting its deficit from $576m^2$ gives

$$576m^2 - 54m^2 - 189(m^2 - s) - 189(s - t_2) = 333m^2 + 189t_2.$$

Thus the displayed corollary $\ell(24m) \geq 333m^2 + 189t_2$ is correct and requires only the stated seed's private-intercalate count. For the certified order-15 seed with $t_2 = 132$, it gives $\ell(360) \geq 99873$.

The standalone independent local checker is `check_lcs_mixed_local.py`. Run `python3\check_lcs_mixed_local.py` from this directory. It imports no project witness or recurrence code and reproduces the finite local and exhaustive relaxed-deletion checks above.

6.10 Elementary seed extension

I independently verified the two added records in `lcs_mixed_small_seeds.json`. Exhaustive enumeration of every labelled Latin square of orders two and three confirms their unique completions. All four recorded clue-deletion alternatives are Latin squares and disagree with the partial square in exactly the designated clue. Direct enumeration of the seed intercalates gives the exact profiles

$$(m, s, t_2, t_3) = (2, 1, 1, 0), \qquad (3, 3, 0, 3).$$

For the second seed, each deletion witness uses the entire order-three square, so every clue is 3-essential. These checks were added to the standalone independent checker and passed.

The order-three seed has state $(N, D, u_2, u_3) = (9, 6, 3, 0)$. Independent calculation gives:

Forward word	Coefficients (a, b, c, d)	Order	Guaranteed size
2333	(294, 1029, 49, 882)	162	17277
23333	(2744, 7203, 0, 8232)	486	168282
233333	(24696, 50421, 0, 74088)	1458	1600974
2333333	(222264, 352947, 0, 666792)	4374	15013818

Each row uses $n^2 - 9a - 6b - 3c$. In particular the last row supersedes the earlier size 11958717 bound at order 4374. The extension introduces no new mathematical assumption: both elementary seeds satisfy the same hypotheses as the other certified seeds, and pure lift words are already included by the recurrence.

The updated implementation checks also passed: pure binary and ternary formulas on all 16 seed profiles, **6798 comparisons** against exhaustive words of length at most ten, and the expanded numerical envelope. No gap was found in this extension.

7 A one-clue improvement of the Fu–Fu–Liao construction

The following construction improves the Fu–Fu–Liao lower bound by one. It is a lower bound; neither sharpness nor a corresponding upper bound is asserted. The original bound is due to Fu, Fu and Liao, *Congressus Numerantium* **110** (1995), 161–166. The construction below was developed from the bitwise mask recorded by Richard Bean in OEIS A080572. No claim of literature priority is made.

Theorem. For every integer $m \geq 2$,

$$\operatorname{lcs}(2^m - 1) \geq 4^m - 3^m - 2^{m+1} + 4.$$

7.1 The square and its clue mask

Put $n = 2^m - 1$. Rows and columns have labels $0, \ldots, n-1$, and symbols have labels $1, \ldots, n$. Write $\oplus$ and $\&$ for binary exclusive-or and binary AND. Define

$$L(r, c) = \begin{cases} r \oplus c, & r \neq c, \\ n \oplus r, & r = c. \end{cases}$$

This is a Latin square. Indeed, deleting row and column n from the XOR square of order $n+1$ leaves row r missing the symbol $n \oplus r$ and containing zero at its diagonal. Replacing that zero by $n \oplus r$ makes the row a permutation of $1, \ldots, n$. The same argument applies to columns.

Write $r = 4u + i$ and $c = 4v + j$, where $0 \leq i, j < 4$, and put

$$K = \{(1,1), (1,2), (1,3), (2,1), (2,2), (3,1), (3,3)\}.$$

Let P retain the entry of L at (r, c) precisely when

$$\big(u \,\&\, v \neq 0 \text{ or } (i, j) \in K\big) \quad \text{and} \quad (r, c) \neq (2, 2).$$

The usual Fu–Fu–Liao mask in this square retains $r \,\&\, c \neq 0$. When $u \,\&\, v = 0$, the present mask replaces low-coordinate positions $(2,3), (3,2)$ by $(1,2), (2,1)$ and finally deletes $(2,2)$. It is therefore an exchange of clues, not a strict enlargement of an existing critical set.

7.2 The exact number of clues

First apply the high-AND/low-K mask to the full square of order $q = 2^m$. There are 3^{m-2} ordered high-coordinate pairs (u, v) with $u \,\&\, v = 0$. For each such pair, exactly nine of the sixteen low-coordinate positions are holes. The mask therefore has

$$q^2 - 9 \cdot 3^{m-2} = 4^m - 3^m$$

clues. Its last row and last column each have $q-2$ clues, and their intersection is a clue. Deleting them leaves $4^m - 3^m - 2q + 5$ clues. Deleting $(2,2)$ leaves

$$|P| = 4^m - 3^m - 2q + 4.$$

Equivalently, the exchange relative to the original mask inserts $2 \cdot 3^{m-2}$ clues and deletes $2 \cdot 3^{m-2} - 1$ clues.

7.3 Unique completion: all forcing steps

For $m = 2$, the retained order-three clues are $(1,1;2), (1,2;3), (2,1;3)$. Row and column singles fill, in order,

$$(1,0), (0,1), (0,2), (0,0), (2,0), (2,2),$$

so this base case has unique completion.

Suppose henceforth that $m \geq 3$. Set $s = 2^{m-2}$ and $A = s-1$, so $n = 4A+3$. Every hole belongs to a high-coordinate block (u, v) with $u \,\&\, v = 0$. Except for the top-left block, a full such block contains the seven K clues of an XOR square of order four on the symbols $4(u \oplus v) + \{0,1,2,3\}$.

Step I: the two truncated blocks. Consider block $(A, 0)$, with rows $4A + i$ for $i = 0, 1, 2$ and columns $j = 0, 1, 2, 3$. Every entry in these rows outside these four columns is already a clue: if its high column is $v > 0$, then $A \,\&\, v = v \neq 0$. Consequently every missing entry in these rows must use a symbol in $4A + \{0,1,2,3\}$.

The low symbols of the K clues in local row 1, columns 1, 2, 3 are $0, 3, 2$; those in local row 2, columns 1, 2 are $3, 0$. The separate diagonal clues $(1,1)$ and $(3,3)$ contain $4A+2$ and $4A+0$ in columns 1 and 3. These row and column constraints force the following local cells, where the displayed values are low symbols:

(i,j)	$(1,0)$	$(0,1)$	$(2,3)$	$(2,0)$	$(0,2)$	$(0,3)$	$(0,0)$
low symbol	1	1	1	2	2	3	0

The notation in this table refers to the local block coordinates. Only ordinary Latin row and column constraints are being used; no condition requiring all diagonal symbols to differ is assumed. Transpose this argument to complete block $(0, A)$.

Step II: all full blocks, in decreasing high bitwise OR. For $z = A, A-1, \ldots, 1$, complete every block (u, v) satisfying $u \,\&\, v = 0$ and $u \mid v = z$, except the two truncated blocks already completed. We first show that every hole in such a block is confined to the four symbols $4z + \{0,1,2,3\}$.

Let $w = 4a + k$ be a candidate at a hole (r, c) in this block. All references to the true occurrence of w in the following argument mean its occurrence in the explicitly displayed square L. A candidate is excluded only by an initial clue or a previously forced cell. Thus the argument does not presume unique completion.

If a fails to contain a 1-bit of u, the ordinary XOR occurrence of w in row r has high column $u \oplus a$, and

$$u \,\&\, (u \oplus a) = u \,\&\, (\neg a) \neq 0.$$

That occurrence is an initial clue. If the occurrence in L is instead diagonal, the diagonal entry is also a clue: the only missing diagonal clues have row labels 0 and 2, hence $u = 0$, which is impossible in the present missing-bit case. Therefore a contains every bit of u. The column argument shows that it contains every bit of v as well, so it contains z.

If a properly contains z, then $a > z$. The ordinary occurrence of w in row r has high column $u \oplus a$, high AND zero and high OR a, so it lies in a block completed earlier. A diagonal occurrence is an initial clue except when $r = 0$ or 2. In that exception $u = 0$ and the current column has $v = z > 0$. Use the occurrence in column c instead: its ordinary occurrence is in a completed OR-a block; if diagonal it would be a clue. In fact this latter diagonal exception cannot occur, since $w = n \oplus r \neq n \oplus c$. Thus $a > z$ is impossible, leaving $a = z$.

The seven K clues now force the remaining low XOR-four entries by row and column singles in the order

(i,j)	$(1,0)$	$(0,1)$	$(2,3)$	$(2,0)$	$(0,2)$	$(0,3)$	$(0,0)$	$(3,0)$	$(3,2)$
$i \oplus j$	1	1	1	2	2	3	0	3	1

because all their candidates have already been confined to that four-symbol class. Blocks with the same value of z may be filled in any order.

Step III: the top-left corner. After Step II, only the top-left four-by-four corner has holes. For $0 \le i < 4$, the symbols left in row i and column i are

$$\{1, 2, 3, n \oplus i\}.$$

For distinct i, j, the special symbols $n \oplus i$ and $n \oplus j$ differ and are outside $\{1, 2, 3\}$, so every off-diagonal hole has candidates only in $\{1, 2, 3\}$. The corner clues are

$$(1, 1; n \oplus 1), \quad (1, 2; 3), \quad (1, 3; 2), \quad (2, 1; 3),$$
$$(3, 1; 2), \quad (3, 3; n \oplus 3).$$

They force the following sequence:

$$(1, 0) = 1, \quad (2, 3) = 1, \quad (2, 0) = 2, \quad (2, 2) = n \oplus 2,$$
$$(3, 0) = 3, \quad (3, 2) = 1, \quad (0, 0) = n, \quad (0, 1) = 1, \quad (0, 2) = 2, \quad (0, 3) = 3.$$

Every hole has been forced. Since L is a completion, it is the unique completion of P.

7.4 Essentiality: a private trade for every clue

A private trade for a clue changes that clue and otherwise only holes, while preserving each affected row's and column's symbol multiset. Substituting its mate gives an alternative completion after the clue is removed. We now exhibit such a trade for every clue of P.

Class 1: ordinary off-diagonal clues. Let (r, c) be a clue with $r \ne c$, excluding for now $(2, n-2)$ and $(n-2, 2)$. Define

$$t = 4(u \,\&\, v) \oplus \delta(i, j),$$

where $\delta = 0$ outside K, and on K it is given by

(i, j)	$(1,1)$	$(1,2)$	$(1,3)$	$(2,1)$	$(2,2)$	$(3,1)$	$(3,3)$
$\delta(i, j)$	1	2	3	2	2	3	3

Use the rectangle with rows $r, r \oplus t$ and columns $c, c \oplus t$, and exchange its two symbols. Because (r, c) is a clue, $t \ne 0$. We verify that it is an intercalate of the contracted square rather than merely of its parent XOR square.

The starting and opposite cells are not diagonal, since $r \ne c$. A cross cell could be diagonal only if $u \oplus v = u \,\&\, v$. The disjoint bit supports of XOR and AND force both sides to be zero, hence $u = v = 0$. The clue would then be an off-diagonal member of K, but none of its four off-diagonal pairs satisfies $i \oplus j = \delta(i, j)$. Thus no cell of the rectangle is diagonal, and all four symbols retain the XOR rule.

No row or column of this rectangle is the deleted index n. For example, $r \oplus t = n$ forces $u \,\&\, (\neg v) = A$, hence $u = A$ and $v = 0$. The only applicable low-coordinate change to 3 is then $i = 1$, $\delta = 2$, $j = 2$. This gives $(r, c) = (n - 2, 2)$, which was excluded. Transposing gives the other excluded position.

At each of the other three rectangle corners the high AND is zero. If $(i, j) \notin K$, the low-coordinate pair is unchanged, and remains outside K. If $(i, j) \in K$, the displayed δ table shows that all three other low-coordinate pairs are outside K. Those three cells are therefore holes. The extra deletion of $(2, 2)$ cannot spoil this property. Hence this intercalate is private to (r, c).

Class 2: diagonal clues. Every retained diagonal clue (r, r) has $r \ne 0, 2$. Set

$$a = \begin{cases} 0, & r \bmod 4 \in \{0, 3\}, \\ 2, & r \bmod 4 \in \{1, 2\}, \end{cases} \qquad b = n \oplus r \oplus a.$$

The rows a, r, b are distinct valid row labels: $a \ne r$ by the excluded diagonal cases; $b = a$ would give $r = n$; $b = r$ would give $a = n$; and $b = n$ would give $a = r$. All are impossible. Consider the six cells

$$(a, a), \ (a, b), \ (r, r), \ (r, b), \ (b, a), \ (b, r).$$

They use only the two distinct symbols $n \oplus a$ and $n \oplus r$. Each affected row and column contains one occurrence of each, so exchanging these symbols gives a six-cell trade.

Only (r, r) is retained. If $a = 0$, all cells in row 0 or column 0 are holes. The remaining cross pairs have complementary binary coordinates and low pair $(0, 3)$ or $(3, 0)$, both outside K. If $a = 2$, $(2, 2)$ is the expressly deleted clue. The other five cells have high AND zero and their relevant low pairs lie among

$$(2, 0), (0, 2), (1, 0), (0, 1), (2, 3), (3, 2),$$

all outside K. Thus the six-cell trade is private to (r, r).

Class 3: the two exceptional off-diagonal clues. Put $b = n \oplus 2 = n - 2$. For the clue $(2, b)$, interchange the entries of rows 0 and 2 in columns $0, 2, b$. In those columns the two rows have respective triples

$$(n, 2, b) \qquad \text{and} \qquad (2, b, n).$$

The swap preserves each row's symbol multiset and exchanges the two symbols in each affected column, so it is a six-cell row-cycle trade. All three row-zero cells are holes, as are $(2, 0)$ and $(2, 2)$. Its sole clue is $(2, b)$. Transposition gives the private trade for $(b, 2)$. These statements also cover $m = 2$, when $n = 3$ and $b = 1$.

Every clue of P has a private trade. Together with the forcing proof and the clue count, this proves the theorem.

7.5 Independent audit and finite certificates

The original independent audit first verified the order-31 example using an independently written integer AllDifferent model: one different-completion query was infeasible, and all 721 deletion queries returned valid alternative completions. It then checked solver-free forcing and private-trade certificates at orders 31 and 63, and audited the uniform construction above. A second proof coverage audit for this review checked the deleted-index exceptions, the candidate exclusion in Step II, and all three trade classes. The certificate counts are

n	$\lvert P\rvert$	forcing steps	private intercalates	private six-cell trades
31	721	240	690	31
63	3243	726	3180	63

The implementation relabels symbols by subtracting one. Its memory cap $2 \le m \le 8$ is not a restriction of the mathematical theorem. The source proof, constructor, verifier and certificates are in `outputs/lcs_ffl_plus_one/`. The uniform proof, rather than agreement of a finite sequence of sizes, establishes the claim for every $m \ge 2$.

8 When a full block is essential: a converse and its extension

This result explains why the unrestricted product formula cannot be obtained merely by counting the entries in the usual defining template. The numerical product conjecture itself is not disproved by the failure of that template.

Theorem 8.1 (Full-block criterion)**.** *Let A, B, C, D be Latin squares of order m, with uniquely completing partial squares Q_B, Q_C, Q_D in the last three. Use disjoint symbol groups and form*

$$M = \begin{pmatrix} A & B + m \\ C + m & D \end{pmatrix}, \qquad P = \begin{pmatrix} A & Q_B + m \\ Q_C + m & Q_D \end{pmatrix}.$$

Then P defines M. A clue $(r, c; s)$ in its full block is essential if and only if there are i, j, t such that

$$A(r, c) = D(i, j) = s, \qquad B(r, j) = C(i, c) = t,$$

and $(r, j), (i, c), (i, j)$ are holes of Q_B, Q_C, Q_D, respectively.

Proof. The full block confines both adjacent partial blocks to the high symbol group. Their unique completions determine them, and then confine the last block to the low group, where its unique completion determines it.

For necessity, delete the full-block clue and suppose an alternate completion changes it. Its retained row and column in A already contain every low symbol except s, so its new value is a high symbol $m + t$. Block B has precisely one low entry, namely s, at (r, j): every other row of B is entirely high, and row r must supply the lost low symbol. Similarly C has precisely one low entry s at (i, c). The exceptional row of B and column of C each lack $m + t$.

Every ordinary column of B contains all m high symbols. Counting the occurrences of each high symbol throughout B, using its ordinary rows and its exceptional row, proves that its exceptional column j also lacks $m + t$. The transposed argument proves that the exceptional row i of C lacks $m + t$. The global row and column conditions therefore put exactly one high entry in D, at (i, j), equal to $m + t$; its exceptional row and column lack the low symbol s.

Repair the three crossed entries by setting

$$B(r, j) = m + t, \qquad C(i, c) = m + t, \qquad D(i, j) = s.$$

Each repaired block is Latin on its intended symbol group. Each contains its prescribed partial square, because a crossed-group cell could not have been a fixed clue. Unique completion identifies the repaired blocks with the original B, C, D. This proves the stated equalities and hole conditions.

Conversely, the four global positions $(r,c),(r,m+j),(m+i,c),(m+i,m+j)$ contain $s,m+t,m+t,s$. Switching these two symbols in that intercalate changes the designated clue and only holes elsewhere, giving a deletion witness. The outer blocks distinguish rows, columns and symbol groups, so no additional conditions $i \neq r$, $j \neq c$ or $t \neq s$ are needed. □

Corollary 8.2 (Identical blocks)**.** *For a defining set $Q \subseteq L$, in the identical-block template a full-block position corresponding to a clue $e \in Q$ is essential if and only if e has a private intercalate in L. A position outside Q always has an essential full-block copy.*

Proof. With $A = B = C = D = L$, the criterion gives the four cells of a base intercalate. When (r,c) is a clue, the other three are holes, so the Latin property forces distinct inner rows, columns and symbols. Conversely, that private intercalate supplies the required rectangle. For a position outside Q, take $i = r, j = c, t = s$ in the general criterion. □

This is a criterion for individual essentiality. It does not permit simultaneously deleting every individually redundant clue.

8.1 The order-17 seed and the failed order-68 template

The verified order-17 size-200 seed has exactly 78 clues with private intercalates. For the retained clue $(1,3;2)$, using zero-based coordinates, the only other-row/other-column pairs of intercalates are

$$(4,6),(5,7),(6,4),(7,5),(8,10),(9,11),$$
$$(10,8),(11,9),(12,14),(13,15),(14,12),(15,13).$$

For other rows 4 through 14 the cell in column 3 is another clue; in the last case $(1,13)$ is another clue. Thus none is private. In the orientation with the full block bottom right, clue $(18,20;2)$ of the 889-entry order-34 template is redundant by the corollary.

Double that defining template again. Its nine seed blocks and seven full order-17 blocks have $9 \cdot 200 + 7 \cdot 17^2 = 3823$ clues. Delete the same redundant clue in the top-left order-34 block. The full bottom-right block confines the other blocks to their intended symbol groups. Each still defines its own completion, including the top-left block after the deletion. Consequently the order-68 template is defining but not critical. This is not a nonexistence proof for a different size-3823 critical set.

The independent finite check replays the seed's exhaustive uniqueness tree (5954 nodes, 236 dead leaves and one completion), all 200 deletion witnesses, the entire intercalate list and the nested product coordinates. These are solver-free certificate checks. Additional AllDifferent infeasibility queries at orders 34 and 68 corroborate the redundancy but are not needed for the deduction.

8.2 Conjugate blocks and the protected size 769

Apply the theorem to all $6^4 = 1296$ choices of coordinate conjugates of the given completion and seed, independently in the four blocks. A separate checker joins equal-symbol holes in B, C, tests the corresponding hole in D, and compares its symbol with A. It reproduces the full histogram: the maximum number of essential full-block clues is 169, attained by 18 choices. Thus none of this finite class of 889-entry templates is critical. Independent inner relabellings are not exhausted by this experiment.

One maximizing choice has $A(r, L(r,c)) = c$ and $B = C = D = L$. There are 169 explicitly checked private intercalates through full-block clues. All $3 \cdot 200 = 600$ clues in the other blocks remain essential, using the seed's deletion trades inside their individual subsquares. Every minimal defining subset therefore has at least $600 + 169 = 769$ clues. Moreover, its private-intercalate clues include the $3 \cdot 78$ inherited ones and those 169 full-block clues. Thus one coherent extracted seed has

$$s \geq 769, \qquad t_2 \geq 403.$$

The safe binary and two-step binary bounds, whose coefficients of s, t_2 are positive, give

$$\mathrm{lcs}(34) \geq 769, \quad \mathrm{lcs}(68) \geq 34^2 + 2 \cdot 769 + 403 = 3097,$$

$$\mathrm{lcs}(136) \geq 7 \cdot 34^2 + 3 \cdot 769 + 6 \cdot 403 = 12817.$$

No critical subset of exactly 769 entries is asserted. In particular, this lower size cannot be substituted for an unknown actual size in a mixed formula having a negative size coefficient. The order-68 result remains below the thesis bound 3247. Reproducible certificates and independent checkers are in `lcs_order68_product_audit/`.

9 Prolonging the thesis constructions: shared seed

The two constructions proved below give, for every **odd integer** $m \geq 3$,

$$\boxed{\operatorname{lcs}(4m+1) \geq \frac{21m^2 - 7m - 22}{2}}, \qquad \boxed{\operatorname{lcs}(16m+1) \geq 182m^2 - 18m - 44}. \tag{9.1}$$

Both start from an explicit defining set, preserve a specified collection of private trades under prolongation, and then take an inclusion-minimal defining subset. The sizes of the augmented defining sets themselves are not used as critical-set lower bounds. The common coordinate lemmas are given in full here.

The order-$4m$ square. Put $p = 4m$. Rows are (R, i) and columns (C, j), with $R, C \in \{0, 1, 2, 3\}$ and $0 \leq i, j < m$. The symbol block is $R \oplus C$, where $\oplus$ is bitwise XOR. Within a block, residues are modulo m. Write $A(i, j) = i - j$, $A^{\mathsf{T}}(i, j) = j - i$ and $B(i, j) = i + j$. The complete square L and clue mask of the thesis partial square P are

	0	1	2	3
0	A	A^{T}	A^{T}	B
1	A	A^{T}	B	A^{T}
2	A	B	A^{T}	A^{T}
3	B	A	A	A

	0	1	2	3
0	H	H	H	G
1	J	F	G	F
2	J	G	F	F
3	G	F	F	F

.

Here F prescribes every entry of the block; the other masks prescribe

$$H:\ 1 \leq j \leq i \leq m-1, \qquad J:\ 1 \leq i \leq j \leq m-1, \qquad G:\ i + j \geq m.$$

Each complete block is Latin, and its symbol block varies bijectively across each block row and column, so L is Latin. The established triangular critical-set construction, used in the thesis, says that each displayed H, J or G partial block is critical within its complete m-square. Thus every triangular clue has a private trade wholly within that block. This standard seed fact is an input; the product and prolongation arguments below do not assume a general critical-product theorem.

Unique completion of P. The full blocks in block row 3 force block 30 to use only symbol block 3; the full blocks in block column 3 do the same for block 03. Their G partials therefore complete uniquely to their prescribed copies of B. In block 12, its full row blocks $11, 13$ exclude symbol blocks $0, 2$, and its full column blocks $22, 32$ exclude $0, 1$. Only symbol block 3 remains. Similarly block 21 is restricted to symbol block 3. Their G partials complete uniquely. Now the other three blocks in each relevant row or column force blocks $10, 20, 01, 02$ into symbol blocks $1, 2, 1, 2$, respectively. Their J or H partials complete uniquely. Finally block 00 is restricted to symbol block 0 and its H partial completes. Hence P defines L.

There are seven full blocks and nine triangles, giving

$$s = |P| = 7m^2 + 9\frac{m(m-1)}{2} = \frac{23m^2 - 9m}{2}. \tag{9.2}$$

The private intercalates constructed on the next pages prove that every full-block clue is essential as well. Together with the triangular private trades, this recovers the criticality needed from the thesis seed.

9.1 Transversal, contraction and private rectangles

A shifted thesis transversal. Let t be the following column permutation of the inner order-$4m$ square:

Row	Column	Symbol block and residue
$(0,i)$	$(0,m-1-i)$	$(0,2i+1)$
$(1,i)$	$(2,i)$	$(3,2i)$
$(2,i)$	$(3,-i)$	$(1,-2i)$
$(3,i)$	$(1,-i)$	$(2,2i)$

All residues are modulo m. Rows and columns occur once each; the four symbol blocks differ, and multiplication by 2 is a permutation because m is odd. Thus t is a transversal. It meets triangular blocks only in 00 and 12, in $(m-1)/2$ clues each, and meets full blocks $23, 31$ in m clues each. Therefore

$$|t \cap P| = 3m-1, \qquad |t \setminus P| = m+1. \tag{9.3}$$

Safe prolongation and minimal pruning. For any Latin square M with transversal T, replace its transversal entries by a new symbol ∞, move each displaced symbol into its row's new column and its column's new row, and put ∞ in the new corner. If Q defines M, prescribe its transformed entries, all of T, and the entire new row and column; call this partial square $U(Q,T)$. In any completion, the fixed new border and the fixed ∞ transversal permit contraction: delete the border and restore its known displaced symbols. The result is a Latin completion of Q, so $U(Q,T)$ is defining.

A private trade for a clue of Q which avoids T survives unchanged in $U(Q,T)$: it meets neither the added transversal clues nor the new border. Call such a clue *protected*. Every defining subset of $U(Q,T)$ must contain it, since otherwise switching that trade gives a second completion. An inclusion-minimal defining subset exists because $U(Q,T)$ is finite; it is a critical set and contains every protected clue.

Private rectangles in the seven full inner blocks. For target $(R,i;C,j)$, take other row (S,x) and other column (D,y). The four positions, in order, are target, horizontal, vertical and diagonal:

$$(R,i;C,j), \quad (R,i;D,y), \quad (S,x;C,j), \quad (S,x;D,y).$$

Target block (R,C)	Other block coordinates (S,D)	Residue relation
$11, 22$	$(0,0)$	$x \equiv j-i+y \pmod m$
33	$(0,0)$	$x \equiv i-j+y \pmod m$
13	$(0,2)$	$x \equiv i-j+y \pmod m$
23	$(0,1)$	$x \equiv i-j+y \pmod m$
31	$(2,0)$	$x \equiv i-j+y \pmod m$
32	$(1,0)$	$x \equiv i-j+y \pmod m$

Opposite symbols agree under these relations; the two symbol blocks differ since $S \neq R$. Thus each rectangle is an intercalate.

For blocks $11, 22$, use $(x,y) = (0,0)$ if $i = j$; otherwise use $((j-i) \bmod m, 0)$ if $j = 0$ or $0 < i < j$, and $(0, (i-j) \bmod m)$ if $i = 0$ or $0 < j < i$. For every other full block use $(i-j, 0)$ if $i \geq j$ and $(0, j-i)$ if $i < j$. These are private witnesses for *all* full-block clues. In $11, 22$, the two zero-index cells are holes and the third is a hole by $j > x$ or $i > y$, with zero target indices covered by the masks. For the other five blocks the two G sums are i or j, below m, and the remaining H or J cell is in its zero row or column.

9.2 Which private witnesses avoid the transversal?

For later products it matters which corner of an inner witness must avoid t. The following counts describe *explicit sufficient sets* of full-block clues, without claiming that all possible witnesses were exhausted:

Required avoidance	Number of selected full-block clues
Target only	$7m^2 - 2m$
Horizontal corner only	$7m^2 - 2$
Vertical corner only	$7m^2 - 2$
Diagonal corner only	$7m^2 - 6$
All four corners	$7m^2 - 2m - 10$

Target. Among full blocks, t meets exactly the m cells $i + j \equiv 0 \pmod m$ in each of 23 and 31. Omit those targets and use any of the preceding unconstrained private rectangles.

Horizontal corner. The horizontal other-position can lie in a block containing t only for target block 13, when it lies in 12 and its transversal condition is $y = i$. In block 13 omit

$$(i, j) = (0, 0), \quad \big((m-1)/2, m-1\big),$$

and choose

$$(x, y) = \begin{cases} (i - j, 0), & i \geq j, \\ (0, j - i), & i < j,\ j \neq 2i, \\ (1, i + 1), & i < j,\ j = 2i < m - 1. \end{cases}$$

The first case has $y = i$ only at the omitted $(0, 0)$; the second avoids it by $j \neq 2i$; the last has $y = i + 1$. In the last case the H corner is a hole since $y > x$, and both G sums equal $j + 1 < m$. The only missing endpoint is $j = 2i = m - 1$, exactly the second excluded target. All other full blocks retain their unconstrained witness, losing only these two clues.

Vertical corner. The corresponding restriction occurs only for target block 32, whose vertical other-position is in 12 with condition $x = j$. Omit

$$(i, j) = (0, 0), \quad \big(m - 1, (m-1)/2\big),$$

and choose $(x, y) = (0, j - i)$ if $i \leq j$, $(i - j, 0)$ if $i > j$ and $i \neq 2j$, and $(j + 1, 1)$ if $i > j$ and $i = 2j < m - 1$. The modified case has $x > y$, so its J corner is a hole; both G sums are $i + 1 < m$, and $x \neq j$. The two exclusions are the zero case and the sole unhandled endpoint. This again loses two selected clues.

Diagonal corner. Only targets in $11, 22, 33$ have the opposite corner in a block containing t, namely 00, where the condition is $x + y \equiv m - 1 \pmod m$. The unconstrained witnesses in $11, 22$ meet it only at target coordinates $(0, 1), (1, 0)$; in 33 only at $(0, m - 1), (m - 1, 0)$. For positive unequal indices the former nonzero coordinate is $|i - j| \leq m - 2$; a zero index gives $m - j$ or $m - i$ and precisely the listed exceptions. In 33 the sum is $|i - j|$. Thus six clues are omitted.

All four corners. The exclusions just described lie in different full blocks: two each in $11, 22, 33, 13, 32$, and m each in $23, 31$. The modified witnesses have no other constrained corner. Hence all four positions avoid t for $7m^2 - 2m - 10$ selected full-block clues. Adding the seven unaffected triangular blocks gives a protected inner set F_0, disjoint from blocks $00, 12$ and from the new border, of size

$$f = |F_0| = 7\frac{m(m-1)}{2} + 7m^2 - 2m - 10 = \frac{21m^2 - 11m - 20}{2}. \tag{9.4}$$

9.3 Order $4m+1$: the complete proof

The additional intercalate families. Prolong L along t and use $U(P,t)$. Extend each of the two omitted finite blocks 00 and 12 by the new row and column. Each extension is a Latin subsquare of order $m+1$: its restricted transversal contains all old symbols of that block, which move into its border, and the subsquare gains ∞. The two extensions have different finite row blocks and different finite column blocks, so intersect only at the new corner. Neither contains a clue of F_0.

In extended block 00, label a finite diagonal position by row i, paired with column $m-1-i$, and label the new corner by $*$. All $m+1$ diagonal positions contain ∞. For distinct finite labels i,j, the crossing cells have equal old residue $i+j+1 \pmod m$. For the pair $\{i,*\}$ they have the displaced residue $2i+1$. In extended block 12 the finite pairing is row i with column i; the corresponding residues are $i+j$ and $2i$ in symbol block 3. Thus every pair of diagonal positions is the opposite pair of an intercalate, whose other repeated value is an old symbol.

Complete-graph cover lemma. These intercalates form a K_{m+1} family: diagonal clues are vertices, and each off-diagonal cell belongs to exactly one pair-intercalate. Suppose a defining set selects k diagonal clues in one such subsquare. Each pair among the other $m+1-k$ vertices must be met by an off-diagonal clue, since switching an unmet intercalate gives another completion. Distinct pairs require distinct off-diagonal clues. Consequently

$$\left|C \cap S\right| \geq k + \binom{m+1-k}{2} \geq m. \tag{9.5}$$

Indeed, on putting $q = m+1-k$, the difference from m is $(q-1)(q-2)/2 \geq 0$ for every integer $0 \leq q \leq m+1$. The argument applies to any defining set, including one restricted to $U(P,t)$; restrictions on available clues cannot weaken it.

Critical subset and count. Choose an inclusion-minimal defining subset $C \subseteq U(P,t)$. Its surviving private trades force $F_0 \subseteq C$. Each of the two extended subsquares requires at least m further clues, by (9.5). Their only common cell is the new corner, so together they require at least $2m-1$ clues outside F_0. Therefore

$$\begin{aligned} |C| &\geq f + 2m - 1 = \frac{21m^2 - 11m - 20}{2} + 2m - 1 \\ &= \frac{21m^2 - 7m - 22}{2}, \end{aligned} \tag{9.6}$$

proving the first bound in (9.1). This includes $m=3$: the core has 68 clues and the two covers force five more. Equivalently,

$$\operatorname{lcs}(n) \geq \frac{21n^2 - 70n - 303}{32} \qquad (n \geq 13,\ n \equiv 5 \pmod 8).$$

Initial values and interpretation.

m	Output order $4m+1$	Protected core f	Proved family bound
3	13	68	73
5	21	225	234
7	29	466	479
9	37	791	808

These are guaranteed sizes of some critical subsets, not exact values of lcs or asserted records. Other verified examples can be stronger. The thesis's original prolongation discussion counted intercalates; the protected-clue and cover argument here supplies the critical-set bound.

9.4 Order $16m + 1$: an explicit defining template

Two binary layers without an assumed critical product. Keep $p = 4m$, and form $M = C_2^2 \times L$ of order $4p = 16m$: rows (a, r), columns (b, c), and symbols

$$M((a, r), (b, c)) = (a \oplus b, L(r, c)), \qquad 0 \le a, b < 4.$$

Define W by filling the seven outer blocks with $a \,\&\, b \neq 0$ completely and putting a copy of P in each of the other nine blocks. The symbol & means bitwise AND.

Why W is defining. Split the outer row and column labels by their high bit. The bottom-right binary quadrant is full, so it forces the high-bit symbol classes in the top-right and bottom-left quadrants. Each of these is itself a binary template: its bottom-right sub-block is full and the other three are copies of P. That full sub-block first fixes the low-bit symbol classes of its two adjacent sub-blocks. These complete uniquely from their copies of P, and then fix the class of the remaining sub-block, which also completes uniquely. After the top-right and bottom-left quadrants have completed, they force the high-bit class of the top-left quadrant. The same inner argument completes it. Thus W uniquely defines M; criticality of this unpruned template has not been assumed.

The product transversal. Take the affine outer mapping

$$\tau = (1, 3, 2, 0), \qquad (a \oplus \tau(a))_{a=0}^{3} = (1, 2, 0, 3).$$

Both lists are permutations, so

$$T : (a, r) \longmapsto (\tau(a), t(r))$$

is a transversal of M. It meets the partial outer blocks $01, 30$ and the full outer blocks $13, 22$. Prolong along T and use the defining set $U(W, T)$ from page 25.

Protected clues in the nine partial blocks. Seven copies of P are completely disjoint from T, and retain all s of their private-trade clues. Each affected partial block $01, 30$ retains the f clues of F_0 whose inner trades avoid t. Their total is

$$7s + 2f = \frac{203m^2 - 85m - 40}{2}. \tag{9.7}$$

Private outer rectangles. Let $K = \{(a, b) : a \,\&\, b \neq 0\}$. Each of its seven cells has a rectangle meeting K only at that target. The following table specifies the other outer row A and column B, and the rectangle's unique intersection with τ:

Outer target	(A, B)	Transversal corner and position
11	$(0, 0)$	vertical, 01
13	$(0, 2)$	target, 13
22	$(0, 0)$	target, 22
23	$(0, 1)$	diagonal, 01
31	$(2, 0)$	horizontal, 30
32	$(1, 0)$	horizontal, 30
33	$(0, 0)$	horizontal, 30

Direct XOR substitution verifies each intercalate and its sole full-block position. The table also verifies the sole transversal intersection. In general a transversal meets an intercalate in at most one cell: adjacent cells share a row or column, and opposite cells share a symbol.

9.5 Order $16m+1$: protected count and pruning

Inner holes in full outer blocks. For an inner hole (r,c) of P outside t, use its four identical inner-coordinate copies at the outer rectangle's positions. Only the target outer block is full, so this is a private product intercalate. It avoids T because $(r,c) \notin t$. Every full outer block retains

$$d_0 = p^2 - s - (m+1) = \frac{9m^2+7m-2}{2} \tag{9.8}$$

such clues, using (9.3).

Full inner-block clues in full outer blocks. Let a private inner intercalate for target (r,c) have other row r' and column c'. Combine it with the target outer rectangle (a,b), other row A and other column B, to obtain

$$\begin{array}{ll} ((a,r),(b,c)), & ((a,r),(B,c')), \\ ((A,r'),(b,c)), & ((A,r'),(B,c')). \end{array}$$

Opposite product symbols agree. The other three outer blocks are partial, and their inner positions are holes of P, so this is private in W. Exactly one outer position lies on τ; hence it avoids T precisely when the corresponding inner corner avoids t. Applying the counts on page 26 gives

Outer block(s)	Inner condition	Selected clues per block
$11, 31, 32, 33$	vertical or horizontal	$d_0 + 7m^2 - 2$
$13, 22$	target	$d_0 + 7m^2 - 2m$
23	diagonal	$d_0 + 7m^2 - 6$

The inner-hole class and full-inner-clue class are disjoint. Summing the seven full blocks and adding (9.7) gives

$$\begin{aligned} \text{full-block total} &= 7d_0 + 49m^2 - 4m - 14 = \frac{161m^2+41m-42}{2}, \\ |F_{16}| &= \frac{203m^2-85m-40}{2} + \frac{161m^2+41m-42}{2} = 182m^2 - 22m - 41. \end{aligned} \tag{9.9}$$

Every defining subset of $U(W,T)$ contains this explicit protected core.

Four further covers. In each affected partial outer block $01, 30$, the inner blocks $00, 12$ were omitted entirely from F_{16}. Extend these four finite blocks with the global new row and column. They are Latin subsquares of order $m+1$: each restricted transversal supplies exactly its own old symbol block to the border, and the subsquare gains ∞. The crossing residues are $i+j+1$ in inner block 00 and $i+j$ in inner block 12, with displaced residues $2i+1$ and $2i$. Thus each has exactly the complete-graph intercalate family used in (9.5) and requires at least m clues of a defining subset.

Their finite block labels are

$$\begin{aligned} \text{rows: } & (0,0),(0,1),(3,0),(3,1), \\ \text{columns: } & (1,0),(1,2),(0,0),(0,2), \end{aligned}$$

where each pair records outer and inner block coordinates. Each list has four distinct members. Consequently the four enlarged subsquares intersect pairwise only at the new corner. None meets F_{16}, so they force at least $4m-3$ further clues, subtracting at most three repeated counts of that corner. An inclusion-minimal defining subset $C \subseteq U(W,T)$ is therefore critical and satisfies

$$|C| \geq |F_{16}| + 4m - 3 = 182m^2 - 18m - 44. \tag{9.10}$$

This proves the second bound in (9.1) for every odd $m \geq 3$.

9.6 Prolongation audit, checks and scope

Equivalent order form and initial values. The order-$16m+1$ result is equivalently

$$\mathrm{lcs}(n) \geq \frac{91n^2 - 326n - 5397}{128} \qquad (n \geq 49,\ n \equiv 17 \pmod{32}).$$

m	Output order $16m+1$	Protected core $\lvert F_{16}\rvert$	Final bound
3	49	1531	1540
5	81	4399	4416
7	113	8723	8748
9	145	14503	14536

Independent symbolic review. The order-$4m+1$ proof was independently reconstructed from the thesis coordinates, including the shifted transversal, every exceptional coordinate, the seven unaffected triangular blocks, and the two cover families. The order-$16m+1$ derivation received a second independent symbolic review. That review checked the $2m, 2, 2, 6$ corner-specific losses, the seven outer rectangles, the protected total, and the four enlarged subsquares. Both verdicts were **pass**; the endpoint $m = 3$ is included. A wording clarification made the binary unique-completion argument explicitly sequential: complete the two adjacent quadrants before forcing the remaining top-left quadrant. It changes no construction or count.

Supplementary finite checks. Standalone standard-library checkers implement the coordinate formulas without importing the parent construction. They verify the selected private intercalates, agreement of opposite symbols, absence of other prescribed clues, avoidance of the product transversal, and all cover-family incidences and intersections. Their saved runs report

Check	Order-$4m+1$ family	Order-$16m+1$ family
Odd parameters checked	$3, 5, \ldots, 51$	$3, 5, \ldots, 31$
Number of parameters	25	15
Private intercalates checked	162375	786385
New cover-family intercalates	24100	11420

The second reviewer audited the $16m+1$ proof symbolically and did not rerun its 15-parameter computation. These finite checks support the formulas; the uniform conclusions follow from the proofs above. The established triangular critical-set theorem remains a stated dependency.

Separate finite pruning examples. Independent integer-variable Latin-square models excluded alternative completions of four thesis-based pruned examples of sizes $94, 166, 267, 539$ at orders $13, 17, 21, 29$, respectively. All 1066 saved deletion alternatives were explicitly replayed as Latin squares agreeing with every retained clue except the designated deleted clue. These are individual critical sets, not claimed records or the basis of the uniform formulas. The order-17 example uses even $m = 4$ with a separately found transversal; it does not extend either odd-m theorem. Finite uniqueness is solver-dependent; the symbolic constructions above do not depend on that solver.

What has and has not been transferred. The earlier two sequential binary lifts give $\mathrm{lcs}(16m) \geq (377m^2 - 27m)/2$, but their chosen critical subset is not an explicit uniform mask. The proof here instead starts from the explicit seven-full-block, nine-P defining template. Its direct protected count before prolongation is $184m^2 - 9m$, smaller by $s - 7m^2 = 9m(m-1)/2$ than that sequential guarantee. No equality between these masks is assumed, and no claim that prolongation preserves criticality is needed. The theorems supply guaranteed lower bounds, not optimal values, even-m extensions, or claims of literature priority.

Reproduction. The order-$4m+1$ witness checker and audit are in `lcs_odd_prolongation/independent_audit/`; run `python3check_thesis_shifted.py--max-m51` there. The order-$16m+1$ files are in `lcs_other_prolongation/independent_audit/`; run `python3check_order16m.py--max-m31` there. Its second review is `lcs_other_prolongation/ternary_audit/order16m_second_review.txt`. The seed is Bean's thesis, Chapter 5, sections 5.3–5.4; the original prolongation section concerns intercalates rather than critical-set sizes.

10 Ternary products followed by prolongation

This section proves a transfer lemma and an unconditional family at orders $3 \cdot 2^k + 1$. Its input is a lower bound for *every defining subset of one specified template*. A large critical set elsewhere at the same order does not supply that input. A defining set has a unique Latin completion; an inclusion-minimal defining set is critical. Throughout, a trade means positions whose entries can be replaced by a different partial Latin square with exactly the same symbol sets in each affected row and column.

Ternary transfer lemma. Let Q be a critical set of size s in a Latin square L of order q, and let T be a transversal. Set $h = |T \setminus Q|$. Prolong L along T, and let U_T prescribe the transformed Q, all transformed transversal positions, and the entire new row and column. Suppose that every defining subset of this particular U_T contains at least b clues. Then

$$\boxed{\operatorname{lcs}(3q+1) \geq 2q^2 + 4s - 2h + b.} \tag{10.1}$$

Proof: a defining template. Use the outer square $C_3(i,j) = i + j \pmod 3$. Its clues $(0,0;0)$ and $(1,1;2)$ form a critical set: the missing entries are forced, successively, at

$$(0,1), (0,2), (1,0), (1,2), (2,0), (2,1), (2,2).$$

Deleting the first clue permits the global symbol interchange $0 \leftrightarrow 1$; deleting the second permits $1 \leftrightarrow 2$. Each interchange is a six-cell trade fixing the other clue.

In $M = C_3 \times L$, index rows by (i,r), columns by (j,c), and symbols by $(i + j \pmod 3, L(r,c))$. Prescribe full blocks $(0,0), (1,1)$ and a copy of Q in each of the other seven blocks; call the result P. The same seven-block order proves uniqueness. For example, block $(0,1)$ can use only outer symbol class 1, since class 0 is already exhausted in its rows and class 2 in its columns. Its Q clues then force its internal copy of L. Continue in the displayed order.

The outer diagonal has the three distinct symbols $0, 2, 1$, so its product with T is a transversal T^* of M. Prolong along T^* and prescribe the transformed P, all T^* positions, and the whole new border; call this defining template U. To prove that it defines the prolonged square, delete the known border in any completion and replace each new-symbol entry on T^* by its known displaced border symbol. The result is Latin and contains P, hence is M. This fixes the original completion too.

Three disjoint contributions. The six off-diagonal blocks are unchanged Latin subsquares and meet U in exactly Q. For each clue of Q, its criticality supplies an alternative completion after that clue is removed. Substitution inside the corresponding block changes no other clue of U. Thus all $6s$ such clues belong to every defining subset of U.

There are $q^2 - s - h$ base positions outside $Q \cup T$. Their nine outer copies form unchanged C_3 subsquares, each meeting U precisely in the two full-block positions $(0,0), (1,1)$. The outer symbol interchanges above privately protect both clues. This contributes another $2(q^2 - s - h)$ clues, all in the two full old blocks.

Finally, block $(2,2)$ together with its new row and column is an actual Latin subsquare S of order $q+1$: its symbols are outer class 1 and the new symbol. Under that relabeling, $U \cap S = U_T$. If a global defining subset $D \subseteq U$ did not define S on restriction, an alternative completion of $D \cap S$ could replace S, preserving every global row and column symbol set. This would contradict uniqueness of D. Consequently $|D \cap S| \geq b$. These cells are disjoint from both earlier contributions, including the new corner, which is counted only here. Therefore

$$|D| \geq 6s + 2(q^2 - s - h) + b = 2q^2 + 4s - 2h + b.$$

Taking an inclusion-minimal defining subset of the finite set U proves (10.1). The template U itself need not be critical.

10.1 The binary factor and its explicit transversal

Fix $q = 2^k$, $k \geq 2$, and a split $1 \leq a < k$. Vectors have coordinates $x_0, \ldots, x_{k-1}$ over $\mathbb{F}_2$, with x_0 the least significant bit. Write $\oplus$ for bitwise addition and & for bitwise intersection. Define the complementary masks

$$u = 2^a - 1, \qquad v = q - 2^a, \qquad M = q - 1 = u \oplus v.$$

The complete mapping used throughout this section is $y = \tau x$, where

$$\begin{aligned} y_i &= x_i + x_{i+1} && (0 \leq i < a-1), \\ y_{a-1} &= x_0 + x_{a-1} + \sum_{j=a}^{k-1} x_j, \\ y_i &= x_0 + \sum_{j=a}^{i} x_j && (a \leq i < k). \end{aligned} \tag{10.2}$$

All sums in the mapping formulas are in $\mathbb{F}_2$. When $a = 1$, the first line is empty and the repeated x_0 in the second line cancels.

Both τ and $p = I + \tau$ are invertible. For $y = \tau x$, put $D = \sum_{i=0}^{a-2} y_i$ and $Z = y_{a-1} + D$. Then $Z = \sum_{j=a}^{k-1} x_j$, and an explicit inverse is

$$\begin{aligned} x_0 &= y_{k-1} + Z, & x_a &= y_a + x_0, \\ x_{i+1} &= x_i + y_i \quad (0 \leq i < a-1), & x_i &= y_i + y_{i-1} \quad (a < i < k). \end{aligned}$$

For $w = px$, the inverse is

$$\begin{aligned} x_0 &= w_a, & x_{k-1} &= w_{a-1} + w_{k-1}, \\ x_i &= w_{i-1} \quad (1 \leq i < a), & x_i &= w_{i+1} + w_i \quad (a \leq i < k-1). \end{aligned}$$

These formulas also cover one-bit heads or tails, with empty index ranges.

Exactly two transversal positions are holes of the AND set. We claim

$$\{x : x \,\&\, \tau x = 0\} = \{0, u\}, \qquad \tau u = v. \tag{10.3}$$

For disjoint $x, \tau x$, any 1 in the first $a-1$ positions forces the next input bit to be 1, because its own output bit must vanish. If $x_0 = 0$, a first 1 in the tail would have output 1 at the same position, so the tail is zero. A first 1 in the head would propagate to $x_{a-1} = 1$, giving $y_{a-1} = 1$, again impossible. Hence $x = 0$. If $x_0 = 1$, the head is all ones. The tail has at most one 1: a second one has output 1 by the prefix-parity formula. But $y_{a-1} = 0$ requires even tail parity, so the tail is zero. This gives $x = u$; substitution yields $\tau u = v$.

The critical set in the XOR square. Let $G(r, c) = r \oplus c$, and take

$$Q = \{(r, c) : r \,\&\, c \neq 0\}, \qquad |Q| = q^2 - 3^k.$$

There are 3^k holes, from the three allowable bit pairs $(0,0), (1,0), (0,1)$. Complete holes in decreasing numerical order of $z = r \mid c$. If a candidate symbol t omits a bit of r, its true row position $(r, r \oplus t)$ is an original AND clue and excludes it; the column argument likewise forces t to contain every bit of c. Thus t contains z. If it properly contains z, its true row position is a hole of OR value $t > z$, already filled. Only $t = z$ remains. This proves uniqueness. For every clue put $d = r \,\&\, c \neq 0$. The rectangle on rows $r, r \oplus d$ and columns $c, c \oplus d$ is an intercalate whose other three positions are holes. It privately witnesses that clue's necessity. Thus Q is critical. Since τ and p are permutations, $T = \{(r, \tau r) : 0 \leq r < q\}$ is a transversal, and (10.3) proves $h = |T \setminus Q| = 2$ exactly.

10.2 Protected clues in the prolonged XOR template

Use $\infty = q$ for the new index and new symbol. The prolongation G^+ is given explicitly by

$$G^+(r,c) = \begin{cases} q, & c = \tau r, \\ r \oplus c, & c \neq \tau r, \end{cases} \qquad r, c < q, \tag{10.4}$$
$$G^+(r,\infty) = p(r), \qquad G^+(\infty,c) = \tau^{-1}(c) \oplus c, \qquad G^+(\infty,\infty) = q.$$

Each displaced transversal symbol appears once in its new row and column, so this is Latin. Let U_T contain the Q positions, all T positions, and the full new border, with their values in G^+. This is exactly the template required in the ternary lemma. It is defining: deleting the known border and restoring its known displaced symbols on T gives a completion of Q, hence the XOR square. No optimality or product conjecture is used.

For a clue $(r,c) \in Q$, its canonical private intercalate has corners

$$(r,c), \quad (r \oplus d, c), \quad (r, c \oplus d), \quad (r \oplus d, c \oplus d), \qquad d = r \,\&\, c. \tag{10.5}$$

If these avoid T, the intercalate survives in G^+ and meets U_T only at (r,c). Let F be the set of these protected clues. Every defining subset of U_T must contain F: otherwise swapping the corresponding intercalate gives a second completion.

Exact count of F. A transversal meets an intercalate in at most one position. Two selected corners in distinct rows and columns would be opposite corners with the same symbol. Consequently the four possible corner-hit classes in (10.5) are disjoint when $d \neq 0$. First count them allowing $d = 0$. Write $\mathcal{S} = \{x : x \,\&\, \tau x = 0\} = \{0, u\}$ and let wt denote bit weight.

corner hit	number	parameterization
(r,c)	q	$c = \tau r,$
$(r \oplus d, c)$	$1 + 2^{k-a}$	$x = r \oplus d \in \mathcal{S},\ c = \tau x,\ d \subseteq \tau x,$
$(r, c \oplus d)$	$1 + 2^a$	$r = x \in \mathcal{S},\ c = \tau x \oplus d,\ d \subseteq x,$
$(r \oplus d, c \oplus d)$	$q + 1$	$x \in \mathcal{S},\ d \subseteq \overline{x \mid \tau x}.$

Here $d \subseteq x$ means inclusion of bit supports, and the complement is within the k coordinates. The last three counts are respectively

$$\sum_{x \in \mathcal{S}} 2^{\mathrm{wt}(\tau x)}, \qquad \sum_{x \in \mathcal{S}} 2^{\mathrm{wt}(x)}, \qquad \sum_{x \in \mathcal{S}} 2^{k - \mathrm{wt}(x \mid \tau x)}.$$

In every class $d = 0$ occurs exactly at $(0,0)$ and (u,v). Subtracting these two cases four times, then using disjointness, gives $2q + 2^a + 2^{k-a} - 5$ unprotected AND clues. Therefore

$$|F| = q^2 - 3^k - 2q - 2^a - 2^{k-a} + 5. \tag{10.6}$$

How trades force additional clues. If a trade avoids F, every defining $D \subseteq U_T$ must meet its projection onto $U_T \setminus F$. Otherwise the trade's mate gives a different completion of D. Pairwise disjoint projections consequently force distinct additional clues, even if the full trades intersect at holes. We next give all the trade families used for the stronger bound. This is a bound for *every* defining subset of U_T, exactly the quantifier needed for b in (10.1).

10.3 Three families of pairs and the prolongation trades

The four corner parameterizations also identify all old AND clues outside F. Use the following position names:

$$\begin{aligned} D_r &= (r, r) && (r \neq 0), \\ H_d &= (u \oplus d, v) && (0 \neq d \subseteq v), \\ J_e &= (u, v \oplus e) && (0 \neq e \subseteq u), \\ T_r &= (r, \tau r) && (r \notin \{0, u\}). \end{aligned}$$

These types are disjoint and exhaust $Q \setminus F$. In addition, U_T contains T_0, T_u, so it contains every transversal position. For example, the fourth-corner hit with $x = 0$ gives exactly the nonzero diagonals D_r; with $x = u$, no nonzero d is possible. The second and third corner hits give the H and J arms; the first gives the remaining T_r.

1. Complementary diagonal pairs. For every unordered pair $\{r, s\}$ with $r \oplus s = M$, $r, s \neq 0$, except $\{u, v\}$, use rows r, s and columns r, s. In the XOR square this is

$$\begin{pmatrix} 0 & M \\ M & 0 \end{pmatrix}.$$

The off-diagonal positions are AND holes. If one lay on T, its row would belong to $\mathcal{S}$; since both coordinates are nonzero, it must be (u, v), excluded here. The diagonal positions do not lie on T, because $p = I + \tau$ has no nonzero kernel. Thus the intercalate survives, and its projection onto U_T is exactly $\{D_r, D_s\}$. There are $q/2 - 2$ pairs.

2. Complementary tail-arm pairs. For nonzero $d, e \subseteq v$ with $d \oplus e = v$, take rows $u \oplus d, u \oplus e$ and columns $0, v$. Their differences in both coordinates are v, so this is an intercalate. Its column-v positions are H_d, H_e; its column-0 positions are AND holes. No position lies on T: the unique transversal row in column 0 is 0, and that in column v is u, neither of which is used. The projection is $\{H_d, H_e\}$. There are $2^{k-a-1} - 1$ such pairs.

3. Complementary head-arm pairs. For nonzero $e, f \subseteq u$ with $e \oplus f = u$, use rows $0, u$ and columns $v \oplus e, v \oplus f$. The coordinate differences are u. The row-u entries are J_e, J_f and the row-0 entries are AND holes. The transversal columns in these rows are $0, v$, both absent. Thus this intercalate survives and projects to $\{J_e, J_f\}$. There are $2^{a-1} - 1$ such pairs.

The three families have mutually disjoint projections in the distinct D, H, J types, and together force

$$\eta = \left(\frac{q}{2} - 2\right) + \left(2^{k-a-1} - 1\right) + \left(2^{a-1} - 1\right) = \frac{q}{2} + 2^{a-1} + 2^{k-a-1} - 4 \tag{10.7}$$

clues outside F. The counts include zero when a head or tail has one bit; for $k = 2$ all three families are empty.

4. The q prolongation intercalates. For every $r < q$, rows r, ∞ and columns $\tau r, \infty$ contain

$$\begin{pmatrix} q & p(r) \\ p(r) & q \end{pmatrix}.$$

All four positions belong to $U_T \setminus F$. Put $Z = (\infty, \infty)$ and

$$E_r = \{T_r, (r, \infty), (\infty, \tau r), Z\}. \tag{10.8}$$

Distinct E_r intersect only at Z. Let V consist of all transversal and border positions. The pair projections above avoid V entirely. The next three families lie in V on projection and strengthen the requirement to hit all q sets E_r.

10.4 Cycle trades and the integer covering argument

Put $A = \tau p^{-1}$ and $B = p^{-1}$. For a permutation X fixing 0, let $c(X)$ be its number of cycles on the nonzero vectors, and set $c = c(A) + c(B) + c(\tau)$.

5. Bottom-border cycles. In rows $0, \infty$, at a nonzero column z, the entries are z and $p\tau^{-1}(z) = A^{-1}(z)$. For each nonzero orbit of A, swap these two rows' entries in the columns of that orbit. Both row symbol sets are preserved because the orbit is invariant under A^{-1}, and each column merely exchanges two symbols. This is a trade. Its row-0 positions are AND holes outside T, so its projection consists exactly of the bottom-border cells in that orbit. There are $c(A)$ disjoint projections.

6. Right-border cycles. In columns $0, \infty$, at nonzero row r, the entries are $r, p(r)$. Swapping these entries in every row of a nonzero orbit of $B = p^{-1}$ preserves column symbol sets and gives a trade. Its column-0 positions are holes outside T. The projections are exactly the nonzero right-border orbits, giving $c(B)$ disjoint sets.

7. Pure transversal cycles. For a nonzero orbit O of τ, use rows $R = \{u \oplus w : w \in O\}$ and swap symbols q, M in the two positions

$$(r, \tau r), \qquad (r, r \oplus M) \qquad (r \in R).$$

Their column sets coincide, since

$$\tau(u \oplus w) = v \oplus \tau w, \qquad (u \oplus w) \oplus M = v \oplus w.$$

The two positions are distinct and the M entry has not been displaced: otherwise $p(r) = M = p(u)$ would give $r = u$, excluded by $w \neq 0$. The M positions are AND holes, because $r \oplus M$ is the complement of r, and they are outside T. Thus the projection consists only of T_r for $r \in R$. Include every nonzero τ orbit, including the one containing u. These $c(\tau)$ projections partition the transversal positions other than T_u and include T_0.

The projections in families 5–7 are mutually disjoint: they occupy the nonzero bottom border, nonzero right border, and transversal, respectively. They avoid F, the pair projections, and Z. Every defining subset $D \subseteq U_T$ must therefore take at least c positions from V.

One more position is forced. The maps τ, A, B have no nonzero cycles of lengths one or two. Indeed

$$I + \tau = p, \qquad I + A = B, \qquad I + B = A$$

are invertible. A fixed vector lies in the kernel of $I + X$; a vector of period two lies in the kernel of $(I + X)^2 = I + X^2$, also impossible. Every nonzero orbit has at least three elements. Consequently

$$3 \leq c \leq 3 \left\lfloor \frac{q-1}{3} \right\rfloor \leq q - 1 < q. \tag{10.9}$$

If D used exactly c cells of V, it would use one from each cycle projection and none elsewhere in V, in particular none at Z. Each selected cell then hits at most one E_r, since the E_r intersect only at Z. It cannot hit all q of them because $c < q$. Fewer than c cells already miss a cycle. Hence $|D \cap V| \geq c + 1$.

Combining this with the protected core and disjoint projected pairs gives the required universal restriction bound

$$\begin{aligned} |D| \geq b_{k,a} &= |F| + \eta + c + 1 \\ &= q^2 - 3^k - \frac{3q}{2} - 2^{a-1} - 2^{k-a-1} + 2 + c, \qquad D \subseteq U_T \text{ defining.} \end{aligned} \tag{10.10}$$

This proof uses explicit trades and integer counting, without numerical optimization. Minimal pruning produces a critical set with at least this many clues, but the stronger displayed statement is what enables transfer.

10.5 The resulting family and exact-cycle refinements

Substitute $s = q^2 - 3^k$, $h = 2$, and (10.10) in the ternary lemma. For every $k \geq 2$ and $1 \leq a < k$, with the mapping in (10.2), the unconditional result is

$$\boxed{\begin{aligned}\operatorname{lcs}(3q+1) \geq 7q^2 - 5 \cdot 3^k - \frac{3q}{2} - 2^{a-1} - 2^{k-a-1} - 2 \\ + c(\tau p^{-1}) + c(p^{-1}) + c(\tau), \qquad q = 2^k.\end{aligned}} \tag{10.11}$$

Since each cycle count is at least one, the balanced choice $a = \lfloor k/2 \rfloor$ yields the uniform formula

$$\boxed{\begin{aligned}\operatorname{lcs}(3 \cdot 2^k + 1) \geq 7 \cdot 4^k - 5 \cdot 3^k - 3 \cdot 2^{k-1} \\ - 2^{a-1} - 2^{k-a-1} + 1, \qquad a = \lfloor k/2 \rfloor.\end{aligned}} \tag{10.12}$$

The powers of two penalizing the split are minimized by a balanced split. The exact-cycle expression may favor another split: every term must then be computed using that same a and that same τ. Cycle enumeration is an exact finite permutation calculation, not a fitted estimate.

q	$3q+1$	Balanced uniform	Balanced exact	Best exact over splits
4	13	60	60	60
8	25	299	299	299
16	49	1360	1366	1366
32	97	5900	5900	5903
64	193	24924	24938	24938
128	385	103550	103550	103550
256	769	425548	425614	425614

At $q = 32$, split $a = 1$ gives the value 5903. The last column maximizes only the proved expression over the finitely many splits, not over Latin squares or critical sets. The small values are not record claims; the separate $16m + 1$ construction, for example, gives the stronger 1540 at order 49. For a combined lower envelope, take the maximum of all valid constructions and verified records.

Scope of the construction. The proof prescribes a finite defining template and a lower bound that survives any pruning to an inclusion-minimal defining subset. It is an existence construction with a finite algorithm: test deletions, retaining a clue exactly when deletion destroys uniqueness. Every protected clue remains essential, and every retained clue remains essential under further deletions. There is no assertion that the initial template is critical, that this pruning is efficient at arbitrary order, or that it produces a set of exactly the displayed size.

The proof assumes neither dyadic optimality $\operatorname{lcs}(2^k) = 4^k - 3^k$ nor a conjectured product inequality. The bitwise set Q is an explicit critical set of that size, which is all that is needed. The two places where criticality matters are its private intercalates and the six unchanged product blocks. The border contribution uses a universal trade-cover bound, not optimality of the prolonged square.

In particular, a verified large critical set at order $q + 1$ cannot simply replace $b_{k,a}$ in (10.1). Such a record is an existence statement. The transfer requires a statement about every defining subset of the specific U_T induced in the enlarged $(2, 2)$ subsquare. This distinction is why the full protected-core and trade-cover argument has been included above.

The exact-cycle computations provide individual rigorous improvements to (10.12). They do not establish a closed formula for the cycle counts in all dimensions, or the sharper proposed family $\operatorname{lcs}(2^k + 1) \geq 4^k - 3^k + 2^{k+1} - 7$. The ternary transfer remains valid independently of the status of that proposal.

10.6 Which other completed products can be prolonged?

A sharp parity obstruction for the cyclic-factor $3m$ route. The product of cyclic tables $C_3 \times C_m$ has a transversal if and only if m is odd. For odd m, the diagonal gives symbols

$$(2i \bmod 3,\ 2r \bmod m), \qquad 0 \le i < 3, \quad 0 \le r < m,$$

which are distinct because multiplication by 2 is a permutation in both coordinates. Thus the diagonal is a transversal.

For even m, suppose any transversal existed and put $S = \sum_{r=0}^{m-1} r = m(m-1)/2$. In the row labels of a transversal, each second coordinate occurs three times. The same is true of its column labels and, by the transversal property, of its symbols. Summing the second-coordinate addition law gives

$$3S \equiv 3S + 3S \pmod m, \qquad \text{hence } 3S \equiv 0 \pmod m.$$

But for even m, $S \equiv m/2 \pmod m$, and therefore $3S \equiv m/2 \not\equiv 0 \pmod m$, a contradiction. This excludes *all* transversals, not only the diagonal. The notation refers to the product of the two cyclic tables; its group need not itself be cyclic when 3 divides m.

It follows that this cyclic-factor construction cannot reach an odd order $3m+1$ by ordinary transversal prolongation: the required even m is exactly the obstructed case. There is no conflict with (10.12). There the even-order factor is the elementary abelian XOR square, not the cyclic table C_{2^k}, and its explicit complete mapping supplies the needed transversal for $k \ge 2$.

Two successive prolongations for odd abelian groups. Let G be an abelian group of odd order greater than one, with addition table $L(x, y) = x + y$. Multiplication by 2 is a permutation, so the diagonal (x, x) is a transversal. Prolong along it, using a new index and symbol ∞.

Choose a nonzero $a \in G$. In the prolonged square the cells

$$\{(x, x + a) : x \in G\} \ \cup\ \{(\infty, \infty)\} \tag{10.13}$$

form another transversal. The old cells avoid the first diagonal because $a \neq 0$, and so their entries remain $2x + a$. They meet every old row and column once and contain every old symbol once. The new corner supplies the remaining row, column and symbol. A second prolongation therefore produces a completed Latin square of order $|G| + 2$.

Taking $G = C_3^k$ gives orders $3^k + 2$, including $11, 29, 83, \ldots$ for $k \ge 2$; the same argument covers the smaller case $|G| = 3$. Taking $G = C_3 \times C_m$ for odd m gives orders $3m + 2$. These statements concern completed Latin squares and explicit transversals. They do not, by themselves, supply critical-set sizes after the second prolongation.

The reason for separating these conclusions is concrete. Prolongation changes entries on a transversal and introduces a new border. A private trade witnessing an old clue's necessity may meet that transversal and cease to be a trade of the new square. Prescribing all changed positions and the border can preserve unique completion, but a large defining set can still have a much smaller critical subset. The protected-trade and restricted-subsquare counts in the ternary proof supply the additional argument needed for a lower bound on lcs, beyond the existence of a transversal.

10.7 Finite $8m$ transversals and independent checks

The $8m$ observations are finite certificates. The absence of a transversal in the outer C_2 table does not imply its product with another Latin square lacks one. Here are explicit transversals in $C_2 \times L_m$ for $m = 1, 3, 5$, where L_m is the thesis square of order $4m$. To fix the indexing completely, write

$$L_m((R,i),(C,j)) = m(R \oplus C) + f_{R,C}(i,j), \quad 0 \le R, C < 4, \quad 0 \le i, j < m,$$

where residues are taken modulo m and

$$(f_{R,C}) = \begin{pmatrix} A & t & t & B \\ A & t & B & t \\ A & B & t & t \\ B & A & A & A \end{pmatrix}, \qquad A = i - j, \quad t = j - i, \quad B = i + j.$$

For the product, rows and columns are indexed by $a(4m) + Rm + i$, with $a \in \{0, 1\}$, and entries are $(a \oplus b)(4m) + L_m((R,i),(C,j))$. Each vector below gives the column selected in successive rows.

$$\begin{aligned} t^{(1)} &= (6,5,1,2,3,0,4,7), \\ t^{(3)} &= (14,16,8,19,4,17,15,13,5,7,11,3, \\ &\qquad 1,12,22,20,9,21,2,23,10,18,6,0), \\ t^{(5)} &= (4,5,20,3,9,31,27,15,38,36,6,19,28,7, \\ &\qquad 33,30,21,10,35,16,14,22,2,29,39,34,17, \\ &\qquad 11,37,1,32,24,25,23,18,13,8,26,12,0). \end{aligned}$$

In each case both the selected columns and the selected symbols are permutations of $0, \ldots, 8m-1$. These are directly checkable certificates for prolongations $8 \to 9$, $24 \to 25$, and $40 \to 41$. They are not an all-m transversal formula and do not assert new critical-set sizes.

Independent audit of the ternary transfer. A separate agent reconstructed the transfer proof and found no gap under its stated universal hypothesis on b. The audit checked the outer two-clue set and its explicit deletion witnesses, the block completion order, both protected-clue classes, the exact enlarged $(2,2)$ subsquare, and the disjointness of all three counted contributions. It also checked that the already audited projected-trade-cover template is exactly U_T, so its universal restriction bound has the required scope. The coordinate proofs reproduced here establish the all-order statements; the following finite checks supplement them.

Check	Verified scope
Ternary product and private trades	$q = 4, 8, 16, 32, 64$: Latin property, product transversal, explicit completion order, every private witness in both protected classes, and entrywise agreement of the enlarged subsquare and its partial template.
Projected trade cover	All 28 dimension/split pairs for $2 \le k \le 8$: the actual trade mates, avoidance of F, exact projections, disjointness and incidence constraints.
Exact cycles and bound arithmetic	All 66 dimension/split pairs for $2 \le k \le 12$.
Other prolongations	The three $8m$ vectors above; two successive prolongations of $C_3 \times C_m$ for all twelve odd $m = 1, 3, \ldots, 23$.

The delivered files under `lcs_other_prolongation/` include `ternary_audit/independent_audit.txt`, `ternary_audit/check_ternary_transfer.py`, its JSON results, and `check_transversals.py`. The previously audited binary dependency proofs and audit comments are in `dependency_proofs/`; their explicit trade checker is `lcs_dyadic_plus_sharp_progress/check_projected_cover.py` in the delivered outputs. These checks use exact arithmetic; the ternary and transversal scripts require only the Python standard library. No claim of literature priority, optimal critical-set size, or a general unproved product-preservation principle is made.

11 Three simultaneous prolongations: orders $4m+3$

For every **odd** $m \geq 7$, the following new family and its safe lifts are proved:

$$\boxed{\operatorname{lcs}(4m+3) \geq S_m := \frac{21m^2 - 19m - 98}{2}.} \tag{11.1}$$

The same critical subset has at least $T_m = 7m^2 - 8m - 52$ clues with private intercalates. Consequently the previously audited binary lifts give

$$\operatorname{lcs}(8m+6) \geq 44m^2 - 3m - 141, \qquad \operatorname{lcs}(16m+12) \geq \frac{371m^2 + 183m - 792}{2}.$$

These follow from $q^2 + 2S_m + T_m$ and $7q^2 + 3S_m + 6T_m$, respectively, where $q = 4m+3$. All coefficients are positive: the unknown actual size of the extracted critical subset may exceed S_m. The family covers $n \equiv 7 \pmod 8$. It **does not cover 11 or 19**. In terms of the final order, (11.1) is $(21n^2 - 202n - 1151)/32$, for $n \equiv 7 \pmod 8$, $n \geq 31$.

Seed and three transversals. Use the order-$4m$ thesis square L and defining set P described on page 24, with masks H, J, G, F as there. For each $b \in \{0, -1, -2\}$ take the cells

row	column	symbol block and residue
$(0, i)$	$(0, m-1-i+b)$	$(0, 2i+1-b)$
$(1, i)$	$(2, i+b)$	$(3, 2i+b)$
$(2, i)$	$(3, -i+b)$	$(1, -2i+b)$
$(3, i)$	$(1, -i+b)$	$(2, 2i-b)$

with residues modulo m. Multiplication by two is invertible, so each set is a transversal. Different b give different columns in every row, so the three transversals are disjoint. They meet only blocks $00, 12, 23, 31$.

Simultaneous prolongation and contraction. For each transversal T_a, introduce one new row, column and symbol ∞_a. Replace its old entry s at (r, c) by ∞_a, and put s at (r, ∞_a) and (∞_a, c). Fill the new 3×3 corner with a Latin square on the three new symbols. The result L^+ is Latin: each old row/column exchanges three distinct old symbols for the three new ones, and recovers exactly the displaced symbols on the border; each new row/column receives every old symbol once.

Prescribe transformed P, all transversal cells and the entire border; call this defining set U. Indeed, any completion of U has all new symbols already fixed in each old row and column. Deleting the border and restoring the displaced symbols produces a Latin completion of P, necessarily L. Thus the original completion is L^+. An old private trade avoiding the three transversals survives unchanged and meets U only at its target clue. Every defining subset $C \subseteq U$ must therefore contain every clue protected by such a trade.

11.1 Counting the surviving private trades

Seven triangular blocks are untouched, and contribute $7m(m-1)/2$ protected clues. For the seven full blocks, a target $(R,i;C,j)$ has a private intercalate with other row (S,x) and other column (D,y):

(R,C)	(S,D)	residue condition
$11,22$	00	$x=j-i+y$
33	00	$x=i-j+y$
13	02	$x=i-j+y$
23	01	$x=i-j+y$
31	20	$x=i-j+y$
32	10	$x=i-j+y$

All residues here are modulo m. For $11,22$, take $(x,y)=(0,0)$ when $i=j$; take $((j-i) \bmod m, 0)$ when $j=0$ or $0<i<j$; otherwise take $(0,(i-j) \bmod m)$. In every other block the canonical choice is $(i-j,0)$ for $i \geq j$, and $(0,j-i)$ for $i<j$. Substitution in the square and mask shows that the opposite symbols agree and precisely the target is in P.

The only potentially affected corners are the diagonal in 00 for $11,22,33$; the horizontal in 12 for 13; the vertical in 12 for 32; and the target itself for $23,31$. In $11,22$, the forbidden 00 condition is $x+y \equiv m-1, m-2, m-3$. Six targets have exactly one coordinate zero; six more have positive unequal coordinates with difference $m-2$ or $m-3$. Hence each loses 12. In 33, differences $m-1, m-2, m-3$ contribute $2+4+6=12$. Each of $23,31$ loses exactly $3m$ targets.

Flexible witnesses in block 13. Put $h=(m-1)/2$. When $i \geq j$, the private choice is $y=0, x=i-j$; it fails at the six targets $i \in \{0,1,2\}$, $0 \leq j \leq i$. When $i<j$, all admissible choices are

$$j-i \leq y \leq m-1-i, \qquad x=y-(j-i).$$

They are private because the two G corners have sum $i+y<m$ and the H corner has $y>x$. A choice fails only if $y-i \equiv 0,-1,-2$. The interval has $m-j$ consecutive choices, so all can fail only for $j \geq m-3$. The complete failure list is

$j=m-1$	$i=0,h,h+1$
$j=m-2$	$i=0,h$
$j=m-3$	$i=h$

obtained by substituting the one-, two- and three-element intervals into the three forbidden residues. For $m \geq 7$ these are six distinct targets. Including the first six, block 13 loses exactly 12.

11.2 Completing the protected count

Flexible witnesses in block 32. The third corner is a J hole, requiring $x = 0$ or $x > y$. For $i < j$, this forces $x = 0, y = j - i$; failures have $j = m - 1, m - 2$, giving $2m - 3$ targets. For $i = j$, it forces $x = y = 0$ and fails for $j = 0, m - 1, m - 2$, giving three more. For $i > j$, every private choice is

$$0 \le y \le m - 1 - i, \qquad x = i - j + y.$$

The affected vertical corner is forbidden when $x - j \equiv 0, 1, 2$. The interval has $m - i$ consecutive choices; all fail precisely in the following four cases:

$$i = m - 1 : \ j = h, h - 1; \qquad i = m - 2 : \ j = h - 1; \qquad i = m - 3 : \ j = h - 1.$$

For example, the first row follows by solving $m - 1 - 2j \equiv 0, 1, 2$ and excluding $j = i$. The two- and three-choice intervals must start at residue 0 or 1, and at 0, respectively. Thus block 32 loses $(2m - 3) + 3 + 4 = 2m + 4$ targets.

Full block	Excluded targets	Protected targets
$11, 22, 33$ (each)	12	$m^2 - 12$
13	12	$m^2 - 12$
32	$2m + 4$	$m^2 - 2m - 4$
$23, 31$ (each)	$3m$	$m^2 - 3m$
Total	$8m + 52$	$T_m = 7m^2 - 8m - 52$

Together with the untouched triangular clues, the protected set F has

$$|F| = 7m^2 - 8m - 52 + \frac{7m(m-1)}{2} = \frac{21m^2 - 23m - 104}{2}. \tag{11.2}$$

All these clues belong to every defining subset of U; the T_m full-block clues retain their private intercalates in any such subset.

Canonical fallback and the small exception. Using only canonical witnesses gives losses $(3m + 9)/2$ in block 13 and $(7m - 3)/2$ in block 32, hence total full-block loss $11m + 39$. The same extraction and cover argument below gives $(21m^2 - 25m - 72)/2$ for every odd $m \ge 5$. For $m = 5$, flexible block 13 loses 11 rather than 12; the protected count is 154 and the final guarantee is $154 + 13 = 167$ at order 23. These smaller values are retained as valid functions but are dominated by existing bounds at their first orders.

11.3 Why the border contributes at least $2m + 3$

The prolonged copies of old blocks 00 and 12, including their new rows and columns, are Latin subsquares S_0, S_1 of order $m + 3$. They intersect exactly in the new 3×3 corner. Their union is disjoint from F. A defining subset $C \subseteq U$ restricts to a defining set on each S_i: an alternative filling of a subsquare on the same symbol alphabet could otherwise replace it in the whole square.

Auxiliary lemma. Every defining set of a Latin square of order $t \geq 4$ has at least t clues. Fewer than $t - 1$ leave two rows empty, which can be swapped. Exactly $t - 1$ clues must occupy distinct rows, columns and symbols, since two empty labels of any type can be swapped. Relabel these clues as (i, i, i) for the $t - 1$ old labels, with one unused label ∞. A unique completion would be invariant under every simultaneous permutation of the old labels. If $t \geq 5$, the stabilizer of distinct old i, j permutes at least two other old labels. Thus the entry at (i, j) can only be i, j or ∞. The diagonal excludes i, j, forcing ∞ in every off-diagonal old cell, which repeats within a row. For $t = 4$, symmetry forces all off-diagonal old cells uniformly to use ∞, or uniformly the third old label. The first repeats ∞ in a row; the second forces ∞ in the last column of all three old rows. Both contradict Latinness. This proves the lemma.

Write $D_i = C \cap S_i$, $D = C \cap (S_0 \cup S_1)$ and $K = C \cap (S_0 \cap S_1)$. Each D_i covers all but at most one of its rows, columns and symbols. The union consequently covers at least $2m + 1$ of its $2m + 3$ rows, so $|D| \geq 2m + 1$. If $|D| = 2m+1$, there is one clue per occupied row, so $|K| \leq 3$. But the auxiliary lemma gives $|K| = |D_0|+|D_1|-|D| \geq 2(m + 3) - (2m + 1) = 5$, a contradiction.

If $|D| = 2m + 2$, at most one clue repeats an occupied row, so $|K| \leq 4$. The lemma gives the reverse inequality. Therefore $|K| = 4$ and $|D_0| = |D_1| = m + 3$, with exactly $m - 1$ clues outside the corner in each. To cover at least $m + 2$ rows, columns and symbols, the corner must cover all three new labels of each type, and the $m - 1$ other clues must have distinct old rows, columns and symbols. This excludes both off-corner border cells and transformed transversal cells (the latter have new symbols). In S_0, since $C \subseteq U$, these remaining clues must therefore lie in the unchanged mask $1 \leq j \leq i \leq m - 1$. Distinct rows and columns force both label sets to be $\{1, \ldots, m - 1\}$; equality of their sums forces $j = i$ throughout. All entries then have old symbol zero, contradicting distinctness. Hence $|D| \geq 2m + 3$.

Choose an inclusion-minimal defining subset of finite U. It is critical, contains F, and has at least $2m + 3$ further clues in the disjoint union $S_0 \cup S_1$. Adding this to (11.2) proves (11.1). The condition $C \subseteq U$ is essential to the triangular-mask argument; no claim about arbitrary defining sets is used.

11.4 Audit, gains and the unresolved orders 11 and 19

Independent review. One agent independently reconstructed the Latin square, transversals, private rectangles and all-parameter counting argument. A second independently reconstructed the construction and checked the count and subsquare cover. The second proposed the stronger $2m + 3$ cover; the first then independently audited that new argument, including the order-four case of the auxiliary lemma and the exclusion of border and transversal clues. The parent also checked the proof and safe-lift arithmetic. Both independent coordinate programs passed for every odd $m = 3, 5, \ldots, 41$ on the applicable domains. The symbolic argument, rather than those finite checks, establishes the all-parameter theorem. No independently verified *individual* critical set of exactly S_m clues is asserted: the proof guarantees a critical subset of at least that size.

The audit files are retained under `lcs_four_m_plus_three/independent_audit/` and `lcs_four_m_plus_three/second_audit/`. In particular, `subsquare_union_2m_plus_three_audit.txt` audits the final refinement. Earlier intermediate bounds in those audit files are superseded by the final formula, with their historical scope preserved.

Order	Previous proved	Earlier proved	Gain	Source
47	1081	1117	36	Three-transversal family
55	1485	1602	117	Three-transversal family
71	2485	2824	339	Three-transversal family
79	3081	3561	480	Three-transversal family
142	12496	12524	28	Private-intercalate binary lift
174	18792	19200	408	Lift of the order-87 protected profile
247	31578	38442	6864	Three-transversal family

Relative to the report immediately before this update, this odd-m family and its safe lifts improve the proved lower bound at 30 orders through 257. The order-87 family size is below the previous proved bound, but its guaranteed private-intercalate count still improves the lift at 174. Thus retaining profiles of nonmaximal seeds can matter.

Targeted tests at 11 and 19. Three parallel transversals in XOR squares of orders 8 and 16 were also prolonged and pruned with an exact completion oracle. The tested order-19 sets reached 195, below 245. A different explicit near-transversal repair extends Donovan's order-$2s$ square to order $2s + 1$ for odd s: a partial transversal missing two rows, columns and symbols is completed by a 3×3 Latin repair. Its Latinness and defining-set contraction were independently audited, but the tested pruned sizes reached only 55 at 11 and 186 at 19. These experiments do not improve 74 or 245 and do not prove that better choices are impossible. They are not added to the record curve.

For even $m = 2h$ with odd h, a Chinese-remainder construction produces three thesis transversals by using an outer XOR square of order eight. For $h \geq 3$ it supplies three distinct inner shifts and touches more triangular blocks. Its all-parameter protected-count pattern remains unproved; the independently replayed finite instances on the next page are included as discrete bounds. The resulting orders satisfy $4m + 3 = 8h + 3 \equiv 11 \pmod{16}$ and are at least 27. Order 11 needs distinct outer maps because $h = 1$ has only one inner shift; order 19 has $m = 4$ and requires a different even-parameter template.

11.5 Independently checked finite constructions for even m

Write $m = 2h$, with $h \geq 3$ odd. Identify an inner residue by its parity and its residue modulo h. Index the eight outer rows by $r = 2R + p$, where $R \in \{0, 1, 2, 3\}$ and p is the inner row parity. Take the permutation

$$\pi = (1, 6, 2, 5, 7, 0, 4, 3).$$

Both $\pi(r)$ and $r \oplus \pi(r)$ are permutations. If $\pi(2R + p) = 2C + p'$, choose the column parity p' and column residue modulo h to be $-i + b$ in an A or A^{T} block, and $i + b$ in a B block. The Chinese remainder theorem supplies a unique column residue modulo m. For each $b = 0, -1, -2$ this is a transversal: columns are bijective, the outer symbols form XOR of the outer labels, and the inner symbol residue has slope ± 2, invertible modulo odd h. The shifts give three distinct cells per row because $h \geq 3$.

These transversals meet blocks $00, 03, 11, 12, 20, 23, 31, 32$, leaving five triangular blocks $01, 02, 10, 21, 30$ untouched. Apply simultaneous prolongation and the same defining set U. Count the canonical private intercalates that avoid the three transversals, and add the five unchanged triangular critical sets. The following are **exact finite counts**, independently reconstructed and checked against the actual prescribed U.

m	Final order	Private intercalate clues	Protected total	Bound K_{CRT}
6	27	141	216	218
10	43	527	752	754
14	59	1137	1592	1594
18	75	1971	2736	2738
22	91	3029	4184	4186
26	107	4311	5936	5938
30	123	5817	7992	7994
34	139	7547	10352	10354
38	155	9501	13016	13018
42	171	11679	15984	15986
46	187	14081	19256	19258
50	203	16707	22832	22834

The protected set is disjoint from the new 3×3 Latin subsquare. Every defining subset of U restricts to a defining set there and must use at least two corner clues: zero or one would leave two rows empty, allowing a swap. This proves the final column by adding two to the protected count.

The first auditor constructed and checked the transversals; a second reconstructed them directly from the two congruences, checked the prolonged Latin square, and verified every counted private rectangle and the untouched triangles. The second also audited the two-clue corner addition. The independent replay is `second_audit/check_even_crt.py`, with its full counts in `second_audit/even_crt_checks.json`. The protected totals agree with $(19m^2 - 36m - 36)/2$ at the twelve listed parameters. That polynomial has **not** been proved for all m; K_{CRT} is zero outside this finite table. The bound at 91 improves 4095 to 4186, and that at 107 improves 5671 to 5938. These finite cases improve seven further orders. Together with the odd-m family and its safe lifts, this update improves 37 proved lower bounds through order 257 relative to the preceding report.

11.6 A further cover refinement from the minimum-critical-set bound

Horak, Aldred and Fleischner, *Completing Latin Squares: Critical Sets*, *Journal of Combinatorial Designs* 10 (2002), 419–432, Theorem 1.1 (printed p. 420; DOI `10.1002/jcd.10028`), prove that every critical set of order $t \geq 8$ has at least $\lfloor (4t - 8)/3 \rfloor$ clues. The statement uses a *floor*. Every defining set contains a critical subset, so the same lower bound holds for defining sets. This published theorem is an external input to the following new refinement.

For the two enlarged subsquares S_0, S_1 of order $m + 3$, put

$$B_m = \max\left\{ m + 3, \left\lfloor \frac{4m + 4}{3} \right\rfloor \right\} \qquad (m \geq 5).$$

The first term follows from the preceding defining-set lemma; the second from the cited theorem. For any defining subset $C \subseteq U$, write $D_i = C \cap S_i$, $D = D_0 \cup D_1$, $K = D_0 \cap D_1$, and $d = |D|$, $k = |K|$. Each D_i defines its subsquare, so $|D_i| \geq B_m$. The union occupies at least $2m + 1$ rows. Its k corner clues occupy at most three rows, and the other $d - k$ clues at most $d - k$ further rows. Consequently

$$k \leq d - 2m + 2, \qquad k \leq 9, \qquad 2B_m \leq d + k,$$

and hence

$$d \geq B_m + m - 1, \qquad d \geq 2B_m - 9.$$

Equality in the first inequality is impossible. If $d = B_m + m - 1$, both restrictions have size B_m, the corner has $B_m - m + 1$ clues, and each restriction has exactly $m - 1$ clues outside the corner. Those $m - 1$ clues must occupy $m - 1$ distinct *old* rows: otherwise two old rows would be empty in that defining restriction. The same argument applies to old columns and old symbols, since corner clues supply none of those labels. Thus all non-corner clues have old coordinates and old symbols, with each type pairwise distinct. In S_0 they are unchanged H clues, $1 \le j \le i \le m - 1$, because $C \subseteq U$; changed transversal cells have new symbols and border cells have a new coordinate. Distinct rows and columns force all indices $1, \dots, m - 1$ to occur. Equality of their sums then forces $j = i$ in every clue. All these clues have old symbol zero, contradicting distinct symbols. Therefore

$$\boxed{|C \cap (S_0 \cup S_1)| \ge \max\{B_m + m,\, 2B_m - 9\}.}$$

Because $B_m \ge m + 3$, this includes the previous $2m + 3$ estimate. The proof also applies when the hypothetical equality would require more than nine corner clues: that equality is then already impossible.

Combining with the flexible protected core gives the refined family

$$\boxed{\operatorname{lcs}(4m + 3) \ge \frac{21m^2 - 23m - 104}{2} + \max\{B_m + m,\, 2B_m - 9\}} \qquad (m \ge 7 \text{ odd}).$$

For example, this gives 717, 1119 and 1604 at orders 39, 47 and 55, respectively; other constructions or records may be stronger at a particular order. The same cover may be added to the canonical core for odd $m \ge 5$. The independently counted $m = 5$ flexible core still gives $154 + 13 = 167$ at order 23.

An independent agent checked the published statement directly in the local PDF and reconstructed the counting and equality argument above; its complete review is saved as `horak_subsquare_cover_independent_audit.txt`. The review does not claim a new proof of the cited published theorem or literature priority.

12 The Donovan transversal obstruction and a near-transversal repair

Write the Donovan square of order $2m$ with row (a, i) and column (b, j), where $a, b \in \{0, 1\}$ and $0 \le i, j < m$, as

$$L((a, i), (b, j)) = (a \oplus b,\ f_b(i, j)), \qquad f_0(i, j) = i - j, \quad f_1(i, j) = j - i \pmod m.$$

The seed P prescribes the full block 01 and the cells $i < j$ in each of the other three blocks. Donovan's published theorem says that P is critical, with $(5m^2 - 3m)/2$ entries. This is the precise seed used here.

12.1 Exactly when the raw seed square has a transversal

If a transversal has x entries in block 00, its quadrant counts must be $x, m - x, m - x, x$ by row and column coverage. It would use $2x$ symbols of class zero, whereas that class has m symbols. Hence m must be even. In particular the raw squares of orders 10 and 18 cannot be prolonged along a transversal to orders 11 and 19.

Conversely let $m = 2h$. For $0 \le i < h$ select

$$((0, i), (0, 2i)), \quad ((0, h + i), (1, 2i)), \quad ((1, i), (1, 2i + 1)), \quad ((1, h + i), (0, 2i + 1)).$$

Rows and columns occur once. Their symbols, respectively, have residues $-i$, $i - h$, $i + 1$, and $h - i - 1$; the first and third classes have outer symbol zero and together cover all residues, while the second and fourth have outer symbol one and together do the same. This is a transversal. Thus the criterion m even is both necessary and sufficient.

Keeping only the old clue positions after prolongation is not defining: the old last row in block row 1 and the added row are both empty. Swapping these rows gives a second completion. Adding the transversal and the full border restores a defining set, but its size is not a critical-set lower bound. Switching the old corner intercalate can enable transversals in particular odd-m cases; no all-odd-m theorem follows from that experiment. At $m = 3$ the corner-switched square has no transversal, as checked by all $6! = 720$ column permutations. Independently checked pruned examples include sizes 54 at order11 and187 at order19; neither is a record.

12.2 A repair that is Latin for every odd parameter

Now put $m = s \ge 3$ odd, $h = (s - 1)/2$, and $q = 2s$. Use integer row and column labels $as + i$, $bs + j$ for the preceding square. Define

$$T = \{(as + i,\ bs + (2i \bmod s)) : a \in \{0, 1\},\ 1 \le i < s, \quad b = a\ (i \le h),\ b = 1 - a\ (i > h)\}.$$

It uses precisely the old rows and columns other than $0, s$. For $1 \le i \le h$ its entries lie in blocks $00, 11$ and have symbol residues $-i, +i$ in class zero. For $h < i < s$ they lie in blocks $01, 10$ and have residues $+i, -i$ in class one. These are exactly the old symbols other than $0, s$, without repetition. Thus T is a near-transversal missing two rows, two columns and two symbols.

Replace each selected entry by the new symbol q, and put its displaced symbol into the corresponding new row and new column. On rows and columns $0, s, q$, in that order, prescribe the special array

$$\begin{pmatrix} 0 & s & q \\ s & q & 0 \\ q & 0 & s \end{pmatrix}.$$

All other old entries stay unchanged; call the result M. Let U prescribe the values of M at the position set

$$P \cup T \cup (\{0, s, q\} \times \{0, s, q\}) \cup (\{q\} \times \{0, \dots, q\}) \cup (\{0, \dots, q\} \times \{q\}).$$

Here P denotes the old clue positions, with their values now read from M.

For an old row or column other than $0, s$, the missing displaced symbol is restored at its new border and q is added once. In the special old corner only (s, s) changes, from0 to$q$. Row and column0 receive$q$ at their border; row and column$s$ receive their missing0 there. The new row and column receive all old symbols except0 and$s$ from T, and their special positions supply0,s, q. Every row and column is therefore Latin.

12.3 Why the prescribed set defines the repaired square

In any completion of U, all the new-symbol positions are fixed: those of T, (s, s), $(0, q)$ and $(q, 0)$. Every row and column already has its prescribed q, so no additional cell can take that symbol. Delete the new row and column, restore each T-cell from its fixed displaced border value, and restore (s, s) to0. In every old row and column this restores precisely the symbol lost at the deleted border. The result is Latin and extends P: unchanged clues are fixed, changed transversal clues are restored, and (s, s) was not a clue. The published uniqueness of P determines the old square. Reintroducing the prescribed changes and border determines M. Thus U is defining for every odd $s \ge 3$.

12.4 Exact private-intercalate count and its limitation

Exactly $(s-1)^2$ clues of U have private intercalates, all in the old full block 01. For a target with local coordinates (i, j) there, let x, y be the other row and column residues. The old intercalate equations give $x - y = j - i \pmod{s}$. Its other corners are holes of P precisely when $y \le i$, $x \ge j$ and $y \le x$. If $i \le j$, these conditions force $x = j, y = i$. If $i > j$, they give exactly

$$y = 0, \dots, i - j - 1, \qquad x = s + j - i + y.$$

Every target with $i = 0$ fails, because its sole old witness meets the newly fixed corner $(0, 0)$. For $i = j > 0$ the witness survives. For $0 < i < j$ the only possible obstruction is its block-10 corner on T, exactly when $j > h$ and $i = 2j - s$; there are h such targets. For $i > j$ the target itself lies on T exactly when $i > h$ and $j = 2i - s$, another h targets. Otherwise the horizontal and vertical corners are safe. At most one of the $i - j$ choices has a forbidden diagonal corner; that exception requires $i - j > h$, so at least two choices exist and one survives. There are $s + h + h = 2s - 1$ failures among the s^2 targets, proving the count.

This exhausts all private intercalates of the repaired square. Any intercalate meeting its new row or column contains two prescribed border entries. Any old-area intercalate using q contains two prescribed q-entries. A remaining intercalate is unchanged from L. Since s is odd, no cyclic s-block has an intercalate; the outer symbol equations force an old intercalate to meet all four blocks. It meets full block 01, so it can be private only for a target already counted.

The three old triangular blocks are touched by T or the corner repair, so their previous private trades cannot all be counted automatically. The guaranteed $(s-1)^2$ protected clues occupy $s - 1$ rows. A defining set at order $2s + 1$ must occupy at least $2s$ rows, since two empty rows could be swapped. It therefore needs at least $s + 1$ further clues. The resulting valid bound is only

$$\operatorname{lcs}(2s + 1) \ge s^2 - s + 2,$$

below the ordinary triangle bound $2s^2 + s$. The defining size $(5s^2 + 6s + 7)/2$ must not be substituted for this guarantee. Finite pruned outputs55 at order11 and186 at order19 are separate computational examples, not proofs of a general stronger formula.

A separate checker rebuilds the entire square and repair without importing the constructor. For odd $s = 3, \dots, 41$ it checks Latinness, contraction, the failed-target classification and every intercalate. These finite checks corroborate

the all-parameter proof. The checker's source and saved outputs are in `lcs_four_m_plus_three/second_audit/`; they do not independently re-certify criticality of every pruned finite example.

13 Obstructions to direct tensor lifts

The results in this section concern precisely specified constructions. They do not assert nonexistence of large critical sets obtained in other ways. In particular, the verified order-33 example of size 844 already exceeds the former target 838. The failed size-838 templates below remain useful structural results about the proposed lifting mechanisms. Coordinates and symbols are zero based, and $\oplus$ denotes binary XOR.

13.1 Why one-bit literal embedding is impossible

Proper-subsquare obstruction. If a Latin square of order N has a proper Latin subsquare of order k, then $N \geq 2k$. To prove this, choose a column outside the subsquare. In each of the k subsquare rows, its entry must be outside the subsquare's k-symbol set, because every symbol of that set already occurs in that row inside the subsquare. These k entries in one column are distinct, so there must be at least k symbols outside the subsquare. Thus $N - k \geq k$. Consequently a completed order-$(q+1)$ Latin square cannot occur literally as a subsquare of an order-$(2q+1)$ square. This excludes literal completion-preserving embeddings from 9 to 17 and from 17 to 33.

Index-two complete-mapping obstruction. Let G be an elementary abelian 2-group and H an index-two subgroup. A complete mapping of G is a permutation τ for which $x \mapsto x \oplus \tau(x)$ is also a permutation. No such τ can satisfy $\tau(H) = H$. Indeed, a permutation preserving H also preserves its other coset. Thus x and $\tau(x)$ always belong to the same coset, so $x \oplus \tau(x) \in H$ for every $x \in G$. This cannot be a permutation of G. Linearity of τ is not assumed.

Two-bit extension behaves differently. Identify the two-bit group with $\{0,1,2,3\}$ and let

$$\omega = (0,2,3,1).$$

Both ω and $I + \omega = \omega^2$ are permutations. If τ is a complete mapping of $\mathbb{F}_2^m$, then

$$\widehat{\tau}(4r+i) = 4\tau(r) + \omega(i)$$

is a complete mapping of $\mathbb{F}_2^{m+2}$. This follows directly because its two coordinates, and the two coordinates of $I + \widehat{\tau}$, are permutations. If $\tau(0) = 0$, the prolonged square contains the old prolonged square on finite indices divisible by four, together with the new infinity label, with finite symbols multiplied by four. The two-bit extension therefore has no analogous Latin-embedding obstruction; its difficulty is the essentiality of its clues.

13.2 The repaired two-bit template and its exact count

For a complete mapping σ of the XOR group of order Q, denote its prolonged square by L^σ, of order $Q+1$, where

$$L^\sigma(r,c) = \begin{cases} Q, & c = \sigma(r), \\ r \oplus c, & c \neq \sigma(r), \end{cases} \qquad r, c < Q,$$

$$L^\sigma(r,Q) = r \oplus \sigma(r), \qquad L^\sigma(Q,c) = \sigma^{-1}(c) \oplus c,$$

$$L^\sigma(Q,Q) = Q.$$

Here Q serves as the new row, column and symbol label ∞. The formula is Latin because prolongation moves each transversal entry to the new row and column and replaces it by the new symbol.

Start with a normalized seed in the prolonged XOR square of order $q+1$, $q = 2^m$, and put $Q = 4q$ and $\sigma = \widehat{\tau}$ as above. In every finite high-coordinate block (r,c), prescribe all sixteen low-coordinate cells if the old core cell is a clue, and otherwise prescribe precisely

$$K = \{(1,1),(1,2),(1,3),(2,1),(2,2),(3,1),(3,3)\}.$$

Prescribe every cell of the high block $(0,0)$ except global $(0,0)$. Finally prescribe the new border except for the old border holes embedded with low coordinate zero: an old border position (r,c) is embedded as $(4r,4c)$, with the old infinity q mapped to Q. All prescribed values come from $L^{\widehat{\tau}}$.

For the two seeds tested, the old core has $3^m + 1$ holes and the old border has seven holes. Each core hole initially produces nine low holes, and the repair at the origin removes eight. Thus the new core has $9(3^m+1) - 8 = 3^{m+2} + 1$ holes; the whole template has $3^{m+2} + 8$ holes. It consequently has 838 clues at order 33 and 3488 at order 65. These are mask counts, not criticality assertions.

13.3 A uniform Hall-set obstruction

Theorem. Let $q = 2^m$, let τ be any complete mapping with $\tau(0) = 0$, and let $Q = 4q$. Suppose a partial square $P \subseteq \widehat{L^\tau}$ contains every low-K core position, every cell of the top-left four-by-four block except $(0,0)$, and every new-row and new-column position whose finite index is not divisible by four. It may contain arbitrary further clues. Then the clue $(1,2) = Q$ is redundant: deleting it does not change the set of completions. No assumption of unique completion is needed.

Proof. Delete $(1,2)$ and set

$$S = \{4h + 1 : 1 \le h < q\}.$$

Row 1 contains every symbol congruent to 0 or 2 modulo four at columns $4v + 1$ and $4v + 3$. At columns $4v + 2$, $v > 0$, it contains every symbol congruent to 3 except 3 itself; its new-column clue is 3. Its repaired cell $(1,0)$ is 1. Therefore the domain of $(1,2)$ is contained in $S \cup \{Q\}$.

Column 2 contains every symbol congruent to 0 modulo four at rows $4h + 2$. At rows $4h + 1$, $h > 0$, it contains every symbol congruent to 3 except 3; its new-row clue is 3. It also contains the repaired clues $(0,2) = 2$ and $(3,2) = 1$. Hence every unfilled cell of column 2 has domain contained in

$$\{Q\} \cup \{4h + 1, 4h + 2 : 1 \le h < q\}.$$

For each $h = 1, \dots, q - 1$, consider $A_h = (4h + 3, 2)$. The clues of its row at columns $4v + 1$ have values $4(h \oplus v) + 2$, except that at $v = \tau(h)$ the value is Q. The omitted value $4(h \oplus \tau(h)) + 2$ is exactly the clue in the new column of that row. Thus that row contains every symbol congruent to 2 modulo four and also Q. Combining these row exclusions with the column exclusions shows that the value at A_h must belong to S.

The $q - 1$ cells A_h lie in one column and are confined to the $q - 1$ symbols of S. In every completion they use all of S, once each. Hence $(1,2)$ cannot use any member of S and must use Q. If some A_h are already clues, their prescribed distinct values simply occupy part of S; the argument is unchanged. Restoring $(1,2) = Q$ therefore excludes no completion, as claimed.

Translated copies and simultaneous deletions. If the entire high transversal block $(u, \tau(u))$ is prescribed, the same proof makes $(4u + 1, 4\tau(u) + 2) = Q$ redundant. Translate high row labels by u, high column labels by $\tau(u)$ and high symbol labels by $u \oplus \tau(u)$. The new complete mapping is

$$\tau'(a) = \tau(a \oplus u) \oplus \tau(u), \qquad \tau'(0) = 0.$$

The full high block supplies the three repaired cells required at the new origin. The proof for one such target does not use any other transversal clue of low type $(1,2)$: its other low-$(1,2)$ positions are off the transversal, and its auxiliary-row infinity clues have low type $(3,1)$. All these targets can therefore be deleted simultaneously without changing completions. The normalized order-9 seed has six full old transversal blocks and the normalized order-17 seed has twelve. Together with the repaired origin, the two tested templates have seven and thirteen such targets. This is not an upper bound on every critical subset of the template: removing other clues can make a previously redundant clue essential.

An elementary certificate at order 33. For the fixed order-33 template, after removing $(1,2) = 32$, the following are successive cell singles:

$$(1,12) = 13,\ (1,24) = 25,\ (1,20) = 21,\ (1,8) = 9,$$
$$(1,16) = 17,\ (1,28) = 29,\ (7,2) = 5,\ (1,2) = 32.$$

Their domains before any of these deductions are, in order,

$$\{13\},\ \{13,25\},\ \{21,25\},\ \{9,21\},\ \{9,17\},$$
$$\{13,29\},\ \{5\},\ \{5,9,13,17,21,25,29,32\}.$$

Each preceding deduction reduces the next domain to the displayed singleton. The saved certificate checks this directly against the original mask. At order 65 this particular singles chain does not start, but the uniform Hall proof still applies.

13.4 The exact finite class of 6,435 low masks

The wider finite exclusion below fixes the normalized order-9 seed explicitly. Its complete mapping is

$$\tau = (0, 6, 7, 1, 5, 3, 2, 4).$$

The seed completion is L^τ. Its clue positions are the 1s of the following nine-by-nine mask, whose last row and column are the border:

000000000
001111110
010111110
001010111
011101101
011110011
011101010
010111101
011010111

This mask has 46 clues, including 36 in the eight-by-eight core. The seven border holes are

$$(0, 8), (1, 8), (2, 8), (6, 8), (8, 0), (8, 3), (8, 5).$$

In the two-bit template, replace K by every seven-element subset of the fifteen low cells other than $(0, 0)$. The enumeration is lexicographic in the seven-element subsets of $\{1, \ldots, 15\}$, where j denotes low cell $(\lfloor j/4 \rfloor, j \bmod 4)$. All other choices—the seed, complete mapping, full old clue blocks, origin repair and scaled border holes—remain fixed. There are $\binom{15}{7} = 6435$ masks. Every template has

$$36 \cdot 16 + 28 \cdot 7 + 8 = 780$$

core clues and 58 border clues, hence 838 clues in total.

Complete finite classification. All 576 labelled Latin squares of order four are enumerated by choosing four row permutations and retaining just those for which every column is a permutation. Compare each mask's XOR-four entries against all 576 squares. Exactly 4320 masks have a second completion; 2115 uniquely complete XOR-four. Testing each of the seven deletions in the same list finds 27 critical low masks.

For every one of the 4320 nonunique low masks, substitute its saved alternative Latin square into high block $(0, 1)$, rows 0–3 and columns 4–7. This is the untouched Latin subsquare $4 + \mathrm{XOR}_4$: it is outside the origin repair and the prolongation transversal, and its clues are exactly the chosen low mask. Substituting another completion preserves the symbol set of every affected row and column and every clue, giving a different order-33 completion.

Of the other 2115 masks, 108 have a saved full order-33 alternative completion. Each is specified by a list of changed cells; independent replay checks the old values, distinct positions, nonempty change, all rows and columns, and agreement with every one of the 838 clues. For the remaining 2007 masks, deleting $(1, 2) = 32$ leaves a uniquely completable partial square. These 2007 exclusions are solver-dependent: independent integer-cell AllDifferent models fix the remaining clues and forbid the supplied full completion, and every query is infeasible. The supplied completion remains valid after deletion, so infeasibility of a different completion proves redundancy. Thus the final partition is

4320	explicit local alternative completions,
108	explicit full-square alternative completions,
2007	uniqueness after target deletion,
6435	templates excluded, with no unresolved case.

The 4428 explicit alternatives are solver-free to replay. The 2007 negative queries were independently repeated with OR-Tools 9.14.6206, one worker: the first bounded run confirmed 479, leaving two UNKNOWN and 1526 unattempted cases; a fresh-model follow-up confirmed all 1528 remaining cases. No UNKNOWN was interpreted as infeasible. These are completed exact solver queries, not standalone UNSAT certificates. The initial search's time limits do not leave a gap in the final combined coverage.

The finite result varies only the low mask in this fixed construction. It does not quantify over other seeds, complete mappings, nonuniform block masks, border patterns, repairs, or all order-33 critical sets. The complete data and independent reconstruction are in `outputs/lcs_dyadic_plus_sharp_progress/finite_mask_audit/`.

13.5 No single exchange repairs the fixed size-838 template

For the original seven-cell mask K, let $e = (1, 2)$ and let $\mathcal{H}$ be the set of other clues explicitly used in the Hall proof: the clues of row 1 at $4v + 1, 4v + 3$, and at $4v + 2$ for $v > 0$, together with $(1, Q), (1, 0)$; the clues of column 2 at $4h + 2$, and at $4h + 1$ for $h > 0$, together with $(Q, 2), (0, 2), (3, 2)$; and all $(4h + 3, 4v + 1), (4h + 3, Q)$ for $h > 0$, $0 \le v < q$. At $q = 8$ this set has 106 distinct clues and excludes e.

Deleting a clue outside $\mathcal{H} \cup \{e\}$ leaves the Hall proof intact. For each $f \in \mathcal{H}$, an exact integer-cell AllDifferent query deletes both e and f and imposes $L(e) \neq 32$. All 106 queries finished infeasible, with no UNKNOWN; their recorded per-query cap was two seconds and one worker. Thus, after deleting any one other clue, e is still entailed by the remaining clues. Adding a clue cannot invalidate an entailment. Every one-for-one exchange retaining e therefore remains noncritical whenever it is completable.

If the exchange deletes e itself, the remaining 837 clues already complete uniquely, as checked independently. An added compatible clue is then redundant; an incompatible clue makes the partial square incompletable. Hence no one-deletion/one-insertion exchange in this fixed labelled template is critical. The 106 negative queries are solver evidence rather than portable UNSAT certificates; the unchanged-Hall cases require no solver. Exchanges involving two or more deletions are outside this conclusion.

13.6 A sufficient private-trade transfer criterion

There is nevertheless a precise sufficient condition for transferring an individual old private bitrade. Let (T, T') be a Latin bitrade with T in an old prolonged XOR square and intersecting its partial square only in the designated target. For every finite row r used by the bitrade choose $a_r \in \mathbb{F}_2^2$; choose b_c for finite columns and d_s for finite symbols. Embed labels as $(r, a_r), (c, b_c), (s, d_s)$ and send each infinity label to new infinity. These maps are injective because their high labels are unchanged. The first half lies in the two-bit prolonged square provided each triple of T satisfies the appropriate condition

finite or infinite coordinates	low-coordinate condition
r, c, s finite	$d_s = a_r \oplus b_c$
r, c finite, $s = \infty$	$b_c = \omega(a_r)$
$r = \infty$, c, s finite	$d_s = \omega(b_c)$
r, s finite, $c = \infty$	$d_s = \omega^2(a_r)$

The all-infinity triple imposes no condition. The border formulas use $\omega^{-1} + I = \omega$ and $I + \omega = \omega^2$. The high-coordinate conditions already hold in the old prolonged square.

Injective coordinate relabelling preserves the bitrade's row and column symbol balances and disjointness. If every cell of the image support other than the target is a hole of the new partial square, replacing the first half by the second is a private trade for that target. For a low-K block the additional hole condition is $(a_r, b_c) \notin K$; in the repaired origin only low cell $(0, 0)$ is a hole; a scaled border hole requires low coordinate zero and an old border hole. Both the ambient-square equations and these mask-avoidance conditions are required. This lemma neither protects every lifted clue nor supplies private trades for new low-mask or border clues automatically.

14 The order-15 size-148 seed: exact profile and failed extensions

The size-148 order-15 seed is critical: its exhaustive uniqueness tree has 3762 nodes, 190 dead leaves and one solution, and all 148 exact-cover and all 148 independently generated AllDifferent deletion witnesses replay as valid Latin completions. Its full matrix pair is included in the atlas. The finite profile comparisons below refer to these precise archived seeds, not arbitrary sets having the same size. The later size-150 record is a different seed and is not covered by these profile claims.

14.1 Exact local profiles

Independent enumeration gives

seed	s	t_2	t_3	intercalates	order-three subsquares
new order-15 seed	148	58	4	542	19
older binary seed	147	132	15	630	35
older ternary seed	147	130	19	630	35
existing order-30 seed	654	606	45	5265	140

Here t_2, t_3 count distinct clues, not witnesses with multiplicity. To enumerate intercalates, inspect every pair of rows and columns. For order-three subsquares, inspect each row triple and group columns by their unordered sets of three symbols. Any qualifying three-column group is exactly a subsquare; a group cannot have more than three columns, since each of its rows has every symbol once. Enumerate all twelve Latin squares on its three symbols, check the induced clues' unique completion, and test each local clue deletion. Substitute every saved local trade into the full square to check Latinness and retention of all other clues. The new seed's four 3-essential positions are $(7,8),(7,12),(9,5),(9,8)$.

The safe lifts give 579, 2367 and 1198 from the new seed at orders 30, 60 and 45, respectively. The older binary seed gives 651, 2808 and 1215; the older ternary seed gives 1223 at order 45. These comparisons already show why a larger critical set need not be a better seed for this particular family of protected lifts.

14.2 Domination for every nonempty mixed word

For a seed let $N = m^2$, $D = N - s$, $u_2 = s - t_2$, $u_3 = s - t_3$. For a forward word of binary and ternary lifts, the backward coefficient tuple starts at $(a, b, c, d) = (0, 1, 0, 0)$ and transforms by

$$\begin{aligned} B(a,b,c,d) &= (4a + d, 3b, 4c + h, 3d), \\ T(a,b,c,d) &= (9a + 2c, 7b, 7c, 9d + 2h), \\ h &= \max(0, b - c - d). \end{aligned}$$

The guaranteed final deficit is $aN + bD + cu_2 + du_3$. The new and old binary seed states are respectively

$$(225, 77, 90, 144), \qquad (225, 78, 15, 132).$$

Thus the new guaranteed deficit minus the old is $\Delta = -b + 75c + 12d$.

Every nonempty word satisfies

$$b \le 3c + \frac{7}{2}d,$$

with at least one of c, d positive. After the first transformation, (b, c, d) is $(3, 1, 0)$ or $(7, 0, 2)$, giving equality. Set $F = 3c + \frac{7}{2}d - b$. A binary transformation gives $F' = 3F + 3c + 3h \ge 0$, and a ternary transformation gives $F' = 7F + 7d + 7h \ge 0$. This proves the invariant by induction at every word length. Therefore

$$\Delta \ge 72c + \frac{17}{2}d > 0.$$

The old binary seed gives a strictly larger lower bound for the same nonempty forward word. At each destination order there are finitely many factor words, so maximizing over them cannot reverse this domination. The empty word is the exception: 148 improves 147 at order 15 itself.

For the old ternary seed, the deficit difference is $-b + 73c + 16d \ge 70c + \frac{25}{2}d > 0$, proving the same conclusion. Pure iterations from the new seed give the valid but dominated bounds

$$\begin{aligned} 225 \cdot 4^k - 77 \cdot 3^k - 90\beta_k, \quad \beta_0 = 0, \quad \beta_{k+1} = 4\beta_k + \max(0, 3^k - \beta_k), \\ 225 \cdot 9^k - 77 \cdot 7^k - 144\chi_k, \quad \chi_0 = 0, \quad \chi_{k+1} = 9\chi_k + 2\max(0, 7^k - \chi_k). \end{aligned}$$

An independent arithmetic check enumerated all 2047 words of length zero through ten. The all-length claim follows from the invariant, not that finite enumeration. No conclusion is made about other products, relabellings, private trades or prolongations of the seed.

14.3 A Latin-square extension whose clue mask is not critical

Let L be the size-148 seed's completion and let t be any transversal, written as a column permutation. Define $Q(r, c) = L(r, t(c))$. The diagonal symbols $d_r = Q(r, r)$ are all distinct. Replace that diagonal by the new symbol 15, append d_r at the end of row r and at the bottom of column r, and put 15 in the new corner. This gives a Latin square M of order 16: each old row and column has its displaced symbol restored on the border, and the border row and column contain every old symbol once. Its diagonal is constantly 15, and $M(r, 15) = M(15, r)$.

For every integer $a \ge 0$, take its direct product with the XOR square on a bits:

$$V((h, r), (k, c)) = 16(h \oplus k) + M(r, c).$$

This square has order $q = 16 \cdot 2^a$, constant diagonal 15 and identical last row and last column. Delete its last row and column. At every remaining diagonal cell replace 15 by the value deleted from the same row's last column.

Each row removes 15 and restores its other missing symbol; equality of the last row and column makes the same true in each column. The result is Latin on the symbols other than 15. Relabel every symbol above 15 down by one to obtain order $n = 2^{a+4} - 1$. When $a = 0$, this reverses the initial prolongation and recovers the column-normalized old square.

Prolong the clue mask by retaining the 148 core clue positions, with their new values, and prescribing the whole border. It has exactly 77 holes, all in low coordinates below 15. Make an outer block full when $h \,\&\, k \neq 0$, and otherwise give it precisely those 77 holes. There are 3^a outer pairs with $h \,\&\, k = 0$, so the mask has $77 \cdot 3^a$ holes. Deleting the final row and column removes no holes because low coordinate 15 was fully prescribed. Hence the contracted mask has exactly

$$(2^{a+4} - 1)^2 - 77 \cdot 3^a$$

clues. This proves Latinness and the raw clue count for all a, without asserting uniqueness or criticality of the mask.

The first order-31 candidate has a redundant clue. Take the first saved transversal

$$t = (0, 2, 8, 7, 12, 14, 1, 5, 11, 10, 3, 9, 4, 6, 13)$$

in the atlas's order-15 size-148 seed and set $a = 1$. The resulting mask has 730 clues. Its clue $(0, 0) = 29$ is redundant, as proved by a standalone exhaustive certificate. Delete this clue and forbid symbol 29 there. An exact-cover tree has 2529 nodes, 93 dead leaves and zero solutions. A distinct grid-domain verifier checks it as follows: each internal node selects an unsatisfied cell, row-symbol or column-symbol constraint; recompute every legal placement satisfying it from the current grid; check that the node has exactly one branch for each such placement and no others; assign it, recurse and undo; at a dead leaf check that the selected unsatisfied constraint has no legal placement. No solution leaf is permitted. The forbidden target value is excluded throughout. Induction on the finite tree proves that no completion can change the deleted clue, while the displayed completion supplies satisfiability. Thus the clue is entailed by the others and the 730-clue mask is not critical.

The replay needs no solver. The exact inputs and tree are `candidate31.in`, `seed15.in` and `redundancy_tree.json` in `outputs/lcs_order15_generalisation/audit/`; its independent checker is `replay_redundancy.py`. Independent AllDifferent queries also verified uniqueness of the full mask and inability to change the target, but those solver verdicts are not required by the negative certificate. Twelve sampled transversals were tested at orders 15 and 31; this portable redundancy tree concerns the first candidate specifically. Checks at orders 63 and 127 confirmed Latin squares with masks of sizes 3276 and 14050, without certifying those masks critical.

14.4 Conjugate-block and initial-singles checks

The full-block criterion proved earlier can be applied to all $6^4 = 1296$ independent coordinate-conjugate choices of the size-148 seed in four blocks. Each template has $225 + 3 \cdot 148 = 669$ clues. An independent join-by-symbol enumeration finds at most 135 essential full-block clues, attained by six choices. For each choice enumerate holes (r, j) of the upper-right block and (i, c) of the lower-left block with equal symbols; retain them when (i, j) is a hole of the last block and its symbol equals the full block's symbol at (r, c). Count each full-block position once. This is the exact necessary and sufficient criterion, so fewer than 225 essential full-block clues proves all 1296 initial templates noncritical.

Protected reduction still gives a critical subset of size at least $3 \cdot 148 + 135 = 579$, with at least $3 \cdot 58 + 135 = 309$ private-intercalate clues. This is below the existing order-30 size-654 record. The scan does not prove that every critical reduction of the 669-clue templates has size at most 654.

Finally, a direct exhaustive scan of all initially unsatisfied cell, row-symbol and column-symbol constraints of the original size-148 seed finds no singleton in any of the three classes: all counts are zero. Thus an argument using only repeated elementary singles cannot begin from that exact mask. This is compatible with its certified unique completion, which was established by a different method. The independent profile, comparison, tensor and certificate checks are archived under `outputs/lcs_order15_generalisation/audit/`.

15 Balanced counting and amplification by defining-set supersets

This section establishes an existence lower bound for all sufficiently large orders and a finite factorial criterion for every order. It does not construct a critical set explicitly. Let $\mathcal{L}_n$ be the labelled Latin squares of order n, let $N = n^2$, and let $\mathcal{P}_{n,N-h}$ be the partial Latin squares with exactly h holes. For a Latin square L, write $\mathrm{scs}(L)$ for the smallest size of a critical set in L, and set

$$T(n) = \max_{L \in \mathcal{L}_n} \mathrm{scs}(L).$$

Thus $T(n) \le \mathrm{lcs}(n)$. The argument below bounds $T(n)$, which is stronger than bounding only the maximum critical-set size. All logarithms in this section are natural.

15.1 The two permanent inputs

We use the van der Waerden permanent lower bound and Bregman's permanent upper bound as published inputs. The former implies

$$L_0(n) := \frac{(n!)^{2n}}{n^{n^2}} \le |\mathcal{L}_n|.$$

For completeness, when i full rows have been filled, the matrix of allowed symbols for the next row is $(n-i)$-regular. Dividing by $n-i$ gives a doubly stochastic matrix, so its permanent is at least $n!((n-i)/n)^n$. Multiplying this bound for $i = 0, \dots, n-1$ gives exactly $L_0(n)$.

For $0 \le a \le n$, define

$$F_n(a) = \prod_{t=a+1}^{n} (t!)^{1/t}, \qquad F_n(n) = 1.$$

Fix a shape of holes, with u_i holes in row i and v_j holes in column j. The number of Latin fillings of this fixed shape is at most

$$(n!)^{h/n} \prod_{i=1}^{n} \frac{1}{u_i!} \prod_{j=1}^{n} F_n(v_j). \tag{15.1}$$

Here is the full row-filling argument. Fill successive rows. For a row with u_i holes, form a bipartite matching problem between all n symbols and n positions: the $n-u_i$ actual filled positions, together with u_i labelled dummy positions. Each dummy position permits all n symbols. If a real position is in a column containing k previously filled entries, it permits exactly $n-k$ symbols. Every assignment of distinct allowed symbols to the actual positions has precisely $u_i!$ extensions to the dummy positions. Bregman's inequality, applied with positions as matrix rows, therefore bounds the number of assignments by

$$\frac{(n!)^{u_i/n}}{u_i!} \prod_{\text{actual positions in this row}} ((n-k)!)^{1/(n-k)}.$$

The bound is independent of the particular previous filling. Across all filled positions of column j, the previous occupancy k runs through $0, 1, \dots, n-v_j-1$. The product of that column's factors is consequently $F_n(v_j)$. Multiplying over rows proves the fixed-shape bound.

15.2 Balancing both margins

Write $h = an + b$, where $0 \le b < n$. Both the row-hole counts and the column-hole counts sum to h. The product $\prod_i 1/u_i!$ is maximized when its arguments differ by at most one: if $x \ge y+2$, replacing (x, y) by $(x-1, y+1)$ multiplies it by $x/(y+1) > 1$. For the column factors,

$$\log F_n(v+1) - \log F_n(v) = -\frac{\log((v+1)!)}{v+1}.$$

The geometric means $(t!)^{1/t}$ increase with t, so these differences decrease with v. Hence $\log F_n$ is discretely concave, and the column-factor product is also maximized by the balanced counts $a, a+1$. We maximize the two products separately; their maxima need not be attained by the same shape for this to yield an upper bound. There are $\binom{N}{h}$ shapes. Thus

$$|\mathcal{P}_{n,N-h}| \le U(n,h) := \binom{N}{h} (n!)^{h/n} \frac{F_n(a)^{n-b} F_n(a+1)^b}{(a!)^{n-b}((a+1)!)^b}. \tag{15.2}$$

At $h = N$, the factor with exponent $b = 0$ is omitted, so there is no undefined $F_n(n+1)$ term.

15.3 Counting all supersets, with the integer endpoint

Finite criterion. If $0 \le h \le d \le N$ are integers and

$$L_0(n)\binom{d}{h} > U(n,h),$$

then

$$T(n) \ge N - d + 1, \qquad \mathrm{lcs}(n) \ge N - d + 1.$$

Proof. Suppose instead that every Latin square L contains a critical set C_L of size $s_L \le N-d$. Choose one such set for each L. Every $(N-h)$-entry subset of L containing C_L is uniquely completable to L. There are

$$\binom{N-s_L}{h} \ge \binom{d}{h}$$

such subsets, since one chooses its h holes outside C_L. Subsets arising from different Latin squares cannot coincide: any coincidence would give two completions of a uniquely completable set. It follows that

$$|\mathcal{P}_{n,N-h}| \ge |\mathcal{L}_n|\binom{d}{h} \ge L_0(n)\binom{d}{h},$$

contrary to the strict hypothesis and the bound $U(n,h)$. Thus some Latin square has no critical set of size at most $N-d$; the integer conclusion is $N-d+1$. Taking $h=d$ recovers the unamplified counting argument.

15.4 The asymptotic constant, with error terms

Fix $c>0$ and let $h \sim cn^{5/3}$. Stirling's formula gives

$$\log \frac{U(n,h)}{L_0(n)} = 3h\log\!\left(\frac{en^{5/3}}{h}\right) + o(n^{5/3}).$$

We include the expansion to make its coefficient reproducible. Put $a_0 = h/n$. Ignoring integer interpolation for a moment, the column logarithm satisfies

$$\begin{aligned} n\sum_{t=a_0+1}^{n}\frac{\log(t!)}{t} &= n\sum_{t=a_0+1}^{n}\left(\log t - 1 + O\!\left(\frac{\log(t+1)}{t}\right)\right)\\ &= n\log(n!) - n\log(a_0!) - n^2 + h + O(n\log^2 n). \end{aligned}$$

The factorial at a noninteger argument can be read as the gamma function; balancing the actual integer counts changes this estimate by at most $O(n\log n)$, which is absorbed in the displayed error. Consequently

$$\begin{aligned} \log\frac{U(n,h)}{L_0(n)} = {}& \log\binom{N}{h} + (h/n-n)\log(n!) - 2n\log(a_0!)\\ & - N + h + N\log n + O(n\log^2 n). \end{aligned}$$

Use

$$\log\binom{N}{h} = h\log(N/h) + h + O(h^2/N + \log N)$$

and the usual factorial expansion in the remaining terms. The result is

$$5h\log n - 3h\log h + 3h + O(h^2/N + n\log^2 n).$$

Since $h^2/N = O(n^{4/3})$, the remainder is $o(n^{5/3})$.

Now put $d \sim Dn^{5/3}$ and $h \sim zd$, where $D>0$ and $0<z<1$ are fixed. With

$$\mathcal{H}(z) = -z\log z - (1-z)\log(1-z),$$

the logarithm of $U(n,h)/(L_0(n)\binom{d}{h})$, divided by d, tends to

$$3z\log\frac{e}{zD} - \mathcal{H}(z).$$

It is negative exactly when

$$D > \exp\!\left(1 - \frac{2}{3}\log z + \frac{1-z}{3z}\log(1-z)\right).$$

The derivative of the exponent is

$$\frac{-3z - \log(1-z)}{3z^2}.$$

The numerator starts negative and, after its minimum at $z=2/3$, increases to infinity. It has exactly one zero in $(0,1)$:

$$-\log(1-z_*) = 3z_*, \qquad z_* = 0.940479790707359631134397110098\ldots.$$

The exponent tends to infinity as $z \downarrow 0$, and its derivative has the indicated single sign change, so this zero gives the global minimum. Substituting the zero's defining relation simplifies the minimum to

$$c_* = e^{z_*} z_*^{-2/3} = 2.668162063280737765060257237774\ldots.$$

For every fixed $\varepsilon > 0$, take $D = c_* + \varepsilon$, $d = \lfloor Dn^{5/3} \rfloor$ and $h = \lfloor z_* d \rfloor$. For all sufficiently large n they satisfy the strict finite criterion. Therefore

$$T(n) \geq n^2 - (c_* + o(1))n^{5/3}, \qquad \mathrm{lcs}(n) \geq n^2 - (c_* + o(1))n^{5/3}.$$

The $o(1)$ is essential. This proof does not assert a uniform finite bound with coefficient exactly c_*.

15.5 Independent finite audit

A separate implementation evaluates the factorial criterion with 90-digit interval arithmetic, canceling the common $h!$ terms before evaluation and treating the two column tails separately. It imports none of the original counting code. It certifies positive intervals for $\log(L_0(n)\binom{d}{h}/U(n,h))$ for all the following pairs:

n	d	h	$n^2 - d + 1$	logarithmic margin (approximately)
295	32087	30523	54939	0.00179779489
500	78599	74679	171402	0.00508366369
1000	253695	240636	746306	0.00344870791
2000	815295	772181	3184706	0.00281491529
10000	12130857	11456950	87869144	0.00240041413

The full interval endpoints and a certified bracket for z_* and c_* are saved in `outputs/lcs_full_review/counting_independent_results.json`. These are certified choices of integer parameters, not a claim that the choices globally optimize every counting method.

16 Transversal stripping: an exact inverse-prolongation criterion

Here *stripping* reduces the order by one. Let L be a Latin square of order $N \geq 2$, choose a cell (a, b) with symbol z, and, for each $r \neq a$, let c_r be the column of z in row r. The c_r are precisely the columns other than b, each occurring once. Delete row a and column b, and replace each remaining occurrence $L(r, c_r) = z$ by $L(r, b)$. Then the following are equivalent:

1. The resulting array is Latin on the $N - 1$ symbols other than z.
2. $L(r, b) = L(a, c_r)$ for every $r \neq a$.
3. The cell (a, b) belongs to $N - 1$ intercalates.

When these conditions hold, the replaced cells form a transversal T of the contracted square, and ordinary prolongation along T recovers L.

Proof of the criterion. The surviving part of row r has lost $L(r, b)$, which is distinct from z. Replacing its z by this missing symbol always makes the row a permutation of the remaining symbols. Column c_r has instead lost $L(a, c_r)$; its only replacement is the cell (r, c_r). Hence this column is Latin exactly when $L(r, b) = L(a, c_r)$. This proves the equivalence of the first two conditions. The replaced positions occupy distinct rows and columns, and their new symbols $L(r, b)$ are all distinct, proving that they form a transversal.

For each $r \neq a$, the four positions

$$(a, b), \quad (a, c_r), \quad (r, b), \quad (r, c_r)$$

form an intercalate exactly when the displayed equality holds. Conversely, an intercalate through (a, b) must use the unique z in its other row, so these are all its possible intercalates. There are at most $N - 1$. This proves the final equivalence and supplies an exact test, rather than a heuristic search for a suitable transversal.

Defining-set transfer. Let P be any defining subset of L. Contract its surviving entries as above and prescribe every cell of T, whether or not it came from P. Denote the resulting partial square by U. Then U is defining in the contracted square.

Indeed, in any completion K of U, the prescribed positions T have distinct rows, columns and symbols, so are a transversal. Prolong K using a, b, z. Outside T, surviving clues agree with P; all cells of T become z, also agreeing with P. The new column receives the prescribed $L(r, b)$, and the new row receives the same displaced values $L(r, b) = L(a, c_r)$. Thus all clues of P on the deleted row and column are restored. The corner is z. The prolonged completion extends P and must equal L, so K is its original contraction. This argument requires P to be defining, but not necessarily critical. Prescribing T is essential unless its necessary values and transversal property are established in every completion by a separate argument.

16.1 Critical subsets and products under stripping

Survival of private trades. Suppose a clue x of P has a private Latin trade whose support avoids row a, column b, and the changed positions T in the old coordinates. It contains no z, since these excluded positions contain every old occurrence of z. Its mate likewise contains no z, by equality of the per-row and per-column symbol multisets. The trade therefore survives unchanged in the contracted square and meets U precisely in x. Every defining subset of U contains x.

Consequently, if F is a collection of distinct clues with such witnesses, an inclusion-minimal defining subset of finite U is a critical set of size at least $|F|$. New private trades found directly after contraction and disjoint cover arguments can increase this guarantee. The size $|U|$ itself is not a critical-set lower bound. There is no automatic transfer depending only on the old critical-set size: a private witness may meet the removed row, removed column, or the extra prescribed cells.

Exact direct-product formula. Suppose corresponding corners of Latin squares A, B lie in t_A, t_B intercalates. Their corner in $A \times B$ lies in

$$(t_A + 1)(t_B + 1) - 1 \tag{16.1}$$

intercalates. To prove this, include a corner's own row among its compatible rows in condition 2 of the criterion. There are $t + 1$ such rows in a factor. Compatibility in the product holds exactly when it holds in both coordinates. Multiply these counts and remove the product corner's own row. This proves (16.1). In particular a product corner is strippable exactly when both factor corners are strippable.

Every corner of the XOR square of order $k = 2^j$ has $k - 1$ intercalates; its product with a corner having t has $k(t + 1) - 1$. The cyclic order-three square has no intercalates, so its product with a square of order p has at most $p - 1$ at any corner, fewer than $3p - 1$. Thus ordinary ternary products have no stripping corner. These statements concern direct products, not arbitrary arrangements of conjugate blocks. Isotopy and conjugation preserve the intercalate criterion and cannot remove an obstruction.

A valid prolong–double–strip route. If M is an ordinary prolongation of an order-q square, its added corner has q intercalates. The corresponding corner in $C_2 \times M$ has $2(q + 1) - 1$ by (16.1); stripping produces order $2q + 1$. For odd $q = 2m + 1$ this is $4m + 3$, including 11 and 19 when $q = 5, 9$. One must therefore start with an odd-order square before prolongation. Starting with a Donovan seed of even order $2s$ gives order $4s + 1$ under these three operations and does not address $11, 19$.

For any defining subset Q of M, the binary template consisting of one full block and three copies of Q is defining: the full block forces the symbol sets in its adjacent blocks, those copies of Q complete uniquely, and they then force and complete the fourth block. The preceding transfer applies to this template. It gives a rigorous construction mechanism; a large critical subset still requires protection or exact pruning certificates.

16.2 Which existing squares can be stripped?

Obstruction for the raw thesis squares. In the printed order-$4m$ thesis square the symbol block at outer position (R, C) is $R \oplus C$, and every inner block has residue rule $i - j$, $j - i$, or $i + j$ modulo m. Consider two rows in the same outer row block. An intercalate between them must use columns in the same outer column block, because equality of opposite symbol blocks forces those column blocks to agree. Within such a block, the intercalate equations give $2(i - x) = 0 \pmod{m}$ for the two row residues. Thus a corner has no such partner row when m is odd, and at most one when m is even. The other three row blocks can supply at most $3m$ partners. The corner therefore lies in at most $3m$ or $3m + 1$ intercalates, respectively, strictly below $4m - 1$ when $m \geq 3$.

Hence **the raw thesis squares have no stripping corner for** $m \geq 3$. In particular orders 12 and 20 cannot be stripped to 11 and 19 by ordinary inverse prolongation. The exceptional $m = 1, 2$ squares are XOR squares and are strippable throughout. The same argument with two outer row blocks excludes the raw Donovan order-$2m$ squares for $m \geq 3$. Binary products inherit these obstructions by the exact product formula. This says nothing against the different prolong–double–strip construction above.

Exact scans of supplied completions. The following are properties of the specified record squares, not universal assertions about their orders. Coordinates are zero-based.

Order	Critical-set size	Stripping corners	A corner
9	47	1	$(8, 0)$
11	74	1	$(10, 10)$
13	107	1	$(6, 6)$
17	200	1	$(16, 16)$
19	245	1	$(18, 18)$
33	844	1	$(32, 32)$
16	175	256	Every cell
22	303	4	$(10, 10), (10, 21), (21, 10), (21, 21)$

The scanned record completions of sizes $224, 304, 367, 449, 624$ at orders $18, 21, 23, 25, 29$ also each have one corner. The scanned completions 15/150, 20/281, 22/340, 24/405, 26/482, 27/447, 28/536, and 30/654 have none. Here n/s means order n, critical-set size s. The larger size-340 order-22 square need not share the structural property of the size-303 product square.

Independent checking. The solver-free program `lcs_full_review/stripping/check_stripping.py` implements the criterion directly, checks every candidate cell, constructs each accepted contraction and verifies its Latin property and transversal. It checked 29 model constructions and 32 supplied record/partial pairs. The detailed sources, completion hashes and results are in `lcs_full_review/stripping/checks.json`. This is a structural audit; it does not repeat the previous uniqueness and criticality certification of the input records.

16.3 Stripped binary templates: certificates and limits

Normalize a product corner to outer block 00, with base corner (a, b) and symbol z. A private trade confined to a seed block survives the strip if it avoids the following positions in its inner coordinates:

Block	Positions that must be avoided
00	Row a, column b, and all z-cells
01	Row a
10	Column b
11	All z-cells, including the base corner

This follows directly from which product row and column are deleted and which diagonal-block occurrences of the stripped symbol change. It can retain more witnesses than requiring every copied trade to avoid all three types of position. Additional private intercalates can be enumerated directly in the contracted square against its actual defining set U.

Using the certified order-17 size-200 seed and its corner $(16, 16)$, the four choices for the full block give the following order-33 results. The third column counts all private-intercalate clues of U. The last column also includes surviving supplied seed-deletion witnesses, counting each protected clue once.

Full block	Defining size $\lvert U\rvert$	Private intercalates	Combined protected core
00	833	393	464
01	841	393	439
10	840	393	447
11	840	393	447

These exact templates cannot yield a critical set exceeding the existing size 844, because every extracted critical subset is contained in U. The protected cores are lower guarantees, not estimates of the largest critical subset that could be extracted. A different seed or different compatible partial-block arrangement could behave better.

The same construction from the record 9/47 gives order-17 defining supersets of sizes $196, 202, 202, 202$; from 11/74 it gives order-21 sizes $309, 315, 315, 315$; and from 13/107 it gives order-25 sizes $449, 455, 456, 456$, in full-block order $00, 01, 10, 11$. These are reviewable search inputs, not new critical-set records.

The program `lcs_full_review/stripping/strip_binary_template.py` constructs the products and contractions, checks all counted private intercalates against U, and replays the supplied order-17 deletion alternatives as Latin squares before transferring their trades. The resulting 16 completions and partials are retained alongside `binary_template_checks.json`. Their status is explicitly `DEFINING_SUPERSET_NOT_CLAIMED_CRITICAL`. Their unique completion follows from the proved transfer and certified seed; no minimality is inferred from that fact. No new small-order record or all-order lower-bound formula follows from these template tests.

16.4 Direct contractions of the latest records

The record squares themselves can also be contracted at their verified corners, without first taking a product. Applying the defining-set transfer to the supplied critical sets gives the following explicit partial squares.

Input order/size	Contracted order	Defining size $\lvert U\rvert$	Private-intercalate core
17/200	16	178	84
19/245	18	239	23
33/844	32	801	380

The contracted completions and defining partials are saved as `direct_strip_n17_size200_to16.json`, `direct_strip_n19_size245_to18.json`, and `direct_strip_n33_size844_to32.json` in the stripping directory. These are defining supersets, not critical sets of the displayed sizes. In particular $801 > 781$ makes the last partial a possible search input; it does not establish any improvement at order 32 or refute the dyadic equality conjecture.

A single bounded exploratory pruning pass produced sizes 149 at order 16 and 196 at order 18, with an alternative completion for each retained clue. These two outputs are search-certified pending independent verification and are below the existing records. At order 32, the 20-second pass stopped at defining size 752 with 596 retained clues still lacking a completed essentiality check. That output is **not certified critical**, and its size already lies below 781. No new record follows from these passes. Timeouts were retained as unresolved clues; they were not interpreted as uniqueness results.

The pruning program is `lcs_full_review/stripping/prune_direct.py`. It accepts a clue deletion only after an exhaustive no-alternative result from the existing exact-cover oracle. Every retained clue reported as essential has a saved alternative completion. Unique completion of the starting partial follows from the transfer theorem, and accepted deletions preserve it. The statuses and witnesses are saved in `pruned_direct_n16_seed0.json`, `pruned_direct_n18_seed0.json`, and `pruned_direct_n32_seed0.json`.

17 Earlier structural proof and the all-orders threshold $n \geq 8$

The following earlier arguments are retained in full for their threshold and structural dependencies. Their square-root bounds are superseded by the constant-minus-four theorem for $n \geq 14$, but the constant-minus-six theorem below applies for all $n \geq 8$.

17.1 2. Definitions and elementary facts

Let P be a critical set in its unique Latin-square completion M of order n. A **hole** is an unfilled cell of P. Let H=n^2-|P|. An **m-row** or **m-column** has exactly m holes. A symbol has m holes when it occurs n-m times in P. Write RS(i) and CS(j) for the sets of symbols missing from row i and column j. A **conjugation** permutes the three coordinates of the triples (row, column, symbol).

For every filled cell x choose a Latin square W_x completing P minus x and different from M. Necessarily W_x changes x: otherwise it would complete P and equal M. We call this the **witness property**. All other filled cells retain their symbols in W_x.

Intersection fact, corresponding to Černý's Lemma 1. For all i,j, RS(i) intersects CS(j). At a hole, use M(i,j). At a filled cell x=(i,j), use W_x(i,j). This proof does not require non-singularity.

Covering fact, corresponding to Lemma 4. If J is the set of hole columns of row i, then the columns in J contain at least n holes. In fact their missing-symbol sets together contain every symbol. For a symbol missing from row i, its position in M supplies a column in J that misses it. For a symbol k present at x in row i, W_x must move k into a hole of that row. Since the deleted cell is outside J, the destination column already missed k in P. Summing the missing-symbol counts over J gives the result. The argument applies to all conjugates.

The intersection fact gives sum_s d_s^2 at least n^2, where d_s is the number of holes of symbol s: symbol s belongs to exactly d_s^2 row/column intersections.

A 1-row has its only hole in a completely empty column, by the covering fact. Its missing symbol is absent from P, by the intersection fact. Conjugation gives the analogous statements for columns and symbols. There are no 0-lines, by the witness property. There is at most one 1-line of each type: a 1-symbol forces an empty row and column, whose union contains two holes of every other symbol; then conjugate.

There is at most one completely empty row, one completely empty column and one absent symbol. Otherwise swapping two empty rows, columns or absent symbols in M gives another completion of P.

17.2 3. The frame case

Let the minimum hole counts in rows, columns and symbols be ρ,γ,σ. For a pair of these minima, Černý's case 3 occurs exactly when one is 1 and the other is at most 2. All three pairs have this property exactly when at least two minima are 1 and all are at most 2. Conjugate so that a row and a column have one hole each.

Let i0 be the 1-row. Its sole hole lies in an empty column c and carries an absent symbol w. Let j_r be the 1-column. Its sole hole lies in an empty row r and carries an absent symbol w'. The preceding uniqueness observation gives w'=w. Also r is different from i0, and M(i0,c)=w, so M(r,c) is not w.

Call the union of row r, column c and all positions of w in M a **frame**. These three sets have three distinct pairwise intersections and no triple intersection, so the frame contains 3n-3 holes. Write E for the other holes. Let π(i) be the column of w in row i. Then i0=π inverse(c), j_r=π(r), and row i0 meets no E-hole.

Frame lemma. |E| is at least ceil((n-4)/2).

To prove it, let R_E and C_E be the rows and columns meeting E. If every row other than r,i0 belongs to R_E, then |E| is at least n-2 and we are done. Otherwise choose such a row i outside R_E. Choose a column j outside C_E and outside {c,j_r,π(i)}. Then x=(i,j) is filled.

Put s=M(i,j), u=M(i,c), v=M(r,j), and k=π inverse(j). By the excluded columns, k is different from i, i0 and r. Row i has only its two frame holes, so W_x(i,j) belongs to {u,w}. Column j has only its two frame holes, at r and k, so W_x(i,j) belongs to {v,w}.

If u is different from v, the new entry at x is w. The w of row k must consequently leave (k,j). All changeable cells of that row are its two frame holes and its E-holes. It cannot move to (k,c), because row i0 has only one changeable cell, (i0,c), which must retain w. Thus row k belongs to R_E.

There are at least n-3-|C_E| eligible columns j. At most one satisfies M(r,j)=M(i,c), and all the others give distinct rows k in R_E. Therefore |R_E|+|C_E| is at least n-4. Since each is at most |E|, the assertion follows.

Hence the frame case has H at least 7n/2-5. This exceeds the right side of (1)'s hole bound for n at least 8.

For completeness, in any such frame the only possible 1-lines are row i0, column j_r and symbol z=M(r,c). Swapping row and symbol coordinates sends the frame to an empty row w, empty column c and absent symbol r. Its special row is z, because the triple (r,c,z) becomes (z,c,r). Thus the conjugation claim in the old handoff is also correct.

17.3 4. Reduction to the common row and column configuration

Assume some pair of minima avoids case 3 and orient it as rows and columns.

If either minimum is at least 4, H is at least 4n. If one is 3 and the other is at most 3, apply the covering fact to a line with at most three holes. This gives H at least n+3(n-3)=4n-9, which is at least 7n/2-5 for n at least 8. Thus only minimum row and column hole counts both equal to 2 remain.

Let R and C be the sets of 2-rows and 2-columns, and m=|R|, q=|C|. If m is at most n/2, choose a 2-column. Its two hole rows contain at least n holes together. All remaining rows contain at least 3 holes except for at most m 2-rows. Consequently H is at least 4n-m-6, hence at least 7n/2-6. Apply the transpose if q is at most n/2.

Now m,q are at least floor(n/2)+1, hence at least 5. Consider the 2-element sets of hole columns of the rows in R. If two are disjoint, applying the covering fact to both gives H at least 4n-8. Otherwise this is a pairwise intersecting family. Such a family either has a common member or consists of subsets of a 3-element set. To see this, take {u,v},{u,w}; any set avoiding u must be {v,w}, and any set meeting all three lies in {u,v,w}.

In the latter case without a common member, all three pairs occur. If the three column hole counts are h1,h2,h3, the covering fact gives h1+h2, h2+h3 and h3+h1 each at least n. Thus H is at least 3n/2+2(n-3)=7n/2-6. The same argument applies to the 2-columns.

The remaining case has a column j0 meeting a hole of every row in R and a row i0 meeting a hole of every column in C. They each have at least five holes, so j0 is not in C and i0 is not in R.

There is a symbol a missing from all rows in R and all columns in C. Here is the missing detail in the use of Černý's Lemma 8. The distinct symbols M(i,j0), i in R, belong to the union of the RS(i), so that union has at least five members. Similarly the union of CS(j), j in C, has at least five members. These two families of 2-element sets intersect across families by the intersection fact. If the column family has a disjoint pair, the row union has size at most four. If it contains a triangle, the row union has size at most three. Otherwise it is a star, and its union of size at least five supplies at least three distinct leaves. Every row set must contain its center a. This proves the assertion.

17.4 5. Duplicate pairs

We need to exclude all four duplication types: two 2-rows with the same hole columns or the same missing symbols, and the two transposed versions.

Correction accepted from Claude's review. The former argument only proved that column d_t misses b_s when s is different from t. It did not prove that d_t misses b_t, and that claim is false in general (example below). The former estimate H at least 2n+(m-2)^2-1 is withdrawn. The proof below uses only the off-diagonal claim and treats the two small orders separately.

Duplicate lemma, revised. Suppose m at least 5 2-rows all have a hole in column j0 and all miss symbol a. If two of these rows have the same missing-symbol pair or the same hole-column pair, then

$$H \geq 2n + m^2 - 5m + 5. \tag{2}$$

Moreover, if n is at least 8 and m is at least floor(n/2)+1, then H is at least 7n/2-6. In particular, the duplicate cases have at least 22 holes for n=8 and at least 26 holes for n=9.

First suppose rows u,v both miss {a,b}. Relabel them so M(u,j0)=a and M(v,j0)=b. Write d_u,d_v for their other hole columns, so M(u,d_u)=b and M(v,d_v)=a. Let T=R minus {u,v}. Every t in T misses {a,b_t}, with b_t different from a,b, and has M(t,j0)=b_t and M(t,d_t)=a at its other hole. The b_t are pairwise distinct. The columns d_t, t in T, and d_v are pairwise distinct because their entries a occupy different rows of M. They all differ from j0.

For s,t in T with s different from t, column d_t misses b_s. Otherwise b_s occurs in a filled cell x of row t. In W_x, that b_s must move to (t,j0), since column d_t already contains an unchanged b_s. Then W_x(s,j0)=a. But one of W_x(u,j0), W_x(v,j0) must also be a, since those two entries belong to {a,b} and differ. This is impossible. No assertion is made about b_t in column d_t, and none is possible from this configuration alone.

In the following critical set of order 7 and size 24, filled cells are on the left and the completion M on the right, 1-indexed. Rows u=3 and v=5 both miss {1,7} and both have a hole in column j0=4, with a=7. Row t=4 misses {5,7} and has its other hole in column d_t=6, so b_t=5. Yet column 6 contains 5 in row 3.

```
1 2 3 . 5 6 .     1 2 3 4 5 6 7
2 . . . 6 . .     2 7 5 3 6 1 4
3 4 6 . 2 5 .     3 4 6 7 2 5 1
4 6 2 . 1 . 3     4 6 2 5 1 7 3
5 3 . . 4 2 6     5 3 7 1 4 2 6
6 . 4 . . . .     6 1 4 2 7 3 5
. . . . . . .     7 5 1 6 3 4 2
```

Among the order-7 census critical sets, taken in all six conjugates, this local configuration occurs 217,541 times and the claim fails in 14,468 of them (`sec5test_results.txt`). Whether it could hold under the full hypotheses of the lemma, with every 2-row sharing j0 and a and m at least 5, is not needed and is not examined.

Each of the m-2 columns d_t therefore misses a and the m-3 symbols b_s with s different from t, giving at least m-2 holes. Column j0 has at least m holes. Every other column has at least two holes with at most one exception of one hole, by section 2. Adding these disjoint columns proves

$$H \geq (m-2)^2 + m + 2(n-m+1) - 1 = 2n + m^2 - 5m + 5.$$

This is increasing in integer m at least 5. With m at least floor(n/2)+1, its difference from 7n/2-6 is at least k^2-6k+7 for n=2k, and k^2-6k+11/2 for n=2k+1. Both are positive for k at least 5. Thus n at least 10 is settled. For n=8 or 9, m at least 6 is also settled by (2). Only m=5 remains.

Orders 8 and 9 with m=5. Put h equal to the number of holes in j0, so 5 is at most h and h is at most n. None of d_v and the three columns d_t can be a 1-column: its hole in a 2-row would force that row to be completely empty, by section 2. The covering fact on each of those four rows gives

$$c(d_t) \geq \max(3, n-h), \qquad c(d_v) \geq \max(2, n-h),$$

where c(j) denotes the number of holes in column j. Set ϵ=1 if a 1-column exists, and ϵ=0 otherwise. It is outside these five distinguished columns. Counting the other n-5 columns gives

$$H \geq K_n(h, \varepsilon) := h + 3\max(3, n-h) + \max(2, n-h) + 2(n-5) - \varepsilon. \tag{2a}$$

n	h values	K_n(h,1), in the same order
8	5, 6, 7, 8	22, 22, 23, 24
9	5, 6, 7, 8, 9	28, 25, 25, 26, 27

Thus n=8 is finished. For n=9 and ϵ=0 every entry increases by one, so H is at least 26. For ϵ=1 only h=6 or 7 needs an additional hole.

Let ℓ be the 1-column and r its sole hole row. Row r is completely empty, and w=M(r,ℓ) is absent from P. Because w is absent from P, it is missing from every row. In particular w lies in RS(u)={a,b} and in every RS(t)={a,b_t}. Since each b_t differs from b, w=a. Hence M(r,ℓ)=a. Also r is outside R, and d_u is different from ℓ because (u,d_u) is a hole in a 2-row.

We also have d_u different from d_v. If they were equal, the four holes in rows u,v and columns j0,d_u would carry the pattern (a,b; b,a). Interchanging a and b in those four cells would give another Latin-square completion of P. This contradicts uniqueness. Such a four-cell pattern is called an **intercalate**.

There are now two possibilities for d_u:

- If d_u=d_t for some t in T, that column misses b as well as a and the two b_s with s different from t. Those four symbols are distinct, so it has at least four holes instead of the three counted in (2a).
- Otherwise d_u is outside {j0,d_v} and the three columns d_t. It has a hole at (u,d_u) carrying b, a hole at its a-position because a is globally absent, and a hole at (r,d_u) because row r is empty. These are three distinct cells: the a-position is not in row u, and it is not in row r because that row's a is in ℓ. This gives at least three holes instead of the two counted in (2a); d_u is not the exceptional 1-column.

In both cases the count gains one, giving H at least 26. This finishes the missing-symbol version for every n at least 8.

Interchanging columns and symbols proves the hole-column version: the common missing a becomes a common hole column, and the common hole column j0 becomes a common missing symbol. The entire argument allowed a 1-column, so it remains valid after this conjugation. Transposition gives the two column versions. Thus all four duplicate cases have been handled.

Henceforth none of the four duplication types occurs.

17.5 6. The remaining forcing arguments

Each 2-row has holes in j0 and one other column, and the other columns are pairwise distinct. Let x be the number of these other holes lying in 2-columns. Such a 2-column has its other hole in i0. Index these pairs (i_t,j_t), t=1,. . . ,x. Set

$$RS(i_t) = \{a, b_t\}, \qquad CS(j_t) = \{a, c_t\}.$$

The b_t are pairwise distinct, as are the c_t. Counting the other hole columns of rows not paired with C gives

$$m - x \le n - q - 1, \qquad m + q \le n + x - 1. \tag{3}$$

In particular x is at least 2. Transpose if necessary so q is the smaller of m,q. The covering fact on any paired row gives at least n-2 holes in j0. Therefore

$$H \ge (n-2) + 2q + 3(n-q-1) = 4n - q - 5 \ge \frac{7n}{2} - \frac{x}{2} - \frac{9}{2}. \tag{4}$$

If x is at most 3, this is at least 7n/2-6, and the case is finished. This handles the odd-order x=2 possibility and avoids the second half of the printed proof of Lemma 15.

Now suppose x is at least 4.

Pair agreement. For every t, b_t=c_t. Suppose instead b_t is different from c_t. Choose distinct p,q different from t so that b_t differs from c_p,c_q; this is possible because the c values are distinct and x is at least 4. Delete the filled cell (i_t,j_p), whose original value is v. In its witness b_t can occupy neither j_t nor j_p, because these columns do not miss b_t. Thus b_t moves to j0. The changed entry at j_p must then be a, and the entry at j_t must be v=c_t. Repeating with j_q gives M(i_t,j_q)=c_t also, contradicting row distinctness. Hence b_t=c_t.

No b entries in the paired submatrix. If a filled cell (i_1,j_2) contains b_3, use its witness W. The indices are distinct, since row i_1 misses b_1 and column j_2 misses b_2. The following chain is forced:

$$W(i_1, j0) = b_3, \quad W(i_3, j0) = a, \quad W(i_3, j_3) = b_3, \quad W(i0, j_3) = a,$$
$$W(i0, j_1) = b_1, \quad W(i_1, j_1) = a, \quad W(i_1, j_2) = b_1.$$

The last value is impossible: column j_2 did not miss b_1 and the deleted value was b_3. This proves the analogue of Lemma 16.

Forcing a missing symbol in any other row. Suppose a filled cell (i,j_1) contains b_2. The preceding paragraph excludes paired rows, and i is not i0 because that cell is a hole. In its witness W the chain is

$$W(i0, j_1) = b_2, \quad W(i0, j_2) = a, \quad W(i_2, j_2) = b_2, \quad W(i_2, j0) = a,$$
$$W(i_1, j0) = b_1, \quad W(i_1, j_1) = a, \quad W(i, j_1) = b_1.$$

Thus row i misses b_1 in P. If i were another 2-row, it would miss {a,b_1}, duplicating i_1, which has been excluded. Therefore no row in R contains any b_t in any paired column. Transposition gives the corresponding column assertion. This proves Lemma 17 and Corollary 18 under the repaired hypotheses.

Fix s. In column j_s, each b_t with t different from s occurs in P, since the missing symbols there are exactly {a,b_s}. These x-1 entries occupy distinct rows outside R and outside i0. The preceding forcing rule makes b_s missing in each of those rows. Transposition shows that b_s is missing in at least x-1 distinct columns outside C **and outside j0**. Summing over s gives at least x(x-1) distinct missing (column,symbol) pairs in those columns. A missing (column,symbol) pair corresponds to a distinct hole of M. This proves the strengthened, disjoint form of Corollary 19.

17.6 7. Stronger final calculation

Keep q in both counts. From (4) and the preceding disjoint holes,

$$H \geq A := 4n - q - 5, \qquad H \geq B := n - 2 + 2q + x(x-1). \tag{5}$$

The term n-2 counts j0; the term 2q counts C; the last term counts holes outside both. No hole is counted twice.

Use (3) in A, and take the weighted average (2A+B)/3:

$$H \geq f(x) := \frac{7n}{2} - \frac{x}{2} - \frac{9}{2}, \qquad H \geq g(x) := 3n - 4 + \frac{x(x-1)}{3}. \tag{6}$$

For x at least 1, f decreases and g increases. Their positive intersection is

$$x_* = \frac{\sqrt{24n-23}-1}{4}, \qquad f(x_*) = \frac{7n}{2} - \frac{\sqrt{24n-23}+35}{8}.$$

For every x at least 4, max(f(x),g(x)) is at least this value: use f when x is at most x_* and g otherwise. For n at least 8 the subtracted quantity (sqrt(24n-23)+35)/8 is at least 6, with equality at n=8. All the previously dispatched cases give H at least 7n/2-6 or better. This completes the proof of (1).

The old bound follows because sqrt(40n-31)-sqrt(24n-23) is at least 2 for n at least 8 (already 4 at n=8 and increasing).

An optional further improvement is to retain the integer parameters in (5), with x at least 4, q at least max(x,floor(n/2)+1), and q at most floor((n+x-1)/2), then take the minimum of this branch and ceil(7n/2-6). This note uses the simpler closed form (1); no claim of optimal constants is made.

Theorem. For every integer $n \geq 8$,

$$\boxed{\mathrm{lcs}(n) \leq n^2 - \left\lceil \frac{7n}{2} - 6 \right\rceil.} \tag{T}$$

The point is to use the difficult configuration in all three coordinates, and to count the lines having exactly two holes separately from the remaining lines. In the earlier proof, the remaining symbols were only charged two holes each. Most of them must instead be charged three.

17.7 1. Structural input and notation

Let P be a critical set in its unique completion M, and let $H = n^2 - |P|$. A line means a row, a column, or the set of occurrences of one symbol. Its degree is its number of holes. Conjugating a Latin square permutes these three kinds of lines.

The audit proves the following facts.

1. Every line has positive degree; in each coordinate there is at most one line of degree one.
2. A degree-one symbol has its sole hole at the intersection of a completely empty row and a completely empty column.
3. If $H < 7n/2 - 6$, we can orient the square so that the minimum row and column degrees are both two. Write R, C for **all** degree-two rows and columns, and $m = |R|, q = |C|$. Then

   $$m, q \geq \lfloor n/2 \rfloor + 1.$$

   There are a column j_0, a row i_0, and a symbol a such that every row in R has a hole in j_0 and misses a, and every column in C has a hole in i_0 and misses a. The four duplicate-pair types are absent: neither hole locations nor missing symbols coincide for two degree-two rows, and likewise for columns.
4. Let x count the holes incident with both R and C. They form a matching (i_v, j_v). Then

   $$x \geq 4, \qquad m + q \leq n + x - 1. \tag{1}$$

   The degrees of i_0, j_0 are at least $n - 2$.
5. The paired rows and columns miss matching symbol pairs $\{a, b_v\}$, with distinct b_v. For every v, there are at least $x - 1$ holes of b_v outside $C \cup \{j_0\}$, and the analogous row assertion holds.

Items 3–5 are audit sections 4–6. The alternatives rejected there already prove (T). The arguments establishing pair agreement and item 5 use only the star configuration, the absence of the four duplication types, and at least four paired holes. They do not require that every other line have degree at least two. This observation matters in section 3 below.

Put $T = \{s : d_s = 2\}$, $t = |T|$. Define y to be the number of holes incident with a row of R and a symbol of T, and z the number incident with a column of C and a symbol of T. These definitions do not depend on choosing an orientation for the conjugates.

We will prove that the same star configuration is available for the row–symbol and column–symbol pairs, with paired counts $y, z \geq 4$.

17.8 2. When no symbol has degree one

The minimum symbol degree cannot be at least three: the audit's covering argument, applied after conjugation to a degree-two row, gives at least $4n - 9 \geq 7n/2 - 6$ holes. Thus the minimum symbol degree is two.

Apply audit sections 4–6 to the row–symbol coordinate pair. Under the assumed strict inequality $H < 7n/2 - 6$, none of its alternative branches can occur. Thus $t > n/2$, there is a star configuration without duplicated pairs, and its paired count is $y \geq 4$. Applying the same argument to columns and symbols gives $z \geq 4$.

The shared symbol-line of the degree-two rows is necessarily the original a: their distinct missing pairs already have exactly that common member. Therefore the high-degree column in the row–symbol conjugate is a, and

$$d_a \geq n - 2. \tag{2}$$

The same uniqueness of the common member identifies the other shared lines as the original i_0, j_0. In particular, the three applications describe the same sets of lines and the same numbers m, q, t, x, y, z.

17.9 3. The exceptional degree-one symbol

Suppose instead that a symbol w has degree one. Its hole forces an empty row and column. These must be i_0, j_0. Indeed, another empty column would give every row of R the same two hole columns, contradicting the excluded duplication; the argument for the empty row is identical. Consequently

$$w = M(i_0, j_0), \qquad h(i_0) = h(j_0) = n. \tag{3}$$

Each degree-two symbol has exactly one hole in i_0 and one in j_0. These are distinct because the symbol at their intersection is w, which has degree one. The other hole rows of these symbols are distinct, as are their other hole columns, by the Latin property.

This directly supplies the two additional star configurations:

- In row–symbol coordinates, degree-two rows share the symbol-line a; degree-two symbols share row i_0; both families miss the common column-label j_0.
- In column–symbol coordinates, degree-two columns share a; degree-two symbols share column j_0; both families miss the common row-label i_0.

The original degree-two rows and columns have no duplicated pairs by the audit. The degree-two symbols have no duplicated hole-row or hole-column pairs by the preceding Latin-property observation. Also $a \notin T$, since $d_a \geq m \geq 5$. Thus the incidence counts apply even before we know that y, z are large:

$$m + t \leq n + y - 1, \qquad q + t \leq n + z - 1. \tag{4}$$

Explicitly, the m distinct second missing symbols of the rows in R consist of y members of T and $m - y$ members outside $T \cup \{a\}$. Hence $m - y \leq n - t - 1$, even when t or y is zero. The other inequality follows by transposition. This directly constructs the stars and incidence bounds; the global two-minimum reduction is not applied to these conjugates.

For clarity, the row–symbol matrix has rows R, columns T, shared hole column a, shared hole row i_0, and common missing value j_0. Its entries are original column labels. The column–symbol matrix instead has rows C, columns T, shared hole column a, shared hole row j_0, and common missing value i_0; its entries are original row labels.

It remains to exclude $y \le 3$ and $z \le 3$. For the original row–column pairs, let $\epsilon_v = 1$ when the paired hole carries b_v, and zero when it carries a. Inclusion–exclusion and the disjoint holes in structural item 5 give

$$d_a \ge m + q - x + \sum_v \epsilon_v, \qquad d_{b_v} \ge x + 1 - \epsilon_v.$$

Hence

$$d_a + \sum_v d_{b_v} \ge m + q + x^2. \tag{5}$$

All these $x + 1$ symbols have degree at least four, so they are different from w and the t degree-two symbols. Every remaining symbol has degree at least three. Therefore

$$H \ge m + q + x^2 + 2t + 3(n - x - t - 2) + 1 = 3n + m + q - t + x^2 - 3x - 5. \tag{6}$$

If $y \le 3$, use (4) to obtain

$$H \ge 2n + 2m + q + x^2 - 3x - y - 4 \ge \frac{7n}{2} - \frac{3}{2}. \tag{7}$$

The last step uses $m, q \ge n/2 + 1/2$, $x \ge 4$, and $y \le 3$. This contradicts our assumption. Interchanging rows and columns excludes $z \le 3$.

Now $y, z \ge 4$, so the audit's local pair-agreement and forcing arguments apply to these conjugates too. A paired row–symbol hole has a degree-two row missing a and a degree-two symbol. The covering fact on that row gives $d_a + 2 \ge n$, proving (2). The three common lines therefore all have degree at least $n - 2$ in this exceptional case as well.

17.10 4. Counting all three coordinates

We now have all three star configurations, with $x, y, z \ge 4$, and

$$m + q \le n + x - 1, \quad m + t \le n + y - 1, \quad q + t \le n + z - 1. \tag{8}$$

In the original row–column configuration, at most one ϵ_v is one. If $\epsilon_v = 1$, both (i_v, j_0) and (i_0, j_v) carry a; two such indices would repeat a in j_0. Thus

$$\sum_v d_{b_v} \ge x(x + 1) - 1. \tag{9}$$

Count symbols in four disjoint classes: the high-degree symbol a; the x distinct symbols b_v; the t symbols of degree two; and all others. The last class has degree at least three, except for at most one degree-one symbol. Allowing that exception, even when it does not occur, yields

$$\begin{aligned} H &\ge (n - 2) + x(x + 1) - 1 + 2t + 3(n - x - t - 1) - 2 \\ &= 4n - t + x^2 - 2x - 8. \end{aligned} \tag{10}$$

The same argument in the other two coordinate pairs gives

$$H \ge 4n - q + y^2 - 2y - 8, \qquad H \ge 4n - m + z^2 - 2z - 8. \tag{11}$$

Let $S = m + q + t$ and $V = x + y + z$. Adding (8) gives $2S \le 3n + V - 3$. Adding (10)–(11), then substituting this bound, gives

$$\begin{aligned} 3H &\ge 12n - S + x^2 + y^2 + z^2 - 2V - 24, \\ H &\ge \frac{7n}{2} - \frac{15}{2} + \frac{(2x^2 - 5x) + (2y^2 - 5y) + (2z^2 - 5z)}{6}. \end{aligned} \tag{12}$$

For every integer $v \ge 4$, $2v^2 - 5v \ge 12$. Consequently

$$H \ge \frac{7n}{2} - \frac{3}{2} > \frac{7n}{2} - 6,$$

the required contradiction. Therefore $H \ge 7n/2 - 6$, and integrality proves (T).

18 Earlier symbol-counting and integer-parameter refinements

18.1 A stronger upper bound from counting symbols

18.1.1 The structural input

Use the notation of the supplied audit. Let P be a critical set with unique completion M, and let $H = n^2 - |P|$. Its sections 2–6 show that, for $n \geq 8$, either

$$H \geq \frac{7n}{2} - 6, \tag{1}$$

or the following configuration can be used, after conjugating and transposing if necessary.

There are sets R, C of all 2-rows and all 2-columns, with sizes m, q, where

$$m \geq q \geq \lfloor n/2 \rfloor + 1.$$

All rows in R have a hole in column j_0; all columns in C have a hole in row i_0. The row i_0 is outside R and column j_0 outside C. Every row in R, and every column in C, misses the same symbol a.

No two 2-rows have the same pair of hole columns or the same pair of missing symbols; the analogous column duplications are also excluded. There are $x \geq 4$ paired holes (i_t, j_t), with $i_t \in R$, $j_t \in C$, and

$$RS(i_t) = CS(j_t) = \{a, b_t\}, \qquad t = 1, \ldots, x,$$

where the b_t are distinct. Furthermore,

$$m + q \leq n + x - 1, \tag{2}$$

$$H \geq 4n - q - 5, \tag{3}$$

and, for each t, there are at least $x - 1$ holes of symbol b_t in columns outside $C \cup \{j_0\}$.

These are precise structural inputs, not consequences of the audit's numerical upper bound alone. The last assertion is its strengthened Corollary 19; pair agreement requires $x \geq 4$. All branches with $x \leq 3$ have already been covered by (1).

18.1.2 The additional inequality

Let d_s be the number of holes carrying symbol s in M. Put

$$\epsilon_t = \begin{cases} 0, & M(i_t, j_t) = a, \\ 1, & M(i_t, j_t) = b_t, \end{cases} \qquad \epsilon = \sum_t \epsilon_t.$$

Each row in R and each column in C has an a-hole. Their common a-holes are exactly the paired positions for which $\epsilon_t = 0$: the only holes at intersections of R and C are the paired positions. Inclusion–exclusion therefore gives

$$d_a \geq m + q - x + \epsilon. \tag{4}$$

If $\epsilon_t = 0$, the two holes (i_t, j_0) and (i_0, j_t) both carry b_t. They are in columns j_0 and C, so neither is among the $x - 1$ holes supplied by Corollary 19. If $\epsilon_t = 1$, the paired hole itself carries b_t and is likewise outside that count. Thus, in both cases,

$$d_{b_t} \geq x + 1 - \epsilon_t. \tag{5}$$

Adding (4) and (5) gives the especially useful cancellation

$$\boxed{d_a + \sum_{t=1}^{x} d_{b_t} \geq m + q + x^2.} \tag{6}$$

A critical set has no symbol with zero holes, and at most one symbol with just one hole. These are the conjugated elementary facts from audit section 2. Consequently the remaining $n - x - 1$ symbols contribute at least $2(n - x - 1) - 1$ holes. We obtain

$$\boxed{H \geq 2n + m + q + x^2 - 2x - 3.} \tag{7}$$

This is the extra inequality missing from the earlier final calculation. It counts disjoint symbol classes, so there is no overlap between its main terms. The orientation of a paired diagonal is not assumed; the ϵ terms cancel.

18.1.3 A first closed form

Since $m \ge q$, (7) yields $H \ge 2n + 2q + x^2 - 2x - 3$. Combining this with two copies of (3) yields

$$H \ge \frac{10n + x^2 - 2x - 13}{3}. \tag{8}$$

Meanwhile (2) and (3) give

$$H \ge \frac{7n}{2} - \frac{x}{2} - \frac{9}{2}. \tag{9}$$

The right side of (9) decreases in x, and that of (8) increases for $x \ge 1$. They intersect at

$$x_* = (1 + \sqrt{8n - 7})/4.$$

Therefore the difficult branch has

$$H \ge \frac{7n}{2} - \frac{\sqrt{8n - 7} + 37}{8}. \tag{10}$$

Together with (1), this already replaces the audit's $\sqrt{24n - 23}$ by a square root with leading coefficient $\sqrt{8}$.

18.1.4 Removing another small loss

Let $\eta = 1$ if a symbol has exactly one hole, and $\eta = 0$ otherwise. The distinguished symbols $a, b_1, \ldots, b_x$ cannot have just one hole: (4) gives $d_a \ge \max(m, q)$, and (5) gives $d_{b_t} \ge x$. Hence (7) improves to

$$H \ge 2n + m + q + x^2 - 2x - 2 - \eta. \tag{11}$$

If $\eta = 1$, its sole hole forces its row and column to be empty. That empty column must be j_0: otherwise all 2-rows would have the same two hole columns, contradicting the exclusion of duplicates. Similarly the empty row is i_0. Thus column j_0 has n holes, rather than merely $n - 2$, and (3) strengthens to

$$H \ge 4n - q - 5 + 2\eta. \tag{12}$$

For $\eta = 0$, the decreasing and increasing estimates are now

$$H \ge \frac{7n}{2} - \frac{x}{2} - \frac{9}{2}, \qquad H \ge \frac{10n + x^2 - 2x - 12}{3}.$$

Their intersection gives

$$H \ge \frac{7n}{2} - \frac{\sqrt{8n - 23} + 37}{8}. \tag{13}$$

For $\eta = 1$, the corresponding estimates give the stronger result

$$H \ge \frac{7n}{2} - \frac{\sqrt{8n + 25} + 21}{8}.$$

It is stronger than (13) for $n \ge 8$, since $\sqrt{8n + 25} - \sqrt{8n - 23} < 16$.

Finally, (1) and (13), followed by integer rounding, prove

$$\boxed{H \ge \min\left\{\left\lceil \frac{7n}{2} - 6 \right\rceil, \left\lceil \frac{7n}{2} - \frac{\sqrt{8n - 23} + 37}{8} \right\rceil\right\} \quad (n \ge 8).} \tag{14}$$

For $n \ge 18$, the second expression is the smaller one, proving (U). This improves the supplied audit's closed form for every order in the weak sense, and strictly at many orders. It also dominates the Horák–Dejter bound throughout $n \ge 8$, including asymptotically.

18.1.5 Retaining integer parameters

A slightly sharper computable hole bound is

$$\begin{gathered} K(n) = \min\{\lceil 7n/2 - 6 \rceil, \min_{(\eta,x,q)\in\mathcal{A}_n} \max(A, B, C)\}, \\ A = 4n - q - 5 + 2\eta, \quad B = n - 2 + 2\eta + 2q + x(x - 1), \\ C = 2n + 2q + x^2 - 2x - 2 - \eta, \\ \mathcal{A}_n = \{(\eta, x, q) : \eta \in \{0, 1\},\ 4 \le x < n, \\ \max(x, \lfloor n/2 \rfloor + 1) \le q \le \lfloor (n + x - 1)/2 \rfloor\}. \end{gathered} \tag{15}$$

Empty inner feasible sets are ignored. Then $\ell(n) \le n^2 - K(n)$. The middle term retains the audit's column count. Replacing m by q in the last term is valid because $m \ge q$. For fixed x, η, the maximum is one decreasing affine function and one increasing affine function of q; the endpoints and the two integers around their intersection suffice. The supplied implementation checks this shortcut against exhaustive integer optimization for orders 8 through 150.

n	Audit upper	New upper from (15)	Verified lower or standard construction
8	42	42	37
9	55	55	46
10	71	71	59
11	88	88	74
12	108	108	90
13	130	129	107
14	153	153	126
15	179	178	147
16	206	206	175
17	236	235	200
32	919	918	781
50	2333	2332	—
100	9660	9658	—
1000	996523	996515	746306 from counting below
10000	99965065	99965039	87869144 from counting below

A dash does not mean absence of a lower bound; the column is deliberately limited to the listed constructions and selected counting examples. The powers-of-two construction supplies the entries at 8 and 32. The entries from 9 through 17 were reverified here.

19 Independent review notes and coverage

20 Independent audit of the constant-minus-four upper bound

14 October 2026. Audit of the complete constant-minus-four proof, read after it was saved. The initial working draft was incomplete at odd order with three paired holes; that draft was not accepted as a complete proof. The saved proof replaces all positive-small-overlap cases with a different witness argument, and closes the gap.

20.1 Verdict and scope

No mathematical gap was found in the complete proof. It establishes

$$\boxed{\operatorname{lcs}(n) \le n^2 - \left\lceil \frac{7n}{2} - 4 \right\rceil \qquad (n \ge 14).}$$

In particular, its order-17 upper bound is **233**; together with the previously verified size-200 example, this gives $200 \le \operatorname{lcs}(17) \le 233$.

The review independently reconstructed the changed arguments, rather than assuming that the earlier reduction remained valid at the new threshold. It checked the saved document against the original repaired audit, the three-coordinate proof, and the previous threshold audit. The earlier independent audit remains the detailed check of the final forcing chains and the exceptional one-hole-symbol case.

During review, the reviewer suggested the stronger duplicate estimate and a shorter zero-overlap argument; these were incorporated into the final proof and then checked in that saved form. Thus this is an independent agent review with mathematical feedback and repairs, not a claim that the reviewer had no involvement in the final argument. It is an AI proof check, not human refereeing or an assessment of publication priority.

20.2 1. Threshold and early reduction

Put $H = n^2 - |P|$, assume $H < 7n/2 - 4$, and write $n = 2k$ or $2k+1$. The correct integer caps are respectively

$$H \le 7k - 5, \qquad H \le 7k - 1.$$

For $n \ge 14$, $k \ge 7$. The improved frame lemma already reaches the desired ceiling and excludes the all-exceptional case. The alternatives $4n$ and $4n - 9$ exclude admissible coordinate pairs with a minimum at least three. The surviving pair has both minima two.

The covering inequalities $H \ge 4n - m - 6$, $H \ge 4n - q - 6$ then give $m, q \ge k - 1 \ge 6$, without assuming either family exceeds $n/2$. Disjoint hole pairs give $4n - 8$; non-star triangles give $4n - 6$. The latter uses the full set-covering statement: each symbol belongs to at least two of the three missing sets. Both estimates exceed the contrary threshold in this range.

The common missing symbol is justified by the cross-intersection argument, with unions of size at least six. The proof therefore obtains stars before trying to exclude duplicate pairs. No argument here presupposes a positive overlap.

20.3 2. Strengthened duplicate estimate, including its exception

The new bound

$$H \ge 3n + m^2 - 6m + 4$$

was checked directly from the repaired duplicate configuration. Its distinguished columns are disjoint: j_0, d_v, and the $m - 2$ columns d_t. Covering row v gives a combined contribution at least n from the first two; the d_t contribute at least $(m-2)^2$. The remaining $n - m$ columns contribute $2(n - m)$, with a possible deficit of one if a one-hole column exists. This gives the claimed constant 4 immediately when there is no exception.

With an exceptional one-hole column λ, its hole lies in an empty row r, and its missing symbol is globally absent. The distinct row missing pairs identify that symbol as a. The additional column d_u repairs the lost unit:

- It cannot equal d_v, because that would leave an intercalate entirely in holes and violate uniqueness.
- It cannot equal λ, whose unique hole is in the empty row rather than the degree-two row u.

- If it equals a d_t, it adds the distinct missing symbol b to the $m-2$ symbols already counted there.
- Otherwise it is an undistinguished column with three distinct holes: its b-position in row u, its absent-a position, and its empty-row position. The last two differ because row r's a is in λ.

This argument is not restricted to the original small-order use of this repair. Allowing the exceptional one-hole column is precisely what makes the column–symbol conjugation legitimate when excluding repeated hole pairs. Transposition supplies the two column cases.

At $m = k-1$, the excess over $7n/2 - 4$ is

$$(k-7)(k-2)+1 \quad \text{or} \quad (k-7)(k-2)+\tfrac{1}{2},$$

according to parity. Both are positive for $k \geq 7$, and the polynomial increases thereafter. In particular the borderline order 15 is covered; no unjustified $n \geq 18$ restriction remains.

20.4 3. The new witness argument and positive overlap

Suppose the common missing symbol a were retained in the shared column j_0. For a degree-two row p, its a must occur in its second-hole column d_p. For another degree-two row i, distinct second-hole columns ensure that (i, d_p) is a clue.

If d_p does not miss b_i, the changed value of this clue in a deletion witness is necessarily a: the row misses $\{a, b_i\}$, and the column misses a but not b_i. This displaces the a of row p. Only its two original holes can change, so the displaced a must move to j_0, where the retained a-clue blocks it. The deleted clue is in $d_p \neq j_0$, so that blocker is indeed retained.

Thus every d_p misses a and all $m-1$ other b_i, giving degree at least m. Counting disjoint columns yields

$$H \geq 2n + m^2 - m - 2,$$

which contradicts the threshold for $m \geq k-1$, $k \geq 7$. The transpose proves that a is missing from both shared lines. This proof requires distinct pairs but does not require pair agreement or four overlaps.

With one paired hole, covering now gives shared-line degree at least $n-1$, because a is already missing there. The count $H \geq 4n - q - 4$ and its transpose give $m, q > n/2$. Incidence consequently supplies at least two paired holes. Their distinct missing pairs intersect only in a; the complement of the shared line's missing set is contained in that intersection and excludes a. The shared line is therefore empty.

Empty centers improve the count to $H \geq 4n - q - 3$ and its transpose. The integer caps give $m, q \geq k+2$, hence at least five overlaps for even order and four for odd order. This correctly handles the formerly unresolved odd three-overlap case. There is no invocation of the pair-agreement lemma during this bootstrap.

20.5 4. Zero overlap

The proof of zero-overlap impossibility was checked separately; it does not use the positive-overlap conclusion.

Let $h = d(j_0)$ and $B = [n] \setminus CS(j_0)$. Every selected second-hole column misses B. Since a is missing from j_0, it is outside B. At most one degree-two row places its a in j_0, so at least $m-1$ distinct second-hole columns miss a in addition to B. All selected columns are outside $C \cup \{j_0\}$ when the overlap is zero. The resulting count is therefore valid without double counting:

$$H \geq h + 3n - q - 3 + (m-1)\max(0, n-h-2) + \max(0, n-h-3).$$

Its unique minimum is $4n - q - 5$ at $h = n-2$. In the variable $e = n-h$, the increments stated in the proof are correct: the expression decreases up to $e = 2$, then rises by $m-2$ and subsequently by $m-1$.

Consequently $m, q \geq k$. Incidence immediately excludes even order. In odd order it forces $m = q = k$; equality in the minimum then gives $h = n-2$ and $H = 7k-1$. The complement $B = \{b, c\}$ avoids a. Both b, c have degree at least k, and the disjoint row/column a-holes give $d_a \geq 2k$. Charging every other symbol two holes, except for at most one charged only one, gives

$$H \geq 4n - 9 = 8k - 5 > 7k - 1.$$

The three high-degree symbols are distinct and cannot be the exceptional degree-one symbol, so this count is valid. This closes the zero-overlap branch for the stated range.

20.6 5. Third coordinate and conclusion

If no symbol has degree one, a minimum at least three is excluded by covering. The new structural reduction can then be applied independently to the other two minimum-two coordinate pairs. The identification of their shared lines and their three incidence counts is the same as in the previously audited three-coordinate proof.

If a symbol has degree one, the proof instead constructs the other stars directly from its empty row and column. It does not apply a minimum-two reduction to a coordinate pair of minimum one. The degree-two symbols have distinct second hole rows and columns; their incidence bounds remain valid even when their number or an overlap is small. The previously checked small-overlap calculation gives $H \geq 7n/2 - 3/2$ if either additional overlap is at most three, contradicting the new threshold as well.

Once all three overlaps are at least four, the original local forcing lemmas and the final three-coordinate count apply unchanged. Their lower estimate

$$H \geq 7n/2 - 3/2$$

is stronger than the contrary threshold. Integrality therefore establishes the stated ceiling.

No finite computation is substituted for these structural arguments. This audit does not assert the -4 bound for orders 8–13, does not settle the superlinear-deficit conjectures, and does not establish novelty in the literature.

20.7 Integrated LaTeX exposition

The reviewer subsequently read the complete upper-bound chapter in `UPPER_TEX` of `lcs_review_full_sections.py` and checked its emitted form in `lcs_review_8_33.tex`. The emitted chapter is an exact copy of that source. The expanded frame witness argument, duplicate repair, both local forcing chains, identification of the conjugate centers, complete one-hole-symbol count, and final three-coordinate count are mathematically faithful to the proof audited above. No omitted mathematical dependency or new gap was found in the integrated chapter. Before acceptance, the final subsection was clarified to define the paired-hole indicators in both branches and to establish the leaf-degree inequality independently of the existence of a one-hole symbol; this also explicitly separates those leaves from the degree-two symbols. The clarification is present in both the source module and the regenerated LaTeX document. This is an audit of the mathematical exposition and its generation, not of PDF typography, and claims no additional theorem or assessment of novelty.

INDEPENDENT AUDIT OF BALANCED AND SUPERSET COUNTING Auditor: /root/audit_three_prolongations; 17 October 2026. Source audited: `outputs/lcs_bounds.md`, “A stronger finite and asymptotic lower bound”, equations (16)–(21), plus recorded finite pairs. Verdict: PASS for the finite criterion, integer endpoint, T(n) conclusion, and asymptotic constant 2.668162063280737765... . No claim of novelty or globally optimal finite parameter choice is assessed here.

1. Published permanent inputs. The Latin-square lower count follows from the van der Waerden permanent bound applied to each regular allowed-symbol matrix. The product is exactly $(n!)^{(2n)/n}(n^2)$. I independently derived the fixed-shape upper count using Bregman: add u_i labelled dummy positions allowing all n symbols to each row; each valid filling of its actual positions has u_i! dummy extensions. The allowed degree in a column is n minus its prior occupancy. Across the column the degrees run n,...,v_j+1, giving F_n(v_j). This checks the source formula without relying on the transcription of a numbered theorem.
2. Balancing. The reciprocal factorial product and F_n product are separately discretely concave in their integer hole counts. Both totals are h. Balancing each separately is legitimate for an upper bound even when their maxima do not come from a common shape. Boundary h=N is handled by omitting zero-power F_n(n+1); a definition of F_n(n+1) is unnecessary.
3. Superset amplification. Choosing one small critical set for each Latin square produces at least binom(d,h) distinct (N-h)-entry defining sets per Latin square. A defining set cannot be counted for two Latin squares. The contradiction proves that some square has no critical set of size <=N-d, hence its minimum critical set has size >=N-d+1. The +1 and strict comparison are correct. This yields T(n), max over squares of their minimum critical size, and consequently lcs.
4. Asymptotics and errors. For h~c*n^(5/3), a=h/n~c*n^(2/3). Expanding the column log product gives n log(n!)-n log(a!)-n^2+h+O(n log^2 n). Together with the row denominator, the (n!)^(h/n) factor, the Latin lower count and log binom(N,h), this gives 5h log n-3h log h+3h+O(h^2/N+n log^2 n). Both errors are o(n^(5/3)). Integer balancing changes only the lower-order remainder. For h~zd, d~D*n^(5/3), the strict criterion has normalized log ratio 3z log(e/(zD))-H(z). Its threshold is the claimed exponential expression. The derivative numerator -3z-log(1-z) has exactly one interior zero, with negative-to-positive sign change; the endpoint z=0 is excluded. At that zero the threshold reduces to e^z z^(-2/3). Rounding h,d preserves the limit. The theorem requires +epsilon or o(1); dropping this is not authorized.
5. Independent numerics. `counting_independent_check.py` imports none of the original scripts. It uses a canceled-factorial expression (h! cancels) and separate column tails, with 90-digit interval arithmetic. All five finite rows pass with positive lower endpoints. It brackets the root and constant by interval sign tests. The output is

`counting_independent_results.json`. This validates the listed parameters, not their optimality. The mathematical proof, including the full Bregman padding argument, is typeset in `counting_full.tex`.

Audit limitations: published permanent inequalities are external inputs; their original proofs are not repeated. This is an independent agent audit, not a claim of external peer review.

INDEPENDENT AUDIT: HORAK–ALDRED–FLEISCHNER SUBSQUARE COVER

VERDICT: PASS. The parent's proposed refinement is valid, including exclusion of equality in the row-count estimate. It may replace the earlier2m+3 cover by max(B_m+m,2B_m-9), with the definition and domains below.

20.8 Published input checked directly

I extracted text from and visually inspected PDF page2, printed page420, of /Users/richardbean/Documents/maths/papers/horak-critical-sets-i.pdf. Theorem1.1 of P.Horak, R.E.L.Aldred and H.Fleischner, "Completing Latin Squares: Critical Sets", Journal of Combinatorial Designs10(2002),419–432, DOI10.1002/jcd.10028, says: every critical set of order t>=8 has at least floor((4t-8)/3) entries. The rounding is FLOOR, not ceiling. The condition is t>=8. Any defining set contains an inclusion-minimal defining subset, so the same size lower bound holds for every defining set of that order. The published theorem is an external input here; this audit does not claim to reprove its13-page proof.

20.9 Setup

Use the established triple prolongation, with the two enlarged subsquares S_0,S_1 of order t=m+3, intersecting in the3-by-3 new corner. For a defining subset C of the prescribed U, put D_i=C intersect S_i, D=D_0 union D_1, K=D_0 intersect D_1, and d=|D|,k=|K|. Let B_m=max(m+3,floor((4m+4)/3)), m>=5. The first term follows from the independently proved t>=4 defining-set lemma; the second is the published theorem at t=m+3>=8. Each restriction D_i defines S_i, so |D_i|>=B_m.

20.10 Counting and equality exclusion

The union has at least2m+1 occupied rows by the existing row-cover argument. Its corner clues occupy at most3 rows. The other d-k clues add at most d-k rows, so 2m+1<=3+d-k, or k<=d-2m+2. Also k<=9. Since 2B_m<=|D_0|+|D_1|=d+k, these inequalities give d>=B_m+m-1 and d>=2B_m-9.

Suppose d=B_m+m-1. Then both sides of the first derivation are equalities: k=B_m-m+1 and |D_0|=|D_1|=B_m. Thus each D_i has exactly m-1 clues outside the corner. Those clues must occupy at least m-1 of the m OLD rows, else two old rows would have no clue in D_i. Therefore all m-1 non-corner clues have distinct old rows. The identical argument applies to OLD columns and OLD symbols. The corner can supply none of these old labels. Consequently all non-corner clues have old coordinates and old symbols, with each type pairwise distinct. This directly excludes every off-corner border cell and every changed transversal cell. Row/column/symbol coverage additionally forces the corner to meet all3 new labels of each type, though that extra observation is not needed for the H argument.

In S_0 these m-1 unchanged clues lie in the H mask1<=j<=i<=m-1, because C is a subset of U. Distinct old rows and columns force them to use all indices1,. . . ,m-1; summing j<=i forces j=i for every clue. All have old symbol0, contradicting the distinct-symbol conclusion when m>=3. Thus d>=B_m+m. Combining gives d>=max(B_m+m,2B_m-9). Since B_m>=m+3 this automatically includes the older2m+3 bound. If the hypothetical k exceeds9, equality is already impossible; the H argument remains a valid conditional contradiction and needs no upper restriction on m.

20.11 Final consequence and checked examples

For odd m>=7, add the protected count F_m=(21mˆ2-23m-104)/2, which is disjoint from the union, to obtain lcs(4m+3)>=F_m+max(B_m+m,2B_m-9). The earlier canonical count can be combined with exactly the same cover for odd m>=5. At m=5 the separately counted flexible core154 still gives167.

m order B_m protected F_m union cover new total 7 31 10 382 17 399 9 39 13 695 22 717 11 47 16 1092 27 1119 13 55 18 1573 31 1604

No assertion is made that these construction values exceed the record or best other bound at every displayed order. The odd-m condition on the seed transversals remains essential. This new cover argument is independently audited; the published lower bound is cited with its exact stated domain.

20.12 Coverage of the collected proofs

An independent reviewer inventoried22 proof and obstruction families against the archived sources. The review led to full expansions of the ternary local lemma, Fu–Fu–Liao exchange, permanent-counting derivation and

near-transversal repair, plus the explicit product and contraction arguments. Historical intermediate bounds are marked as such; unsupported all-parameter CRT fits, the unrestricted product formula, and a sharp838 at order33 are not promoted to theorems. The complete source inventory and remaining scope limitations are archived in `lcs_full_review/proof_coverage_audit.txt`.

20.13 Published inputs and exact provenance

The principal input sources are Donovan (1998), *Critical sets for families of Latin squares*, Utilitas Mathematica53,3–16, Theorem3.5; Bean's thesis, Chapters4–5; Gower (2000), Ars Combinatoria55,293–317; Cao–Du (2002), Utilitas Mathematica62,217–236; and Horák–Aldred–Fleischner (2002), *Completing Latin Squares: Critical Sets*, Journal of Combinatorial Designs10,419–432, Theorem1.1. The counting inputs are the permanent estimates in Ghandehari–Hatami–Mahmoodian. The upper-bound audit starts from Černý's premature-partial-square framework but proves the required critical-set structural claims explicitly; no verbatim transfer is assumed.

Primary online versions include Donovan's paper, Bean's thesis, Gower's paper, Horák–Aldred–Fleischner, and Ghandehari–Hatami–Mahmoodian. The companion manifest records exact local proof sources and hashes.

21 Explicit lower-bound functions: complete catalogue

The following tables collect *all lower-bound families used or discussed in this report*, including the seed-dependent and recursive families. They are a catalogue of the present results, not a claim to exhaust the literature. Each function contributes zero outside its stated domain; maximize over the admissible parameters and available certified seeds. All entries in the following construction, product and counting tables are **proved**. Conjectural terms are listed separately on page 81.

1. Closed families and finite records.

Function	Orders/domain	Value	Source or restriction
$A(n)$	All $n \geq 1$	$n(n-1)/2$	Triangular critical set
$Q_2(n)$	$n = 2m$, $m \geq 1$	$(5m^2 - 3m)/2$	Donovan (1998), Thm. 3.5; $m = 1$ direct
$Q_4(n)$	$n = 4m$, $m \geq 1$	$(23m^2 - 9m)/2$	Bean thesis, Thm. 5.3.4
$Q_{16}(n)$	$n = 16m$, $m \geq 1$	$(377m^2 - 27m)/2$	Two safe binary lifts of the thesis seed
$T_9(n)$	$9 \mid n$	$(29n^2 - 27n)/54$	Cao–Du, Thm. 4.1
$P_2(n)$	$n = 2^k$, $k \geq 1$	$4^k - 3^k$	Stinson–van Rees; explicit XOR set
$P_3(n)$	$n = 3^k$, $k \geq 1$	$9^k - 6 \cdot 7^{k-1}$	Gower; size-three seed at order three
$P_3^{(0)}(n)$	$n = 3^k$, $k \geq 1$	$9^k - 7^k$	Size-two ternary seed; dominated by P_3
$M_-^{(0)}(n)$	$n = 2^k - 1$, $k \geq 2$	$4^k - 3^k - 2^{k+1} + 3$	Historical Fu–Fu–Liao; dominated by M_-
$M_-(n)$	$n = 2^k - 1$, $k \geq 2$	$4^k - 3^k - 2^{k+1} + 4$	Audited one-clue improvement
$J(n)$	$n = 2^a m$, $a \geq 1$	$n^2 - 3^a m^2$	Even cyclic seed, then $a - 1$ binary lifts
$E_0(n)$	$1 \leq n \leq 7$	$0, 1, 3, 7, 11, 18, 25$ in order	Exact small-order values
$R(n)$	Certified/published record orders	Largest available verified or published example size	Zero elsewhere; includes exact small orders

For J, the strongest admissible a is the largest exponent such that $2^a \mid n$. The P_2 row is retained even though it is a special case of J. Likewise E_0 is included in R, and $M_-^{(0)}$ is dominated by M_-. These redundancies make the catalogue explicit without changing its maximum. No formula in this table assumes that a general direct product preserves criticality.

Attribution. The Q_2 bound is Donovan (1998), Theorem 3.5, *Critical sets for families of Latin squares*, *Utilitas Mathematica* 53, 3–16. It is cited, not originated, in Bean's thesis. The Q_4 seed is Bean's theorem. Other primary sources are Bean's thesis, Chapters 4–5, Gower, Ars Combinatoria 55 (2000), 293–317, and Cao–Du, *Utilitas Mathematica* 62 (2002), 217–236.

Record status. R retains 47, 150, 447 and 844 at orders 9, 15, 27 and 33, respectively, and the other independently checked records in the record table. The older unverified 445 at order 27 is superseded. $M_-(15) = 147$ is a lower bound, not an equality: the verified record is 150.

21.1 Explicit lower-bound functions: prolongations

Put $q = 2^k$, $k \geq 2$, $1 \leq a < k$, and $a_0 = \lfloor k/2 \rfloor$. Define

$$g_{k,a} = q^2 - 3^k - \frac{3q}{2} - 2^{a-1} - 2^{k-a-1}, \qquad \gamma_{k,a} = c(\tau p^{-1}) + c(p^{-1}) + c(\tau),$$

where τ is the explicit map in (10.2), $p = I + \tau$, and c counts cycles on nonzero vectors. These are finite computable permutations, and $\gamma_{k,a} \geq 3$. All terms for a split use the same map.

Function	Orders/domain	Value	Source or restriction
M_+^{unif}	$n = q + 1$	$g_{k,a_0} + 5$	Uniform projected trade cover
M_+^{cyc}	$n = q + 1$	$\max_{1 \leq a < k} (g_{k,a} + 2 + \gamma_{k,a})$	Exact-cycle refinement; includes the uniform value
$P_{4,+}$	$n = 4m + 1$ odd $m \geq 3$	$(21m^2 - 7m - 22)/2$	Thesis prolongation; full proof p. 27
$P_{16,+}$	$n = 16m + 1$ odd $m \geq 3$	$182m^2 - 18m - 44$	Explicit binary template; p. 28
$P_{3,+}^{\mathrm{unif}}$	$n = 3q + 1$	$7q^2 - 5 \cdot 3^k - 3q/2$ $- 2^{a_0 - 1} - 2^{k - a_0 - 1} + 1$	Uniform ternary transfer
$P_{3,+}^{\mathrm{cyc}}$	$n = 3q + 1$	$2q^2 + 4(q^2 - 3^k) - 4$ $+ \max_{1 \leq a < k} (g_{k,a} + 2 + \gamma_{k,a})$	Exact-cycle ternary transfer; p. 31
$\mathcal{T}_+$	$n = 3q + 1$; eligible (L, Q, T, b)	$2q^2 + 4s - 2h + b$	General transfer; hypothesis below is essential
K_{34}	$n = 34$	769	Audited conjugate-block protected set
K_{68}	$n = 68$	$34^2 + 2 \cdot 769 + 403 = 3097$	Safe binary lift of the same profile
K_{136}	$n = 136$	$7 \cdot 34^2 + 3 \cdot 769 + 6 \cdot 403 = 12817$	Safe two-step lift of that profile

General transfer hypothesis. Q must be a critical set of size s in a specified order-q square L, with transversal T and $h = |T \setminus Q|$. After prolongation, U_T prescribes transformed Q, all transversal positions and the complete new border. The input b must bound the size of *every defining subset of that particular* U_T. An arbitrary lower bound on $\mathrm{lcs}(q + 1)$ does not suffice. Thus the two $P_{3,+}$ rows are valid specializations of $\mathcal{T}_+$; unrelated record sizes are not substituted for b.

Which values are evaluated. The numerical report includes the stored audited M_+^{cyc} instances and their uniform fallback. It computes all splits for each applicable $P_{3,+}^{\mathrm{cyc}}$ order shown. At order 33 the exact-cycle guarantee is 735, while $R(33) = 844$. At 49 the ternary value 1366 is dominated by $P_{16,+}(49) = 1540$. The K profile guarantees a critical subset of size at least 769 with at least 403 private-intercalate clues. Its two displayed lifts have positive coefficients, so these lower counts are sufficient; no exact size is assumed for arbitrary mixed lifts.

21.2 Explicit lower-bound functions: three prolongations

These additional functions come from three disjoint transversals of the thesis square. Their full proof and independent audit details start on page 39. The oddness restriction on m is essential.

Function	Orders/domain	Value	Source or restriction
$P^{\mathrm{can}}_{4,+3}$	$n = 4m+3$, odd $m \geq 5$	$(21m^2 - 25m - 72)/2$	Canonical private rectangles; dominated below
$P_{4,+3}$	$n = 4m+3$, odd $m \geq 7$	$(21m^2 - 19m - 98)/2$	Flexible private rectangles and $2m+3$ cover
$P^{(5)}_{4,+3}$	$n = 23$	167	Exact protected count at $m = 5$
K_{CRT}	$n = 27, 43, \ldots, 203$	Explicit finite table on page 44	Independently replayed protected cores, plus two corner clues
$H_2[P_{4,+3}]$	$n = 8m+6$, odd $m \geq 7$	$44m^2 - 3m - 141$	Same seed's private-intercalate profile
$H_4[P_{4,+3}]$	$n = 16m+12$, odd $m \geq 7$	$(371m^2 + 183m - 792)/2$	Two safe binary lifts of that profile

The new refinement in Section 11.6 adds one further function:

$$P^{\mathrm{H}}_{4,+3}(4m+3) = \frac{21m^2 - 23m - 104}{2} + \max\{B_m + m, 2B_m - 9\}, \quad B_m = \max\{m+3, \lfloor (4m+4)/3 \rfloor\},$$

for odd $m \geq 7$. It dominates $P_{4,+3}$. Its private-intercalate profile is the same T_m below, so the general positive-coefficient H_2, H_3, H_4 rows also apply with this stronger size. These are instances of the generic functions already displayed. The earlier values remain as valid historical specializations.

The extracted order-$q = 4m+3$ critical set has size at least $S_m = (21m^2 - 19m - 98)/2$ and at least $T_m = 7m^2 - 8m - 52$ clues with private intercalates. The last two rows substitute these lower counts into $q^2 + 2s + t_2$ and $7q^2 + 3s + 6t_2$, which have positive coefficients. They do not assume an explicit critical set of *exactly* S_m clues, or justify substituting S_m for an unknown actual size in an arbitrary mixed-word expression with a negative size coefficient.

The general H_3 row on the next page also gives

$$\mathrm{lcs}(12m+9) \geq 2(4m+3)^2 + 5S_m = \frac{169m^2 + m - 454}{2} \qquad (m \geq 7 \text{ odd}),$$

and all recursive lifts of established sizes remain available. The numerical envelope includes that consequence through its recursive H_3 calculation. The canonical row is explicitly retained although the flexible family, together with its exceptional value at 23, dominates it throughout its domain. These functions do not assert a new lower bound at 11 or 19.

21.3 Explicit lower-bound functions: certified seed products

Use one coherent certified seed tuple (m, s, t_2, t_3): order m, actual critical-set size s, and lower counts t_2, t_3 of clues with private intercalate and order-three-subsquare witnesses. These count clues, not trades. Maximize each row over the available tuples satisfying its domain. Put $D = m^2 - s$, $u_2 = s - t_2$, $u_3 = s - t_3$.

Function	Orders/domain	Value for one eligible seed	Source or restriction
H_2	$n = 2m$	$m^2 + 2s + t_2$	Protected binary product
H_3	$n = 3m$	$2m^2 + 5s + 2t_3$	Protected ternary product
H_4	$n = 4m$	$7m^2 + 3s + 6t_2$	Two binary lifts
H_{24}	$n = 24m$	$333m^2 + 189t_2$	Mixed word 3222; explicit useful specialization
F_2	$n = 2^a m$, $a \geq 1$	$n^2 - 3^a D - \beta_a u_2$	Repeated binary lifts
F_3	$n = 3^a m$, $a \geq 1$	$n^2 - 7^a D - \chi_a u_3$	Repeated ternary lifts
F_{23}	$n = m\prod_i w_i$, $w_i \in \{2, 3\}$	$n^2 - A_w m^2 - B_w D - C_w u_2 - D_w u_3$	All finite mixed words; coefficients defined below
$\Psi_{3,2}$	$q = 3^a m$, $a \geq 1$, $n = 2q$	$q^2 + 2F_3(q) + 7^a t_2$	Ternary run followed by one binary lift
$\Psi_{3,4}$	$q = 3^a m$, $a \geq 1$, $n = 4q$	$7q^2 + 3F_3(q) + 6 \cdot 7^a t_2$	Ternary run followed by two binary lifts

The $F_3(q)$ in the last two rows is evaluated using the *same seed* as t_2. Its inherited private-intercalate count is at least $7^a t_2$. The specialized rows are retained even when a mixed-word calculation duplicates or improves them.

Coefficients for repeated lifts.

$$\beta_0 = 0, \quad \beta_1 = 1, \quad \beta_2 = 6, \quad \beta_a = 27\,4^{a-3}\ (a \geq 3), \qquad \chi_0 = 0, \quad \chi_{a+1} = 9\chi_a + 2\max(0, 7^a - \chi_a).$$

Coefficients for mixed words. List the factors of w in forward construction order. Start at $(A, B, C, D) = (0, 1, 0, 0)$ and process the word in reverse, using

$$\begin{aligned} \mathcal{B}(A, B, C, D) &= (4A + D,\ 3B,\ 4C + z,\ 3D), \\ \mathcal{T}(A, B, C, D) &= (9A + 2C,\ 7B,\ 7C,\ 9D + 2z), \\ z &= \max(0, B - C - D). \end{aligned}$$

The resulting tuple is (A_w, B_w, C_w, D_w). The D in this tuple is a coefficient, distinct from the seed deficit $m^2 - s$. These explicit rules cover every finite word, so the catalogue does not hide an unnamed "additional lifts" term.

Seed restrictions. The largest size and largest witness counts from different squares cannot be combined. Unknown witness counts may be set to zero. For F_2, F_3, F_{23}, use the actual seed size: its coefficient can be negative after expansion. The positive-size formulas H_2, H_3, H_4 also admit established lower sizes, as used in the next table.

21.4 Explicit lower-bound functions: recursion and counting

The recursive functions use already established values at *smaller* orders, with witness counts set to zero. They therefore introduce no circular assumption about lcs.

Function	Orders/domain	Value	Source or restriction
$\mathcal{S}_-$	$n = N - 1$; eligible defining set and corner	$\lvert F\rvert$	Protected contraction; hypotheses below
$G_2(n)$	$n = 2m$	$m^2 + 2L_{\text{proved}}(m)$	H_2, taking $t_2 = 0$
$G_3(n)$	$n = 3m$	$2m^2 + 5L_{\text{proved}}(m)$	H_3, taking $t_3 = 0$
$G_4(n)$	$n = 4m$	$7m^2 + 3L_{\text{proved}}(m)$	H_4, taking $t_2 = 0$
$K_{\text{HM}}(n)$	$n \geq 2$	$\max\{0, \lceil f_{\text{HM}}(n)\rceil\}$	Hatami–Mahmoodian, Theorem 1
$K_0(n;h)$	$0 \leq h \leq n^2$; $U(n,h) < L_0(n)$	$n^2 - h + 1$	Unamplified finite counting; certified strict inequality
$K_\Sigma(n;d,h)$	$0 \leq h \leq d \leq n^2$; condition below	$n^2 - d + 1$	Superset-amplified finite counting

Contraction input. For $\mathcal{S}_-$, take a specified defining set $P \subseteq L$ of order N and a corner satisfying the criterion in Section 16. Form its contracted defining set U with all transversal cells prescribed. Every clue in $F \subseteq U$ must have a certified private trade in U; surviving old trades are one way to provide these certificates. Maximize $|F|$ over such verified inputs. Neither $|P|$ nor $|U|$ alone can replace $|F|$. The tested instances give valid lower guarantees but no improvement of the best known bound.

Published all-order counting function. With natural logarithms,

$$f_{\text{HM}}(n) = n^2\left(1 - \frac{2 + \ln 2}{\ln n}\right) + n\left(1 + \frac{\ln(8\pi)}{\ln n}\right) - \frac{\ln 2}{\ln n}.$$

See Hatami–Mahmoodian, *A lower bound for the size of the largest critical sets in Latin squares*, Theorem 1. Integrality permits the ceiling. This function improves the previous proved envelope at 15 orders from 199 to 251; orders 8–33 are unchanged.

The exact counting condition. Put $N = n^2$ and define

$$L_0(n) = \frac{(n!)^{2n}}{n^{n^2}}, \qquad F_n(a) = \prod_{t=a+1}^{n} (t!)^{1/t}.$$

Write $h = an + b$, $0 \leq b < n$, and set

$$U(n,h) = \binom{N}{h}(n!)^{h/n}\frac{F_n(a)^{n-b}F_n(a+1)^b}{(a!)^{n-b}((a+1)!)^b}.$$

At $h = N$, omit the factor with exponent $b = 0$, so $F_n(n+1)$ is not needed. The admissibility condition for K_Σ is

$$\boxed{L_0(n)\binom{d}{h} > U(n,h).}$$

K_0 is the case $d = h$, retained separately to show the original counting function. Maximize over certified pairs (d,h); a rounded numerical approximation without a certified positive margin is not sufficient. The balanced partial-square count and superset argument are proved in full in Section 15.

Finite evaluation versus asymptotics. Stored certificates give $K_\Sigma = 988, 4967, 23462$ at $n = 50, 100, 200$ (below constructions there), and 54939, 171402, 746306 at $n = 295, 500, 1000$. The proved maximum includes K_{HM} and stored certified pairs, separately from the construction curve; it does not exhaustively optimize all counting pairs. Asymptotically,

$$\text{lcs}(n) \geq n^2 - (2.668162063280737765\ldots + o(1))n^{5/3}.$$

It is *not* an additional finite-order function with the $o(1)$ omitted.

21.5 Seed coverage and the complete maximum

Coverage of the seed families. Let $\theta(m) = m(m+2)/8$ for even m and $\theta(m) = 0$ for odd m. The special seed profiles used for the repeated binary rows are

seed order	s	t_2
$2m$	$(5m^2 - 3m)/2$	$m^2 + 3\theta(m)$
$4m$	$(23m^2 - 9m)/2$	$7m^2 + 9\theta(m)$.

Thus the generator's separately labelled Donovan and thesis iterates are instances of F_2, rather than omitted lower-bound functions. The stored finite private-witness profiles likewise instantiate the H, F, Ψ rows.

With each seed or parameter family maximized over its admissible, established inputs, the full proved maximum in this report is

$$\begin{aligned} L_{\text{proved}}(n) = \max\{&A, Q_2, Q_4, Q_{16}, T_9, P_2, P_3, P_3^{(0)}, M_-^{(0)}, M_-, J, E_0, R, \\ &M_+^{\text{unif}}, M_+^{\text{cyc}}, P_{4,+}, P_{16,+}, P_{3,+}^{\text{unif}}, P_{3,+}^{\text{cyc}}, \mathcal{T}_+, \\ &P_{4,+3}^{\text{can}}, P_{4,+3}, P_{4,+3}^{(5)}, K_{\text{CRT}}, H_2[P_{4,+3}], H_4[P_{4,+3}], \\ &P_{4,+3}^{\text{H}}, K_{34}, K_{68}, K_{136}, H_2, H_3, H_4, H_{24}, F_2, F_3, F_{23}, \\ &\mathcal{S}_-, \Psi_{3,2}, \Psi_{3,4}, G_2, G_3, G_4, K_{\text{HM}}, K_0, K_\Sigma\}(n). \end{aligned}$$

This specifies all 46 proved families and special cases in the catalogue; the numerical implementation uses the documented finite seed and cycle tables, the stated closed formulas, stored counting certificates, and the safe recursive closure. An unevaluated general transfer instance is not assigned a guessed numerical value.

The currently retained *conjectural* left-hand side is

$$\boxed{\max\{L_{\text{proved}}(n), B(n), C(n)\} \overset{?}{\leq} \operatorname{lcs}(n).}$$

Only the additional B, C terms are conjectural. They are included in the orange curve; the proved construction curve and verified records retain their separate meanings.

Function	Orders/domain	Proposed lower value	Status in the displayed LHS
$B(n)$	All $n \geq 1$	$\max_{pq=n} \left(n^2 - \frac{p(p+1)q(q+1)}{4} \right)$	Conjectural; included
$C(n)$	All $n \geq 1$	$\#\{(i,j) : 0 \leq i, j < n,\ i \,\&\, j \neq 0\}$	Conjectural for general n; included
$D_{s,t}(n)$	$n = mp$ seed sizes s, t	$n^2 - (m^2 - s)(p^2 - t)$	Unrestricted product proposal; unproved, not included
$S_+(n)$	$n = 2^k + 1$, $k \geq 3$	$4^k - 3^k + 2^{k+1} - 7$	Former sharp-plus lower proposal; unproved in general, not included

$D_{s,t}$ is the numerical product proposal, not a theorem that its standard template is critical. Its proposed order-68 value 3823 remains uncertified. $S_+(33) = 838$ has been surpassed by the verified 844; that proves this one lower instance, not the all-k formula, and rules out equality at 33. At powers of two, $C = P_2$ is proved; its general-order lower claim is the remaining conjectural part. The dyadic equality proposal adds an upper claim and supplies no additional LHS function beyond P_2.

The old $E(n)$ and $\lceil E(n) \rceil$ are upper proposals, not LHS terms; both are refuted by the order-33 record. The catalogue does not restore either as an upper bound. The audited proved upper curve remains separate.

Reproducibility. Function domains and witness restrictions are in these tables; the detailed proofs and independent audits follow in this document or the cited earlier review. Numeric sources are recorded in `lcs_dyadic_conjectures_data.json`; the generator uses exact powers at dyadic orders and 80-digit arithmetic for the refuted reference curve.

22 Explicit examples: current best at each order

Each following landscape page gives both arrays in full. A dot is a hole; $0, \ldots, 9$ represent themselves and A,B,...,W represent10,11,...,32. Rows and columns are in increasing zero-based order. Reading the partial alone therefore reconstructs the critical-set candidate; the completion specifies its intended Latin square. The full SHA256 fingerprint uses compact JSON `[n,completion,partial]` with flat integer arrays and no spaces. A certificate claim is limited to the status printed on that page.

E001: order 9, 47 clues

Status I. Independently verified: two encodings and all deletion witnesses; exhaustive uniqueness tree replayed

Source: `outputs/lcs_order9_verification/bunya_record.json`

SHA256: `b2b9172988cbb8d4329a9ce6cb0e7cb1f94629a6ee1a7e49bbf3150b73ac3156`

Completion

```
1 4 8 3 0 6 7 5 2
5 7 2 0 8 3 4 1 6
4 8 0 2 3 1 5 6 7
7 6 3 1 5 2 8 0 4
8 0 1 6 7 4 3 2 5
3 2 7 8 1 5 6 4 0
0 1 6 5 4 8 2 7 3
6 5 4 7 2 0 1 3 8
2 3 5 4 6 7 0 8 1
```

Partial square

```
. 4 8 3 . . 7 5 2
5 7 . 0 . 3 . 1 .
4 8 0 2 . 1 5 . .
7 . 3 1 . . 8 0 4
8 0 1 . . 4 3 2 .
3 2 7 . . 5 . 4 0
. 1 . 5 . 8 2 7 3
. . . . . . . . .
2 . 5 4 . 7 0 . 1
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E002: order 10, 59 clues

Status I. Previously verified construction/example or published record

Source: `outputs/lcs_examples.json:1`

SHA256: `f697422c13c9a4e3c9b59cd72cb9b39c61d144fe7c6c6262d482d93fe1cbcec4`

Completion

```
0 1 2 3 4 5 6 7 8 9
5 9 7 6 1 2 0 4 3 8
7 4 6 9 0 3 2 8 1 5
4 8 1 2 9 7 3 0 5 6
9 3 8 1 2 4 7 5 6 0
3 0 5 7 8 9 1 6 4 2
2 6 9 8 7 1 5 3 0 4
8 5 4 0 3 6 9 2 7 1
6 2 3 4 5 0 8 1 9 7
1 7 0 5 6 8 4 9 2 3
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

Partial square

```
. 1 2 . 4 5 6 7 . 9
5 . 7 . 1 2 0 4 . 8
7 4 6 9 . . 2 8 . 5
4 8 1 2 9 . . 0 . 6
9 . . 1 2 4 7 . . 0
. 0 5 7 8 9 1 6 . .
. . . . . . . . . .
. 5 . 0 . 6 9 2 . 1
6 2 . 4 5 0 . . . 7
1 7 . 5 . 8 4 9 . .
```

E003: order 11, 74 clues

Status I. Previously verified construction/example or published record

Source: `outputs/lcs_examples.json:2`

SHA256: `cdd11a917e77e05eecf049cf108fa8f253d8d68e8d613f6b2d9d4bfb725c2287`

Completion

```
8 9 1 2 3 4 5 6 7 A 0
0 8 9 3 4 2 6 7 5 1 A
9 A 8 4 2 3 7 5 6 0 1
2 3 4 8 9 7 A 0 1 5 6
3 4 2 6 8 9 0 1 A 7 5
4 2 3 9 5 8 1 A 0 6 7
5 6 7 A 0 1 8 9 4 2 3
6 7 5 0 1 A 3 8 9 4 2
7 5 6 1 A 0 9 2 8 3 4
A 1 0 5 7 6 2 4 3 9 8
1 0 A 7 6 5 4 3 2 8 9
```

Partial square

```
. 9 1 2 3 4 . 6 . . 0
0 . . 3 4 2 6 . 5 . A
9 A . . 2 3 7 5 6 . 1
. 3 4 . 9 7 A 0 1 . 6
3 4 2 6 . 9 . 1 A . 5
4 . 3 9 5 . 1 A 0 . 7
5 6 7 A 0 1 . . 4 . .
6 7 5 0 . . 3 . 9 . 2
7 5 . . A 0 9 2 . . .
. . . . . . . . . . .
. 0 A 7 . 5 4 3 2 . 9
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E004: order 12, 90 clues

Status I. Previously verified construction/example or published record

Source: `outputs/lcs_examples.json:3`

SHA256: `ee28be4a641731c84d9a7b0653e5b7f4d1a2ded85c1cf20cf4d47cc24303ac56`

Completion

```
9 0 1 2 3 4 5 6 7 8 A B
0 1 9 4 2 3 6 7 5 A B 8
1 9 0 3 4 2 7 5 6 B 8 A
2 3 4 9 0 1 A 8 B 6 5 7
3 4 2 1 9 0 B A 8 7 6 5
4 2 3 0 1 9 8 B A 5 7 6
5 7 6 A B 8 9 1 0 4 2 3
6 5 7 8 A B 0 9 1 2 3 4
7 6 5 B 8 A 1 0 9 3 4 2
8 B A 6 7 5 4 2 3 9 1 0
A 8 B 5 6 7 2 3 4 0 9 1
B A 8 7 5 6 3 4 2 1 0 9
```

Partial square

```
. 0 . 2 . . . 6 7 . A B
. 1 9 . 2 3 . 7 5 A B .
1 9 . . 4 2 . 5 6 B . A
2 3 4 . . 1 . . . 6 5 .
3 4 2 . 9 . . A . 7 . 5
4 2 3 0 1 9 . B A 5 7 6
5 7 . A B . . 1 0 4 2 .
6 5 7 . A B . 9 1 2 3 4
. 6 5 B . A . 0 9 3 . 2
. B . 6 7 5 . 2 . 9 1 0
. . . . . . . . . . . .
B A . 7 5 6 . 4 2 1 0 9
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E005: order 13, 107 clues

Status I. Previously verified construction/example or published record

Source: `outputs/lcs_examples.json:4`

SHA256: `1f99d634fd05ff8cf67a4fcb3aac05ad7083a116cfad0d84b55941b4a11ad6b1`

Completion

```
9 0 1 2 3 4 5 6 7 8 A B C
0 1 9 3 2 5 4 7 8 6 C A B
5 2 4 1 0 9 8 C B A 6 3 7
C 3 B 8 6 7 A 5 4 2 0 1 9
3 C A 6 7 8 B 2 5 4 1 9 0
7 6 8 A C B 9 0 3 1 2 4 5
B A C 4 5 2 3 1 9 0 7 8 6
4 5 2 0 9 1 6 A C B 8 7 3
1 9 0 5 4 3 2 8 6 7 B C A
2 4 5 9 1 0 7 B A C 3 6 8
6 8 7 B A C 0 9 1 3 5 2 4
A B 3 7 8 6 C 4 2 5 9 0 1
8 7 6 C B A 1 3 0 9 4 5 2
```

Partial square

```
9 0 . 2 3 4 . 6 7 8 A B .
0 1 . . 2 5 4 7 8 . . A B
5 2 . 1 0 9 . . B A 6 3 7
. 3 . 8 6 . A 5 4 2 . 1 9
3 . . 6 . . B 2 5 4 1 9 .
. 6 . A . . 9 0 . 1 2 4 5
B A . 4 . 2 3 . 9 . 7 8 6
4 5 . 0 9 1 6 A . . . 7 3
1 9 . 5 . 3 2 8 6 7 B . A
2 4 . . . 0 7 B A . 3 . 8
6 8 . B A . 0 9 1 3 5 . 4
. . . . . . . . . . . . .
. . . . . . 1 3 0 9 4 5 2
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E006: order 14, 126 clues

Status I. Previously verified construction/example or published record

Source: `outputs/lcs_examples.json:5`

SHA256: `af24233defdb4e203c07425c3d4f1c06297b6725afea3077a8c6e2b927fa7607`

Completion

```
0 2 1 4 3 6 5 7 9 8 B A D C
2 1 0 5 6 3 4 9 8 7 C D A B
1 0 2 6 5 4 3 8 7 9 D C B A
4 5 6 3 0 1 2 B C D A 7 8 9
3 6 5 0 4 2 1 A D C 7 B 9 8
6 3 4 1 2 5 0 D A B 8 9 C 7
5 4 3 2 1 0 6 C B A 9 8 7 D
7 9 8 B A D C 0 2 1 4 3 6 5
9 8 7 C D A B 2 1 0 5 6 3 4
8 7 9 D C B A 1 0 2 6 5 4 3
B C D A 7 8 9 4 5 6 3 0 1 2
A D C 7 B 9 8 3 6 5 0 4 2 1
D A B 8 9 C 7 6 3 4 1 2 5 0
C B A 9 8 7 D 5 4 3 2 1 0 6
```

Partial square

```
0 . . 4 3 6 5 . 9 8 B A D .
. 1 . 5 6 3 4 9 . 7 C D A B
1 0 . 6 5 4 3 8 7 9 . . B A
4 5 . 3 . 1 . B C D . 7 8 9
3 . . 0 . . 1 A D C 7 . 9 8
6 . . 1 . 5 0 D . B . 9 . 7
5 4 . . . . . C B A . . 7 .
. 9 . B A D C 0 . . 4 3 6 5
9 8 . C D A . . 1 0 5 6 . 4
. . . . . .
B C . A 7 . 9 4 . 6 3 0 1 .
. D . . B 9 8 3 . 5 0 4 . 1
D A . 8 9 C 7 6 3 . 1 . 5 0
C B . 9 8 7 D . 4 3 . 1 0 6
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E007: order 15, 150 clues

Status I. Independently verified: two encodings and all deletion witnesses; exhaustive uniqueness tree replayed

Source: `outputs/lcs_bunya_verification_round2/n15/bunya_record.json`

SHA256: `0329af95a3b4458070adde2beafdc2264689516d25967c3ff9736d3e4b590c57`

Completion

```
E C D 5 B 4 0 1 6 2 A 3 7 9 8
7 8 A 2 9 1 6 3 0 D 5 4 E B C
C E 2 A 4 B 3 6 1 5 D 9 8 0 7
6 1 8 B C A 7 5 E 9 4 D 0 2 3
2 D C 7 0 3 B 4 9 E 8 1 5 6 A
A 5 7 C 1 6 9 B 4 8 E 0 D 3 2
9 A 1 0 7 8 5 D 2 6 3 C B E 4
8 7 5 D 6 9 1 0 3 A 2 B C 4 E
4 B 0 1 5 E D C A 3 6 7 2 8 9
D 2 E 8 3 0 4 9 B C 7 6 A 1 5
B 4 3 6 E C 2 A 5 0 1 8 9 7 D
1 6 9 4 A 5 8 E C 7 B 2 3 D 0
5 9 6 3 8 7 A 2 D 1 0 E 4 C B
3 0 B E D 2 C 7 8 4 9 5 1 A 6
0 3 4 9 2 D E 8 7 B C A 6 5 1
```

Partial square

```
E . D 5 . 4 . . . 2 A 3 7 9 8
7 8 A 2 . . 6 3 0 D . 4 E B .
. . 2 A . B 3 6 1 . D 9 8 0 7
6 1 . B . A 7 5 E 9 4 D . 2 .
2 D . 7 . 3 B . 9 E 8 . 5 6 .
. 5 7 . . 6 9 B 4 8 E 0 D 3 2
9 . 1 0 . 8 5 . . 6 3 . B E 4
8 7 5 D . 9 . 0 3 . . B . 4 E
. . . . . . . . . . . . . . .
D 2 E 8 . . 4 9 B . 7 . A 1 5
B 4 3 . . . 2 A 5 0 1 8 9 . D
1 6 . 4 . . 8 E . . B 2 3 D 0
5 9 6 . . 7 A 2 D 1 0 E 4 . .
3 0 B . . 2 . 7 . 4 9 5 1 A 6
0 . 4 9 . D . 8 7 B . A 6 5 1
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E008: order 16, 175 clues

Status I. Previously verified construction/example or published record

Source: `outputs/lcs_examples.json:7`

SHA256: `ca80cb56508b6d1ae088f2217b5bb1505dc3df3a343ca8651b003b62ff74ecb6`

Completion

```
0 1 2 3 4 5 6 7 8 9 A B C D E F
1 0 3 2 5 4 7 6 9 8 B A D C F E
2 3 0 1 6 7 4 5 A B 8 9 E F C D
3 2 1 0 7 6 5 4 B A 9 8 F E D C
4 5 6 7 0 1 2 3 C D E F 8 9 A B
5 4 7 6 1 0 3 2 D C F E 9 8 B A
6 7 4 5 2 3 0 1 E F C D A B 8 9
7 6 5 4 3 2 1 0 F E D C B A 9 8
8 9 A B C D E F 0 1 2 3 4 5 6 7
9 8 B A D C F E 1 0 3 2 5 4 7 6
A B 8 9 E F C D 2 3 0 1 6 7 4 5
B A 9 8 F E D C 3 2 1 0 7 6 5 4
C D E F 8 9 A B 4 5 6 7 0 1 2 3
D C F E 9 8 B A 5 4 7 6 1 0 3 2
E F C D A B 8 9 6 7 4 5 2 3 0 1
F E D C B A 9 8 7 6 5 4 3 2 1 0
```

Partial square

```
. 1 . 3 . 5 . 7 . 9 . B . D . F
1 . . 2 . 4 7 . . 8 B . D . . E
2 3 . 1 . 7 4 5 . B 8 9 E F . D
. . . . . . . . . . . . . . . .
4 5 . 7 0 1 2 3 . D E F 8 9 A B
. . . . 1 0 3 2 . . . . 9 8 B A
. 7 . 5 2 3 0 1 . F . D A B 8 9
7 . . 4 3 2 1 0 . E D . B A 9 8
8 9 . B . D E F 0 1 2 3 4 5 6 7
. . . . . . . . 1 0 3 2 5 4 7 6
. B . 9 . F . D 2 3 0 1 6 7 4 5
B . . 8 . E D . 3 2 1 0 7 6 5 4
. D . F 8 9 A B 4 5 6 7 0 1 2 3
D . . E 9 8 B A 5 4 7 6 1 0 3 2
E F . D A B 8 9 6 7 4 5 2 3 0 1
. . . . B A 9 8 7 6 5 4 3 2 1 0
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E009: order 17, 200 clues

Status I. Previously verified construction/example or published record

Source: `outputs/lcs_examples.json:8`

SHA256: `b9642f346d5959e31b635102722c527609794018422f8b703a44567270488cd9`

Completion

```
G 1 2 3 4 5 6 7 8 9 A B C D E F 0
1 0 G 2 5 4 7 6 9 8 B A D C F E 3
2 3 0 G 6 7 4 5 A B 8 9 E F C D 1
3 G 1 0 7 6 5 4 B A 9 8 F E D C 2
4 5 6 7 0 1 2 3 G D E F 8 9 A B C
5 4 7 6 1 0 3 2 D C G E 9 8 B A F
6 7 4 5 2 3 0 1 E F C G A B 8 9 D
7 6 5 4 3 2 1 0 F G D C B A 9 8 E
8 9 A B C D E F 0 1 2 3 G 5 6 7 4
9 8 B A D C F E 1 0 3 2 5 4 G 6 7
A B 8 9 E F C D 2 3 0 1 6 7 4 G 5
B A 9 8 F E D C 3 2 1 0 7 G 5 4 6
C D E F G 9 A B 4 5 6 7 0 1 2 3 8
D C F E 9 8 G A 5 4 7 6 1 0 3 2 B
E F C D A B 8 G 6 7 4 5 2 3 0 1 9
F E D C B G 9 8 7 6 5 4 3 2 1 0 A
0 2 3 1 8 A B 9 C E F D 4 6 7 5 G
```

Partial square

```
. . . . . . . . . . . . . . . . .
. . . 2 5 . 7 6 9 . B A . C F E 3
. 3 . . 6 7 4 5 A . . 9 E F C D 1
. G 1 . 7 6 5 4 B A . 8 F E D C 2
. 5 6 7 . 1 2 3 G . E F 8 9 . B C
. 4 . 6 1 . . 2 D C G E 9 8 B . .
. . 4 5 2 3 . 1 E F . G A B 8 . D
. 6 5 4 3 2 1 . F G D C B . 9 8 .
. 9 A B C D E F . 1 2 3 G . 6 7 4
. 8 B A D C F E . . 3 2 5 . G 6 .
. B 8 9 E . C . 2 3 . . 6 7 4 G 5
. . 9 8 F . D C . 2 1 . 7 G 5 4 6
. D E F . 9 A B 4 5 6 7 . 1 2 3 8
. C . E 9 8 G A 5 4 7 . 1 . . 2 B
. F C D A B 8 G . 7 . 5 . 3 . 1 9
. E D . . G 9 . 7 6 5 4 3 2 1 . A
. 2 3 . 8 A . 9 C E F D 4 6 7 5 G
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E010: order 18, 224 clues

Status I. Independently verified: two encodings and all deletion witnesses; exhaustive uniqueness tree replayed

Source: `outputs/lcs_bunya_verification_2026_10_15/n18/bunya_record.json`

SHA256: `6c439bde09b96b405619ad96c1dc0675f58da055901fd4d2734045b291b32be7`

Completion

```
G 1 2 3 4 5 6 7 8 9 A B C D H F 0 E
1 0 G 2 5 4 7 6 9 8 B A D H F E 3 C
2 3 0 G 6 7 4 5 A B 8 H E F C D 1 9
H G 1 0 7 6 5 4 B A 9 8 F E D C 2 3
4 5 6 7 0 1 2 3 G D E F 8 9 A H C B
5 4 7 6 1 0 3 H D C G E 9 8 B A F 2
6 7 4 5 2 3 H 1 E F C G A B 8 9 D 0
7 H 5 4 3 2 1 0 F G D C B A 9 8 E 6
8 9 A B C D E F 0 H 2 3 G 5 6 7 4 1
9 8 B A D C F E 1 0 3 2 H 4 G 6 7 5
A B 8 9 E H C D 2 3 0 1 6 7 4 G 5 F
B A 9 H F E D C 3 2 1 0 7 G 5 4 6 8
C D E F G 9 A B H 5 6 7 0 1 2 3 8 4
D C F E 9 8 G A 5 4 H 6 1 0 3 2 B 7
E F C D H B 8 G 6 7 4 5 2 3 0 1 9 A
F E H C B G 9 8 7 6 5 4 3 2 1 0 A D
0 2 3 1 8 A B 9 C E F D 4 6 7 5 H G
3 6 D 8 A F 0 2 4 1 7 9 5 C E B G H
```

Partial square

```
. 1 2 3 4 5 6 7 8 . A B . D . F . .
1 0 . 2 5 4 7 6 9 . B A D H . E . C
2 3 0 . 6 7 4 5 A B 8 . . F C D . .
H . 1 0 . 6 5 4 . A 9 8 F E D C . 3
4 5 6 7 0 1 2 . . D E F 8 9 A . . .
5 4 7 . 1 . 3 . D C . E 9 8 B A . 2
6 7 4 . . 3 H 1 E F C . A B 8 9 . .
7 H 5 4 3 2 1 0 . . D C B A 9 8 . 6
8 . A B C D E F 0 . 2 . . . 6 7 . .
9 . B A D C F E 1 0 3 2 . 4 . 6 . 5
A B . 9 E . C D 2 3 0 1 6 7 4 . . F
. A 9 . F E . C 3 . 1 0 7 . 5 4 . 8
C D E F . 9 A B H 5 6 7 0 1 2 3 . .
D C F E . . . A 5 4 . 6 1 0 3 2 . .
E F C D H B 8 . 6 7 4 5 2 3 0 1 . A
F E . C B . 9 8 7 6 5 4 3 . 1 0 . .
. . . . . . . . . . . . . . . . . .
. . D 8 . F 0 2 4 1 . . 5 . E . . H
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E011: order 19, 245 clues

Status I. Independently verified: two encodings and all deletion witnesses; exhaustive uniqueness tree replayed

Source: `outputs/lcs_bunya_verification_2026_10_15/n19/bunya_record.json`

SHA256: `6d2d548313032cdd158573f1ed14e3c2b6c6fe62c2fbacbd8aee7da241ade126`

Completion

```
8 1 2 3 4 5 7 6 0 H A B C D E G F I 9
1 H 3 9 F G E 7 B A I C 0 6 D 5 4 2 8
2 3 0 1 7 6 4 8 5 B C 9 A G F I H E D
3 8 1 0 6 7 5 4 2 I H A 9 F G E D B C
4 5 I 6 0 8 2 3 1 D E G F 9 H B C A 7
5 4 6 8 1 0 3 I 7 E D F H A 9 C B G 2
7 6 8 5 2 3 I 1 4 G F H E B C 9 A D 0
6 7 5 4 8 2 1 0 3 F G I D H B A 9 C E
0 2 4 7 3 I 6 5 8 9 B D G C A F E H 1
H A B C D E G F 9 8 1 2 3 4 5 7 I 0 6
A 9 C B E D H G F 2 0 3 1 I 4 8 7 6 5
I C 9 A G F D H E 1 3 0 2 7 6 4 8 5 B
C 0 A 2 5 B 8 D 6 3 9 1 I E 7 H G F 4
D E G I 9 H B C A 4 5 7 6 0 8 2 3 1 F
E D F H I 9 C B G 5 4 6 8 1 0 3 2 7 A
G F H E B C 9 A D 7 6 8 5 2 I 0 1 4 3
F I E D H 4 A 9 C 6 7 5 B 8 2 1 0 3 G
9 B D G C A F E I 0 2 4 7 3 1 6 5 8 H
B G 7 F A 1 0 2 H C 8 E 4 5 3 D 6 9 I
```

Partial square

```
8 1 2 3 4 5 7 6 . . A B C D E G F . 9
. . 3 9 F G E 7 . A . C 0 6 . 5 . . 8
2 3 0 1 7 . 4 8 . B C . . G F . . E .
3 8 1 0 6 7 5 . . . . A 9 F G E D B C
4 5 . 6 0 8 2 3 . D E G F 9 . . C A 7
5 4 6 8 1 0 3 . . E . F . A 9 C B G .
7 6 . 5 2 3 I 1 . G F . E . C 9 A D .
6 7 5 4 8 2 1 0 . F . I D . B A 9 C .
0 2 4 7 3 I 6 5 . 9 B D G C . F . . .
. A B . . E G . . 8 1 . . 4 5 7 . 0 .
A 9 C B E D . G . . 0 3 . . 4 8 7 6 .
I C 9 A G F D . . . 3 0 2 7 6 4 8 5 B
C 0 A . 5 . 8 D . . 9 1 . E 7 . G F .
D E G . 9 . B C . 4 5 7 6 . 8 2 3 1 F
E D F . I 9 C B . 5 4 6 . 1 0 3 2 7 A
G F . E B C 9 A . 7 6 8 5 2 I 0 1 4 .
F . E D . . . 9 . 6 7 5 . . . 1 . 3 .
. . . . . . . . . . . . . . . . . . .
B . 7 . A . 0 . . . 8 E . 5 . D 6 9 I
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E012: order 20, 281 clues

Status I. Independently verified: two encodings and all deletion witnesses; exhaustive uniqueness tree replayed

Source: `outputs/lcs_bunya_verification_2026_10_15/n20/bunya_record.json`

SHA256: `d98ee4114b94ff06fd6b3f380a353b98cc588468ec681b3f88d180d0ece2d9b2`

Completion

```
E C 9 3 7 I J 4 H 0 B 1 G A 5 D 2 6 F 8
B F C 6 J 2 9 1 8 G 0 D A E 3 H 5 I 7 4
3 H 8 E 1 A 4 J C 6 2 7 5 I G F B 0 D 9
9 0 E 8 B D G 5 6 C 7 2 J F 4 I 1 H A 3
8 5 3 9 I 7 6 0 G 4 D A H 1 C B F J 2 E
0 9 J 5 F 6 7 C D A G 4 E B 2 1 I 3 8 H
1 I 2 7 3 C 5 G A D 8 E 4 H J 0 9 F 6 B
2 4 1 A H G D 8 7 5 6 9 I 3 0 J E B C F
A J 7 2 C 3 F H 1 B E 8 0 G I 4 6 5 9 D
H 3 6 C 2 J I A E 8 1 B D 4 F G 7 9 5 0
F G B D 0 4 A I 5 7 9 6 C J H 3 8 1 E 2
6 D H G 4 0 8 7 9 I 5 F 3 2 B C A E 1 J
7 B A 1 E 8 0 6 2 J C 3 F 9 D 5 H 4 G I
G 7 F I 9 5 C D 4 E A H B 0 6 8 3 2 J 1
C E G H A 1 B 2 3 F J I 9 7 8 6 4 D 0 5
4 2 5 J 6 F 3 E B 1 H 0 8 D 9 A C 7 I G
I 1 D B 8 E H F J 2 3 C 6 5 A 9 0 G 4 7
J A 0 4 G H E 3 I 9 F 5 7 C 1 2 D 8 B 6
5 8 4 0 D B 1 9 F 3 I J 2 6 E 7 G A H C
D 6 I F 5 9 2 B 0 H 4 G 1 8 7 E J C 3 A
```

Partial square

```
E . 9 3 7 I J . H . B 1 . A 5 D 2 6 . 8
B F C 6 . 2 9 1 8 . 0 . . E 3 H 5 I . .
3 . 8 . 1 A 4 . C . 2 7 5 I . F B 0 D 9
9 0 E 8 . D . 5 . . 7 2 J . 4 I . H . 3
8 5 3 9 I 7 6 0 . . D A H 1 . B F J . E
0 9 J . F . 7 C . . . 4 E B 2 1 I 3 . .
1 I 2 7 3 . 5 . A . . E 4 H J 0 . F 6 .
2 . 1 A H . . 8 7 . 6 9 I . . J E B C .
A J 7 2 C 3 F H 1 . E 8 . . I . 6 5 9 D
H . 6 C 2 J I A E . 1 B D 4 . . 7 9 5 0
F . B D 0 4 A . 5 . 9 6 C J H 3 8 1 E .
6 D H . 4 0 8 7 9 . 5 F 3 2 B C A . . J
7 B A 1 E . 0 6 2 . C . F 9 D 5 H 4 . I
. . . . . . . . . . . . . . . . . . . .
. E . H A 1 B 2 3 . J I . 7 8 6 4 D 0 5
4 2 5 J . F 3 E B . H . 8 D 9 A C . I .
I 1 D . 8 E H F J . 3 . 6 . A 9 0 . 4 7
J A 0 4 . H . 3 I . F 5 7 C . 2 D 8 B 6
. 8 . 0 D . 1 9 F . I J . 6 E 7 . . H C
D 6 . F . 9 2 . 0 . 4 . 1 . . E J C 3 A
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E013: order 21, 304 clues

Status I. Independently verified: two encodings and all deletion witnesses; exhaustive uniqueness tree replayed

Source: `outputs/lcs_bunya_verification_2026_10_15/n21/bunya_record.json`

SHA256: `0d54b05fdd69b7705172767466bd5bbc07121d246db131c08d7acfb27ab15685`

Completion

```
K 4 3 2 1 5 6 7 8 9 A B C D E F G H I J 0
1 0 4 3 K 9 5 6 7 8 E A B C D G H I J F 2
2 1 0 K 3 8 9 5 6 7 D E A B C H I J F G 4
3 2 K 0 4 7 8 9 5 6 C D E A B I J F G H 1
4 K 2 1 0 6 7 8 9 5 B C D E A J F G H I 3
5 9 8 7 6 0 1 2 3 4 K G H I J A B C D E F
6 5 9 8 7 4 0 1 2 3 G K I J F E A B C D H
7 6 5 9 8 3 4 0 1 2 H I K F G D E A B C J
8 7 6 5 9 2 3 4 0 1 I J F K H C D E A B G
9 8 7 6 5 1 2 3 4 0 J F G H K B C D E A I
A E D C B F G H I J 0 1 2 3 4 K 6 7 8 9 5
B A E D C G H I J F 4 0 1 2 3 9 5 6 7 K 8
C B A E D H I J F G 3 4 0 1 2 8 9 5 K 7 6
D C B A E I J F G H 2 3 4 0 1 7 8 K 5 6 9
E D C B A J F G H I 1 2 3 4 0 6 K 8 9 5 7
F G H I J K E D C B 5 9 8 7 6 0 4 3 2 1 A
G H I J F B A E D K 6 5 9 8 7 1 0 4 3 2 C
H I J F G C B A K D 7 6 5 9 8 2 1 0 4 3 E
I J F G H D C K A E 8 7 6 5 9 3 2 1 0 4 B
J F G H I E K C B A 9 8 7 6 5 4 3 2 1 0 D
0 3 1 4 2 A D B E C F H J G I 5 7 9 6 8 K
```

Partial square

```
. 4 3 2 1 5 6 . . 9 A B . D E . G H I J .
1 0 4 3 . . 5 6 . 8 E A B C D G H I J F .
2 . . . 3 8 9 5 6 7 D E A B C . I . F G .
3 2 . 0 4 7 8 . . . C . E . B I J F G H .
4 . 2 1 0 6 7 8 9 5 B . D E A J F G . . .
5 9 . 7 6 0 . 2 3 4 . G H I J A B C . E .
. 5 9 . 7 . 0 . 2 3 G . I J F E A B C D .
7 6 5 9 8 3 4 . 1 2 H I . F G . E A B . .
. . 6 5 9 2 3 4 . 1 I J F . H . D E A B .
9 . 7 6 5 1 2 3 . 0 J F G H . . C . E A .
A E D C . . G H I J 0 1 2 3 . . . 7 8 9 .
B A . . C . H I J F . 0 1 2 3 9 5 6 7 . .
C . A E D H . J F G 3 4 0 1 2 8 9 5 . 7 .
D . B . E I J F G . 2 3 . 0 1 7 . . 5 6 .
. D C B A J F . H I . 2 3 . 0 6 . 8 9 5 .
F . H . . . E D C B . 9 8 7 6 0 4 3 2 1 .
. . I J F B A E D . 6 . 9 8 7 1 0 4 3 . .
H . . F . C B A . D 7 6 5 . . 2 1 0 . 3 .
I J F G H D C . A . . 7 6 5 9 3 2 . 0 4 .
. F . H I E . C B A 9 8 7 6 5 4 . . . 0 .
. . . . . . . . . . . . . . . . . . . . K
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E014: order 22, 340 clues

Status I. Independently verified: two encodings and all deletion witnesses; exhaustive uniqueness tree replayed

Source: `outputs/lcs_bunya_verification_2026_10_15/n22/bunya_record.json`

SHA256: `802f2b0ef80274208823100b1338d898385de87ed6c4c2c9af3dc685fe765e51`

Completion

```
8 0 9 2 3 4 5 6 7 A 1 J K C D E F G H I L B
9 8 A 3 4 2 6 7 5 1 0 B J K E F D H I G C L
1 9 8 4 2 3 7 5 6 0 A K L J F D E I G H B C
2 3 4 8 6 9 A 0 1 5 7 D E F J K I L B C G H
3 4 2 9 8 5 0 1 A 7 6 E F D H J K B C L I G
4 2 3 7 9 8 1 A 0 6 5 F D E K G J C L B H I
5 6 7 A 0 1 8 3 9 2 4 G H I L B C J K F D E
6 7 5 0 1 A 9 8 2 4 3 H I G B C L E J K F D
7 5 6 1 A 0 4 9 8 3 2 I G H C L B K D J E F
A 1 0 5 7 6 2 4 3 9 8 L C B G I H D F E K J
0 A 1 6 5 7 3 2 4 8 9 C B L I H G F E D J K
L D E F G H I J B C K 8 9 1 2 3 4 5 6 7 A 0
B K G E F J H I C D L 0 8 9 3 4 2 6 7 5 1 A
K L F G E I J H D B C 9 A 8 4 2 3 7 5 6 0 1
I J B C D K L G E F H 2 3 4 8 9 7 A 0 1 5 6
H I D B C L E K F G J 3 4 2 6 8 9 0 1 A 7 5
J H C D B F K L G E I 4 2 3 9 5 8 1 A 0 6 7
F G K L J B C D H I E 5 6 7 A 0 1 8 9 4 2 3
E F L H K D B C I J G 6 7 5 0 1 A 3 8 9 4 2
G E I K L C D B J H F 7 5 6 1 A 0 9 2 8 3 4
D C H J I E G F L K B A 1 0 5 7 6 2 4 3 9 8
C B J I H G F E K L D 1 0 A 7 6 5 4 3 2 8 9
```

Partial square

```
. 0 9 . 3 4 5 6 7 . . . K C D E F . H . . B
9 . A 3 4 . 6 7 5 . 0 B . . E F D H . G . L
1 . . 4 2 3 7 5 . . A K L . . D E I G H . C
2 3 . . 6 9 A 0 . . 7 . E F . K I L B C . H
3 4 2 9 . 5 0 . A . . E F D H . K . C L . G
4 2 3 7 9 . 1 . 0 . 5 F . E K G . C L B . I
. 6 7 A . 1 . 3 9 . 4 G H I L B C . . F . .
6 . 5 0 1 A . . 2 . 3 H I G B . . E . K . D
. 5 6 1 A 0 4 9 . . 2 I G . . L B K D . . .
. . . . . . . . . . . . . . . . . . . . . .
0 A 1 6 5 7 . 2 . . 9 . B L I . G F E D . K
L D E F G H I J B C K . 9 1 2 3 4 . 6 . . 0
B K G E F J H I C . L 0 . . 3 4 2 6 . 5 . A
K L F G E I J H D B C 9 A . . 2 3 7 5 6 . 1
I J B C D K L G E F H . 3 4 . 9 7 A 0 1 . 6
H I D B C L E K F G J 3 4 2 6 . 9 . 1 A . 5
J H C D B F K L G E I 4 . 3 9 5 . 1 A 0 . 7
F G K L . B C D H I E 5 6 7 A 0 1 . . 4 . .
E F L H K D B C I . G 6 7 5 0 . . 3 . 9 . 2
G E I K L C D B J H F 7 5 . . A 0 9 2 . . .
D C H J I E G F L K B . . . . . . . . . . .
C B J I H G F E K L D . 0 A 7 . 5 4 3 2 . 9
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E015: order 23, 367 clues

Status I. Independently verified: two encodings and all deletion witnesses; exhaustive uniqueness tree replayed

Source: `outputs/lcs_bunya_verification_round3/n23/bunya_record.json`

SHA256: `368318dd591371fdf545aafaa7195e748d0f8fbba024cb2cd15a10e7e5c2d273`

Completion

```
8 9 1 2 3 4 5 6 7 A 0 J K C D E F G M I L B H
M K C 3 4 2 6 7 5 L B 8 9 1 E F D H I G A 0 J
9 A 8 4 2 3 7 5 6 0 M K L J F D E I G H B C 1
2 3 4 8 9 7 A 0 1 5 6 D E F J K I L B C M H G
3 4 2 6 8 9 0 1 A M 5 E F D H J K B C L I G 7
4 2 3 9 5 8 1 A 0 6 7 F D E K G J C L B H M I
5 6 7 M 0 1 F 9 D J 3 G H I L B C 4 K 2 8 E A
6 7 5 0 1 A 3 8 9 4 2 H I G B C M E J K F D L
7 5 6 1 A M 9 2 8 3 4 I G H C L B K D J E F 0
A 1 0 5 7 6 2 4 M 9 8 L C B G I H D F E K J 3
1 0 A B 6 5 4 3 2 8 9 C 7 L I H G F E M J K D
0 8 9 D E F G H I 1 A B J K M 3 4 5 6 7 C L 2
B J K E M D H I G C L 0 8 9 3 4 2 6 7 5 1 A F
K L J F D E I G H B C M A 8 4 2 3 7 5 6 0 1 9
D E F J K I L M C G H 2 3 4 8 9 7 A 0 1 5 6 B
E F D H J K B C L I G 3 M 2 6 8 9 0 1 A 7 5 4
F D M K G J C L B H I 4 2 3 9 5 8 1 A 0 6 7 E
G H I L B C J K F D E 5 6 7 A 0 1 M 9 4 2 3 8
H I G 7 C L E J K F D 6 B M 0 1 A 3 8 9 4 2 5
I G H C L B M D J E F 7 5 6 1 A 0 9 2 8 3 4 K
L M B G I H D F E K J A 1 0 5 7 6 2 4 3 9 8 C
C B L I H G 8 E 4 2 K 1 0 A 7 M 5 J 3 F D 9 6
J C E A F 0 K B 3 7 1 9 4 5 2 6 L 8 H D G I M
```

Partial square

```
8 9 . 2 3 4 5 6 7 A 0 J K . D E F G . . L B .
. . . . . . . 6 7 . . . . . 1 E . . . I . . . .
9 . 8 4 2 3 . 5 . 0 . K . J F D E . G H B C .
2 3 4 8 9 . A 0 1 5 6 D E F J K . L B C . H .
3 4 2 . 8 9 0 1 A . 5 E . D . J K B C L I G .
4 2 3 9 5 8 1 A 0 6 7 F D E K . J C L B H . .
5 6 7 . 0 1 . 9 . . 3 G H I L . C . K . . E .
6 7 5 0 1 A . 8 9 4 2 H I G B C . . J K F D .
7 5 6 1 A . 9 . 8 3 4 . G H C . B K . J E F .
A 1 0 5 7 6 2 4 . . . L C B G I H D F E K J .
1 0 A B 6 5 4 3 2 8 9 C . L I H G F E . . K .
. . 9 . E . . . . . A B . . . . 4 5 . . . . .
. J K E . D H I G C L . 8 9 3 4 2 6 . 5 1 A .
K . J F . E I G . B C . A 8 4 2 3 7 5 6 0 1 .
D E F J K . L . C G H 2 3 4 8 9 7 A 0 1 . 6 .
E F D H J K B C . I G . . 2 . 8 9 0 1 A 7 5 .
F D . K . J C L B H I 4 2 3 9 . 8 1 A 0 6 7 .
G H I L B C J K . D E 5 6 7 A 0 1 . 9 . 2 3 .
H I G . C L E J K F D 6 . . 0 1 A . 8 9 4 2 .
I G H C L B . D J E . 7 5 6 1 A 0 . . 8 3 4 .
L . B G I . D F . . . A 1 0 . 7 6 2 4 . . . .
C B L I H G . E 4 . K 1 0 A 7 . 5 . 3 . . 9 .
. . . . . . . . . . . . . . . . . . . . . . M
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E016: order 24, 405 clues

Status I. Previously verified construction/example or published record

Source: `outputs/lcs_examples.json:10`

SHA256: `27e2cf0eb6248f0432830ae0055d0090b003facccad840083e6bda938dcd479e`

Completion

```
9 0 1 2 3 4 5 6 7 8 A B L C D E F G H I J K M N
0 1 9 4 2 3 6 7 5 A B 8 C D L G E F I J H M N K
1 9 0 3 4 2 7 5 6 B 8 A D L C F G E J H I N K M
2 3 4 9 0 1 A 8 B 6 5 7 E F G L C D M K N I H J
3 4 2 1 9 0 B A 8 7 6 5 F G E D L C N M K J I H
4 2 3 0 1 9 8 B A 5 7 6 G E F C D L K N M H J I
5 7 6 A B 8 9 1 0 4 2 3 H J I M N K L D C G E F
6 5 7 8 A B 0 9 1 2 3 4 I H J K M N C L D E F G
7 6 5 B 8 A 1 0 9 3 4 2 J I H N K M D C L F G E
8 B A 6 7 5 4 2 3 9 1 0 K N M I J H G E F L D C
A 8 B 5 6 7 2 3 4 0 9 1 M K N H I J E F G C L D
B A 8 7 5 6 3 4 2 1 0 9 N M K J H I F G E D C L
L C D E F G H I J K M N 9 0 1 2 3 4 5 6 7 8 A B
C D L G E F I J H M N K 0 1 9 4 2 3 6 7 5 A B 8
D L C F G E J H I N K M 1 9 0 3 4 2 7 5 6 B 8 A
E F G L C D M K N I H J 2 3 4 9 0 1 A 8 B 6 5 7
F G E D L C N M K J I H 3 4 2 1 9 0 B A 8 7 6 5
G E F C D L K N M H J I 4 2 3 0 1 9 8 B A 5 7 6
H J I M N K L D C G E F 5 7 6 A B 8 9 1 0 4 2 3
I H J K M N C L D E F G 6 5 7 8 A B 0 9 1 2 3 4
J I H N K M D C L F G E 7 6 5 B 8 A 1 0 9 3 4 2
K N M I J H G E F L D C 8 B A 6 7 5 4 2 3 9 1 0
M K N H I J E F G C L D A 8 B 5 6 7 2 3 4 0 9 1
N M K J H I F G E D C L B A 8 7 5 6 3 4 2 1 0 9
```

Partial square

```
9 0 1 2 3 4 5 6 7 8 A B . C . E . . . I J . M N
0 1 9 4 2 3 6 7 5 A B 8 . D L . E F . J H M N .
1 9 0 3 4 2 7 5 6 B 8 A D L . . G E . H I N . M
2 3 4 9 0 1 A 8 B 6 5 7 E F G . . D . . . I H .
3 4 2 1 9 0 B . 8 7 6 5 F G E . L . . M . J . H
4 2 3 0 1 9 8 . . 5 7 6 G E F C D L . N M H J I
5 7 6 A B 8 9 1 0 4 2 3 H J . M N . . D C G E .
6 5 7 8 A B 0 9 1 2 3 4 I H J . M N . L D E F G
7 6 5 B 8 A 1 0 9 3 4 2 . I H N . M . C L F . E
8 . A 6 7 5 4 . 3 9 1 0 . N . I J H . E . L D C
A 8 B 5 6 7 2 3 4 0 9 1 . . . . . .
. . 8 7 5 6 3 . . 1 0 9 N M . J H I . G E D C L
. C . E . . . I J . M N . 0 . 2 . . . 6 7 . A B
. D L . E F . J H M N . . 1 9 . 2 3 . 7 5 A B .
D L . . G E . H I N . M 1 9 . . 4 2 . 5 6 B . A
E F G . . D . . . I H . 2 3 4 . . 1 . . . 6 5 .
F G E . L . . M . J . H 3 4 2 . 9 . . A . 7 . 5
G E F C D L . N M H J I 4 2 3 0 1 9 . B A 5 7 6
H J . M N . . D C G E . 5 7 . A B . . 1 0 4 2 .
I H J . M N . L D E F G 6 5 7 . A B . 9 1 2 3 4
. I H N . M . C L F . E . 6 5 B . A . 0 9 3 . 2
. N . I J H . E . L D C . B . 6 7 5 . 2 . 9 1 0
. . . . . .
N M . J H I . G E D C L B A . 7 5 6 . 4 2 1 0 9
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E017: order 25, 449 clues

Status I. Independently verified: two encodings and all deletion witnesses; uniqueness relies on two completed solver checks

Source: `outputs/lcs_bunya_verification_2026_10_15/n25/bunya_record.json`

SHA256: `63977a0ea9bb540c3c3312f463354abc4a02173157da3fc5020c168f66f18068`

Completion

```
0 2 1 3 4 5 6 7 8 9 A B C E D F G H I J K L O N M
1 0 2 5 3 4 8 6 7 A B 9 D C E H F G K I J M N O L
2 1 0 4 5 3 7 8 6 B 9 A E D C G H F J K I O L M N
3 5 4 0 1 2 9 A B 6 7 O F H G C D E L M N I J K 8
4 3 5 2 0 1 A B 9 8 6 7 G F H O C D M N L K I J E
5 4 3 1 2 0 B 9 A O 8 6 H G F D E C N L M J K I 7
6 8 7 9 A B 0 1 2 3 4 5 I K O L M N C D E F G H J
7 6 8 A B 9 O 0 1 5 3 4 J I K M N L E C D H F G 2
8 7 6 B 9 A 1 2 0 4 5 3 O J I N L M D E C G H F K
9 O B 6 8 7 3 5 4 0 2 1 L M N I K J F H G C E D A
A B O 7 6 8 4 3 5 1 0 2 M N L J I K G F H D C E 9
O 9 A 8 7 6 5 4 3 2 1 0 N L M K J I H G F E D C B
C E D O G H I J K L M N 0 2 1 3 4 5 6 7 8 9 A B F
D C E H F O K I J M N L 1 0 2 5 3 4 8 6 7 A B 9 G
E D C G O F J K I N L M 2 1 0 4 5 3 7 8 6 B 9 A H
F H G C D E L M N I J K 3 5 4 0 O 2 9 A B 6 7 8 1
G F H E C D M N L K O J 4 3 5 2 0 1 A B 9 8 6 7 I
H G F D E C N L M J K I 5 4 3 1 2 O B 9 A 7 8 6 O
I K J L M N C O E F G H 6 8 7 9 A B 0 1 2 3 4 5 D
J I K M N L E C D H F G 7 O 8 A B 9 2 0 1 5 3 4 6
K J I N L M D E O G H F 8 7 6 B 9 A 1 2 0 4 5 3 C
L M N I K J F H G C E D 9 A B 6 8 7 O 5 4 0 2 1 3
M N L J I K G F H D C E A B 9 7 6 8 4 3 O 1 0 2 5
N L M K J I H G F E D C B 9 A 8 7 6 5 O 3 2 1 0 4
B A 9 F H G 2 D C 7 I 8 K 6 J E 1 0 3 4 5 N M L O
```

Partial square

```
0 2 . 3 4 . 6 . 8 . . B . E D . G H . J K L . N .
. . 2 5 3 . 8 6 7 A B 9 D C E . F G K . . M N . .
2 . . 4 5 3 7 . . B 9 A E D . G . F J K I . L M .
3 . . 0 . . 9 A B 6 7 . F H G C D E L M N I J K .
4 3 5 2 0 1 A . 9 8 6 7 G F . . C D . . L K I J .
5 4 . 1 . 0 B 9 A . 8 6 H G F D E C . L . J K I .
6 8 . 9 A B 0 1 2 3 4 . I K . L M . C D E . . H .
7 6 8 A . 9 . 0 1 5 . 4 J I K M N L E C D H . G .
8 7 6 B 9 A 1 2 . . 5 . . J I . L . D . . G H . .
9 . . . 8 7 3 5 4 0 2 1 L M N I K . . H G C . D .
A B . . 6 8 4 3 . 1 0 . M N L J I K G F H . C E .
. 9 A 8 7 . 5 4 3 2 1 0 N L M K J I H . F E D C .
C E . . G H . J K L M N 0 2 1 3 4 5 . 7 . 9 A B .
D C E H F . . . J M N . 1 . 2 5 3 . 8 6 7 A B 9 .
. . . G . F J . I N . M 2 1 0 4 5 3 . 8 6 B 9 A .
. H G C D E . M N I J K . 5 4 0 . 2 9 A B . 7 8 .
G F H E C D M N L K . J . 3 5 2 0 1 A B . 8 6 7 .
H G F D E C N L M J K I 5 4 . 1 2 . B 9 A . 8 6 .
I K J . . N C . E F G H 6 8 7 9 A B 0 1 2 3 4 . .
J I K . N L E C D H F . 7 . 8 A . 9 2 0 1 5 3 . .
K J I . . M D E . G H F 8 7 6 B . A 1 2 0 . 5 3 .
L M N I K . F H G C E D 9 A B 6 . 7 . 5 4 0 2 . .
M N L J I K G F H D C E A B 9 7 . 8 4 3 . . 0 2 .
N L M . J . H . F E D . . 9 A . 7 6 5 . 3 2 1 0 .
. . . . . . . . . . . . . . . . . . . . . . . . O
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E018: order 26, 482 clues

Status I. Independently verified: two encodings and all deletion witnesses; uniqueness relies on two completed solver checks

Source: `outputs/lcs_bunya_verification_2026_10_15/n26/bunya_record.json`

SHA256: `ff82d75e235a3a98d21ac265d3576d213de9469bc13eff033babcb45427949d8`

Completion

```
1 0 8 7 2 9 A 5 C 6 3 B 4 E D L K F M N I P J G O H
8 1 0 2 9 7 C 6 A 5 4 3 B L E D F M K P J N I H G O
9 2 7 0 1 6 8 4 B 3 5 A C M F K D E J L H O G I N P
4 3 B 1 0 8 6 C 5 A 9 2 7 H G O E D L J P I N M F K
7 9 2 6 8 0 1 B 3 4 C 5 A K M F J L D E O G H P I N
2 7 9 8 6 1 0 3 4 B A C 5 F K M L J E D G H O N P I
A 5 C 4 3 B 7 0 8 1 2 9 6 N I P H G O K D L E F M J
5 C A B 4 3 9 1 0 8 6 7 2 I P N O H G M E D L J K F
C A 5 3 B 4 2 8 1 0 7 6 9 P N I G O H F L E D K J M
0 8 1 9 7 2 5 A 6 C B 4 3 D L E M K F I N J P O H G
B 6 4 5 A C 3 7 9 2 0 1 8 O J H I N P G K M F D E L
6 B 3 A C 5 4 9 2 7 8 0 1 J O G N P I H M F K L D E
3 4 6 C 5 A B 2 7 9 1 8 0 G H J P I N O F K M E L D
E D L K F M N I P J G O H 1 2 3 4 5 6 7 8 9 0 A B C
L E D F M K P J N I H G O 0 1 4 3 6 5 9 7 8 2 B C A
M F K D E J L H O G I N P 4 3 1 B 2 0 A C 6 5 9 8 7
H G O E D L J P I N M F K A B 9 1 8 7 4 5 3 C 6 2 0
K M F J L D E O G H P I N C A 7 0 9 8 3 4 5 B 2 6 1
F K M L J E D G H O N P I 7 9 A 8 B C 1 0 2 6 3 5 4
N I P H G O K D L E F M J 9 7 5 6 3 4 C A B 8 1 0 2
I P N O H G M E D L J K F 3 5 2 C 0 1 6 B A 4 7 9 8
P N I G O H F L E D K J M 2 0 6 5 4 3 8 9 7 1 C A B
D L E M K F I N J P O H G 5 4 0 A 1 2 B 6 C 3 8 7 9
O J H I N P G K M F D E L 6 8 B 9 C A 0 2 1 7 4 3 5
J O G N P I H M F K L D E B C 8 2 7 9 5 3 4 A 0 1 6
G H J P I N O F K M E L D 8 6 C 7 A B 2 1 0 9 5 4 3
```

Partial square

```
1 0 . 7 2 9 A 5 C . . . . E D . K F M N I P . . . .
8 1 . 2 . 7 C . A 5 4 3 . L E . F . K P . N I H G .
9 2 . 0 1 6 8 4 . 3 5 . C M F . D E J L H . G I . P
4 3 . . 0 8 6 C . A 9 2 7 H G . . D L J P . N M F K
7 9 . 6 . 0 1 . 3 4 C 5 A K M . J . D E . G H P I N
2 7 . 8 . 1 . 3 4 . A C 5 F K . L . E . G H . N P I
A 5 . 4 3 . 7 0 8 . 2 . 6 N I . H G . K D L . F . J
. . . . . 9 1 0 8 6 7 2 . . . . . M E D L J K F
. A . 3 . . . 8 1 0 . 6 9 . N . G . . . L E D . J M
0 8 . . 7 2 5 A 6 C . 4 3 D L . . K F I N J P . H G
. 6 . 5 A . 3 7 9 2 0 1 8 . J . I N . G K M F D E L
6 . . A . . 4 9 2 . 8 0 1 J . . N . . H M F . L D E
. . . . . .
E D . K F M N I P . . . . 1 2 3 . 5 6 7 8 9 0 A B C
L E . F . K P . N I H G . . 1 4 3 6 5 9 7 8 2 . C A
M F . D E J L H . G I . P 4 3 1 . 2 0 A . 6 5 9 8 7
H G . . D L J P . N M F K A . 9 1 8 7 4 5 3 C 6 2 0
K M . J . D E . G H P I N C A 7 0 9 8 3 4 5 B 2 6 1
F K . L . E . G H . N P I 7 9 A 8 B C 1 0 2 6 3 5 4
N I . H G . K D L . F . J 9 7 5 6 3 4 C A B 8 1 0 2
. . . . . M E D L J K F 3 5 2 C 0 1 6 B A 4 7 9 8
. N . G . . . L E D . J M 2 0 6 5 4 3 8 9 7 1 C A B
D L . . K F I N J P . H G 5 4 0 A 1 2 . 6 C 3 8 7 9
. J . I N . G K M F D E L 6 8 B 9 C A 0 2 1 7 4 3 5
J . . N . . H M F . L D E B C 8 2 7 9 5 3 4 A 0 1 6
. . . . . 8 6 C 7 A . 2 1 0 9 5 4 3
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E019: order 27, 447 clues

Status I. Independently verified: two solver encodings and two sets of 447 deletion witnesses; uniqueness solver-dependent

Source: `outputs/lcs27_bunya_29909858/independent_audit/record.json`

SHA256: `68611c3006f75066a5da2cede54259bed6e0fb790349e2f2ba56d7e3d79ff1d8`

Completion

```
0 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q
1 0 7 6 5 2 3 8 4 A 9 G F E B C H D J I P O N K L Q M
2 5 0 4 8 1 7 3 6 B E 9 D H A G C F K N I M Q J P L O
8 6 4 0 2 3 1 5 7 H F D 9 B C A E G Q O M I K L J N P
5 2 1 8 0 7 4 6 3 E B A H 9 G D F C N K J Q I P M O L
6 3 5 1 7 0 8 4 2 F C E A G 9 H D B O L N J P I Q M K
3 4 6 7 1 8 0 2 5 C D F G A H 9 B E L M O P J Q I K N
7 8 3 5 6 4 2 0 1 G H C E F D B 9 A P Q L N O M K I J
4 7 8 2 3 6 5 1 0 D G H B C F E A 9 M P Q K L O N J I
I J K L M N O P Q 0 1 2 3 4 5 6 7 8 9 A B C D E F G H
J I P O N K L Q M 1 0 7 6 5 2 3 8 4 A 9 G F E B C H D
K N I M Q J P L O 2 5 0 4 8 1 7 3 6 B E 9 D H A G C F
Q O M I K L J N P 8 6 4 0 2 3 1 5 7 H F D 9 B C A E G
N K J Q I P M O L 5 2 1 8 0 7 4 6 3 E B A H 9 G D F C
O L N J P I Q M K 6 3 5 1 7 0 8 4 2 F C E A G 9 H D B
L M O P J Q I K N 3 4 6 7 1 8 0 2 5 C D F G A H 9 B E
P Q L N O M K I J 7 8 3 5 6 4 2 0 1 G H C E F D B 9 A
M P Q K L O N J I 4 7 8 2 3 6 5 1 0 D G H B C F E A 9
9 A B C D E F G H I J K L M N O P Q 0 1 2 3 4 5 6 7 8
A 9 G F E B C H D J I P O N K L Q M 1 0 7 6 5 2 3 8 4
B E 9 D H A G C F K N I M Q J P L O 2 5 0 4 8 1 7 3 6
H F D 9 B C A E G Q O M I K L J N P 8 6 4 0 2 3 1 5 7
E B A H 9 G D F C N K J Q I P M O L 5 2 1 8 0 7 4 6 3
F C E A G 9 H D B O L N J P I Q M K 6 3 5 1 7 0 8 4 2
C D F G A H 9 B E L M O P J Q I K N 3 4 6 7 1 8 0 2 5
G H C E F D B 9 A P Q L N O M K I J 7 8 3 5 6 4 2 0 1
D G H B C F E A 9 M P Q K L O N J I 4 7 8 2 3 6 5 1 0
```

Partial square

```
. . . . . . . . . . . . . . . . . . . . . . . . . . .
1 . 7 6 . . 3 8 4 A . G F E . C . . . I . O . K L Q M
. 5 0 4 8 1 7 3 . . . 9 D . A . . F K N I . . . P L O
8 6 4 0 . . . . 7 . . D 9 B C A E G Q O . . K L . N P
. 2 . 8 0 7 4 6 3 E . A H 9 G . F C N K . Q I . M . .
6 3 . 1 7 0 8 4 2 F . E A . 9 H D B O L . . P . . . K
3 . 6 7 1 8 . . 5 . . . . A H 9 B . . M O P . Q . K N
7 8 3 5 6 4 2 0 1 . . C E F . B 9 A P Q L N O . K I .
4 7 8 . 3 6 5 1 0 D . H B C . E A 9 . P . K L . N . I
. . K L M N O . Q O . 2 3 4 . 6 . 8 9 A B C D E F G .
. I P O N K L Q . . . . . 5 2 3 8 . . . G F E B . . D
K N . . Q . . L . 2 . 0 4 . 1 . . 6 B E 9 . H A G C .
. . M I . L . N P 8 . . 0 2 3 1 5 7 H . D 9 . C A . G
N . . Q I P . O L 5 . 1 8 . . 4 6 3 . B A H 9 G D . C
. L N . P I Q M K 6 . . 1 7 . 8 4 2 . C . A . . H . B
L . . . . Q I . N 3 . 6 7 . 8 0 2 5 C D F . A H 9 B .
P Q L N O . . I . 7 . 3 5 . . . 0 1 G . C E F D B 9 .
. P . K L . N . I 4 . 8 2 3 6 5 1 0 D . H . C . E A 9
. A B C D . F . . I . K L M N O P Q 0 1 2 3 . . 6 7 8
. 9 G . E . C H D . I P O . K L . . 1 . 7 . 5 2 3 . 4
. . . . H A . . . . . . . . . . . . 2 . . . . . . . .
H F D . . C . E G Q . M I K L . . P . 6 . 0 2 3 1 5 .
. B . . 9 . D F C N . . Q I P . O . 5 2 1 8 0 . . 6 .
. C E A G . . D B . . N . P . . . K 6 3 . . 7 0 8 4 .
C D F G . H . . E L . O . . Q . . N 3 4 6 . 1 . . 2 5
G H C . F . B 9 . . . L . O M K I . . . . 5 6 4 2 0 1
. G . B . F . A 9 . . . K . O N . I . 7 8 2 3 6 5 1 .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E020: order 28, 536 clues

Status I. Independently verified: two encodings and all deletion witnesses; uniqueness relies on two completed solver checks

Source: `outputs/lcs_bunya_verification_2026_10_15/n28/bunya_record.json`

SHA256: `e215fdb929ffc8d448443eba84ed67ccfdadb02a980ea8d19baa2e0fa5b63505`

Completion

```
0 2 1 4 3 6 5 7 9 8 B A D C E G F I H K J L N M P O R Q
2 1 0 5 6 3 4 9 8 7 C D A B G F E J K H I N M L Q R O P
1 0 2 6 5 4 3 8 7 9 D C B A F E G K J I H M L N R Q P O
4 5 6 3 0 1 2 B C D A 7 8 9 I J K H E F G P Q R O L M N
3 6 5 0 4 2 1 A D C 7 B 9 8 H K J E I G F O R Q L P N M
6 3 4 1 2 5 0 D A B 8 9 C 7 K H I F G J E R O P M N Q L
5 4 3 2 1 0 6 C B A 9 8 7 D J I H G F E K Q P O N M L R
7 9 8 B A D C 0 2 1 4 3 6 5 L N M P O R Q E G F I H K J
9 8 7 C D A B 2 1 0 5 6 3 4 N M L Q R O P G F E J K H I
8 7 9 D C B A 1 0 2 6 5 4 3 M L N R Q P O F E G K J I H
B C D A 7 8 9 4 5 6 3 0 1 2 P Q R O L M N I J K H E F G
A D C 7 B 9 8 3 6 5 0 4 2 1 O R Q L P N M H K J E I G F
D A B 8 9 C 7 6 3 4 1 2 5 0 R O P M N Q L K H I F G J E
C B A 9 8 7 D 5 4 3 2 1 0 6 Q P O N M L R J I H G F E K
E G F I H K J L N M P O R Q 0 2 1 4 3 6 5 7 9 8 B A D C
G F E J K H I N M L Q R O P 2 1 0 5 6 3 4 9 8 7 C D A B
F E G K J I H M L N R Q P O 1 0 2 6 5 4 3 8 7 9 D C B A
I J K H E F G P Q R O L M N 4 5 6 3 0 1 2 B C D A 7 8 9
H K J E I G F O R Q L P N M 3 6 5 0 4 2 1 A D C 7 B 9 8
K H I F G J E R O P M N Q L 6 3 4 1 2 5 0 D A B 8 9 C 7
J I H G F E K Q P O N M L R 5 4 3 2 1 0 6 C B A 9 8 7 D
L N M P O R Q E G F I H K J 7 9 8 B A D C 0 2 1 4 3 6 5
N M L Q R O P G F E J K H I 9 8 7 C D A B 2 1 0 5 6 3 4
M L N R Q P O F E G K J I H 8 7 9 D C B A 1 0 2 6 5 4 3
P Q R O L M N I J K H E F G B C D A 7 8 9 4 5 6 3 0 1 2
O R Q L P N M H K J E I G F A D C 7 B 9 8 3 6 5 0 4 2 1
R O P M N Q L K H I F G J E D A B 8 9 C 7 6 3 4 1 2 5 0
Q P O N M L R J I H G F E K C B A 9 8 7 D 5 4 3 2 1 0 6
```

Partial square

```
0 . . 4 3 6 5 . 9 8 B A D . E . . I H K J . N M P O R .
. 1 . 5 6 3 4 9 . 7 C D A B . F . J K H I N . L Q R O P
1 0 . 6 5 4 3 8 7 9 . . B A F E . K J I H M L N . . P O
4 5 . 3 . 1 . B C D . 7 8 9 I J . H . F . P Q R . L M N
3 . . 0 . . 1 A D C 7 . 9 8 H . . E . . F O R Q L . N M
6 . . 1 . 5 0 D . B . 9 . 7 K . . F . J E R . P . N . L
5 4 . . . . . C B A . . 7 . J I . . . . . Q P O . . L .
. 9 . B A D C 0 . . 4 3 6 5 . N . P O R Q E . . I H K J
9 8 . C D A . . 1 0 5 6 . 4 N M . Q R O . . F E J K . I
. . . . . .
B C . A 7 . 9 4 . 6 3 0 1 . P Q . O L . N I . K H E F .
. D . . B 9 8 3 . 5 0 4 . 1 . R . . P N M H . J E I . F
D A . 8 9 C 7 6 3 . 1 . 5 0 R O . M N Q L K H . F . J E
C B . 9 8 7 D . 4 3 . 1 0 6 Q P . N M L R . I H . F E K
E . . I H K J . N M P O R . . 2 1 . 3 6 5 7 . 8 . . . C
. F . J K H I N . L Q R O P 2 . 0 5 6 3 . . 8 . . . A .
F E . K J I H M L N . . P O . . 2 6 5 . 3 8 7 9 D C B A
I J . H . F . P Q R . L M N 4 5 6 3 0 1 2 B C D A . 8 9
H . . E . . F O R Q L . N M . 6 5 0 4 2 1 A D C 7 B . 8
K . . F . J E R . P . N . L 6 3 4 1 2 5 0 . A B 8 9 C 7
J I . . . . . Q P O . . L . 5 4 3 2 1 0 6 C B A 9 8 7 D
. N . P O R Q E . . I H K J 7 9 8 . A D . 0 2 1 4 3 6 5
N M . Q R O . . F E J K . I 9 8 7 C . A B 2 1 0 5 6 3 4
. . . . . 8 7 9 D C B A 1 0 2 6 5 4 3
P Q . O L . N I . K H E F . B C D A . 8 9 . 5 . 3 0 . 2
. R . . P N M H . J E I . F A D C 7 . 9 8 3 6 . 0 . 2 1
R O . M N Q L K H . F . J E . . B 8 9 C 7 . . 4 . 2 5 .
Q P . N M L R . I H . F E K C . A 9 8 7 D 5 4 3 2 1 . 6
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E021: order 29, 624 clues

Status I. Independently verified: two encodings and all deletion witnesses; uniqueness relies on two completed solver checks

Source: `outputs/lcs_bunya_verification_round2/n29/bunya_record.json`

SHA256: `c684036a893367021b17aa1ab298aa047a509faf82a7c628374b0800a14fb32e`

Completion

```
S 6 5 4 3 2 1 7 8 9 A B C D E F G H I J K L M N O P Q R 0
1 0 6 5 4 3 S D 7 8 9 A B C K E F G H I J M N O P Q R L 2
2 1 0 6 5 S 3 C D 7 8 9 A B J K E F G H I N O P Q R L M 4
3 2 1 0 S 5 4 B C D 7 8 9 A I J K E F G H O P Q R L M N 6
4 3 2 S 0 6 5 A B C D 7 8 9 H I J K E F G P Q R L M N O 1
5 4 S 2 1 0 6 9 A B C D 7 8 G H I J K E F Q R L M N O P 3
6 S 4 3 2 1 0 8 9 A B C D 7 F G H I J K E R L M N O P Q 5
7 D C B A 9 8 0 1 2 3 4 5 6 S M N O P Q R E F G H I J K L
8 7 D C B A 9 6 0 1 2 3 4 5 M S O P Q R L K E F G H I J N
9 8 7 D C B A 5 6 0 1 2 3 4 N O S Q R L M J K E F G H I P
A 9 8 7 D C B 4 5 6 0 1 2 3 O P Q S L M N I J K E F G H R
B A 9 8 7 D C 3 4 5 6 0 1 2 P Q R L S N O H I J K E F G M
C B A 9 8 7 D 2 3 4 5 6 0 1 Q R L M N S P G H I J K E F O
D C B A 9 8 7 1 2 3 4 5 6 0 R L M N O P S F G H I J K E Q
E K J I H G F L M N O P Q R 0 1 2 3 4 5 6 S 8 9 A B C D 7
F E K J I H G M N O P Q R L 6 0 1 2 3 4 5 D 7 8 9 A B S C
G F E K J I H N O P Q R L M 5 6 0 1 2 3 4 C D 7 8 9 S B A
H G F E K J I O P Q R L M N 4 5 6 0 1 2 3 B C D 7 S 9 A 8
I H G F E K J P Q R L M N O 3 4 5 6 0 1 2 A B C S 7 8 9 D
J I H G F E K Q R L M N O P 2 3 4 5 6 0 1 9 A S C D 7 8 B
K J I H G F E R L M N O P Q 1 2 3 4 5 6 0 8 S A B C D 7 9
L M N O P Q R S K J I H G F 7 D C B A 9 8 0 6 5 4 3 2 1 E
M N O P Q R L F E K J I H S 8 7 D C B A 9 1 0 6 5 4 3 2 G
N O P Q R L M G F E K J S H 9 8 7 D C B A 2 1 0 6 5 4 3 I
O P Q R L M N H G F E S J I A 9 8 7 D C B 3 2 1 0 6 5 4 K
P Q R L M N O I H G S E K J B A 9 8 7 D C 4 3 2 1 0 6 5 F
Q R L M N O P J I S G F E K C B A 9 8 7 D 5 4 3 2 1 0 6 H
R L M N O P Q K S I H G F E D C B A 9 8 7 6 5 4 3 2 1 0 J
0 5 3 1 6 4 2 E J H F K I G L N P R M O Q 7 9 B D 8 A C S
```

Partial square

```
. . 5 . 3 2 1 7 . . A B C D E F . H I . K . M . O P . . .
1 0 6 5 4 3 . D 7 8 9 A B . K . F G H I J M N . . Q R L .
2 1 . 6 5 . . C D . . . A B J K . . G H I N O P Q R L M .
3 2 1 0 . 5 4 B C D 7 8 9 A . J K E F G H O P Q . L M N .
4 . 2 . 0 6 5 A B . D 7 8 9 H I J K E F G P Q R . M . O .
5 4 . 2 1 0 6 9 A . . . . 8 . H I J K E F Q . L M N O P .
6 . 4 3 . 1 0 8 9 . B . D 7 F G H I J . . R L . N O P Q .
7 D . . . 9 8 0 1 2 3 4 5 6 . M N O P Q R E . G H I J K .
8 7 D . B A 9 6 . 1 2 3 4 5 M . . P . R L K . . . H I J .
9 8 7 . C B A 5 6 0 . 2 . 4 N O . Q R L M . K E F G H . .
A 9 8 7 D C . 4 5 6 . 1 2 3 O . Q . L M N . J K E F G H .
B A . 8 7 D C 3 . 5 6 0 1 2 P Q R L . N O . I J K E F . .
C B A 9 8 7 . 2 3 4 5 6 0 . Q R L M N . P G H I J K E F .
D C . A 9 8 . 1 . 3 4 5 6 . R L M N O P . F G . I J K E .
. K . I . G . L M N O P Q R 0 1 2 3 4 5 6 . 8 9 A B C D .
F . . J I H G . N . P Q R . . 0 1 2 3 4 5 D 7 8 9 A B . .
G . E K J I H N . . Q R L M 5 6 0 . 2 3 4 C D 7 8 9 . . .
. G F E K J I O P Q R L . N 4 . 6 . 1 2 . B C D . . 9 A .
. H G . . K J P Q R L M N O 3 4 5 6 . 1 2 A B C . 7 8 9 .
. I H G . E K . R L M N . . 2 3 4 5 6 0 1 9 A . C D 7 8 .
. J I H G F E R L M N . P Q 1 2 3 4 5 6 0 8 . A B C D 7 .
L M N O P Q . . K J I H G F 7 D C . A 9 . . 6 5 4 . 2 1 .
M N . P Q R L F E K J I H . 8 7 D C B A 9 1 0 6 . . 3 . .
N O P Q R . M G F E K J . H 9 8 . D C B A . 1 0 6 5 . 3 .
O . Q R L M . H G F E . J I A 9 8 7 . . B 3 2 1 . . . 4 .
. Q R L M N O I H G . E K J B A . 8 7 D C 4 3 2 1 0 6 5 .
Q R . M N O P J . . G F E K C B . 9 . 7 D 5 4 3 2 1 . 6 .
R L M . . P Q K . I H G F E D C B A . 8 7 6 5 4 . 2 1 0 .
. . . . . . . . . . . . . . . . . . . . . . . . . . . . S
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E022: order 30, 654 clues

Status I. Previously verified construction/example or published record

Source: `outputs/lcs_examples.json:11`

SHA256: `62337455595e54a1eb99d09465e7ac5e8ba5bdaf98752b8878dac5f70a87e106`

Completion

```
0 2 1 4 3 6 5 8 7 A 9 C B E D F H G J I L K N M P O R Q T S
2 1 0 5 6 3 4 9 A 7 8 D E B C H G F K L I J O P M N S T Q R
1 0 2 6 5 4 3 A 9 8 7 E D C B G F H L K J I P O N M T S R Q
4 5 6 3 0 1 2 B C D E 7 8 9 A J K L I F G H Q R S T M N O P
3 6 5 0 4 2 1 C B E D 8 7 A 9 I L K F J H G R Q T S N M P O
6 3 4 1 2 5 0 D E B C 9 A 7 8 L I J G H K F S T Q R O P M N
5 4 3 2 1 0 6 E D C B A 9 8 7 K J I H G F L T S R Q P O N M
8 9 A B C D E 7 0 1 2 3 4 5 6 N O P Q R S T M F G H I J K L
7 A 9 C B E D 0 8 2 1 4 3 6 5 M P O R Q T S F N H G J I L K
A 7 8 D E B C 1 2 9 0 5 6 3 4 P M N S T Q R G H O F K L I J
9 8 7 E D C B 2 1 0 A 6 5 4 3 O N M T S R Q H G F P L K J I
C D E 7 8 9 A 3 4 5 6 B 0 1 2 R S T M N O P I J K L Q F G H
B E D 8 7 A 9 4 3 6 5 0 C 2 1 Q T S N M P O J I L K F R H G
E B C 9 A 7 8 5 6 3 4 1 2 D 0 T Q R O P M N K L I J G H S F
D C B A 9 8 7 6 5 4 3 2 1 0 E S R Q P O N M L K J I H G F T
F H G J I L K N M P O R Q T S 0 2 1 4 3 6 5 8 7 A 9 C B E D
H G F K L I J O P M N S T Q R 2 1 0 5 6 3 4 9 A 7 8 D E B C
G F H L K J I P O N M T S R Q 1 0 2 6 5 4 3 A 9 8 7 E D C B
J K L I F G H Q R S T M N O P 4 5 6 3 0 1 2 B C D E 7 8 9 A
I L K F J H G R Q T S N M P O 3 6 5 0 4 2 1 C B E D 8 7 A 9
L I J G H K F S T Q R O P M N 6 3 4 1 2 5 0 D E B C 9 A 7 8
K J I H G F L T S R Q P O N M 5 4 3 2 1 0 6 E D C B A 9 8 7
N O P Q R S T M F G H I J K L 8 9 A B C D E 7 0 1 2 3 4 5 6
M P O R Q T S F N H G J I L K 7 A 9 C B E D 0 8 2 1 4 3 6 5
P M N S T Q R G H O F K L I J A 7 8 D E B C 1 2 9 0 5 6 3 4
O N M T S R Q H G F P L K J I 9 8 7 E D C B 2 1 0 A 6 5 4 3
R S T M N O P I J K L Q F G H C D E 7 8 9 A 3 4 5 6 B 0 1 2
Q T S N M P O J I L K F R H G B E D 8 7 A 9 4 3 6 5 0 C 2 1
T Q R O P M N K L I J G H S F E B C 9 A 7 8 5 6 3 4 1 2 D 0
S R Q P O N M L K J I H G F T D C B A 9 8 7 6 5 4 3 2 1 0 E
```

Partial square

```
. 2 1 4 3 6 5 8 7 A 9 C . E D F H G J I L K N M P O . Q T S
2 . 0 5 6 3 4 9 A 7 8 D E B C H G F . . . . O P M N . . . .
1 0 . 6 5 4 3 A 9 8 7 E D C B G F H . K . I P O N M . S . Q
4 5 6 . 0 1 2 B C D E 7 8 9 A J . . I F . . . . S T . . O P
3 6 5 0 . 2 1 C B E D 8 7 A 9 I . K F J . G . Q T S . M P O
6 3 4 1 2 . 0 D E B C 9 A 7 8 L . . . . K F S T . . . . M N
5 4 3 2 1 0 . E D C B A 9 8 7 K . I . G F L T S . Q . O N M
8 9 A B C D E 7 0 1 2 3 4 5 6 N O P . . S T M F G H . . K L
7 A 9 C B E D 0 . 2 1 4 3 6 5 M P O . Q T S F N H G . I L K
A 7 8 D E B C 1 2 9 0 5 6 3 4 P M N S T . . G H O F . . I J
9 8 7 E D C B 2 1 0 . 6 5 4 3 O N M T S . Q H G F P . K J I
C D E 7 8 9 A 3 4 5 6 B 0 1 2 . . . . . .
. E D 8 7 A 9 4 3 6 5 0 C 2 1 Q . S . M . O . I . K . . . G
E B C 9 A 7 8 5 6 3 4 1 2 D 0 T . . O P M N K L I J . . S F
D C B A 9 8 7 6 5 4 3 2 1 0 . S . Q P O N M L K J I . G F T
F H G J I L K N M P O . Q T S 0 2 1 4 3 6 5 8 7 A 9 . B E D
H G F . . . . O P M N . . . . 2 1 0 . . . . 9 A 7 8 . . . .
G F H . K . I P O N M . S . Q 1 0 2 . 5 . 3 A 9 8 7 . D . B
J . . I F . . . . S T . . O P 4 . . 3 0 . . . . D E . . 9 A
I . K F J . G . Q T S . M P O 3 . 5 0 4 . 1 . B E D . 7 A 9
L . . . . K F S T . . . . M N 6 . . . . 5 0 D E . . . . 7 8
K . I . G F L T S . Q . O N M 5 . 3 . 1 0 6 E D . B . 9 8 7
N O P . . S T M F G H . . K L 8 9 A . . D E 7 0 1 2 . . 5 6
M P O . Q T S F N H G . I L K 7 A 9 . B E D 0 8 2 1 . 3 6 5
P M N S T . . G H O F . . I J A 7 8 D E . . 1 2 9 0 . . 3 4
O N M T S . Q H G F P . K J I 9 8 7 E D . B 2 1 0 A . 5 4 3
. . . . . .
Q . S . M . O . I . K . . . G B . D . 7 . 9 . 3 . 5 . . . 1
T . . O P M N K L I J . . S F E . . 9 A 7 8 5 6 3 4 . . D 0
S . Q P O N M L K J I . G F T D . B A 9 8 7 6 5 4 3 . 1 0 E
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E023: order 31, 721 clues

Status T. Explicit theorem construction; proof supplied in this volume.

Source: `outputs/lcs_ffl_plus_one/order31_certificate.json`

SHA256: `71d21ac12f50291b48dbb7e8c281a089c6df4de4908484077170db7d5a9c062e`

Completion

```
U 0 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T
0 T 2 1 4 3 6 5 8 7 A 9 C B E D G F I H K J M L O N Q P S R U
1 2 S 0 5 6 3 4 9 A 7 8 D E B C H I F G L M J K P Q N O T U R
2 1 0 R 6 5 4 3 A 9 8 7 E D C B I H G F M L K J Q P O N U T S
3 4 5 6 Q 0 1 2 B C D E 7 8 9 A J K L M F G H I R S T U N O P
4 3 6 5 0 P 2 1 C B E D 8 7 A 9 K J M L G F I H S R U T O N Q
5 6 3 4 1 2 O 0 D E B C 9 A 7 8 L M J K H I F G T U R S P Q N
6 5 4 3 2 1 0 N E D C B A 9 8 7 M L K J I H G F U T S R Q P O
7 8 9 A B C D E M 0 1 2 3 4 5 6 N O P Q R S T U F G H I J K L
8 7 A 9 C B E D 0 L 2 1 4 3 6 5 O N Q P S R U T G F I H K J M
9 A 7 8 D E B C 1 2 K 0 5 6 3 4 P Q N O T U R S H I F G L M J
A 9 8 7 E D C B 2 1 0 J 6 5 4 3 Q P O N U T S R I H G F M L K
B C D E 7 8 9 A 3 4 5 6 I 0 1 2 R S T U N O P Q J K L M F G H
C B E D 8 7 A 9 4 3 6 5 0 H 2 1 S R U T O N Q P K J M L G F I
D E B C 9 A 7 8 5 6 3 4 1 2 G 0 T U R S P Q N O L M J K H I F
E D C B A 9 8 7 6 5 4 3 2 1 0 F U T S R Q P O N M L K J I H G
F G H I J K L M N O P Q R S T U E 0 1 2 3 4 5 6 7 8 9 A B C D
G F I H K J M L O N Q P S R U T 0 D 2 1 4 3 6 5 8 7 A 9 C B E
H I F G L M J K P Q N O T U R S 1 2 C 0 5 6 3 4 9 A 7 8 D E B
I H G F M L K J Q P O N U T S R 2 1 0 B 6 5 4 3 A 9 8 7 E D C
J K L M F G H I R S T U N O P Q 3 4 5 6 A 0 1 2 B C D E 7 8 9
K J M L G F I H S R U T O N Q P 4 3 6 5 0 9 2 1 C B E D 8 7 A
L M J K H I F G T U R S P Q N O 5 6 3 4 1 2 8 0 D E B C 9 A 7
M L K J I H G F U T S R Q P O N 6 5 4 3 2 1 0 7 E D C B A 9 8
N O P Q R S T U F G H I J K L M 7 8 9 A B C D E 6 0 1 2 3 4 5
O N Q P S R U T G F I H K J M L 8 7 A 9 C B E D 0 5 2 1 4 3 6
P Q N O T U R S H I F G L M J K 9 A 7 8 D E B C 1 2 4 0 5 6 3
Q P O N U T S R I H G F M L K J A 9 8 7 E D C B 2 1 0 3 6 5 4
R S T U N O P Q J K L M F G H I B C D E 7 8 9 A 3 4 5 6 2 0 1
S R U T O N Q P K J M L G F I H C B E D 8 7 A 9 4 3 6 5 0 1 2
T U R S P Q N O L M J K H I F G D E B C 9 A 7 8 5 6 3 4 1 2 0
```

Partial square

```
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
. T 2 1 . 3 6 5 . 7 A 9 . B E D . F I H . J M L . N Q P . R U
. 2 . . . 6 3 . . A 7 . . E B . . I F . . M J . . Q N . . U R
. 1 . R . 5 . 3 . 9 . 7 . D . B . H . F . L . J . P . N . T .
. . . . Q 0 1 2 . . . . 7 8 9 A . . . . F G H I . . . . N O P
. 3 6 5 0 P 2 1 . B E D 8 7 A 9 . J M L G F I H . R U T O N Q
. 6 3 . 1 2 O 0 . E B . 9 A 7 8 . M J . H I F G . U R . P Q N
. 5 . 3 2 1 0 N . D . B A 9 8 7 . L . J I H G F . T . R Q P O
. . . . . . . . . M 0 1 2 3 4 5 6 . . . . . . . . F G H I J K L
. 7 A 9 . B E D 0 L 2 1 4 3 6 5 . N Q P . R U T G F I H K J M
. A 7 . . E B . 1 2 K 0 5 6 3 4 . Q N . . U R . H I F G L M J
. 9 . 7 . D . B 2 1 0 J 6 5 4 3 . P . N . T . R I H G F M L K
. . . . 7 8 9 A 3 4 5 6 I 0 1 2 . . . . N O P Q J K L M F G H
. B E D 8 7 A 9 4 3 6 5 0 H 2 1 . R U T O N Q P K J M L G F I
. E B . 9 A 7 8 5 6 3 4 1 2 G 0 . U R . P Q N O L M J K H I F
. D . B A 9 8 7 6 5 4 3 2 1 0 F . T . R Q P O N M L K J I H G
. . . . . . . . . . . . . . . . . E 0 1 2 3 4 5 6 7 8 9 A B C D
. F I H . J M L . N Q P . R U T 0 D 2 1 4 3 6 5 8 7 A 9 C B E
. I F . . M J . . Q N . . U R . 1 2 C 0 5 6 3 4 9 A 7 8 D E B
. H . F . L . J . P . N . T . R 2 1 0 B 6 5 4 3 A 9 8 7 E D C
. . . . F G H I . . . . N O P Q 3 4 5 6 A 0 1 2 B C D E 7 8 9
. J M L G F I H . R U T O N Q P 4 3 6 5 0 9 2 1 C B E D 8 7 A
. M J . H I F G . U R . P Q N O 5 6 3 4 1 2 8 0 D E B C 9 A 7
. L . J I H G F . T . R Q P O N 6 5 4 3 2 1 0 7 E D C B A 9 8
. . . . . . . . . F G H I J K L M 7 8 9 A B C D E 6 0 1 2 3 4 5
. N Q P . R U T G F I H K J M L 8 7 A 9 C B E D 0 5 2 1 4 3 6
. Q N . . U R . H I F G L M J K 9 A 7 8 D E B C 1 2 4 0 5 6 3
. P . N . T . R I H G F M L K J A 9 8 7 E D C B 2 1 0 3 6 5 4
. . . . N O P Q J K L M F G H I B C D E 7 8 9 A 3 4 5 6 2 0 1
. R U T O N Q P K J M L G F I H C B E D 8 7 A 9 4 3 6 5 0 1 2
. U R . P Q N O L M J K H I F G D E B C 9 A 7 8 5 6 3 4 1 2 0
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E024: order 32, 781 clues

Status T. Explicit theorem construction; proof supplied in this volume.

Source: `generated:XOR(32),clueiffrowANDcolumnisnonzero`

SHA256: `8b1c6db24f5569d71e100c95b68952e1d8f656db681e3b136d59d747c81ac1e2`

Completion

```
0 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V
1 0 3 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U
2 3 0 1 6 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T
3 2 1 0 7 6 5 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S
4 5 6 7 0 1 2 3 C D E F 8 9 A B K L M N G H I J S T U V O P Q R
5 4 7 6 1 0 3 2 D C F E 9 8 B A L K N M H G J I T S V U P O R Q
6 7 4 5 2 3 0 1 E F C D A B 8 9 M N K L I J G H U V S T Q R O P
7 6 5 4 3 2 1 0 F E D C B A 9 8 N M L K J I H G V U T S R Q P O
8 9 A B C D E F 0 1 2 3 4 5 6 7 O P Q R S T U V G H I J K L M N
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O R Q T S V U H G J I L K N M
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P U V S T I J G H M N K L
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U T S J I H G N M L K
C D E F 8 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R K L M N G H I J
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K N M H G J I
E F C D A B 8 9 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L I J G H
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I H G
G H I J K L M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 9 A B C D E F
H G J I L K N M P O R Q T S V U 1 0 3 2 5 4 7 6 9 8 B A D C F E
I J G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D
J I H G N M L K R Q P O V U T S 3 2 1 0 7 6 5 4 B A 9 8 F E D C
K L M N G H I J S T U V O P Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B
L K N M H G J I T S V U P O R Q 5 4 7 6 1 0 3 2 D C F E 9 8 B A
M N K L I J G H U V S T Q R O P 6 7 4 5 2 3 0 1 E F C D A B 8 9
N M L K J I H G V U T S R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8
O P Q R S T U V G H I J K L M N 8 9 A B C D E F 0 1 2 3 4 5 6 7
P O R Q T S V U H G J I L K N M 9 8 B A D C F E 1 0 3 2 5 4 7 6
Q R O P U V S T I J G H M N K L A B 8 9 E F C D 2 3 0 1 6 7 4 5
R Q P O V U T S J I H G N M L K B A 9 8 F E D C 3 2 1 0 7 6 5 4
S T U V O P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 1 2 3
T S V U P O R Q L K N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 2
U V S T Q R O P M N K L I J G H E F C D A B 8 9 6 7 4 5 2 3 0 1
V U T S R Q P O N M L K J I H G F E D C B A 9 8 7 6 5 4 3 2 1 0
```

Partial square

```
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
. 0 . 2 . 4 . 6 . 8 . A . C . E . G . I . K . M . O . Q . S . U
. . 0 1 . . 4 5 . . 8 9 . . C D . . G H . . K L . . O P . . S T
. 2 1 0 . 6 5 4 . A 9 8 . E D C . I H G . M L K . Q P O . U T S
. . . . 0 1 2 3 . . . . 8 9 A B . . . . G H I J . . . . O P Q R
. 4 . 6 1 0 3 2 . C . E 9 8 B A . K . M H G J I . S . U P O R Q
. . 4 5 2 3 0 1 . . C D A B 8 9 . . K L I J G H . . S T Q R O P
. 6 5 4 3 2 1 0 . E D C B A 9 8 . M L K J I H G . U T S R Q P O
. . . . . . . . 0 1 2 3 4 5 6 7 . . . . . . . . G H I J K L M N
. 8 . A . C . E 1 0 3 2 5 4 7 6 . O . Q . S . U H G J I L K N M
. . 8 9 . . C D 2 3 0 1 6 7 4 5 . . O P . . S T I J G H M N K L
. A 9 8 . E D C 3 2 1 0 7 6 5 4 . Q P O . U T S J I H G N M L K
. . . . 8 9 A B 4 5 6 7 0 1 2 3 . . . . O P Q R K L M N G H I J
. C . E 9 8 B A 5 4 7 6 1 0 3 2 . S . U P O R Q L K N M H G J I
. . C D A B 8 9 6 7 4 5 2 3 0 1 . . S T Q R O P M N K L I J G H
. E D C B A 9 8 7 6 5 4 3 2 1 0 . U T S R Q P O N M L K J I H G
. . . . . . . . . . . . . . . . . 0 1 2 3 4 5 6 7 8 9 A B C D E F
. G . I . K . M . O . Q . S . U 1 0 3 2 5 4 7 6 9 8 B A D C F E
. . G H . . K L . . O P . . S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D
. I H G . M L K . Q P O . U T S 3 2 1 0 7 6 5 4 B A 9 8 F E D C
. . . . G H I J . . . . O P Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B
. K . M H G J I . S . U P O R Q 5 4 7 6 1 0 3 2 D C F E 9 8 B A
. . K L I J G H . . S T Q R O P 6 7 4 5 2 3 0 1 E F C D A B 8 9
. M L K J I H G . U T S R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8
. . . . . . . . G H I J K L M N 8 9 A B C D E F 0 1 2 3 4 5 6 7
. O . Q . S . U H G J I L K N M 9 8 B A D C F E 1 0 3 2 5 4 7 6
. . O P . . S T I J G H M N K L A B 8 9 E F C D 2 3 0 1 6 7 4 5
. Q P O . U T S J I H G N M L K B A 9 8 F E D C 3 2 1 0 7 6 5 4
. . . . O P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 1 2 3
. S . U P O R Q L K N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 2
. . S T Q R O P M N K L I J G H E F C D A B 8 9 6 7 4 5 2 3 0 1
. U T S R Q P O N M L K J I H G F E D C B A 9 8 7 6 5 4 3 2 1 0
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E025: order 33, 844 clues

Status I. Independently verified: two solver encodings and all 844 deletion witnesses replayed; uniqueness solver-dependent

Source: `outputs/lcs33_844_bunya_29909858/independent_audit/record.json`

SHA256: `2386ce60543d3c95cd2e68dd4c534d83b9990c226de0fa4c047007b870139c06`

Completion

```
W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
2 3 0 1 W 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 6
3 2 1 0 7 6 W 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 5
4 5 6 7 0 1 2 3 W D E F 8 9 A B K L M N G H I J S T U V O P Q R C
5 4 7 6 1 0 3 2 D C W E 9 8 B A L K N M H G J I T S V U P O R Q F
6 7 4 5 2 3 0 1 E F C D W B 8 9 M N K L I J G H U V S T Q R O P A
7 6 5 4 3 2 1 0 F E D C B A W 8 N M L K J I H G V U T S R Q P O 9
8 9 A B C D E F 0 1 2 3 4 5 6 7 W P Q R S T U V G H I J K L M N O
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O W Q T S V U H G J I L K N M R
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P W V S T I J G H M N K L U
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U W S J I H G N M L K T
C D E F 8 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R W L M N G H I J K
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K W M H G J I N
E F C D A B 8 9 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L W J G H I
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I W G H
G H I J K W M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 9 A B C D E F L
H G J I L K N W P O R Q T S V U 1 0 3 2 5 4 7 6 9 8 B A D C F E M
I W G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D J
J I H W N M L K R Q P O V U T S 3 2 1 0 7 6 5 4 B A 9 8 F E D C G
K L M N G H I J S T U V O W Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B P
L K N M H G J I T S V U P O R W 5 4 7 6 1 0 3 2 D C F E 9 8 B A Q
M N K L I J G H U W S T Q R O P 6 7 4 5 2 3 0 1 E F C D A B 8 9 V
N M L K J I H G V U T W R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 S
O P Q R S T U V G H I J K L M N 8 9 A B C W E F 0 1 2 3 4 5 6 7 D
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
Q R O P U V S T I J G H M N K L A W 8 9 E F C D 2 3 0 1 6 7 4 5 B
R Q P O V U T S J I H G N M L K B A 9 W F E D C 3 2 1 0 7 6 5 4 8
S T U V O P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 W 2 3 1
T S V U P O R Q L K N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 W 2
U V S T Q R O P M N K L I J G H E F C D A B 8 9 6 W 4 5 2 3 0 1 7
V U T S R Q P O N M L K J I H G F E D C B A 9 8 7 6 5 W 3 2 1 0 4
0 J 3 G 6 L 5 M C V F S A P 9 Q O B R 8 U D T E K 7 N 4 I 1 H 2 W
```

Partial square

```
. . . 3 . 5 6 . . . . . C D . F . . . . K L M N . . . . S T U V 0
. . W . 5 4 7 6 . 8 B A . C . E . G J I L K N M . O . Q T . V U 3
2 3 0 1 . . 4 5 A B . . . F . D I . G H M N . . . R O . U V S T .
3 2 1 0 . 6 W 4 B A . 8 F E . C J I H G N M L K R Q . . . U T S 5
4 5 6 7 0 1 2 3 W D E . 8 . . B . L M N G H I J S T U V O P Q R .
. 4 7 6 . . 3 2 D C . E . 8 . A L K . M H G J I . . V U P . R Q F
6 7 4 5 2 3 0 1 E . C D . . . . M N K L I . G H U V S T . R O P A
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
8 . A B C D E F 0 1 2 3 4 5 . 7 W P Q R S T U V G H I J K L M . O
. . B A D C F E 1 0 3 2 5 4 . 6 P O W Q . . V U . G J I L K N M R
A B 8 . E F C D 2 . 0 1 6 7 . 5 Q R O . W V S T I J G H M N K L U
B A . 8 . E D C 3 2 . 0 7 . . . R Q P O V . W S J I H G N M L . T
C D E F 8 . A B . 5 . 7 0 1 . 3 S . U V . P Q R W L M N G H I J K
. C F E . 8 B A 5 4 7 6 1 0 . 2 T S V . P O R Q L K . M H . J I N
E F . . A B 8 . 6 7 4 5 2 3 . 1 U V S T . R O P M N K . W J G . I
F E D . B A . . 7 . 5 4 3 2 . 0 V U T . . Q P O N M L K J I . G H
G H I J K W M N O P Q R S T . V 0 1 2 3 4 5 6 7 8 . . B C D E F .
H G . I L . N W P O R Q T . . U 1 0 3 2 5 . 7 6 . 8 B A D C F E M
I . G H M N K L . R O P U V . T . 3 0 1 6 7 4 5 A B 8 . E F C D J
J I H W N M L K R Q P . V U . S 3 2 1 0 7 6 5 4 B A . 8 F E . C G
K L M N G H I J S . U V O W . R 4 5 6 7 0 1 2 3 C D E F 8 . . B P
L K N M H G J I T S V . P O . . 5 4 7 6 1 0 . 2 D C F E . 8 B A Q
M N K L I J G H U W S T Q . . P 6 7 4 5 2 3 0 1 E F . D A B 8 . V
N M L . . I H G V U T W . Q . O 7 6 5 4 3 2 1 0 F E D C B A . 8 .
O P Q R S T U V G H I J K L . . 8 . A B C W E F 0 1 2 3 4 . 6 7 .
P O . . T S V U H G . . L K . M . 8 B A D C . W 1 0 . 2 5 4 7 6 E
Q R O P U V S T I J G H M N . L A W 8 . E F C . 2 3 0 1 6 7 4 5 B
R . P O V U T S J I H G N M . . . A . W . E D C . . 1 0 7 6 . 4 8
S T U V O P Q R K L M N . H . J . . E . . . A B 4 5 6 . 0 W 2 . .
T S V U P O R Q L K N M H . . I D C F E . 8 B A 5 4 7 6 1 0 3 W 2
U V . T Q R O . . N K L I J . H E F C D A B 8 . 6 W . 5 . 3 0 1 7
V U T S R Q . O N . L K J I . G F E D C . A . . 7 6 5 W 3 . 1 0 .
0 . 3 G 6 L 5 M C V F S A P . . O B . 8 . D T E K 7 N 4 I 1 H 2 W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

23 Additional archived examples and unresolved instances

These pages retain distinct saved matrix pairs, even when a stronger example exists at the same order. Producer-only and unresolved outputs are explicitly marked; they do not inherit the verification of another partial square on the same completion.

E026: order 9, 46 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_dyadic_plus_sharp_progress/seed_order9_normalized.json`

SHA256: `01f1167e565e7a470d97c69a90182ea6a0979221c8065da8228dfe691b0dcdb0`

Completion

```
8 1 2 3 4 5 6 7 0
1 0 3 2 5 4 8 6 7
2 3 0 1 6 7 4 8 5
3 8 1 0 7 6 5 4 2
4 5 6 7 0 8 2 3 1
5 4 7 8 1 0 3 2 6
6 7 8 5 2 3 0 1 4
7 6 5 4 8 2 1 0 3
0 2 4 6 3 1 7 5 8
```

Partial square

```
. . . . . . . . .
. . 3 2 5 4 8 6 .
. 3 . 1 6 7 4 8 .
. . 1 . 7 . 5 4 2
. 5 6 7 . 8 2 . 1
. 4 7 8 1 . . 2 6
. 7 8 5 . 3 . 1 .
. 6 . 4 8 2 1 . 3
. 2 4 . 3 . 7 5 8
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E027: order 9, 46 clues

Status I. Previously independently verified example; see the archived certificate directory.

Source: `outputs/lcs_examples.json:0`

SHA256: `ab33cc0df63bafddab96538c646c3ad0c0541ed2ad60ac128d36441e29e6857a`

Completion

```
8 1 2 3 4 5 7 6 0
1 0 3 2 5 4 8 7 6
2 3 0 1 7 6 4 8 5
3 8 1 0 6 7 5 4 2
4 5 7 6 0 8 2 3 1
5 4 6 8 1 0 3 2 7
7 6 8 5 2 3 0 1 4
6 7 5 4 8 2 1 0 3
0 2 4 7 3 1 6 5 8
```

Partial square

```
8 1 2 . . 5 . 6 0
. 0 . 2 . 4 8 7 .
2 . . 1 . 6 4 8 5
. 8 1 . . 7 5 4 2
4 . 7 6 . . 2 . 1
5 4 6 8 . 0 . 2 .
7 6 8 5 . . 0 1 .
. . . . . . . . .
0 . 4 7 . 1 6 . 8
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E028: order 9, 36 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m4_seed0.json`

SHA256: `63b9e9f331f314eb7eede038ad48a69cabd7d82cfb8352342f6930818f842421`

Completion

```
8 3 2 1 4 5 6 7 0
1 0 8 2 7 4 5 6 3
2 1 0 3 8 7 4 5 6
3 2 1 0 5 6 8 4 7
4 7 6 5 0 8 2 3 1
5 4 7 6 3 0 1 8 2
6 8 4 7 2 3 0 1 5
7 6 5 8 1 2 3 0 4
0 5 3 4 6 1 7 2 8
```

Partial square

```
. 3 2 1 . 5 6 7 .
. . 8 2 7 4 5 6 .
. . . 3 8 7 4 5 6
. . . . 5 6 8 4 .
. 7 6 5 . 8 2 3 .
. . 7 6 . . 1 8 .
. 8 . . . . . 1 .
. . . . . . . . 4
0 . . . . . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E029: order 11, 55 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_four_m_plus_three/near_examples/n11_seed2.json`

SHA256: `cc343f551c41a651db3436f94ade3d5098bf71818c2719968c75eff15e797843`

Completion

```
0 4 3 2 1 5 6 7 8 9 A
1 0 A 3 2 9 5 6 7 8 4
2 1 0 4 A 8 9 5 6 7 3
3 2 1 0 4 7 A 9 5 6 8
4 3 2 1 0 6 7 8 A 5 9
5 9 8 7 6 A 1 2 3 4 0
6 5 9 8 7 4 0 A 2 3 1
7 6 5 9 8 3 4 0 1 A 2
8 A 6 5 9 2 3 4 0 1 7
9 8 7 A 5 1 2 3 4 0 6
A 7 4 6 3 0 8 1 9 2 5
```

Partial square

```
. 4 3 2 1 . 6 7 8 9 A
. . . 3 2 9 5 6 . 8 .
. . . 4 . 8 9 5 6 7 .
. . . . 4 7 A 9 5 . 8
. . . . . 6 7 8 A 5 9
. . 8 7 . . . 2 . 4 .
. . 9 8 7 . . . 2 3 1
. . . . 8 . . . 1 . .
. A . . 9 . . . . 1 .
. . . . . . . . . . 6
A . 4 . 3 . . 1 . 2 5
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E030: order 11, 54 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m5_seed6.json`

SHA256: `3194ccf71dc6ffc2519cdb1e779b328203a96989a83fb22b65afb158817344b3`

Completion

```
5 4 3 2 1 0 6 A 8 9 7
1 0 4 A 2 9 5 6 7 8 3
2 1 0 4 3 8 9 5 A 7 6
3 2 A 0 4 7 8 9 5 6 1
A 3 2 1 0 6 7 8 9 5 4
0 9 8 7 6 A 1 2 3 4 5
6 5 9 8 7 4 A 1 2 3 0
7 6 5 9 8 3 4 0 1 A 2
8 7 6 5 A 2 3 4 0 1 9
9 A 7 6 5 1 2 3 4 0 8
4 8 1 3 9 5 0 7 6 2 A
```

Partial square

```
5 4 3 . 1 . 6 A . 9 .
. . 4 A 2 9 . 6 7 8 3
. . . . 3 8 9 5 A 7 6
. . A . 4 7 8 9 5 6 .
. . . . . 6 7 8 9 5 .
0 9 . . 6 A . . . . .
. . 9 8 7 . A 1 2 3 .
. . . 9 . . . . 1 A .
. . . . A . . . . 1 .
. . . . . . . . . . 8
4 . . 3 . . 0 . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E031: order 11, 54 clues

Status I. Previously independently checked uniqueness and all clue-deletion witnesses; uniqueness is solver-dependent.

Source: `outputs/lcs_odd_prolongation/examples/m5_seed4.json`

SHA256: `31f3a3c58dc2149de16ac78d83ed9f77b4d563a5dfa698d03e29154de68d1c49`

Completion

```
5 4 3 2 1 0 6 A 8 9 7
1 0 4 A 2 9 5 6 7 8 3
2 1 0 4 3 8 9 5 A 7 6
3 2 A 0 4 7 8 9 5 6 1
A 3 2 1 0 6 7 8 9 5 4
0 9 8 7 6 A 1 2 3 4 5
6 5 9 8 7 4 A 1 2 3 0
7 6 5 9 8 3 4 0 1 A 2
8 7 6 5 A 2 3 4 0 1 9
9 A 7 6 5 1 2 3 4 0 8
4 8 1 3 9 5 0 7 6 2 A
```

Partial square

```
5 4 3 . . . 6 A . . .
. . 4 . 2 9 . 6 7 . 3
. . . . 3 8 9 5 . 7 .
. . A . . 7 8 9 5 6 1
A . . . . 6 7 8 . 5 .
. . 8 7 . . 1 2 3 . .
. . 9 8 7 . A 1 2 3 0
. . . 9 . . . . 1 . 2
. . . . . . . . . 1 9
. A . . . . . . . . .
4 8 . . . 5 0 7 6 2 .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E032: order 11, 54 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m5_seed5.json`

SHA256: `a3ac53a3111fd4bc54610aaf8149d552d425cdf241988354108ac5ad87ab8b7b`

Completion

```
5 4 3 2 1 0 6 A 8 9 7
1 0 4 A 2 9 5 6 7 8 3
2 1 0 4 3 8 9 5 A 7 6
3 2 A 0 4 7 8 9 5 6 1
A 3 2 1 0 6 7 8 9 5 4
0 9 8 7 6 A 1 2 3 4 5
6 5 9 8 7 4 A 1 2 3 0
7 6 5 9 8 3 4 0 1 A 2
8 7 6 5 A 2 3 4 0 1 9
9 A 7 6 5 1 2 3 4 0 8
4 8 1 3 9 5 0 7 6 2 A
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

Partial square

```
5 4 . 2 1 0 6 . . . 7
. . 4 A 2 9 . 6 7 . .
. . . 4 3 8 9 5 . 7 6
. . A . . 7 8 9 5 6 1
. . . . . 6 7 8 . 5 .
0 . 8 . . A . . 3 4 .
. . 9 8 7 . . 1 2 3 .
. . . 9 . . . . 1 . 2
. . . . A . . . . 1 9
. A . . . . . . . . .
4 . . . 9 5 0 7 . . .
```

E033: order 11, 53 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m5_seed2.json`

SHA256: `01bc7fd16b17ef58df436ff9bbef7fa3f8c59095a8c29dbdef8ccdbca4aa9110`

Completion

```
5 4 3 2 1 0 6 A 8 9 7
1 0 4 A 2 9 5 6 7 8 3
2 1 0 4 3 8 9 5 A 7 6
3 2 A 0 4 7 8 9 5 6 1
A 3 2 1 0 6 7 8 9 5 4
0 9 8 7 6 A 1 2 3 4 5
6 5 9 8 7 4 A 1 2 3 0
7 6 5 9 8 3 4 0 1 A 2
8 7 6 5 A 2 3 4 0 1 9
9 A 7 6 5 1 2 3 4 0 8
4 8 1 3 9 5 0 7 6 2 A
```

Partial square

```
. 4 3 2 1 . . . 8 . 7
. . 4 A 2 9 5 6 7 . .
. . . 4 3 8 9 5 . 7 6
. . A . . 7 8 9 5 6 1
. . . . . 6 7 8 . 5 .
. 9 . . 6 A . 2 . 4 .
. . 9 8 7 . . 1 2 . 0
. . . 9 . . . . 1 A .
. . . . A . . . . 1 .
. . . . . . . . . . 8
4 . 1 3 . . . . 6 2 .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E034: order 11, 53 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_four_m_plus_three/examples/xor3_a1_seed0.json`

SHA256: `3b674bad1cdb55f6e2b97445d91f1470d4529e30c5574f2c36e9114b57dba2ba`

Completion

```
8 9 A 3 4 5 6 7 0 1 2
1 0 3 2 A 4 8 9 7 6 5
2 3 0 1 6 A 9 8 5 4 7
9 8 1 A 7 6 5 4 2 3 0
4 5 6 7 9 8 2 A 1 0 3
5 A 9 8 1 0 3 2 6 7 4
A 7 8 9 2 3 0 1 4 5 6
7 6 5 4 8 9 A 0 3 2 1
0 2 4 6 3 1 7 5 8 A 9
3 1 7 5 0 2 4 6 A 9 8
6 4 2 0 5 7 1 3 9 8 A
```

Partial square

```
8 9 . . . . . . . . 2
. . . 2 . 4 . . 7 . 5
. . 0 . . A 9 . . 4 .
. . 1 A . 6 . 4 . 3 0
. . . . 9 . . A . 0 3
. A . 8 . . 3 . . . 4
A . 8 9 2 . 0 1 . . .
. 6 5 4 . 9 A 0 . 2 .
. 2 4 . 3 . . 5 8 A .
. 1 7 5 . . . 6 . . .
6 . . . . 7 1 3 9 . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E035: order 11, 53 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_four_m_plus_three/examples/xor3_a2_seed0.json`

SHA256: `6aa6bd97b761eb5b0f8cbccc27426638e22d885cda0e887a5f22826f5b0f2724`

Completion

```
8 9 A 3 4 5 6 7 0 1 2
1 0 3 2 5 A 9 8 6 7 4
2 A 9 8 6 7 4 5 1 0 3
3 2 1 0 8 9 A 4 7 6 5
4 5 6 7 A 1 8 9 2 3 0
9 8 7 A 1 0 3 2 4 5 6
6 7 4 5 9 8 0 A 3 2 1
A 6 8 9 3 2 1 0 5 4 7
0 4 5 1 7 3 2 6 8 A 9
5 1 0 4 2 6 7 3 A 9 8
7 3 2 6 0 4 5 1 9 8 A
```

Partial square

```
8 9 . . . . . . . . 2
. . . 2 . . 9 . . 7 .
. A . . . . 4 5 . 0 3
. . 1 . 8 9 A . . . 5
. . . . A 1 . . 2 3 0
. 8 . . 1 . . 2 4 . .
. . 4 5 9 8 0 . 3 . .
. . . 9 3 . 1 0 . 4 7
0 . 5 1 7 . . . 8 . 9
5 . 0 . 2 6 . 3 . . .
7 3 . . . . 5 . 9 . A
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E036: order 11, 53 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m5_seed1.json`

SHA256: `6cf0937407e6c2a34c951197f9572cf85752750f449f86ca8278e7f29b9d966a`

Completion

```
5 4 3 2 1 0 6 A 8 9 7
1 0 4 A 2 9 5 6 7 8 3
2 1 0 4 3 8 9 5 A 7 6
3 2 A 0 4 7 8 9 5 6 1
A 3 2 1 0 6 7 8 9 5 4
0 9 8 7 6 A 1 2 3 4 5
6 5 9 8 7 4 A 1 2 3 0
7 6 5 9 8 3 4 0 1 A 2
8 7 6 5 A 2 3 4 0 1 9
9 A 7 6 5 1 2 3 4 0 8
4 8 1 3 9 5 0 7 6 2 A
```

Partial square

```
5 . . 2 . . 6 . 8 . 7
. . . . 2 9 5 6 7 8 .
. . . . 3 8 9 5 . 7 6
. . A . 4 7 8 9 5 6 1
A . . . . 6 7 . . 5 4
0 9 . . . A . 2 . . .
. . 9 8 7 . . 1 2 3 .
. . . 9 8 . . . 1 . 2
. . . . . . . . . 1 .
. A . . . . . . . . 8
4 8 . 3 9 5 0 . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E037: order 11, 53 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_four_m_plus_three/near_examples/n11_seed0.json`

SHA256: `9ff92e1e8648df310809af365f730b452ca82f60dcb44b75b2404d2036620ba3`

Completion

```
0 4 3 2 1 5 6 7 8 9 A
1 0 A 3 2 9 5 6 7 8 4
2 1 0 4 A 8 9 5 6 7 3
3 2 1 0 4 7 A 9 5 6 8
4 3 2 1 0 6 7 8 A 5 9
5 9 8 7 6 A 1 2 3 4 0
6 5 9 8 7 4 0 A 2 3 1
7 6 5 9 8 3 4 0 1 A 2
8 A 6 5 9 2 3 4 0 1 7
9 8 7 A 5 1 2 3 4 0 6
A 7 4 6 3 0 8 1 9 2 5
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

Partial square

```
0 . 3 2 1 5 6 . 8 . .
. . . 3 2 9 5 6 7 8 4
. . . 4 . 8 9 5 6 7 .
. . . . 4 7 A 9 5 . .
. . . . . 6 7 8 A 5 .
5 . 8 . . A . . . 4 0
. . 9 8 7 . . . 2 3 .
. . . . . . . . 1 A 2
. A . . 9 . . . . 1 7
. . . . . . . . . . .
. . 4 6 3 . . 1 . 2 .
```

E038: order 11, 53 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_four_m_plus_three/examples/xor3_a1_seed1.json`

SHA256: `cedc9ba48c7b58fb5f5f9356e4b3ca63c2e8f0363031eb2ce111e7f10e5d3249`

Completion

```
8 9 A 3 4 5 6 7 0 1 2
1 0 3 2 A 4 8 9 7 6 5
2 3 0 1 6 A 9 8 5 4 7
9 8 1 A 7 6 5 4 2 3 0
4 5 6 7 9 8 2 A 1 0 3
5 A 9 8 1 0 3 2 6 7 4
A 7 8 9 2 3 0 1 4 5 6
7 6 5 4 8 9 A 0 3 2 1
0 2 4 6 3 1 7 5 8 A 9
3 1 7 5 0 2 4 6 A 9 8
6 4 2 0 5 7 1 3 9 8 A
```

Partial square

```
8 9 A . . . . . 0 . .
. 0 . 2 A . . 9 . 6 5
. . . . . A 9 . . . 7
. . . A . . 5 4 . . 0
. . . . . 8 . . 1 . .
. A 9 . . . . 2 . 7 4
. . . . 2 3 0 1 4 . .
. 6 . 4 8 . A . 3 . 1
. . 4 6 . 1 . 5 8 . 9
3 . 7 . 0 . . . . 9 8
6 4 . 0 5 7 . . . 8 A
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E039: order 11, 53 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m5_seed3.json`

SHA256: `d81f1729f514d2301704e3a244322d6078d35d198d3acee9e116943e503bb817`

Completion

```
5 4 3 2 1 0 6 A 8 9 7
1 0 4 A 2 9 5 6 7 8 3
2 1 0 4 3 8 9 5 A 7 6
3 2 A 0 4 7 8 9 5 6 1
A 3 2 1 0 6 7 8 9 5 4
0 9 8 7 6 A 1 2 3 4 5
6 5 9 8 7 4 A 1 2 3 0
7 6 5 9 8 3 4 0 1 A 2
8 7 6 5 A 2 3 4 0 1 9
9 A 7 6 5 1 2 3 4 0 8
4 8 1 3 9 5 0 7 6 2 A
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

Partial square

```
5 4 3 . 1 . 6 A 8 . .
. . 4 . 2 9 . 6 7 8 3
. . . . 3 8 9 5 A 7 6
. . . . 4 7 8 9 5 6 .
A . . . . 6 7 8 9 5 .
0 . 8 . 6 A . 2 . . .
. . 9 8 7 . . 1 2 3 .
. . . 9 . . . . 1 A .
. . . . . . . . . 1 9
. A . . . . . . . . .
4 . 1 . . . 0 . . . .
```

E040: order 11, 53 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_four_m_plus_three/near_examples/n11_seed3.json`

SHA256: `e733c4ab9594bedab8b3130cce877522298b4bd5a92b77a315bf53e024528c06`

Completion

```
0 4 3 2 1 5 6 7 8 9 A
1 0 A 3 2 9 5 6 7 8 4
2 1 0 4 A 8 9 5 6 7 3
3 2 1 0 4 7 A 9 5 6 8
4 3 2 1 0 6 7 8 A 5 9
5 9 8 7 6 A 1 2 3 4 0
6 5 9 8 7 4 0 A 2 3 1
7 6 5 9 8 3 4 0 1 A 2
8 A 6 5 9 2 3 4 0 1 7
9 8 7 A 5 1 2 3 4 0 6
A 7 4 6 3 0 8 1 9 2 5
```

Partial square

```
0 4 3 . . 5 . . 8 . .
. . . 3 2 9 5 6 . 8 4
. . . 4 . 8 9 5 6 7 .
. . . . 4 7 A 9 5 . .
. . . . . 6 7 8 . 5 9
. . 8 . . . 1 2 3 4 .
. . 9 8 7 . . . 2 3 .
. . . 9 8 . . . 1 . .
. A . . . . . . . 1 7
. . . A . . . . . . 6
A 7 4 . 3 0 8 . . 2 .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E041: order 11, 52 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m5_seed0.json`

SHA256: `2159403e9d7eada6b6e2a855cd40af5309608f816224462ded1a31f96224c233`

Completion

```
5 4 3 2 1 0 6 A 8 9 7
1 0 4 A 2 9 5 6 7 8 3
2 1 0 4 3 8 9 5 A 7 6
3 2 A 0 4 7 8 9 5 6 1
A 3 2 1 0 6 7 8 9 5 4
0 9 8 7 6 A 1 2 3 4 5
6 5 9 8 7 4 A 1 2 3 0
7 6 5 9 8 3 4 0 1 A 2
8 7 6 5 A 2 3 4 0 1 9
9 A 7 6 5 1 2 3 4 0 8
4 8 1 3 9 5 0 7 6 2 A
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

Partial square

```
5 4 . 2 . 0 6 . . . .
. . 4 A 2 9 . 6 7 8 .
. . . 4 3 8 9 5 A 7 .
. . . . . 7 8 9 5 6 1
A . . . . 6 7 . . 5 .
0 . 8 7 6 A . . 3 . .
. . 9 8 7 . . 1 2 3 .
. . . 9 8 . . . 1 A .
. . . . A . . . . 1 .
. A . . . . . . . . 8
4 . . . . . 0 7 . . .
```

E042: order 11, 51 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_four_m_plus_three/examples/xor3_a2_seed1.json`

SHA256: `6ad63fd28ea50da59cd351705063763dceabe386689d6986ac3599eee3ef2ff2`

Completion

```
8 9 A 3 4 5 6 7 0 1 2
1 0 3 2 5 A 9 8 6 7 4
2 A 9 8 6 7 4 5 1 0 3
3 2 1 0 8 9 A 4 7 6 5
4 5 6 7 A 1 8 9 2 3 0
9 8 7 A 1 0 3 2 4 5 6
6 7 4 5 9 8 0 A 3 2 1
A 6 8 9 3 2 1 0 5 4 7
0 4 5 1 7 3 2 6 8 A 9
5 1 0 4 2 6 7 3 A 9 8
7 3 2 6 0 4 5 1 9 8 A
```

Partial square

```
8 9 A . . . . . 0 . .
. 0 . . . A . . 6 7 .
. A 9 . . . 4 5 . . 3
. 2 1 . 8 9 A 4 7 . .
. . . . A . 8 . . . .
. 8 . . . 0 . . 4 . 6
. . 4 5 . . . A . 2 1
. . 8 9 3 2 . . . . .
0 . 5 1 . 3 2 6 8 . .
5 . . 4 . . . 3 . . .
7 . . 6 0 . 5 . 9 8 .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E043: order 11, 51 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_four_m_plus_three/near_examples/n11_seed1.json`

SHA256: `fa3c5f7a70a2b3f728f0064950db1e389ff2753ed54aed3363e3b1689fee2114`

Completion

```
0 4 3 2 1 5 6 7 8 9 A
1 0 A 3 2 9 5 6 7 8 4
2 1 0 4 A 8 9 5 6 7 3
3 2 1 0 4 7 A 9 5 6 8
4 3 2 1 0 6 7 8 A 5 9
5 9 8 7 6 A 1 2 3 4 0
6 5 9 8 7 4 0 A 2 3 1
7 6 5 9 8 3 4 0 1 A 2
8 A 6 5 9 2 3 4 0 1 7
9 8 7 A 5 1 2 3 4 0 6
A 7 4 6 3 0 8 1 9 2 5
```

Partial square

```
0 4 . . . 5 6 7 . . .
. . . 3 2 9 5 6 7 8 .
. . . 4 . 8 9 5 6 7 3
. . . . 4 7 . 9 5 . .
. . . . . 6 7 8 A 5 .
5 . 8 . . A . . . . 0
. . 9 8 7 . . A 2 3 .
. . . . 8 . . . 1 A 2
. . . . 9 . . . . 1 .
. . . . . . . . . . 6
. 7 4 6 3 0 8 . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E044: order 12, 90 clues

Status I. Previously verified construction/example or published record

Source: `outputs/lcs_ternary_extra_examples.json:0`

SHA256: `adc9a76d6d55caa156e405f493c053c207a5c2b16b5661fbaea72b86264c1ddf`

Completion

```
9 0 1 2 3 4 5 6 7 8 A B
0 1 9 4 2 3 6 7 5 A B 8
1 9 0 3 4 2 7 5 6 B 8 A
2 3 4 9 0 1 A 8 B 6 5 7
3 4 2 1 9 0 B A 8 7 6 5
4 2 3 0 1 9 8 B A 5 7 6
5 7 6 A B 8 9 1 0 4 2 3
6 5 7 8 A B 0 9 1 2 3 4
7 6 5 B 8 A 1 0 9 3 4 2
8 B A 6 7 5 4 2 3 9 1 0
A 8 B 5 6 7 2 3 4 0 9 1
B A 8 7 5 6 3 4 2 1 0 9
```

Partial square

```
. . . . . . . . . . . .
0 1 . 4 2 . 6 . 5 A . 8
1 . . . 4 . . . 6 . . A
2 . . 9 0 1 A . . 6 5 7
. . . 1 9 0 . . . 7 6 5
4 2 . 0 1 9 8 . A 5 7 6
. . . . . . 9 1 0 4 2 3
6 5 . 8 A . 0 9 1 2 3 4
. 6 . . 8 . 1 0 9 3 4 2
8 . . 6 7 5 4 2 3 9 1 0
A 8 . 5 6 7 2 3 4 0 9 1
. . . 7 5 6 3 4 2 1 0 9
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E045: order 13, 94 clues

Status I. Previously independently checked uniqueness and all clue-deletion witnesses; uniqueness is solver-dependent.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m3_seed3.json`

SHA256: `41f0b2a36cf1ac8d267591df161f35bea33f2367323ccf76540303372665b5b2`

Completion

```
C 2 1 3 4 5 6 7 8 9 A B 0
1 0 C 5 3 4 8 6 7 A B 9 2
2 C 0 4 5 3 7 8 6 B 9 A 1
3 5 4 0 1 2 C A B 6 7 8 9
4 3 5 2 0 1 A C 9 8 6 7 B
5 4 3 1 2 0 B 9 C 7 8 6 A
6 8 7 9 A B 0 1 2 C 4 5 3
7 6 8 A B 9 2 0 1 5 3 C 4
8 7 6 B 9 A 1 2 0 4 C 3 5
9 A B C 8 7 3 5 4 0 2 1 6
A B 9 7 6 C 4 3 5 1 0 2 8
B 9 A 8 C 6 5 4 3 2 1 0 7
0 1 2 6 7 8 9 B A 3 5 4 C
```

Partial square

```
. . . . . . . . . . . . 0
. 0 C . 3 . . 6 . . . 9 .
. C 0 . 5 3 . 8 6 . 9 A 1
. . . 0 1 2 . . . 6 7 8 9
. 3 5 2 0 1 . C . 8 6 7 .
. . 3 1 2 . . . C 7 8 6 A
. . . . . 0 1 2 C 4 5 .
. 6 8 . . 9 2 . 1 5 3 C .
. . 6 . 9 A 1 2 . 4 C 3 .
. . . C 8 7 3 5 4 0 2 1 .
. . 9 7 6 C 4 3 5 1 . 2 .
. 9 A 8 C 6 5 . 3 2 1 0 7
. 1 . . . . 9 B . . 5 . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E046: order 13, 93 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m3_seed2.json`

SHA256: `21dea1240d95feee5ae10bd74fc94027d79a2887e0e3e3e8f1326685d03104fe`

Completion

```
C 2 1 3 4 5 6 7 8 9 A B 0
1 0 C 5 3 4 8 6 7 A B 9 2
2 C 0 4 5 3 7 8 6 B 9 A 1
3 5 4 0 1 2 C A B 6 7 8 9
4 3 5 2 0 1 A C 9 8 6 7 B
5 4 3 1 2 0 B 9 C 7 8 6 A
6 8 7 9 A B 0 1 2 C 4 5 3
7 6 8 A B 9 2 0 1 5 3 C 4
8 7 6 B 9 A 1 2 0 4 C 3 5
9 A B C 8 7 3 5 4 0 2 1 6
A B 9 7 6 C 4 3 5 1 0 2 8
B 9 A 8 C 6 5 4 3 2 1 0 7
0 1 2 6 7 8 9 B A 3 5 4 C
```

Partial square

```
. . . . . . . . . . . . .
. . C . 3 . . 6 . . . 9 2
. C 0 . 5 3 . 8 6 . 9 A .
. . . 0 1 2 C . . 6 7 8 .
. 3 5 2 0 1 . C 9 8 6 . .
. . 3 1 2 0 . . . 7 8 6 A
. . . . . 0 1 2 . 4 5 .
. 6 8 . . 9 2 . 1 5 3 C 4
. . 6 . 9 A 1 2 0 4 . 3 .
. . . C 8 7 . 5 4 0 2 1 6
. . 9 7 6 . 4 3 5 1 0 2 .
. 9 A 8 . 6 5 . 3 2 1 . 7
0 . 2 . 7 8 9 B . . . . C
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E047: order 13, 93 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m3_seed6.json`

SHA256: `e777dadd5566718232e2f7b0c5305ef14d8c677a8a9f449bba93986be1e240d0`

Completion

```
C 2 1 3 4 5 6 7 8 9 A B 0
1 0 C 5 3 4 8 6 7 A B 9 2
2 C 0 4 5 3 7 8 6 B 9 A 1
3 5 4 0 1 2 C A B 6 7 8 9
4 3 5 2 0 1 A C 9 8 6 7 B
5 4 3 1 2 0 B 9 C 7 8 6 A
6 8 7 9 A B 0 1 2 C 4 5 3
7 6 8 A B 9 2 0 1 5 3 C 4
8 7 6 B 9 A 1 2 0 4 C 3 5
9 A B C 8 7 3 5 4 0 2 1 6
A B 9 7 6 C 4 3 5 1 0 2 8
B 9 A 8 C 6 5 4 3 2 1 0 7
0 1 2 6 7 8 9 B A 3 5 4 C
```

Partial square

```
C . . . . . . . . . . . .
. 0 . . 3 . . 6 . . . 9 2
. C 0 . 5 3 . 8 6 . 9 A .
. . . . 1 2 . . . 6 7 8 .
. 3 5 2 0 1 . C 9 8 6 7 .
. . 3 1 2 . . . C 7 8 6 A
. . . . . 0 1 2 . 4 5 3
. 6 8 . . 9 2 0 1 5 3 C .
. . 6 . 9 A 1 2 0 4 C 3 .
. . . C 8 7 3 5 4 0 2 1 .
. . 9 7 6 C 4 3 5 1 . 2 .
. 9 A 8 . 6 5 . 3 2 1 0 7
. . 2 . . . 9 B A . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E048: order 13, 92 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m3_seed0.json`

SHA256: `121e538d54b1f199861f86e827c2b6062cd4cdec8a4efc110c0d4ec34bc40b35`

Completion

```
C 2 1 3 4 5 6 7 8 9 A B 0
1 0 C 5 3 4 8 6 7 A B 9 2
2 C 0 4 5 3 7 8 6 B 9 A 1
3 5 4 0 1 2 C A B 6 7 8 9
4 3 5 2 0 1 A C 9 8 6 7 B
5 4 3 1 2 0 B 9 C 7 8 6 A
6 8 7 9 A B 0 1 2 C 4 5 3
7 6 8 A B 9 2 0 1 5 3 C 4
8 7 6 B 9 A 1 2 0 4 C 3 5
9 A B C 8 7 3 5 4 0 2 1 6
A B 9 7 6 C 4 3 5 1 0 2 8
B 9 A 8 C 6 5 4 3 2 1 0 7
0 1 2 6 7 8 9 B A 3 5 4 C
```

Partial square

```
C . . . . . . . . . . . .
. 0 . . 3 . . 6 . . . 9 2
. C 0 . 5 3 . 8 6 . 9 A .
. . . 0 1 2 C . . 6 7 8 .
. 3 5 2 0 1 . . 9 8 6 7 .
. . 3 1 2 0 . . . 7 8 6 A
. . . . . 1 2 C 4 5 . . .
. 6 8 . . 9 2 . 1 5 3 . 4
. . 6 . 9 A 1 2 0 4 C 3 .
. . . . 8 7 . 5 4 0 2 1 6
. . 9 7 6 C 4 3 5 1 . 2 .
. 9 A 8 C 6 5 4 3 2 1 0 .
. . 2 . . . 9 B . . 5 . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E049: order 13, 90 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m3_seed4.json`

SHA256: `1b595899fe98f6f0084ebbd4a3982d243e2403c228ab78bc1b3e74808a539c2f`

Completion

```
C 2 1 3 4 5 6 7 8 9 A B 0
1 0 C 5 3 4 8 6 7 A B 9 2
2 C 0 4 5 3 7 8 6 B 9 A 1
3 5 4 0 1 2 C A B 6 7 8 9
4 3 5 2 0 1 A C 9 8 6 7 B
5 4 3 1 2 0 B 9 C 7 8 6 A
6 8 7 9 A B 0 1 2 C 4 5 3
7 6 8 A B 9 2 0 1 5 3 C 4
8 7 6 B 9 A 1 2 0 4 C 3 5
9 A B C 8 7 3 5 4 0 2 1 6
A B 9 7 6 C 4 3 5 1 0 2 8
B 9 A 8 C 6 5 4 3 2 1 0 7
0 1 2 6 7 8 9 B A 3 5 4 C
```

Partial square

```
C . . . . . . . . . . . .
. 0 C . 3 . . 6 . . . 9 .
. C . . 5 3 . 8 6 . 9 A 1
. . . 0 1 2 C . . . 7 8 9
. 3 5 2 0 1 . C . 8 6 . B
. . 3 1 2 . . . . 7 8 6 .
. . . . . 0 1 2 . 4 5 .
. 6 8 . . 9 2 . 1 5 3 . 4
. . 6 . 9 A 1 2 . 4 C 3 .
. . . . 8 7 . 5 4 0 2 1 6
. . 9 7 6 . 4 3 5 1 . 2 8
. 9 A 8 . 6 5 . 3 2 1 . 7
0 1 . . . . 9 B . . 5 4 C
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E050: order 13, 90 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_shifted_examples/thesis_m3_seed0.json`

SHA256: `8b48219dada0735fa1506e28ed07c4dee4e9089a8dcd27a39fe38caa4bbc516a`

Completion

```
0 2 C 3 4 5 6 7 8 9 A B 1
1 C 2 5 3 4 8 6 7 A B 9 0
C 1 0 4 5 3 7 8 6 B 9 A 2
3 5 4 0 1 2 C A B 6 7 8 9
4 3 5 2 0 1 A C 9 8 6 7 B
5 4 3 1 2 0 B 9 C 7 8 6 A
6 8 7 9 A B 0 1 2 C 4 5 3
7 6 8 A B 9 2 0 1 5 3 C 4
8 7 6 B 9 A 1 2 0 4 C 3 5
9 A B C 8 7 3 5 4 0 2 1 6
A B 9 7 6 C 4 3 5 1 0 2 8
B 9 A 8 C 6 5 4 3 2 1 0 7
2 0 1 6 7 8 9 B A 3 5 4 C
```

Partial square

```
. . . . . . . . . . . . . 1
. C . . 3 . . 6 . . . 9 .
. 1 0 . 5 3 . 8 6 . 9 A .
. . . 0 1 2 C . . 6 7 8 .
. 3 5 2 0 1 . . 9 8 6 7 .
. . 3 1 2 0 . . . 7 8 6 A
. . . . . 0 . 2 C 4 5 .
. 6 8 . . 9 . 0 1 5 3 C 4
. . 6 . 9 A 1 2 0 4 . 3 .
. . . . 8 7 . 5 4 0 2 . 6
. . 9 7 6 C 4 3 5 1 0 2 .
. 9 A 8 C 6 5 4 3 . 1 0 .
2 . . . . . 9 B . . 5 . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E051: order 13, 89 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m3_seed1.json`

SHA256: `0ca6cc6e3acf40e6c7eb9c6fa01feffe0ad1510efd59a848a7759b515b54b09d`

Completion

```
C 2 1 3 4 5 6 7 8 9 A B 0
1 0 C 5 3 4 8 6 7 A B 9 2
2 C 0 4 5 3 7 8 6 B 9 A 1
3 5 4 0 1 2 C A B 6 7 8 9
4 3 5 2 0 1 A C 9 8 6 7 B
5 4 3 1 2 0 B 9 C 7 8 6 A
6 8 7 9 A B 0 1 2 C 4 5 3
7 6 8 A B 9 2 0 1 5 3 C 4
8 7 6 B 9 A 1 2 0 4 C 3 5
9 A B C 8 7 3 5 4 0 2 1 6
A B 9 7 6 C 4 3 5 1 0 2 8
B 9 A 8 C 6 5 4 3 2 1 0 7
0 1 2 6 7 8 9 B A 3 5 4 C
```

Partial square

```
C . . . . . . . . . . . 0
. 0 . . 3 . . 6 . . . 9 2
. C 0 . 5 3 . 8 6 . 9 A .
. . . 0 1 2 C . . . 7 8 9
. 3 5 2 . 1 . C . 8 6 . .
. . 3 1 2 . . . . 7 8 6 A
. . . . . 0 1 2 . 4 5 .
. 6 8 . . 9 2 0 1 5 3 . 4
. . 6 . 9 A 1 2 . 4 C 3 .
. . . . 8 7 . 5 4 . 2 1 .
. . 9 7 6 . 4 3 5 1 . 2 .
. 9 A 8 . 6 5 . 3 2 1 . 7
. . 2 6 7 8 9 . A 3 . 4 .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E052: order 13, 89 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_shifted_examples/thesis_m3_seed1.json`

SHA256: `6154fdb1424683b229ac6b3fcd97b6401902c34df05ca8b7226dd84ed5c9da8c`

Completion

```
0 2 C 3 4 5 6 7 8 9 A B 1
1 C 2 5 3 4 8 6 7 A B 9 0
C 1 0 4 5 3 7 8 6 B 9 A 2
3 5 4 0 1 2 C A B 6 7 8 9
4 3 5 2 0 1 A C 9 8 6 7 B
5 4 3 1 2 0 B 9 C 7 8 6 A
6 8 7 9 A B 0 1 2 C 4 5 3
7 6 8 A B 9 2 0 1 5 3 C 4
8 7 6 B 9 A 1 2 0 4 C 3 5
9 A B C 8 7 3 5 4 0 2 1 6
A B 9 7 6 C 4 3 5 1 0 2 8
B 9 A 8 C 6 5 4 3 2 1 0 7
2 0 1 6 7 8 9 B A 3 5 4 C
```

Partial square

```
. . C . . . . . . . . . 1
. . . . 3 . . 6 . . . 9 0
C . . . 5 3 . 8 6 . 9 A .
. . . 0 1 2 C . . . 7 8 9
. 3 5 2 0 1 . C . 8 6 . .
. . 3 1 2 0 . . . 7 8 6 A
. . . . . . 0 . 2 . 4 5 .
. 6 8 . . 9 . 0 1 5 3 . 4
. . 6 . 9 A 1 2 0 4 . 3 5
. . . . 8 7 . 5 4 0 2 . .
. . 9 7 6 . 4 3 5 1 0 2 .
. 9 A 8 . 6 5 . 3 . 1 0 7
2 . 1 6 7 8 9 . A 3 . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E053: order 13, 88 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m3_seed5.json`

SHA256: `886615677758e604f34340d73ba66292fcbc4f60eac7176843d77b0cbe1b8f4d`

Completion

```
C 2 1 3 4 5 6 7 8 9 A B 0
1 0 C 5 3 4 8 6 7 A B 9 2
2 C 0 4 5 3 7 8 6 B 9 A 1
3 5 4 0 1 2 C A B 6 7 8 9
4 3 5 2 0 1 A C 9 8 6 7 B
5 4 3 1 2 0 B 9 C 7 8 6 A
6 8 7 9 A B 0 1 2 C 4 5 3
7 6 8 A B 9 2 0 1 5 3 C 4
8 7 6 B 9 A 1 2 0 4 C 3 5
9 A B C 8 7 3 5 4 0 2 1 6
A B 9 7 6 C 4 3 5 1 0 2 8
B 9 A 8 C 6 5 4 3 2 1 0 7
0 1 2 6 7 8 9 B A 3 5 4 C
```

Partial square

```
C . . . . . . . . . . . 0
. 0 C . 3 . . 6 . . . 9 .
. . . . 5 3 . 8 6 . 9 A 1
. . . 0 1 2 C . . . 7 8 .
. 3 5 2 . 1 . . 9 8 6 . B
. . 3 1 2 . . . . 7 8 6 A
. . . . . 0 1 2 . 4 5 3
. 6 8 . . 9 2 . 1 5 3 . .
. . 6 . 9 A 1 2 . 4 . 3 .
. . . . 8 7 . 5 4 0 2 1 6
. . 9 7 6 C 4 3 5 1 . 2 .
. 9 A 8 . 6 5 4 3 2 1 0 .
0 1 . . 7 . 9 . A . . . C
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E054: order 13, 87 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_shifted_examples/thesis_m3_seed2.json`

SHA256: `8b737b5c9add24dca45c65ba12ef6b123aabde9a82c8ac880b3dc78c88ace3ab`

Completion

```
0 2 C 3 4 5 6 7 8 9 A B 1
1 C 2 5 3 4 8 6 7 A B 9 0
C 1 0 4 5 3 7 8 6 B 9 A 2
3 5 4 0 1 2 C A B 6 7 8 9
4 3 5 2 0 1 A C 9 8 6 7 B
5 4 3 1 2 0 B 9 C 7 8 6 A
6 8 7 9 A B 0 1 2 C 4 5 3
7 6 8 A B 9 2 0 1 5 3 C 4
8 7 6 B 9 A 1 2 0 4 C 3 5
9 A B C 8 7 3 5 4 0 2 1 6
A B 9 7 6 C 4 3 5 1 0 2 8
B 9 A 8 C 6 5 4 3 2 1 0 7
2 0 1 6 7 8 9 B A 3 5 4 C
```

Partial square

```
. . C . . . . . . . . . .
. . . . 3 . . 6 . . . 9 0
C . . . 5 3 . 8 6 . 9 A 2
. . . 0 . 2 C . . 6 7 8 .
. 3 5 2 0 1 . . 9 8 6 7 .
. . 3 1 2 0 . . . 7 8 6 A
. . . . . . 0 1 2 . 4 5 .
. 6 8 . . 9 . 0 1 5 3 . .
. . 6 . 9 A 1 2 0 4 . 3 .
. . . . 8 7 . 5 4 0 2 . 6
. . 9 7 6 . 4 3 5 1 0 2 .
. 9 A 8 . 6 5 . 3 . 1 0 7
2 . 1 . . 8 9 B . . . . C
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E055: order 13, 83 clues

Status I. Previously independently checked uniqueness and all clue-deletion witnesses; uniqueness is solver-dependent.

Source: `outputs/lcs_odd_prolongation/examples/m6_seed3.json`

SHA256: `24cc5961ced42bbc6d7e2163c3d83bf3dca041463faa46b072fd0ad0010e1581`

Completion

```
C 5 4 3 2 1 6 7 8 9 A B 0
1 0 C 4 3 2 B 6 7 8 9 A 5
2 1 0 5 C 3 A B 6 7 8 9 4
3 2 1 0 5 4 C A B 6 7 8 9
4 3 2 1 0 5 8 9 C B 6 7 A
5 4 3 2 1 0 7 8 9 A C 6 B
6 B A 9 8 7 0 C 2 3 4 5 1
7 6 B A 9 8 5 0 1 C 3 4 2
8 7 6 B A 9 4 5 0 1 2 C 3
9 C 7 6 B A 3 4 5 0 1 2 8
A 9 8 C 6 B 2 3 4 5 0 1 7
B A 9 8 7 C 1 2 3 4 5 0 6
0 8 5 7 4 6 9 1 A 2 B 3 C
```

Partial square

```
. 5 4 3 2 1 6 . . 9 A . .
. . C 4 3 2 B 6 7 8 9 A .
. . . 5 C 3 A B 6 7 8 9 .
. . . . 5 4 C A B 6 . 8 .
. . . . . 5 8 9 C B 6 7 .
. . . . . . 7 8 9 A C . B
. B A 9 8 7 . C 2 3 4 5 .
. . B A 9 . . . 1 C 3 4 2
. . . B A . . . . 1 2 C .
. . . . B A . . . . 1 2 8
. . . C . B . . . . . 1 7
. . . . . C . . . . . . .
0 8 . . . . 9 . . . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E056: order 13, 82 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m6_seed6.json`

SHA256: `909d2b85ab38ef5df5f998182cf610a4e2b44ae059a1ba46f8ddaa3a5c0fee44`

Completion

```
C 5 4 3 2 1 6 7 8 9 A B 0
1 0 C 4 3 2 B 6 7 8 9 A 5
2 1 0 5 C 3 A B 6 7 8 9 4
3 2 1 0 5 4 C A B 6 7 8 9
4 3 2 1 0 5 8 9 C B 6 7 A
5 4 3 2 1 0 7 8 9 A C 6 B
6 B A 9 8 7 0 C 2 3 4 5 1
7 6 B A 9 8 5 0 1 C 3 4 2
8 7 6 B A 9 4 5 0 1 2 C 3
9 C 7 6 B A 3 4 5 0 1 2 8
A 9 8 C 6 B 2 3 4 5 0 1 7
B A 9 8 7 C 1 2 3 4 5 0 6
0 8 5 7 4 6 9 1 A 2 B 3 C
```

Partial square

```
. 5 4 3 2 1 6 7 8 9 A B .
. . C 4 3 2 B 6 7 . 9 A .
. . . 5 C 3 A B 6 7 8 9 .
. . . . 5 4 C A B 6 . 8 .
. . . . . 5 8 9 C B 6 7 .
. . . . . 7 8 9 A C . . .
. B A 9 8 7 . C 2 3 4 5 .
. . B A 9 8 . . 1 C 3 4 .
. . . B A . . . . 1 2 C .
. . . . B A . . . . 1 2 8
. . . . . B . . . . . 1 7
. . . . . 6
0 . . . . . 9 . . . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E057: order 13, 82 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m6_seed0.json`

SHA256: `a6c04ef40c7afb0cb05a3592788241d169c515218901fba964dc8068375ad753`

Completion

```
C 5 4 3 2 1 6 7 8 9 A B 0
1 0 C 4 3 2 B 6 7 8 9 A 5
2 1 0 5 C 3 A B 6 7 8 9 4
3 2 1 0 5 4 C A B 6 7 8 9
4 3 2 1 0 5 8 9 C B 6 7 A
5 4 3 2 1 0 7 8 9 A C 6 B
6 B A 9 8 7 0 C 2 3 4 5 1
7 6 B A 9 8 5 0 1 C 3 4 2
8 7 6 B A 9 4 5 0 1 2 C 3
9 C 7 6 B A 3 4 5 0 1 2 8
A 9 8 C 6 B 2 3 4 5 0 1 7
B A 9 8 7 C 1 2 3 4 5 0 6
0 8 5 7 4 6 9 1 A 2 B 3 C
```

Partial square

```
. 5 4 3 2 1 . 7 . 9 A B .
. . C 4 3 2 B 6 7 . 9 A .
. . . 5 C 3 A B 6 7 8 9 .
. . . . 5 4 C A B 6 . 8 .
. . . . . 5 8 9 C B 6 7 .
. . . . . . 7 8 9 A C 6 .
. B A 9 8 7 . C 2 3 4 5 .
. . B A 9 8 . . 1 . 3 4 2
. . . B A 9 . . . 1 2 C .
. . . . B A . . . . 1 2 8
. . . . . B . . . . . 1 .
. . . . . . . . . . . . 6
0 8 . 7 . . . . . . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E058: order 13, 80 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m6_seed2.json`

SHA256: `cc3fc9d04da3f941fa6beee76f4660a90517c7e0f1fa590e7bf9eb91d3d3802f`

Completion

```
C 5 4 3 2 1 6 7 8 9 A B 0
1 0 C 4 3 2 B 6 7 8 9 A 5
2 1 0 5 C 3 A B 6 7 8 9 4
3 2 1 0 5 4 C A B 6 7 8 9
4 3 2 1 0 5 8 9 C B 6 7 A
5 4 3 2 1 0 7 8 9 A C 6 B
6 B A 9 8 7 0 C 2 3 4 5 1
7 6 B A 9 8 5 0 1 C 3 4 2
8 7 6 B A 9 4 5 0 1 2 C 3
9 C 7 6 B A 3 4 5 0 1 2 8
A 9 8 C 6 B 2 3 4 5 0 1 7
B A 9 8 7 C 1 2 3 4 5 0 6
0 8 5 7 4 6 9 1 A 2 B 3 C
```

Partial square

```
. . . 3 2 . 6 7 8 9 A B .
. . C 4 3 2 B 6 7 . 9 A 5
. . . 5 . 3 A B 6 7 8 9 4
. . . . 5 4 . A B 6 . 8 .
. . . . . 5 8 9 . B 6 7 .
. . . . . 7 8 9 A . . . .
. . A 9 8 . . . 2 3 4 5 1
. . B . 9 . . . 1 . 3 4 2
. . . B A . . . . 1 2 . .
. C . . B A . . . . 1 2 8
. . . C . B . . . . . 1 7
. . . . . C . . . . . . 6
. . 5 7 4 6 9 . . . . 3 C
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E059: order 13, 79 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m6_seed4.json`

SHA256: `97165531e91defc505dc6e87a440148bd0854502671727148beae758e55b876c`

Completion

```
C 5 4 3 2 1 6 7 8 9 A B 0
1 0 C 4 3 2 B 6 7 8 9 A 5
2 1 0 5 C 3 A B 6 7 8 9 4
3 2 1 0 5 4 C A B 6 7 8 9
4 3 2 1 0 5 8 9 C B 6 7 A
5 4 3 2 1 0 7 8 9 A C 6 B
6 B A 9 8 7 0 C 2 3 4 5 1
7 6 B A 9 8 5 0 1 C 3 4 2
8 7 6 B A 9 4 5 0 1 2 C 3
9 C 7 6 B A 3 4 5 0 1 2 8
A 9 8 C 6 B 2 3 4 5 0 1 7
B A 9 8 7 C 1 2 3 4 5 0 6
0 8 5 7 4 6 9 1 A 2 B 3 C
```

Partial square

```
C . . . . . 6 7 8 9 A . 0
. . . 4 3 2 B 6 7 . 9 A .
. . . 5 . 3 A B 6 7 8 9 4
. . . . 5 4 . A B 6 . 8 .
. . . . . 5 8 9 . B 6 7 A
. . . . . . 7 8 9 A . . .
. . A 9 8 . . . 2 3 4 5 1
. . B A 9 . . . 1 . 3 4 2
. . . B A . . . . 1 2 . 3
. C . . B . . . . . 1 2 8
. . . C . B . . . . . 1 7
. . . . . C . . . . . . .
0 8 5 7 . 6 9 1 . . B . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E060: order 13, 77 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m6_seed1.json`

SHA256: `afaccfc337854e98097c3ed1e631e664b0633d5d6848fa0a739c32fd10db2af0`

Completion

```
C 5 4 3 2 1 6 7 8 9 A B 0
1 0 C 4 3 2 B 6 7 8 9 A 5
2 1 0 5 C 3 A B 6 7 8 9 4
3 2 1 0 5 4 C A B 6 7 8 9
4 3 2 1 0 5 8 9 C B 6 7 A
5 4 3 2 1 0 7 8 9 A C 6 B
6 B A 9 8 7 0 C 2 3 4 5 1
7 6 B A 9 8 5 0 1 C 3 4 2
8 7 6 B A 9 4 5 0 1 2 C 3
9 C 7 6 B A 3 4 5 0 1 2 8
A 9 8 C 6 B 2 3 4 5 0 1 7
B A 9 8 7 C 1 2 3 4 5 0 6
0 8 5 7 4 6 9 1 A 2 B 3 C
```

Partial square

```
C 5 4 . . . 6 7 8 . . . 0
. . . 4 3 2 B 6 7 . 9 A .
. . . 5 . 3 A B 6 7 8 9 .
. . . . 5 4 . A B 6 . 8 .
. . . . . 5 8 9 . B 6 7 .
. . . . . 7 8 9 A . . B
. . A 9 8 7 . . 2 3 4 5 .
. . B . 9 8 . . 1 . 3 4 2
. . . B A . . . . 1 2 . .
. C . . B . . . . . 1 2 8
. . . . . 1 7
. . . . . C . . . . . 6
0 8 . 7 . 6 9 1 . 2 B . C
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E061: order 13, 77 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m6_seed5.json`

SHA256: `bca6abc05a82aca508e33e908be1e74b1ec13f87ac5910a813fe7ce02440790b`

Completion

```
C 5 4 3 2 1 6 7 8 9 A B 0
1 0 C 4 3 2 B 6 7 8 9 A 5
2 1 0 5 C 3 A B 6 7 8 9 4
3 2 1 0 5 4 C A B 6 7 8 9
4 3 2 1 0 5 8 9 C B 6 7 A
5 4 3 2 1 0 7 8 9 A C 6 B
6 B A 9 8 7 0 C 2 3 4 5 1
7 6 B A 9 8 5 0 1 C 3 4 2
8 7 6 B A 9 4 5 0 1 2 C 3
9 C 7 6 B A 3 4 5 0 1 2 8
A 9 8 C 6 B 2 3 4 5 0 1 7
B A 9 8 7 C 1 2 3 4 5 0 6
0 8 5 7 4 6 9 1 A 2 B 3 C
```

Partial square

```
C 5 . . . . 6 7 8 9 . . 0
. . . 4 3 2 B 6 7 . 9 A .
. . . 5 . 3 A B 6 7 8 9 .
. . . . 5 4 . A B 6 . 8 .
. . . . . 5 8 9 . B 6 7 .
. . . . . . 7 8 9 A . . .
. . A 9 8 . . . 2 3 4 5 1
. . B . 9 . . . 1 . 3 4 2
. . . B A . . . . 1 2 . 3
. C . . B A . . . . 1 2 8
. . . C . B . . . . . 1 7
. . . . . C . . . . . . .
. 8 5 . 4 6 . . . 2 B 3 C
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E062: order 15, 148 clues

Status I. Previously independently verified example; see the archived certificate directory.

Source: `outputs/lcs_order15_verification/bunya_record.json`

SHA256: `1552ed769ae0b806b14c81c01465a9baa12a5f1c54a1826a62d476d19bdfd083`

Completion

```
E 0 6 A 4 1 5 B D 7 8 2 C 9 3
1 2 0 9 8 E B 5 C 3 4 6 D A 7
3 5 B 8 9 7 0 D 6 1 A C 2 4 E
A D C E 7 9 6 2 0 4 3 B 5 1 8
0 E 1 D 2 A 3 7 B 8 6 4 9 C 5
7 B 5 4 A 3 C 0 2 E 9 D 6 8 1
D A 9 0 B C 8 4 E 2 5 7 1 3 6
5 3 7 6 C B E 1 8 A D 9 4 2 0
4 1 E C 6 0 7 3 9 5 2 8 A D B
9 C D 1 3 8 2 6 A B 7 5 0 E 4
8 6 2 3 1 4 D A 5 9 E 0 B 7 C
6 8 4 5 E 2 A 9 3 C 0 1 7 B D
B 7 3 2 D 5 1 E 4 0 C A 8 6 9
C 9 A 7 5 D 4 8 1 6 B 3 E 0 2
2 4 8 B 0 6 9 C 7 D 1 E 3 5 A
```

Partial square

```
E . 6 A 4 . 5 . D . 8 2 C 9 3
1 . 0 9 8 . . 5 . . 4 6 D A 7
3 5 . 8 9 7 0 D 6 . A C . 4 .
A D C E 7 9 6 2 . . 3 . . 1 8
. . . . . . . . . . . . . . .
7 . . 4 A 3 C 0 . . 9 D . . 1
D A 9 0 . C 8 . E . 5 7 1 3 6
5 3 7 6 C . E 1 8 . . 9 4 2 0
4 1 E C 6 0 . 3 9 . . 8 . D .
9 C D . 3 8 2 6 A . . . 0 E 4
. 6 2 3 1 . D . 5 . E 0 . 7 C
6 8 . . . 2 A 9 3 . 0 1 7 . D
. 7 . 2 . . 1 E 4 . C A 8 6 9
C . . 7 5 D 4 8 1 . . 3 E 0 2
2 4 8 . 0 6 . C 7 . . E 3 5 .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E063: order 15, 147 clues

Status I. Previously independently verified example; see the archived certificate directory.

Source: `outputs/lcs_ternary_extra_examples.json:1`

SHA256: `23b18e7fca160d382955485d316ef93129a1885b9db205456ac256154b318dd8`

Completion

```
0 2 1 4 3 6 5 8 7 A 9 C B E D
2 1 0 5 6 3 4 9 A 7 8 D E B C
1 0 2 6 5 4 3 A 9 8 7 E D C B
4 5 6 3 0 1 2 B C D E 7 8 9 A
3 6 5 0 4 2 1 C B E D 8 7 A 9
6 3 4 1 2 5 0 D E B C 9 A 7 8
5 4 3 2 1 0 6 E D C B A 9 8 7
8 9 A B C D E 7 0 1 2 3 4 5 6
7 A 9 C B E D 0 8 2 1 4 3 6 5
A 7 8 D E B C 1 2 9 0 5 6 3 4
9 8 7 E D C B 2 1 0 A 6 5 4 3
C D E 7 8 9 A 3 4 5 6 B 0 1 2
B E D 8 7 A 9 4 3 6 5 0 C 2 1
E B C 9 A 7 8 5 6 3 4 1 2 D 0
D C B A 9 8 7 6 5 4 3 2 1 0 E
```

Partial square

```
0 2 1 4 3 6 5 . . . . . . . .
2 1 0 5 6 3 4 9 . 7 8 . E B C
1 0 2 6 5 4 3 . . 8 7 . . C B
4 5 6 3 0 1 2 . . . . 7 8 9 A
3 6 5 0 4 2 1 C . E . 8 7 . 9
6 3 4 1 2 5 0 . . B C 9 A 7 8
5 4 3 2 1 0 6 E . C B A 9 8 7
. 9 . . C . E 7 . 1 . . 4 . 6
. . . . . . . . . . . . . . .
. 7 8 . E B C 1 . 9 0 . 6 3 4
. 8 7 . . C B . . 0 . . . 4 3
. . . 7 8 9 A . . . . . 0 1 2
. E . 8 7 A 9 4 . 6 . 0 C 2 1
. . C 9 A 7 8 . . 3 4 1 2 . 0
. C B A 9 8 7 6 . 4 3 2 1 0 E
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E064: order 15, 147 clues

Status I. Previously independently verified example; see the archived certificate directory.

Source: `outputs/lcs_examples.json:6`

SHA256: `f31777b20f41ccba7eed13c7c2d4763fc76eb8fefda2cdb40ca6234adad4dc26`

Completion

```
0 2 1 4 3 6 5 8 7 A 9 C B E D
2 1 0 5 6 3 4 9 A 7 8 D E B C
1 0 2 6 5 4 3 A 9 8 7 E D C B
4 5 6 3 0 1 2 B C D E 7 8 9 A
3 6 5 0 4 2 1 C B E D 8 7 A 9
6 3 4 1 2 5 0 D E B C 9 A 7 8
5 4 3 2 1 0 6 E D C B A 9 8 7
8 9 A B C D E 7 0 1 2 3 4 5 6
7 A 9 C B E D 0 8 2 1 4 3 6 5
A 7 8 D E B C 1 2 9 0 5 6 3 4
9 8 7 E D C B 2 1 0 A 6 5 4 3
C D E 7 8 9 A 3 4 5 6 B 0 1 2
B E D 8 7 A 9 4 3 6 5 0 C 2 1
E B C 9 A 7 8 5 6 3 4 1 2 D 0
D C B A 9 8 7 6 5 4 3 2 1 0 E
```

Partial square

```
0 2 1 4 3 6 5 8 7 A 9 . B E D
2 1 0 . . . . 9 A 7 8 . . . .
1 0 2 . 5 . 3 A 9 8 7 . D . B
4 . . 3 0 . . . . D E . . 9 A
3 . 5 0 4 . 1 . B E D . 7 A 9
6 . . . . 5 0 D E . . . . 7 8
5 . 3 . 1 0 6 E D . B . 9 8 7
8 9 A . . D E 7 0 1 2 . . 5 6
7 A 9 . B E D 0 8 2 1 . 3 6 5
A 7 8 D E . . 1 2 9 0 . . 3 4
9 8 7 E D . B 2 1 0 A . 5 4 3
. . . . . . . . . . . . . . .
B . D . 7 . 9 . 3 . 5 . . . 1
E . . 9 A 7 8 5 6 3 4 . . D 0
D . B A 9 8 7 6 5 4 3 . 1 0 E
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E065: order 16, 149 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_full_review/stripping/pruned_direct_n16_seed0.json`

SHA256: `5e1fabc1577faf278419d5fa363b12945de7752caeb9fdf2dbe08b74cac9578a`

Completion

```
0 1 2 3 4 5 6 7 8 9 A B C D E F
1 0 3 2 5 4 7 6 9 8 B A D C F E
2 3 0 1 6 7 4 5 A B 8 9 E F C D
3 2 1 0 7 6 5 4 B A 9 8 F E D C
4 5 6 7 0 1 2 3 C D E F 8 9 A B
5 4 7 6 1 0 3 2 D C F E 9 8 B A
6 7 4 5 2 3 0 1 E F C D A B 8 9
7 6 5 4 3 2 1 0 F E D C B A 9 8
8 9 A B C D E F 0 1 2 3 4 5 6 7
9 8 B A D C F E 1 0 3 2 5 4 7 6
A B 8 9 E F C D 2 3 0 1 6 7 4 5
B A 9 8 F E D C 3 2 1 0 7 6 5 4
C D E F 8 9 A B 4 5 6 7 0 1 2 3
D C F E 9 8 B A 5 4 7 6 1 0 3 2
E F C D A B 8 9 6 7 4 5 2 3 0 1
F E D C B A 9 8 7 6 5 4 3 2 1 0
```

Partial square

```
0 . . . . . . . . . . . . . . . .
. . . 2 5 . 7 6 9 . B A . C F E
. 3 . 1 6 7 4 . A . . 9 E F . D
. 2 1 . 7 6 5 . B A . . F . D C
. 5 6 7 . 1 . 3 . . E . 8 9 . B
. 4 . 6 . . . 2 . C F E 9 8 B .
. . . 5 2 3 . 1 E F . D A . 8 .
. 6 5 4 . 2 1 . F E D C . . . 8
. 9 A B C D . F . 1 2 3 4 . 6 .
. 8 B A D C F E . . 3 2 . . 7 6
. B 8 9 E . C . . . . . 6 7 4 5
. . 9 8 F . D C . 2 1 . 7 6 5 4
. D E F 8 . A B 4 5 6 7 . 1 2 .
. . . E 9 8 B A 5 4 7 . 1 . . 2
. . C D A B 8 9 . . . 5 . 3 . 1
. E D . . A 9 . 7 . 5 4 3 . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E066: order 17, 166 clues

Status I. Previously independently checked uniqueness and all clue-deletion witnesses; uniqueness is solver-dependent.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m4_seed0.json`

SHA256: `4d4d0330a7c79d20800b6665c7449c3fa5175f6c333003097c3235bed3c9cd64`

Completion	Partial square
0 G 2 1 4 5 6 7 8 9 A B C D E F 3	. . . .
1 0 3 2 7 4 5 6 B 8 9 G D E F C A	. 0 . . . 4 . . . 8 . . . . . C .
2 1 0 3 G 7 4 5 A B 8 9 E F C D 6	. 1 0 . . 7 4 . . B 8 . . . C D .
3 2 1 0 5 6 7 4 9 A G 8 F C D E B	. . 1 0 . 6 7 4 . A G . . C D E .
4 7 6 G 0 1 2 3 C D E F 8 9 A B 5	. . . . 0 . 2 3 . . . . 8 9 A B .
5 4 7 6 3 0 1 2 G E F C B 8 9 A D	. 4 7 6 3 0 1 2 . . . . B 8 9 A .
6 5 4 7 2 3 0 1 E F C D A B 8 G 9	. . 4 7 2 3 0 1 . . C D A B 8 G .
7 6 5 4 1 G 3 0 F C D E 9 A B 8 2	. . . 4 1 G 3 0 . C D E 9 A B 8 .
8 B A 9 C D G F 0 1 2 3 4 5 6 7 E	. . . . . 0 1 2 3 4 5 6 7 E
9 8 B A D E F G 3 0 1 2 7 4 5 6 C	. 8 B A . . . G 3 0 1 2 7 4 5 6 C
A 9 G B E F C D 2 3 0 1 6 7 4 5 8	. . G B . . C D 2 . 0 1 6 7 4 5 .
B A 9 8 F C D E 1 2 3 0 5 6 G 4 7	. . . 8 . C D . 1 2 3 0 5 6 G 4 7
C D E F 8 B A 9 4 7 6 5 G 3 2 1 0	. . . . 8 B A 9 4 7 6 5 G 3 2 1 .
D E F C 9 8 B A 5 G 7 6 1 0 3 2 4	. . . C 9 8 B A 5 G 7 6 1 0 3 2 4
E F C D A 9 8 B 6 5 4 7 2 G 0 3 1	. . C D A 9 8 B 6 5 4 7 2 G 0 3 .
G C D E B A 9 8 7 6 5 4 3 2 1 0 F	. C D E B A 9 . 7 . 5 4 3 2 . . .
F 3 8 5 6 2 E C D 4 B A 0 1 7 9 G	. . . . . 2 . C . . . A . . 7 9 G

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E067: order 17, 160 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m4_seed2.json`

SHA256: `92b0b6d02a958b7d632ce90d3338368a78a263a7bfd9bdcaabc2a6ee93cc8ba8`

Completion

```
0 G 2 1 4 5 6 7 8 9 A B C D E F 3
1 0 3 2 7 4 5 6 B 8 9 G D E F C A
2 1 0 3 G 7 4 5 A B 8 9 E F C D 6
3 2 1 0 5 6 7 4 9 A G 8 F C D E B
4 7 6 G 0 1 2 3 C D E F 8 9 A B 5
5 4 7 6 3 0 1 2 G E F C B 8 9 A D
6 5 4 7 2 3 0 1 E F C D A B 8 G 9
7 6 5 4 1 G 3 0 F C D E 9 A B 8 2
8 B A 9 C D G F 0 1 2 3 4 5 6 7 E
9 8 B A D E F G 3 0 1 2 7 4 5 6 C
A 9 G B E F C D 2 3 0 1 6 7 4 5 8
B A 9 8 F C D E 1 2 3 0 5 6 G 4 7
C D E F 8 B A 9 4 7 6 5 G 3 2 1 0
D E F C 9 8 B A 5 G 7 6 1 0 3 2 4
E F C D A 9 8 B 6 5 4 7 2 G 0 3 1
G C D E B A 9 8 7 6 5 4 3 2 1 0 F
F 3 8 5 6 2 E C D 4 B A 0 1 7 9 G
```

Partial square

```
. G . . . . . . . . . . . . . . 3
. . . . . 4 . . . 8 . . . . . C A
. 1 0 . G 7 . . . B 8 . . . C D .
. . 1 0 . 6 7 4 . . . 8 . C D E .
. . . G 0 1 2 3 . . . . 8 9 A B 5
. 4 7 6 3 0 1 2 . . . . B . 9 A D
. . 4 7 2 . 0 1 . . C D A B 8 G .
. . . . 1 . 3 0 . C . E 9 A B 8 .
. . . . . G . 0 1 2 3 4 5 6 7 E
. 8 B A . . . . 3 0 1 2 7 4 5 6 .
. . . B . . C D 2 3 0 . . 7 4 5 8
. . . 8 . C D E 1 2 3 0 5 . G 4 .
. . . . 8 B . 9 4 7 6 5 . 3 2 1 .
. . . C 9 8 . A 5 G 7 6 1 0 3 2 .
. . C D A 9 8 B 6 5 4 7 2 . 0 3 1
G C D . . A 9 . 7 . . 4 3 2 1 . F
F . . . 6 2 . C D . B . 0 . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E068: order 17, 159 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m4_seed3.json`

SHA256: `07d8b0aa7226d9d5c71cd148eca173084324abadf1e46fdb98ed3c7a03a3dacc`

Completion

```
0 G 2 1 4 5 6 7 8 9 A B C D E F 3
1 0 3 2 7 4 5 6 B 8 9 G D E F C A
2 1 0 3 G 7 4 5 A B 8 9 E F C D 6
3 2 1 0 5 6 7 4 9 A G 8 F C D E B
4 7 6 G 0 1 2 3 C D E F 8 9 A B 5
5 4 7 6 3 0 1 2 G E F C B 8 9 A D
6 5 4 7 2 3 0 1 E F C D A B 8 G 9
7 6 5 4 1 G 3 0 F C D E 9 A B 8 2
8 B A 9 C D G F 0 1 2 3 4 5 6 7 E
9 8 B A D E F G 3 0 1 2 7 4 5 6 C
A 9 G B E F C D 2 3 0 1 6 7 4 5 8
B A 9 8 F C D E 1 2 3 0 5 6 G 4 7
C D E F 8 B A 9 4 7 6 5 G 3 2 1 0
D E F C 9 8 B A 5 G 7 6 1 0 3 2 4
E F C D A 9 8 B 6 5 4 7 2 G 0 3 1
G C D E B A 9 8 7 6 5 4 3 2 1 0 F
F 3 8 5 6 2 E C D 4 B A 0 1 7 9 G
```

Partial square

```
. . . . . . . . . . . . . . . . . .
. . . . . 4 . . . 8 . . . . . C A
. 1 0 . . 7 . . . B 8 . . . C D .
. . 1 0 . . 7 4 . . G 8 . C D E .
. . . . 0 . 2 3 . . . . 8 9 A B 5
. 4 7 6 3 0 1 2 . . . C . . 9 A D
. . 4 7 2 3 0 1 . . C D A B 8 G .
. . . 4 1 G 3 0 . C D E 9 A B 8 .
. . . . . . . . 0 1 2 3 4 5 6 7 .
. 8 B A . . . G 3 0 1 2 7 4 5 6 C
. . G B . . . D 2 3 0 1 6 7 4 5 .
. . . 8 . C D E 1 2 3 0 5 . G 4 7
. . . . 8 B A 9 4 7 6 . G 3 2 1 .
. . . C 9 8 B A 5 G 7 6 1 0 3 2 .
. . C D A 9 . B 6 5 4 7 2 G 0 . .
. C D E B . 9 . 7 6 5 4 . . . . F
. 3 . . 6 2 E . . . B . . . . 9 .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E069: order 17, 158 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m4_seed6.json`

SHA256: `eee67b631b305ff3029eb2f51301e247bbda2fc908516172b50a0d495d731e90`

Completion

```
0 G 2 1 4 5 6 7 8 9 A B C D E F 3
1 0 3 2 7 4 5 6 B 8 9 G D E F C A
2 1 0 3 G 7 4 5 A B 8 9 E F C D 6
3 2 1 0 5 6 7 4 9 A G 8 F C D E B
4 7 6 G 0 1 2 3 C D E F 8 9 A B 5
5 4 7 6 3 0 1 2 G E F C B 8 9 A D
6 5 4 7 2 3 0 1 E F C D A B 8 G 9
7 6 5 4 1 G 3 0 F C D E 9 A B 8 2
8 B A 9 C D G F 0 1 2 3 4 5 6 7 E
9 8 B A D E F G 3 0 1 2 7 4 5 6 C
A 9 G B E F C D 2 3 0 1 6 7 4 5 8
B A 9 8 F C D E 1 2 3 0 5 6 G 4 7
C D E F 8 B A 9 4 7 6 5 G 3 2 1 0
D E F C 9 8 B A 5 G 7 6 1 0 3 2 4
E F C D A 9 8 B 6 5 4 7 2 G 0 3 1
G C D E B A 9 8 7 6 5 4 3 2 1 0 F
F 3 8 5 6 2 E C D 4 B A 0 1 7 9 G
```

Partial square

```
. G . . . . . . . . . . . . . . . .
. 0 . . . 4 . . . 8 . . . . . C A .
. 1 0 . . 7 4 . . B 8 . . . C D . .
. 2 1 0 . 6 7 4 . . G . . C D E B .
. . . . 0 1 2 3 . . . . 8 9 A B . .
. 4 7 . 3 0 1 2 G . . C . . 9 A . .
. . 4 . 2 . 0 1 . . C D A B 8 G 9 .
. . . 4 1 G 3 0 . . D E 9 A B 8 . .
. . . . . . . . 0 1 2 3 4 5 6 7 . .
. 8 B A . . . G 3 0 1 2 7 4 5 6 C .
. . G B . . C D 2 3 0 1 . 7 4 5 . .
. . . 8 . C D E 1 2 3 0 5 6 G 4 . .
. . . . 8 B . 9 4 7 6 5 G 3 2 1 . .
. . . C 9 8 . A 5 . 7 6 1 0 3 2 4 .
. . C D A 9 . B 6 5 4 7 2 G 0 3 . .
. . D E B . . 8 7 6 5 . 3 . . . F .
. . . 5 6 . E . . . B . . . . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E070: order 17, 154 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m4_seed4.json`

SHA256: `7ce35697bcf750d9fe3ef1d9f0cf95d1514c55dee0765290b769c85eda9db230`

Completion

```
0 G 2 1 4 5 6 7 8 9 A B C D E F 3
1 0 3 2 7 4 5 6 B 8 9 G D E F C A
2 1 0 3 G 7 4 5 A B 8 9 E F C D 6
3 2 1 0 5 6 7 4 9 A G 8 F C D E B
4 7 6 G 0 1 2 3 C D E F 8 9 A B 5
5 4 7 6 3 0 1 2 G E F C B 8 9 A D
6 5 4 7 2 3 0 1 E F C D A B 8 G 9
7 6 5 4 1 G 3 0 F C D E 9 A B 8 2
8 B A 9 C D G F 0 1 2 3 4 5 6 7 E
9 8 B A D E F G 3 0 1 2 7 4 5 6 C
A 9 G B E F C D 2 3 0 1 6 7 4 5 8
B A 9 8 F C D E 1 2 3 0 5 6 G 4 7
C D E F 8 B A 9 4 7 6 5 G 3 2 1 0
D E F C 9 8 B A 5 G 7 6 1 0 3 2 4
E F C D A 9 8 B 6 5 4 7 2 G 0 3 1
G C D E B A 9 8 7 6 5 4 3 2 1 0 F
F 3 8 5 6 2 E C D 4 B A 0 1 7 9 G
```

Partial square

```
. . . . . . . . . . . . . . . . . 3
. . . . . 4 . . . . . G . . . C . .
. . 0 . G 7 . . . B 8 . . . C D 6 .
. . 1 0 . . 7 4 . A . . . C D E B .
. . . . 0 . 2 3 . . . . 8 9 A B . .
. 4 7 . 3 0 1 2 G . . C . 8 9 A D .
. . 4 . 2 3 0 1 . . C D A B 8 . 9 .
. . . . 1 . 3 0 . . D E 9 A B 8 . .
. . . . . G . 0 1 2 3 4 5 6 7 E . .
. 8 B A . . . . 3 0 1 2 7 4 5 6 . .
. . . B . . . D 2 . 0 1 6 7 4 5 . .
. . . 8 . C D E 1 2 3 0 5 . . 4 7 .
. . . . 8 B A 9 4 7 6 . . 3 2 1 . .
. . . C 9 8 . A 5 . 7 6 1 0 3 2 4 .
. . C D A 9 . B 6 5 4 7 2 . 0 . . .
G . D E B A 9 . 7 6 . . 3 . . . F .
F . 8 5 6 2 . C D . . A 0 1 . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E071: order 17, 153 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m4_seed5.json`

SHA256: `259a530b9aed75a04fa36d54267f05363e21b34c038911848b323b1673b21fe9`

Completion

```
0 G 2 1 4 5 6 7 8 9 A B C D E F 3
1 0 3 2 7 4 5 6 B 8 9 G D E F C A
2 1 0 3 G 7 4 5 A B 8 9 E F C D 6
3 2 1 0 5 6 7 4 9 A G 8 F C D E B
4 7 6 G 0 1 2 3 C D E F 8 9 A B 5
5 4 7 6 3 0 1 2 G E F C B 8 9 A D
6 5 4 7 2 3 0 1 E F C D A B 8 G 9
7 6 5 4 1 G 3 0 F C D E 9 A B 8 2
8 B A 9 C D G F 0 1 2 3 4 5 6 7 E
9 8 B A D E F G 3 0 1 2 7 4 5 6 C
A 9 G B E F C D 2 3 0 1 6 7 4 5 8
B A 9 8 F C D E 1 2 3 0 5 6 G 4 7
C D E F 8 B A 9 4 7 6 5 G 3 2 1 0
D E F C 9 8 B A 5 G 7 6 1 0 3 2 4
E F C D A 9 8 B 6 5 4 7 2 G 0 3 1
G C D E B A 9 8 7 6 5 4 3 2 1 0 F
F 3 8 5 6 2 E C D 4 B A 0 1 7 9 G
```

Partial square

```
. G . . . . . . . . . . . . . . 3
. . . . . 4 . . . 8 . . . . . C A
. 1 0 . . 7 . . . B 8 . . . C D .
. 2 1 0 . 6 7 4 . . . 8 . C D E .
. . . . 0 1 2 3 . . . . 8 9 A B 5
. 4 7 . 3 0 1 2 G . . . B . 9 A .
. . 4 . 2 . 0 1 . . C D A B 8 . 9
. . . . 1 . 3 0 . C D E 9 A B 8 .
. . . . . . G . 0 1 2 3 4 5 6 7 .
. 8 B A . . . . 3 0 1 2 7 4 5 6 C
. . G B . . C D 2 3 0 . . 7 4 5 .
. . . 8 . C D E 1 2 3 0 5 . . 4 .
. . . . 8 B . 9 4 7 6 5 G 3 2 1 .
. . . C 9 8 . A 5 . 7 6 1 0 3 2 .
. . C D A 9 . B 6 5 4 7 2 G 0 3 .
. C D E B A 9 . 7 6 . . . . . . F
F . . 5 . . . . . . . A . . 7 9 G
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E072: order 17, 152 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m4_seed1.json`

SHA256: `949a50a6df95664fee37f37282a779f870acc58fe59494bdc4efdb92cdbc7e3f`

Completion

```
0 G 2 1 4 5 6 7 8 9 A B C D E F 3
1 0 3 2 7 4 5 6 B 8 9 G D E F C A
2 1 0 3 G 7 4 5 A B 8 9 E F C D 6
3 2 1 0 5 6 7 4 9 A G 8 F C D E B
4 7 6 G 0 1 2 3 C D E F 8 9 A B 5
5 4 7 6 3 0 1 2 G E F C B 8 9 A D
6 5 4 7 2 3 0 1 E F C D A B 8 G 9
7 6 5 4 1 G 3 0 F C D E 9 A B 8 2
8 B A 9 C D G F 0 1 2 3 4 5 6 7 E
9 8 B A D E F G 3 0 1 2 7 4 5 6 C
A 9 G B E F C D 2 3 0 1 6 7 4 5 8
B A 9 8 F C D E 1 2 3 0 5 6 G 4 7
C D E F 8 B A 9 4 7 6 5 G 3 2 1 0
D E F C 9 8 B A 5 G 7 6 1 0 3 2 4
E F C D A 9 8 B 6 5 4 7 2 G 0 3 1
G C D E B A 9 8 7 6 5 4 3 2 1 0 F
F 3 8 5 6 2 E C D 4 B A 0 1 7 9 G
```

Partial square

```
. G . . . . . . . . . . . . . . . 3
. . . . . 4 . . . 8 . G . . . C . .
. 1 0 . G 7 . . . B 8 . . . C D . .
. 2 1 0 . . 7 4 . A . . . C D E B .
. . . . 0 1 2 3 . . . . 8 9 A B . .
. 4 7 . 3 0 1 2 G . . C . . 9 A D .
. . 4 7 . 3 0 1 . . C D A B 8 . . .
. . . . 1 . 3 0 . . D E 9 A B 8 . .
. . . . . G . 0 1 2 3 4 5 6 7 E . .
. 8 B A . . . . 3 0 1 2 7 4 5 6 C .
. . . B . . C D 2 . 0 . 6 7 4 5 . .
. . . 8 . C D . 1 2 3 0 5 6 . 4 7 .
. . . . 8 B . 9 4 7 6 . . 3 2 1 . .
. . . C 9 8 . A 5 . 7 6 1 0 3 2 . .
. . C D A 9 . B 6 5 4 7 2 . 0 . . .
G C D E . A 9 . 7 . . 4 . . . . F .
. 3 . . 6 2 E . D 4 B . . . . . G .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E073: order 17, 148 clues

Status I. Previously independently checked uniqueness and all clue-deletion witnesses; uniqueness is solver-dependent.

Source: `outputs/lcs_odd_prolongation/examples/m8_seed5.json`

SHA256: `94f6ce316b24f5638433ccc69dec58515e91128e26b023a55143ebd87271c659`

Completion

```
G 7 6 5 4 3 2 1 8 9 A B C D E F 0
1 0 G 6 5 4 3 2 F 8 9 A B C D E 7
2 1 0 7 G 5 4 3 E F 8 9 A B C D 6
3 2 1 0 7 6 G 4 D E F 8 9 A B C 5
4 3 2 1 0 7 6 5 G D E F 8 9 A B C
5 4 3 2 1 0 7 6 B C G E F 8 9 A D
6 5 4 3 2 1 0 7 A B C D G F 8 9 E
7 6 5 4 3 2 1 0 9 A B C D E G 8 F
8 F E D C B A 9 0 G 2 3 4 5 6 7 1
9 8 F E D C B A 7 0 1 G 3 4 5 6 2
A 9 8 F E D C B 6 7 0 1 2 G 4 5 3
B A 9 8 F E D C 5 6 7 0 1 2 3 G 4
C G A 9 8 F E D 4 5 6 7 0 1 2 3 B
D C B G 9 8 F E 3 4 5 6 7 0 1 2 A
E D C B A G 8 F 2 3 4 5 6 7 0 1 9
F E D C B A 9 G 1 2 3 4 5 6 7 0 8
0 B 7 A 6 9 5 8 C 1 D 2 E 3 F 4 G
```

Partial square

```
G 7 6 5 4 3 . 1 8 9 . . . . . . .
. . . 6 5 4 3 2 F 8 9 A B C D E 7
. . . 7 G 5 4 3 E F 8 9 A B C D .
. . . . 7 6 G 4 D E F 8 9 A B C .
. . . . . 7 6 5 . D E F 8 9 A B C
. . . . . 7 6 B C . E F 8 . A D
. . . . . 7 A B C D . F 8 9 E
. . . . . 9 A B C D E G 8 F
. . E D C B A . . G 2 3 4 5 6 7 .
. . F E D C B A . . 1 . 3 4 5 6 .
. . . F E D C . . . . 1 2 . 4 5 3
. . . . F E D . . . . . 1 2 3 . 4
. G . . . F E D . . . . . 1 2 3 B
. . . . . F E . . . . . 1 2 A
. . . . . G . . . . . 1 9
. . . . . .
0 B . . . 9 . 8 C . D 2 . . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E074: order 17, 147 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m8_seed3.json`

SHA256: `b0f93468076947a788ea99cec0d81eded610cb2d0b25854bf724afb6a338edfc`

Completion

```
G 7 6 5 4 3 2 1 8 9 A B C D E F 0
1 0 G 6 5 4 3 2 F 8 9 A B C D E 7
2 1 0 7 G 5 4 3 E F 8 9 A B C D 6
3 2 1 0 7 6 G 4 D E F 8 9 A B C 5
4 3 2 1 0 7 6 5 G D E F 8 9 A B C
5 4 3 2 1 0 7 6 B C G E F 8 9 A D
6 5 4 3 2 1 0 7 A B C D G F 8 9 E
7 6 5 4 3 2 1 0 9 A B C D E G 8 F
8 F E D C B A 9 0 G 2 3 4 5 6 7 1
9 8 F E D C B A 7 0 1 G 3 4 5 6 2
A 9 8 F E D C B 6 7 0 1 2 G 4 5 3
B A 9 8 F E D C 5 6 7 0 1 2 3 G 4
C G A 9 8 F E D 4 5 6 7 0 1 2 3 B
D C B G 9 8 F E 3 4 5 6 7 0 1 2 A
E D C B A G 8 F 2 3 4 5 6 7 0 1 9
F E D C B A 9 G 1 2 3 4 5 6 7 0 8
0 B 7 A 6 9 5 8 C 1 D 2 E 3 F 4 G
```

Partial square

```
. 7 6 5 . . . 1 8 9 A B . . . . 0
. . . 6 5 4 3 2 F 8 9 A B C D E .
. . . 7 G 5 4 3 E F 8 9 A B C D .
. . . . 7 6 G 4 D E F 8 9 . B C .
. . . . . 7 6 5 G D E F 8 9 A B .
. . . . . 7 6 B C G E F 8 . A .
. . . . . 7 A B C D G F 8 9 .
. . . . . 9 A B C D E G . .
. F E D C B A 9 . G 2 3 4 5 6 7 .
. . F E D C B A . . 1 G 3 4 5 6 .
. . . F E D C B . . . 1 2 G 4 5 .
. . . . F E D . . . . . 1 2 3 G 4
. . . . . F E D . . . . . 1 2 3 .
. . . . . F E . . . . . 1 2 A
. . . . . F . . . . . 1 9
. . . . . 8
. B 7 . 6 . 5 . . . . . 3 . 4 .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E075: order 17, 147 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m8_seed6.json`

SHA256: `ce617a487bfc2f9d48485fb22e560eb580baeb1f922ba4adad7c07ad79653434`

Completion

```
G 7 6 5 4 3 2 1 8 9 A B C D E F 0
1 0 G 6 5 4 3 2 F 8 9 A B C D E 7
2 1 0 7 G 5 4 3 E F 8 9 A B C D 6
3 2 1 0 7 6 G 4 D E F 8 9 A B C 5
4 3 2 1 0 7 6 5 G D E F 8 9 A B C
5 4 3 2 1 0 7 6 B C G E F 8 9 A D
6 5 4 3 2 1 0 7 A B C D G F 8 9 E
7 6 5 4 3 2 1 0 9 A B C D E G 8 F
8 F E D C B A 9 0 G 2 3 4 5 6 7 1
9 8 F E D C B A 7 0 1 G 3 4 5 6 2
A 9 8 F E D C B 6 7 0 1 2 G 4 5 3
B A 9 8 F E D C 5 6 7 0 1 2 3 G 4
C G A 9 8 F E D 4 5 6 7 0 1 2 3 B
D C B G 9 8 F E 3 4 5 6 7 0 1 2 A
E D C B A G 8 F 2 3 4 5 6 7 0 1 9
F E D C B A 9 G 1 2 3 4 5 6 7 0 8
0 B 7 A 6 9 5 8 C 1 D 2 E 3 F 4 G
```

Partial square

```
. 7 6 5 . 3 . 1 8 9 A B C D E F .
. . . 6 5 4 3 2 F 8 9 A . C D E 7
. . . 7 G 5 4 3 E F 8 9 A B C D .
. . . . 7 6 G 4 D E F 8 9 . B C .
. . . . . 7 6 5 G D E F 8 9 A B .
. . . . . 7 6 B C G E F 8 . A .
. . . . . 7 A B C D G F 8 9 .
. . . . . 9 A B C D E . 8 .
. F E D C B A 9 . . 2 3 4 5 6 7 .
. . F E D C B A . . 1 . 3 4 5 6 2
. . . F E D C . . . . 1 2 . 4 5 3
. . . . F E D C . . . . 1 2 3 . 4
. . . . . F E D . . . . . 1 2 3 B
. . . . . F E . . . . . 1 2 A
. . . . . F . . . . . 1 9
. . . . . .
0 B . . . . . 3 . . G
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E076: order 17, 145 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m8_seed2.json`

SHA256: `6c7b90b03f1c376babdbf03f8ef381a9f6e54464b304b92929a96093b0612ec5`

Completion

```
G 7 6 5 4 3 2 1 8 9 A B C D E F 0
1 0 G 6 5 4 3 2 F 8 9 A B C D E 7
2 1 0 7 G 5 4 3 E F 8 9 A B C D 6
3 2 1 0 7 6 G 4 D E F 8 9 A B C 5
4 3 2 1 0 7 6 5 G D E F 8 9 A B C
5 4 3 2 1 0 7 6 B C G E F 8 9 A D
6 5 4 3 2 1 0 7 A B C D G F 8 9 E
7 6 5 4 3 2 1 0 9 A B C D E G 8 F
8 F E D C B A 9 0 G 2 3 4 5 6 7 1
9 8 F E D C B A 7 0 1 G 3 4 5 6 2
A 9 8 F E D C B 6 7 0 1 2 G 4 5 3
B A 9 8 F E D C 5 6 7 0 1 2 3 G 4
C G A 9 8 F E D 4 5 6 7 0 1 2 3 B
D C B G 9 8 F E 3 4 5 6 7 0 1 2 A
E D C B A G 8 F 2 3 4 5 6 7 0 1 9
F E D C B A 9 G 1 2 3 4 5 6 7 0 8
0 B 7 A 6 9 5 8 C 1 D 2 E 3 F 4 G
```

Partial square

```
. 7 6 5 4 3 2 1 . . . . . . E F .
. . . 6 5 4 3 2 F 8 9 A . C D E .
. . . 7 . 5 4 3 E F 8 9 A B C D .
. . . . 7 6 . 4 D E F 8 9 . B C .
. . . . . 7 6 5 G D E F 8 9 A B .
. . . . . 7 6 B C G E F 8 . A D
. . . . . 7 A B C D . F 8 9 .
. . . . . 9 A B C D E . . F
. F E D C B A . . . 2 3 4 5 6 7 1
. . F E D C B . . . 1 . 3 4 5 6 2
. . . F E D C . . . . 1 2 . 4 5 3
. . . . F E D C . . . . 1 2 3 G 4
. G . . . F E D . . . . . 1 2 3 .
. . . G . . F E . . . . . 1 2 A
. . . . . G . . . . . . . . 1 9
. . . . . . . . . . . . . . . 8
0 . 7 . 6 . 5 8 . . . . E 3 . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E077: order 17, 143 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m8_seed0.json`

SHA256: `cc4d408c38ac1aa347d64897e56410a087b7fdae61b9919ece893c620321ad48`

Completion

```
G 7 6 5 4 3 2 1 8 9 A B C D E F 0
1 0 G 6 5 4 3 2 F 8 9 A B C D E 7
2 1 0 7 G 5 4 3 E F 8 9 A B C D 6
3 2 1 0 7 6 G 4 D E F 8 9 A B C 5
4 3 2 1 0 7 6 5 G D E F 8 9 A B C
5 4 3 2 1 0 7 6 B C G E F 8 9 A D
6 5 4 3 2 1 0 7 A B C D G F 8 9 E
7 6 5 4 3 2 1 0 9 A B C D E G 8 F
8 F E D C B A 9 0 G 2 3 4 5 6 7 1
9 8 F E D C B A 7 0 1 G 3 4 5 6 2
A 9 8 F E D C B 6 7 0 1 2 G 4 5 3
B A 9 8 F E D C 5 6 7 0 1 2 3 G 4
C G A 9 8 F E D 4 5 6 7 0 1 2 3 B
D C B G 9 8 F E 3 4 5 6 7 0 1 2 A
E D C B A G 8 F 2 3 4 5 6 7 0 1 9
F E D C B A 9 G 1 2 3 4 5 6 7 0 8
0 B 7 A 6 9 5 8 C 1 D 2 E 3 F 4 G
```

Partial square

```
. 7 6 5 4 3 2 1 . 9 . . . . . F .
. . . 6 5 4 3 2 F 8 9 A B C D E .
. . . 7 . 5 4 3 E F 8 9 A B C D .
. . . . 7 6 . 4 D E F 8 9 A B C .
. . . . . 7 6 5 G D E F 8 9 A B C
. . . . . 7 6 B C G E F 8 9 A D
. . . . . 7 A B C D G F 8 9 E
. . . . . 9 A B C D E G . .
. F E D C B A 9 . . 2 3 4 5 6 7 .
. . F E D C B A . . 1 . 3 4 5 6 .
. . . F E D C . . . . 1 2 G 4 5 .
. . . . F E D . . . . . 1 2 3 . .
. . . . . F E D . . . . . 1 2 3 B
. . . . . F E . . . . . 1 2 A
. . . . . F . . . . . 1 .
. . . . . 8
0 B . . . . . C . . . . . G
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E078: order 17, 140 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m8_seed1.json`

SHA256: `4cdbffc9f0ee5dd4b739aa1051731349c049d5c4e56189d4c1a966a651b08a59`

Completion

```
G 7 6 5 4 3 2 1 8 9 A B C D E F 0
1 0 G 6 5 4 3 2 F 8 9 A B C D E 7
2 1 0 7 G 5 4 3 E F 8 9 A B C D 6
3 2 1 0 7 6 G 4 D E F 8 9 A B C 5
4 3 2 1 0 7 6 5 G D E F 8 9 A B C
5 4 3 2 1 0 7 6 B C G E F 8 9 A D
6 5 4 3 2 1 0 7 A B C D G F 8 9 E
7 6 5 4 3 2 1 0 9 A B C D E G 8 F
8 F E D C B A 9 0 G 2 3 4 5 6 7 1
9 8 F E D C B A 7 0 1 G 3 4 5 6 2
A 9 8 F E D C B 6 7 0 1 2 G 4 5 3
B A 9 8 F E D C 5 6 7 0 1 2 3 G 4
C G A 9 8 F E D 4 5 6 7 0 1 2 3 B
D C B G 9 8 F E 3 4 5 6 7 0 1 2 A
E D C B A G 8 F 2 3 4 5 6 7 0 1 9
F E D C B A 9 G 1 2 3 4 5 6 7 0 8
0 B 7 A 6 9 5 8 C 1 D 2 E 3 F 4 G
```

Partial square

```
G 7 6 5 4 3 2 . . . . . . . . . 0
. . . 6 5 4 3 2 F 8 9 A . C D E .
. . . 7 . 5 4 3 E F 8 9 A B C D 6
. . . . 7 6 . 4 D E F 8 9 . B C 5
. . . . . 7 6 5 . D E F 8 9 A B C
. . . . . 7 6 B C . E F 8 . A D
. . . . . 7 A B C D . F 8 9 .
. . . . . 9 A B C D E . . F
. . E D C B A . . . 2 3 4 5 6 7 .
. . F E D C B . . . 1 . 3 4 5 6 .
. . . F E D C . . . . 1 2 . 4 5 .
. . . . F E D . . . . . 1 2 3 . 4
. G . . . F E . . . . . 1 2 3 .
. . . G . . F . . . . . . 1 2 A
. . . . . F . . . . . . . 1 9
. . . . . G . . . . . . . . 8
0 B 7 A . . . 8 . 1 . 2 E 3 . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E079: order 17, 139 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m8_seed4.json`

SHA256: `c172edada3a7513bf1e5d434315a6351491326d624961378e8a6120254ad7032`

Completion

```
G 7 6 5 4 3 2 1 8 9 A B C D E F 0
1 0 G 6 5 4 3 2 F 8 9 A B C D E 7
2 1 0 7 G 5 4 3 E F 8 9 A B C D 6
3 2 1 0 7 6 G 4 D E F 8 9 A B C 5
4 3 2 1 0 7 6 5 G D E F 8 9 A B C
5 4 3 2 1 0 7 6 B C G E F 8 9 A D
6 5 4 3 2 1 0 7 A B C D G F 8 9 E
7 6 5 4 3 2 1 0 9 A B C D E G 8 F
8 F E D C B A 9 0 G 2 3 4 5 6 7 1
9 8 F E D C B A 7 0 1 G 3 4 5 6 2
A 9 8 F E D C B 6 7 0 1 2 G 4 5 3
B A 9 8 F E D C 5 6 7 0 1 2 3 G 4
C G A 9 8 F E D 4 5 6 7 0 1 2 3 B
D C B G 9 8 F E 3 4 5 6 7 0 1 2 A
E D C B A G 8 F 2 3 4 5 6 7 0 1 9
F E D C B A 9 G 1 2 3 4 5 6 7 0 8
0 B 7 A 6 9 5 8 C 1 D 2 E 3 F 4 G
```

Partial square

```
G 7 6 . . . . . 8 9 A B C . . . 0
. . . 6 5 4 3 2 F 8 9 A . C D E .
. . . 7 . 5 4 3 E F 8 9 A B C D .
. . . . 7 6 . 4 D E F 8 9 . B C .
. . . . . 7 6 5 . D E F 8 9 A B .
. . . . . 7 6 B C . E F 8 . A .
. . . . . 7 A B C D . F 8 9 .
. . . . . 9 A B C D E . . .
. F E D C B A . . . 2 3 4 5 6 7 .
. . F E D C B . . . 1 . 3 4 5 6 .
. . . F E D C . . . . 1 2 . 4 5 .
. . . . F E D . . . . . 1 2 3 . .
. . . . . F E . . . . . 1 2 3 B
. . . G . . F . . . . . 1 2 A
. . . . . 1 9
. . . . . G . . . . . 8
. B 7 A 6 9 5 . C . D . E 3 F 4 G
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E080: order 18, 196 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_full_review/stripping/pruned_direct_n18_seed0.json`

SHA256: `3156c7673c27d2128c01ec30d07bc04a75807e16722f77d724251a085ca097ce`

Completion

```
8 1 2 3 4 5 7 6 0 H A B C D E G F 9
1 H 3 9 F G E 7 B A 8 C 0 6 D 5 4 2
2 3 0 1 7 6 4 8 5 B C 9 A G F D H E
3 8 1 0 6 7 5 4 2 C H A 9 F G E D B
4 5 7 6 0 8 2 3 1 D E G F 9 H B C A
5 4 6 8 1 0 3 2 7 E D F H A 9 C B G
7 6 8 5 2 3 0 1 4 G F H E B C 9 A D
6 7 5 4 8 2 1 0 3 F G E D H B A 9 C
0 2 4 7 3 1 6 5 8 9 B D G C A F E H
H A B C D E G F 9 8 1 2 3 4 5 7 6 0
A 9 C B E D H G F 2 0 3 1 5 4 8 7 6
B C 9 A G F D H E 1 3 0 2 7 6 4 8 5
C 0 A 2 5 B 8 D 6 3 9 1 4 E 7 H G F
D E G F 9 H B C A 4 5 7 6 0 8 2 3 1
E D F H A 9 C B G 5 4 6 8 1 0 3 2 7
G F H E B C 9 A D 7 6 8 5 2 3 0 1 4
F G E D H 4 A 9 C 6 7 5 B 8 2 1 0 3
9 B D G C A F E H 0 2 4 7 3 1 6 5 8
```

Partial square

```
8 . 2 3 4 5 7 6 . . A B C D E . F 9
. . 3 9 F . E 7 . A 8 C 0 6 . 5 . .
2 . . 1 7 . 4 8 . B C . . G F D . E
3 8 . . 6 . 5 . . C . . 9 . G E D B
4 5 . 6 0 8 2 . . D . G F 9 . . . A
5 4 6 8 1 0 3 2 . E . F . A 9 C B G
7 . . . 2 3 0 . . G . . E . C . A D
6 7 5 4 8 2 . 0 . F . . D . B . 9 C
. 2 4 . . 1 6 5 . 9 B D G C . . . .
. A B . . . G . . 8 1 . . 4 5 7 . 0
A 9 C B E . . . . . 0 . . 5 4 8 . 6
B C 9 A G F D . . . 3 . 2 7 6 4 8 .
C 0 A . 5 . 8 . . . 9 1 4 E . . G F
D . G F 9 . B C . 4 5 . 6 . 8 2 3 1
E D . . A 9 C B . 5 4 6 . 1 . 3 2 7
G . . E B C 9 A . 7 6 8 5 2 3 0 1 4
F G E D . . . 9 . 6 . . . . . 1 . 3
. . . . . . . . . H . . . . . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E081: order 19, 218 clues

Status I. Previously independently verified example; see the archived certificate directory.

Source: `outputs/lcs_order19_verification/construction.json`

SHA256: `53aca2ca633342ef2f95798c4a3e951b277bac50ca61964a2b23aa71ad80d32f`

Completion

```
G 1 2 3 4 5 6 7 8 H A I C D E F 0 9 B
1 0 G 2 5 4 7 6 9 8 B A I C H E 3 F D
2 3 0 G 6 7 4 H A B 8 9 E F I D 1 5 C
3 G 1 I 7 6 5 4 B A 9 8 F H D C 2 E 0
H 5 6 7 0 1 2 3 G D I F 8 9 A B C 4 E
I 4 7 6 1 0 3 2 D C G E 9 8 B H F A 5
6 I 4 5 2 3 H 1 E F C G A B 8 9 D 0 7
7 6 5 4 3 H 1 0 F G D C B I 9 8 E 2 A
8 9 A B C D E I 0 1 2 H G 5 6 7 4 3 F
9 8 B A D C F E I 0 3 2 5 4 G 6 7 1 H
A B H 9 E F C D 2 3 0 1 6 7 4 G 5 I 8
B A 9 8 F E D C 3 2 1 0 H G 5 4 I 7 6
C D E F I 9 A B 4 5 H 7 0 1 2 3 8 6 G
D H F E 9 8 G A 5 I 7 6 1 0 3 2 B C 4
E F C H A B 8 G 6 7 4 5 2 3 0 I 9 D 1
F E D C H G I 8 7 6 5 4 3 2 1 0 A B 9
0 2 I 1 8 A B 9 C E F D 4 6 7 5 H G 3
4 C 8 D B I 0 5 1 9 6 3 7 E F A G H 2
5 7 3 0 G 2 9 F H 4 E B D A C 1 6 8 I
```

Partial square

```
G . . 3 4 5 6 7 . H . I C D . F 0 9 B
. 0 . 2 5 4 7 . . . B A I . . E 3 F D
2 3 . G 6 7 4 H . B 8 . E F I D . 5 C
3 G . I 7 6 5 4 . A . . . H D . 2 E .
. 5 6 7 0 . . 3 . D . F 8 9 . B C 4 .
I 4 7 6 . . 3 2 . C . . . . B H F . 5
6 I 4 5 2 3 . . . . . G A . 8 9 D 0 7
7 6 5 4 3 H . 0 . G D C B I 9 . . 2 A
8 9 A . . D . I . . 2 . . 5 . 7 4 3 F
. . . . . .
A B H 9 E . C D . 3 . . 6 7 4 G 5 I 8
. A 9 . F E D C . 2 . 0 H G 5 4 I 7 6
C . E . I 9 A B . . . 7 0 . 2 . . 6 G
D . F E 9 . G A . . . . . 0 3 2 . C 4
E F C H . . . G . . . 5 2 3 0 I 9 D .
F E . . . . . 6 5 . . 2 . . . . .
. 2 I . . A B 9 . E . . 4 . 7 5 H G .
4 . . D B I . 5 . . . 3 7 E F A G H .
5 7 3 . G . 9 F . . . B . A C . 6 . I
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E082: order 19, 195 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_four_m_plus_three/examples/xor4_a3_seed1.json`

SHA256: `4f2c5cfa289990a341ae2bd0059cb1a845107fc3bb86264008df397fed33e980`

Completion

```
G H I 3 4 5 6 7 8 9 A B C D E F 0 1 2
1 0 3 2 5 4 7 6 9 8 B A H G F I C D E
2 I H G 6 7 4 5 A B 8 9 E F C D 1 0 3
3 2 1 0 7 6 5 4 B A 9 8 I E G H D C F
4 5 6 7 I 1 G H C D E F 8 9 A B 2 3 0
5 4 7 6 1 0 3 2 D I H G 9 8 B A E F C
6 7 4 5 H G 0 I E F C D A B 8 9 3 2 1
7 6 5 4 3 2 1 0 G H I C B A 9 8 F E D
8 9 A B C D E F 0 1 2 3 G H I 7 4 5 6
H G B I D C F E 1 0 3 2 5 4 7 6 8 9 A
A B 8 9 E F C D 2 3 0 1 6 I H G 5 4 7
I A G H F E D C 3 2 1 0 7 6 5 4 9 8 B
C D E F 8 9 A B I 5 G H 0 1 2 3 6 7 4
D C F E 9 I H G 5 4 7 6 1 0 3 2 A B 8
E F C D A B 8 9 H G 4 I 2 3 0 1 7 6 5
F E D C G H I 8 7 6 5 4 3 2 1 0 B A 9
0 8 9 1 B 3 2 A F 7 6 E 4 C D 5 G I H
9 1 0 8 2 A B 3 6 E F 7 D 5 4 C I H G
B 3 2 A 0 8 9 1 4 C D 5 F 7 6 E H G I
```

Partial square

```
G H . . . . . . . . . . . . . . . 0 1 2
. . . 2 . 4 . 6 . 8 . A . G . I C . .
. . H . . . 4 5 . . 8 9 . . C D 1 . .
. 2 1 0 . 6 5 4 . A 9 8 . . . . D C F
. . . . I 1 G . . . . . 8 9 A B . 3 .
. 4 . 6 1 0 3 2 . I . . . 8 B . . . C
. . 4 5 H . 0 I . . C D A B 8 9 3 . .
. 6 5 4 3 2 1 0 G H . C . . . . F . D
. . . . . . . . . 0 1 2 3 G . I 7 . . .
H . . I . C . E 1 0 3 2 . 4 7 6 . 9 .
. . 8 9 . . C D 2 3 0 1 6 I H . 5 . 7
I . G H . E D C . . . . . 6 5 4 . . B
. . . . 8 9 A B I 5 G . 0 . 2 3 . 7 4
. C . E . I H . 5 4 7 6 . . 3 2 A . .
. . C D A B 8 9 . . 4 I 2 3 0 . 7 . 5
. E D C . H I 8 7 6 5 4 . 2 1 0 . . 9
. 8 9 1 B . . . F . 6 . . . . 5 G . .
. 1 . 8 . . . 3 6 . F 7 D 5 4 . . . .
B 3 . A 0 . . 1 . C . 5 . 7 6 . H . I
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E083: order 19, 195 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_four_m_plus_three/examples/xor4_a3_seed0.json`

SHA256: `83de37470932700275e02bdd8ea47c9518119302ae0b49f3dc058bd09fc179aa`

Completion

```
G H I 3 4 5 6 7 8 9 A B C D E F 0 1 2
1 0 3 2 5 4 7 6 9 8 B A H G F I C D E
2 I H G 6 7 4 5 A B 8 9 E F C D 1 0 3
3 2 1 0 7 6 5 4 B A 9 8 I E G H D C F
4 5 6 7 I 1 G H C D E F 8 9 A B 2 3 0
5 4 7 6 1 0 3 2 D I H G 9 8 B A E F C
6 7 4 5 H G 0 I E F C D A B 8 9 3 2 1
7 6 5 4 3 2 1 0 G H I C B A 9 8 F E D
8 9 A B C D E F 0 1 2 3 G H I 7 4 5 6
H G B I D C F E 1 0 3 2 5 4 7 6 8 9 A
A B 8 9 E F C D 2 3 0 1 6 I H G 5 4 7
I A G H F E D C 3 2 1 0 7 6 5 4 9 8 B
C D E F 8 9 A B I 5 G H 0 1 2 3 6 7 4
D C F E 9 I H G 5 4 7 6 1 0 3 2 A B 8
E F C D A B 8 9 H G 4 I 2 3 0 1 7 6 5
F E D C G H I 8 7 6 5 4 3 2 1 0 B A 9
0 8 9 1 B 3 2 A F 7 6 E 4 C D 5 G I H
9 1 0 8 2 A B 3 6 E F 7 D 5 4 C I H G
B 3 2 A 0 8 9 1 4 C D 5 F 7 6 E H G I
```

Partial square

```
G . I . . . . . . . . . . . . . . . 2
. 0 . 2 . 4 . 6 . 8 . A . . . . C D E
. I H G . . 4 5 . . 8 9 . . C D 1 . .
. 2 1 . . 6 5 4 . A 9 8 I . . . D C .
. . . . . 1 G H . . . . 8 9 A B . 3 0
. 4 . 6 1 0 3 2 . . H . 9 8 . . . F .
. . 4 5 H . 0 . . . C D A B 8 9 3 . .
. 6 5 4 3 2 1 . G . . C . A . 8 . E .
. . . . . . . . . 0 1 . 3 . H . 7 4 . 6
. . . I . C . E . . 3 2 5 4 7 6 8 . A
. . 8 9 . . C D 2 . 0 1 6 . . G 5 . 7
I . G . . E D C 3 2 1 0 . 6 5 4 . . .
. . . . 8 9 A B . 5 G . 0 . 2 3 . 7 4
. C . E 9 I . . 5 4 . 6 . 0 3 . A B 8
. . C D A B 8 9 . G 4 I 2 3 . 1 . 6 .
. E D C . . . 8 7 6 . 4 3 2 1 . B A .
. 8 . . B 3 . . F 7 6 E 4 . . 5 . I .
9 1 . 8 2 A B . 6 E F 7 D . 4 . . . G
. 3 . A . . 9 1 . C D . . . . E H G .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E084: order 19, 192 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_four_m_plus_three/examples/xor4_a2_seed0.json`

SHA256: `d65982fa86ac8fcb985c65c78cccab094893333bd45bb2114cb8ba781dc636f1`

Completion

```
G H I 3 4 5 6 7 8 9 A B C D E F 0 1 2
1 0 3 2 5 4 7 6 9 8 B A D I H G E F C
2 I H G 6 7 4 5 A B 8 9 E F C D 1 0 3
3 2 1 0 7 6 5 4 B A 9 8 G H I C F E D
4 5 6 7 0 1 2 3 C D E F I 9 G H A B 8
H G 7 I 1 0 3 2 D C F E 9 8 B A 4 5 6
6 7 4 5 2 3 0 1 E F C D H G 8 I B A 9
I 6 G H 3 2 1 0 F E D C B A 9 8 5 4 7
8 9 A B C D E F I 1 G H 4 5 6 7 2 3 0
9 8 B A H G F I 1 0 3 2 5 4 7 6 C D E
A B 8 9 E F C D H G 0 I 6 7 4 5 3 2 1
B A 9 8 I E G H 3 2 1 0 7 6 5 4 D C F
C D E F G H I B 4 5 6 7 0 1 2 3 8 9 A
D C F E 9 8 B A 5 I H G 1 0 3 2 6 7 4
E F C D A I H G 6 7 4 5 2 3 0 1 9 8 B
F E D C B A 9 8 G H I 4 3 2 1 0 7 6 5
0 4 5 1 8 C D 9 7 3 2 6 F B A E G I H
5 1 0 4 D 9 8 C 2 6 7 3 A E F B I H G
7 3 2 6 F B A E 0 4 5 1 8 C D 9 H G I
```

Partial square

```
G . I . . . . . . . . . . . . . . . .
. 0 . . . 4 . 6 . 8 . A . I . . E F C
. I H G . . 4 5 . . 8 9 . . C D . 0 3
. 2 1 0 . 6 5 4 . A 9 8 . . . . F E D
. . . . . 1 2 3 . . . . I 9 G . . B 8
. . . . 1 0 3 . . C . E 9 8 . A 4 5 .
. . 4 5 2 . 0 . . . C D H . 8 . B . .
. . G H . . 1 0 . E D C . A . 8 . 4 7
. . . . . . . . . I 1 . H 4 5 6 7 2 3 .
. 8 . A H . . . 1 . 3 2 5 . 7 6 C . E
. . 8 9 . . C D . G 0 . 6 7 4 5 3 . 1
. A 9 8 I E G H 3 2 1 0 . 6 5 . . . .
. . . . . . I B 4 5 6 7 0 . 2 3 . 9 .
. C . E 9 8 B A . . . . 1 0 . . 6 . 4
. . C D A . . G 6 7 4 5 2 . . 1 . . B
. E D C . A 9 8 G H I 4 3 2 . . 7 . .
. 4 . . 8 C . 9 7 3 2 6 F . . E . I .
5 1 . . . 9 8 . 2 6 7 3 A . F . . . G
. 3 . 6 . . A E . 4 5 . . . . 9 H G .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E085: order 19, 191 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_four_m_plus_three/examples/xor4_a1_seed0.json`

SHA256: `684bf4506474eb08b5cb7f422975ae69430485fec7e52fcf07855780a4bf8f06`

Completion

```
G H I 3 4 5 6 7 8 9 A B C D E F 0 1 2
1 0 3 2 5 4 7 6 9 8 B A I C G H F E D
2 3 0 1 6 7 4 5 A B 8 9 E I H G D C F
H G 1 I 7 6 5 4 B A 9 8 F E D C 2 3 0
4 5 6 7 0 1 2 3 C D E F H G A I 9 8 B
5 I H G 1 0 3 2 D C F E 9 8 B A 6 7 4
I 7 G H 2 3 0 1 E F C D A B 8 9 4 5 6
7 6 5 4 3 2 1 0 F E D C G H I 8 B A 9
8 9 A B C D E F H G 2 I 4 5 6 7 1 0 3
9 8 B A D I H G 1 0 3 2 5 4 7 6 E F C
A B 8 9 I F G H 2 3 0 1 6 7 4 5 C D E
B A 9 8 F E D C G H I 0 7 6 5 4 3 2 1
C D E F G H I B 4 5 6 7 0 1 2 3 8 9 A
D C F E 9 8 B A I 4 G H 1 0 3 2 7 6 5
E F C D A B 8 9 6 I H G 2 3 0 1 5 4 7
F E D C H G 9 I 7 6 5 4 3 2 1 0 A B 8
0 2 4 6 8 A C E 3 1 7 5 B 9 F D G I H
3 1 7 5 B 9 F D 0 2 4 6 8 A C E I H G
6 4 2 0 E C A 8 5 7 1 3 D F 9 B H G I
```

Partial square

```
G . I . . . . . . . . . . . . . 1 2
. 0 . 2 . 4 . 6 . 8 . A . C . H F E .
. . . 1 . . 4 5 . . 8 9 . I H G . . F
H . 1 . . 6 5 4 . A 9 8 . E D C 2 . .
. . . . 0 1 . 3 . . . . H . A . 9 8 .
. . . G 1 0 . . . C . E 9 8 B A 6 . 4
I . G . 2 3 . . . . C D A . 8 9 . . .
. . 5 4 3 2 1 . . E D C G . . 8 B . .
. . . . . . . . . H . 2 . 4 5 6 7 . 0 .
. 8 . A . . . . 1 0 3 2 . 4 . 6 . F C
. . 8 9 I . . H 2 3 0 . . 7 4 5 C D .
. A 9 8 . E D C G . I . 7 6 . 4 3 . .
. . . . . H I B 4 5 6 7 0 . 2 3 8 9 A
. C . E 9 8 B A . 4 . . . 0 . . 7 6 .
. . C D . B 8 9 6 . H G 2 . 0 . . 4 7
. E D C . . . I 7 6 5 4 3 . . . . B 8
0 . 4 6 . A C E . 1 7 5 . 9 F . . . .
3 . 7 . B 9 F D 0 . . . . . . . . H G
6 . . 0 . . A 8 . . 1 . D F 9 . H G I
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E086: order 19, 191 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_four_m_plus_three/examples/xor4_a1_seed1.json`

SHA256: `9dde1d1ef0e966839c69237acdcfdc14ce4375ea61e37200a5e60d5bbba34043`

Completion

```
G H I 3 4 5 6 7 8 9 A B C D E F 0 1 2
1 0 3 2 5 4 7 6 9 8 B A I C G H F E D
2 3 0 1 6 7 4 5 A B 8 9 E I H G D C F
H G 1 I 7 6 5 4 B A 9 8 F E D C 2 3 0
4 5 6 7 0 1 2 3 C D E F H G A I 9 8 B
5 I H G 1 0 3 2 D C F E 9 8 B A 6 7 4
I 7 G H 2 3 0 1 E F C D A B 8 9 4 5 6
7 6 5 4 3 2 1 0 F E D C G H I 8 B A 9
8 9 A B C D E F H G 2 I 4 5 6 7 1 0 3
9 8 B A D I H G 1 0 3 2 5 4 7 6 E F C
A B 8 9 I F G H 2 3 0 1 6 7 4 5 C D E
B A 9 8 F E D C G H I 0 7 6 5 4 3 2 1
C D E F G H I B 4 5 6 7 0 1 2 3 8 9 A
D C F E 9 8 B A I 4 G H 1 0 3 2 7 6 5
E F C D A B 8 9 6 I H G 2 3 0 1 5 4 7
F E D C H G 9 I 7 6 5 4 3 2 1 0 A B 8
0 2 4 6 8 A C E 3 1 7 5 B 9 F D G I H
3 1 7 5 B 9 F D 0 2 4 6 8 A C E I H G
6 4 2 0 E C A 8 5 7 1 3 D F 9 B H G I
```

Partial square

```
. H . . . . . . . . . . . . . . . 0 1 2
. . . 2 . 4 . 6 . 8 . A . C G H . . .
. . 0 . . . 4 5 . . 8 9 . I . G . . F
. G 1 . . 6 5 4 . A 9 8 . E . C 2 3 .
. . . . 0 1 . 3 . . . . H G A . . . .
. I H G 1 . . . . C . E . 8 B A . 7 .
I . G . 2 3 0 . . . C D A B 8 9 . 5 6
. . 5 4 3 2 1 . . E D C G . . 8 . A 9
. . . . . . . . . H G 2 I 4 5 6 7 1 . 3
. 8 . A . I . G 1 . 3 . 5 4 7 6 . F C
. . 8 9 I . . H 2 3 0 . 6 . . 5 C D .
. A 9 8 . . D C . H . . 7 6 5 4 3 2 1
. . . . . . I B 4 5 6 7 0 1 . 3 8 9 .
. C . E 9 8 B A I 4 G . . . 3 2 . . 5
. . C D . B 8 9 6 . . G 2 . . 1 . 4 .
. E D C . G 9 . 7 . 5 4 . . 1 0 . B .
0 2 4 6 . . C E . 1 . . . . . D . . H
3 . 7 5 . . . . 0 . 4 6 8 A C . I . .
6 . . . E . . 8 . 7 . 3 . F 9 . H . I
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E087: order 19, 187 clues

Status I. Previously independently checked uniqueness and all clue-deletion witnesses; uniqueness is solver-dependent.

Source: `outputs/lcs_odd_prolongation/examples/m9_seed0.json`

SHA256: `814ea277e33e927b6fdb9278d1ae18260c247ca85ccc6d84264e6f015cd8b6d5`

Completion

```
9 8 7 6 5 I 3 2 1 0 A B C D E F G H 4
I 0 8 7 6 5 4 3 2 H 9 A B C D E F G 1
2 1 0 8 7 6 5 4 3 G H 9 A B I D E F C
3 2 1 0 I 7 6 5 4 F G H 9 A B C D E 8
4 3 2 1 0 8 7 6 5 E F I H 9 A B C D G
5 4 3 2 1 0 8 7 6 D E F G H 9 I B C A
6 5 4 I 2 1 0 8 7 C D E F G H 9 A B 3
7 6 5 4 3 2 1 I 8 B C D E F G H 9 A 0
8 7 6 5 4 3 2 1 0 A B C D I F G H 9 E
0 H G F E D C B A I 1 2 3 4 5 6 7 8 9
A 9 H G F E I C B 8 0 1 2 3 4 5 6 7 D
B A 9 H G F E D C 7 8 0 1 2 3 4 I 6 5
C I A 9 H G F E D 6 7 8 0 1 2 3 4 5 B
D C B A 9 H G F E 5 I 7 8 0 1 2 3 4 6
E D C B A 9 H G F 4 5 6 I 8 0 1 2 3 7
F E D C B A 9 H G 3 4 5 6 7 8 0 1 I 2
G F E D C B A 9 I 2 3 4 5 6 7 8 0 1 H
H G I E D C B A 9 1 2 3 4 5 6 7 8 0 F
1 B F 3 8 4 D 0 H 9 6 G 7 E C A 5 2 I
```

Partial square

```
9 8 7 6 5 . . . . 0 A B . . . . . H .
. . 8 7 6 5 4 3 2 . 9 A . C D E F G 1
. . . 8 7 6 5 4 3 G H 9 A B . D E F C
. . . . I 7 6 5 4 F G H 9 A B C D E .
. . . . . 8 7 6 5 E F I H 9 A B C D .
. . . . . 8 7 6 D E F G H 9 I B C .
. . . . . 8 7 C D E F G H 9 A B 3
. . . . . I 8 B C D E F G H . A .
. . . . . A B C D I F G H 9 E
0 H G . . . C B A I . 2 . 4 5 6 . . .
. . . G F E . C B . . 1 2 3 4 5 6 7 D
. . . H G F E D C . . . 1 2 3 4 I 6 5
. . . . H G F E D . . . . 1 2 3 4 5 .
. . . . . H G F E . . . . . 1 2 3 4 .
. . . . . H G F . . . . . 1 2 3 7
. . . . . H G . . . . . 1 . 2
. . . . . I . . . . . 1 .
. . I . . . . . F
1 B . . . 4 D . . 9 6 . 7 . . . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E088: order 19, 187 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_four_m_plus_three/examples/xor4_a2_seed1.json`

SHA256: `c8b7ea4c037059d2d53e8dc00c9fe4f3d46dc1479e8fb835fbd5f002c2c1a6ea`

Completion

```
G H I 3 4 5 6 7 8 9 A B C D E F 0 1 2
1 0 3 2 5 4 7 6 9 8 B A D I H G E F C
2 I H G 6 7 4 5 A B 8 9 E F C D 1 0 3
3 2 1 0 7 6 5 4 B A 9 8 G H I C F E D
4 5 6 7 0 1 2 3 C D E F I 9 G H A B 8
H G 7 I 1 0 3 2 D C F E 9 8 B A 4 5 6
6 7 4 5 2 3 0 1 E F C D H G 8 I B A 9
I 6 G H 3 2 1 0 F E D C B A 9 8 5 4 7
8 9 A B C D E F I 1 G H 4 5 6 7 2 3 0
9 8 B A H G F I 1 0 3 2 5 4 7 6 C D E
A B 8 9 E F C D H G 0 I 6 7 4 5 3 2 1
B A 9 8 I E G H 3 2 1 0 7 6 5 4 D C F
C D E F G H I B 4 5 6 7 0 1 2 3 8 9 A
D C F E 9 8 B A 5 I H G 1 0 3 2 6 7 4
E F C D A I H G 6 7 4 5 2 3 0 1 9 8 B
F E D C B A 9 8 G H I 4 3 2 1 0 7 6 5
0 4 5 1 8 C D 9 7 3 2 6 F B A E G I H
5 1 0 4 D 9 8 C 2 6 7 3 A E F B I H G
7 3 2 6 F B A E 0 4 5 1 8 C D 9 H G I
```

Partial square

```
G H . . . . . . . . . . . . . . . 0 1 2
. . . 2 . 4 . 6 . 8 . A . . H G E . .
. I . . . . 4 5 . . 8 9 . . C D 1 0 .
. 2 1 0 . 6 5 4 . A 9 8 . . I . . E D
. . . . 0 1 2 3 . . . . . 9 G H . . .
H G . I . . 3 . . C . E 9 8 B A . . 6
. . 4 5 2 . 0 . . . C D . G 8 I B . .
I . G . . . . 0 . E D C . A . 8 5 . 7
. . . . . . . . . 1 G . 4 5 6 7 . . .
. 8 . A H G . . 1 0 3 . . . . 6 C D E
. . 8 9 . . C D H . 0 I 6 7 4 5 3 . .
. A 9 8 I . . . 3 2 1 0 7 . 5 4 . C F
. . . . . . I B 4 5 6 7 0 . 2 3 . 9 A
. C . E . 8 B A 5 I . . 1 . 3 2 6 . 4
. . C D A . H . 6 7 4 5 2 . 0 . . . B
. E D C B A 9 8 G H . 4 . 2 . 0 7 . .
. 4 5 1 8 . D 9 7 . 2 . . . . . G . .
. 1 . 4 . 9 . . 2 . 7 3 A E . . . . .
. 3 . 6 F . . E . 4 . . . C D . H . I
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E089: order 19, 186 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_four_m_plus_three/near_examples/n19_seed0.json`

SHA256: `d1082bf87ae72a5fb8b435d8b078cd2d040a8d536de70ea43c1a72a3f8283b13`

Completion

```
0 8 7 6 5 4 3 2 1 9 A B C D E F G H I
1 0 I 7 6 5 4 3 2 H 9 A B C D E F G 8
2 1 0 8 I 6 5 4 3 G H 9 A B C D E F 7
3 2 1 0 8 7 I 5 4 F G H 9 A B C D E 6
4 3 2 1 0 8 7 6 I E F G H 9 A B C D 5
5 4 3 2 1 0 8 7 6 D I F G H 9 A B C E
6 5 4 3 2 1 0 8 7 C D E I G H 9 A B F
7 6 5 4 3 2 1 0 8 B C D E F I H 9 A G
8 7 6 5 4 3 2 1 0 A B C D E F G I 9 H
9 H G F E D C B A I 1 2 3 4 5 6 7 8 0
A 9 H G F E D C B 8 0 I 2 3 4 5 6 7 1
B A 9 H G F E D C 7 8 0 1 I 3 4 5 6 2
C B A 9 H G F E D 6 7 8 0 1 2 I 4 5 3
D C B A 9 H G F E 5 6 7 8 0 1 2 3 I 4
E I C B A 9 H G F 4 5 6 7 8 0 1 2 3 D
F E D I B A 9 H G 3 4 5 6 7 8 0 1 2 C
G F E D C I A 9 H 2 3 4 5 6 7 8 0 1 B
H G F E D C B I 9 1 2 3 4 5 6 7 8 0 A
I D 8 C 7 B 6 A 5 0 E 1 F 2 G 3 H 4 9
```

Partial square

```
0 . 7 6 5 . . . . 9 A B C . E F G H I
. . . 7 6 5 4 3 2 H 9 A B C D E F G .
. . . 8 . 6 5 4 3 G H 9 A B C D E F .
. . . . 8 7 I 5 4 F G H 9 A B C D E 6
. . . . . 8 7 6 . E F G H 9 A B C D 5
. . . . . 8 7 6 D I F G H 9 A B C .
. . . . . 8 7 C D E . G H 9 A B F
. . . . . 8 B C D E F I H 9 . .
. . . . . A B C D E F G I 9 .
9 . . F . . . B . I . . . 4 5 . . . 0
. . H G F E D C B . . . 2 3 4 5 6 7 1
. . . H G F E D C . . . 1 . 3 4 5 6 .
. . . . H G F E D . . . . 1 2 I 4 5 .
. . . . . H G F E . . . . . 1 2 3 . 4
. I . . . . H G F . . . . . 1 2 3 D
. . . . . G . . . . . 1 2 .
. . . . . H . . . . . 1 B
. . . . . I . . . . . A
I D 8 . 7 . 6 . . 0 . 1 . 2 . . . 4 .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E090: order 19, 185 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m9_seed3.json`

SHA256: `ecd439aeb50639df7557551409251788343ff8d41d1c7e4f3db169c959def8f0`

Completion

```
9 8 7 6 5 I 3 2 1 0 A B C D E F G H 4
I 0 8 7 6 5 4 3 2 H 9 A B C D E F G 1
2 1 0 8 7 6 5 4 3 G H 9 A B I D E F C
3 2 1 0 I 7 6 5 4 F G H 9 A B C D E 8
4 3 2 1 0 8 7 6 5 E F I H 9 A B C D G
5 4 3 2 1 0 8 7 6 D E F G H 9 I B C A
6 5 4 I 2 1 0 8 7 C D E F G H 9 A B 3
7 6 5 4 3 2 1 I 8 B C D E F G H 9 A 0
8 7 6 5 4 3 2 1 0 A B C D I F G H 9 E
0 H G F E D C B A I 1 2 3 4 5 6 7 8 9
A 9 H G F E I C B 8 0 1 2 3 4 5 6 7 D
B A 9 H G F E D C 7 8 0 1 2 3 4 I 6 5
C I A 9 H G F E D 6 7 8 0 1 2 3 4 5 B
D C B A 9 H G F E 5 I 7 8 0 1 2 3 4 6
E D C B A 9 H G F 4 5 6 I 8 0 1 2 3 7
F E D C B A 9 H G 3 4 5 6 7 8 0 1 I 2
G F E D C B A 9 I 2 3 4 5 6 7 8 0 1 H
H G I E D C B A 9 1 2 3 4 5 6 7 8 0 F
1 B F 3 8 4 D 0 H 9 6 G 7 E C A 5 2 I
```

Partial square

```
9 8 7 6 5 I . 2 . 0 A B . . . . . . .
. . 8 7 6 5 4 . 2 . 9 A B C D E F G 1
. . . 8 7 6 5 4 3 G H 9 A B I D E F C
. . . . I 7 6 5 4 F G H 9 A B C D E .
. . . . . 8 7 6 5 E F I H 9 A B C D .
. . . . . 8 7 6 D E F G H 9 I B C A
. . . . . 8 7 C D E F G H 9 A B 3
. . . . . 8 B C D E F G H 9 A .
. . . . . A B C D I F G H 9 E
0 . G F . D C B . I . . . . . 7 . .
. . H G F E I C B . . 1 2 3 4 5 6 7 .
. . . H G F E D C . . . 1 2 3 4 . 6 .
. . . . H G F E D . . . . 1 2 3 4 5 .
. . . . . H G F E . . . . . 1 2 3 4 6
. . . . . H G F . . . . . 1 2 3 .
. . . . . H G . . . . . 1 . .
. . . . . I . . . . . 1 H
. . . . . .
. . F 3 8 4 D 0 . 9 . . . . . I
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E091: order 19, 184 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m9_seed6.json`

SHA256: `a49e10ae26ec9797295ba320225b9f1b71ef153b4337d8525e890eefe9a9b112`

Completion

```
9 8 7 6 5 I 3 2 1 0 A B C D E F G H 4
I 0 8 7 6 5 4 3 2 H 9 A B C D E F G 1
2 1 0 8 7 6 5 4 3 G H 9 A B I D E F C
3 2 1 0 I 7 6 5 4 F G H 9 A B C D E 8
4 3 2 1 0 8 7 6 5 E F I H 9 A B C D G
5 4 3 2 1 0 8 7 6 D E F G H 9 I B C A
6 5 4 I 2 1 0 8 7 C D E F G H 9 A B 3
7 6 5 4 3 2 1 I 8 B C D E F G H 9 A 0
8 7 6 5 4 3 2 1 0 A B C D I F G H 9 E
0 H G F E D C B A I 1 2 3 4 5 6 7 8 9
A 9 H G F E I C B 8 0 1 2 3 4 5 6 7 D
B A 9 H G F E D C 7 8 0 1 2 3 4 I 6 5
C I A 9 H G F E D 6 7 8 0 1 2 3 4 5 B
D C B A 9 H G F E 5 I 7 8 0 1 2 3 4 6
E D C B A 9 H G F 4 5 6 I 8 0 1 2 3 7
F E D C B A 9 H G 3 4 5 6 7 8 0 1 I 2
G F E D C B A 9 I 2 3 4 5 6 7 8 0 1 H
H G I E D C B A 9 1 2 3 4 5 6 7 8 0 F
1 B F 3 8 4 D 0 H 9 6 G 7 E C A 5 2 I
```

Partial square

```
9 8 7 6 5 I . . . 0 A B C . . . . . .
. . 8 7 6 5 4 3 2 . 9 A . C D E F G .
. . . 8 7 6 5 4 3 G H 9 A B . D E F .
. . . . I 7 6 5 4 F G H 9 A B C D E .
. . . . . 8 7 6 5 E F . H 9 A B C D G
. . . . . 8 7 6 D E F G H 9 I B C .
. . . . . 8 7 C D E F G H 9 A B .
. . . . . B C D E F G H 9 . 0
. . . . . A B C D I F G H 9 E
0 H . . E D . B A . 1 2 3 . 5 . 7 . .
. . . G F E I C B . . 1 2 3 4 5 6 7 D
. . . H G F E D C . . . 1 2 3 4 . 6 5
. . . . H G F E D . . . . 1 2 3 4 5 B
. . . . . H G F E . . . . . 1 2 3 4 .
. . . . . H G F . . . . . 1 2 3 7
. . . . . H G . . . . . 1 I .
. . . . . 1 H
. . I . . . . . .
. B . 3 8 4 . 0 H . . . . . I
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E092: order 19, 181 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m9_seed2.json`

SHA256: `2896bf2afb4ca0adb6b508fe2ceae9b5db89230e01276f8d04f0364605c4b2bf`

Completion

```
9 8 7 6 5 I 3 2 1 0 A B C D E F G H 4
I 0 8 7 6 5 4 3 2 H 9 A B C D E F G 1
2 1 0 8 7 6 5 4 3 G H 9 A B I D E F C
3 2 1 0 I 7 6 5 4 F G H 9 A B C D E 8
4 3 2 1 0 8 7 6 5 E F I H 9 A B C D G
5 4 3 2 1 0 8 7 6 D E F G H 9 I B C A
6 5 4 I 2 1 0 8 7 C D E F G H 9 A B 3
7 6 5 4 3 2 1 I 8 B C D E F G H 9 A 0
8 7 6 5 4 3 2 1 0 A B C D I F G H 9 E
0 H G F E D C B A I 1 2 3 4 5 6 7 8 9
A 9 H G F E I C B 8 0 1 2 3 4 5 6 7 D
B A 9 H G F E D C 7 8 0 1 2 3 4 I 6 5
C I A 9 H G F E D 6 7 8 0 1 2 3 4 5 B
D C B A 9 H G F E 5 I 7 8 0 1 2 3 4 6
E D C B A 9 H G F 4 5 6 I 8 0 1 2 3 7
F E D C B A 9 H G 3 4 5 6 7 8 0 1 I 2
G F E D C B A 9 I 2 3 4 5 6 7 8 0 1 H
H G I E D C B A 9 1 2 3 4 5 6 7 8 0 F
1 B F 3 8 4 D 0 H 9 6 G 7 E C A 5 2 I
```

Partial square

```
9 . 7 6 5 . 3 2 . 0 A . C D E F G . .
I . 8 7 6 5 4 . 2 . 9 A . C D E F G .
. . . 8 7 6 5 4 3 G H 9 A B . D E F .
. . . . . 7 6 5 4 F G H 9 A B C D E .
. . . . . 8 7 6 5 E F . H 9 A B C D .
. . . . . 8 7 6 D E F G H 9 . B C .
. . . I . . . 8 7 C D E F G H 9 A B 3
. . . . . B C D E F G H 9 A 0
. . . . . A B C D I F G H 9 .
0 H . . . . . I . . 3 . 5 . . . 9
. . H G F E . C B . . 1 2 3 4 5 6 7 .
. . . H G F E D C . . . 1 2 3 4 . 6 .
. I . . H G F E D . . . . 1 2 3 4 5 B
. . . . . H G F E . . . . . 1 2 3 4 6
. . . . . H G F . . . . . 1 2 3 .
. . . . . H G . . . . . 1 I 2
. . . . . I . . . . . 1 H
. . . . . .
1 B . 3 8 4 . 0 . . . . . I
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E093: order 19, 181 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m9_seed5.json`

SHA256: `5f5ec69e3dd989ade77b355c5da8c7c214060674bea10e16e31283543e6b4e2e`

Completion

```
9 8 7 6 5 I 3 2 1 0 A B C D E F G H 4
I 0 8 7 6 5 4 3 2 H 9 A B C D E F G 1
2 1 0 8 7 6 5 4 3 G H 9 A B I D E F C
3 2 1 0 I 7 6 5 4 F G H 9 A B C D E 8
4 3 2 1 0 8 7 6 5 E F I H 9 A B C D G
5 4 3 2 1 0 8 7 6 D E F G H 9 I B C A
6 5 4 I 2 1 0 8 7 C D E F G H 9 A B 3
7 6 5 4 3 2 1 I 8 B C D E F G H 9 A 0
8 7 6 5 4 3 2 1 0 A B C D I F G H 9 E
0 H G F E D C B A I 1 2 3 4 5 6 7 8 9
A 9 H G F E I C B 8 0 1 2 3 4 5 6 7 D
B A 9 H G F E D C 7 8 0 1 2 3 4 I 6 5
C I A 9 H G F E D 6 7 8 0 1 2 3 4 5 B
D C B A 9 H G F E 5 I 7 8 0 1 2 3 4 6
E D C B A 9 H G F 4 5 6 I 8 0 1 2 3 7
F E D C B A 9 H G 3 4 5 6 7 8 0 1 I 2
G F E D C B A 9 I 2 3 4 5 6 7 8 0 1 H
H G I E D C B A 9 1 2 3 4 5 6 7 8 0 F
1 B F 3 8 4 D 0 H 9 6 G 7 E C A 5 2 I
```

Partial square

```
9 8 7 6 5 I 3 2 . . . . . . . . . . 4
I . . 7 6 5 4 . 2 H 9 A B C D E F G .
. . . 8 7 6 5 4 3 G H 9 A B . D E . C
. . . . . 7 6 5 4 F G H 9 A B C D E 8
. . . . . 8 7 6 5 E F I H 9 A B C D .
. . . . . 8 7 6 D E F G H 9 . B C .
. . . I . . . 8 7 C D E F G H 9 A B 3
. . . . . B C D E F G H 9 A 0
. . . . . A B C D I F G H 9 .
. . G . . D . . A I 1 . . . . . .
. . . G F E I C B . . 1 2 3 4 5 6 7 .
. . . H G F E D C . . . 1 2 3 4 . 6 5
. . . . H G F E D . . . . 1 2 3 4 5 B
. . . . . H G F E . . . . . 1 2 3 4 6
. . . . . H G . . . . . . . 1 2 3 7
. . . . . H G . . . . . . . 1 . 2
. . . . . 1 H
. . I . . . . . . . . . . . . . . . F
. B . 3 8 4 . 0 H . . . . . A 5 2 .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E094: order 19, 181 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_four_m_plus_three/near_examples/n19_seed3.json`

SHA256: `aaf21e9f605d094cce0056ac15682286a1e8dd7d0f5318a3c51955546e40d09e`

Completion

```
0 8 7 6 5 4 3 2 1 9 A B C D E F G H I
1 0 I 7 6 5 4 3 2 H 9 A B C D E F G 8
2 1 0 8 I 6 5 4 3 G H 9 A B C D E F 7
3 2 1 0 8 7 I 5 4 F G H 9 A B C D E 6
4 3 2 1 0 8 7 6 I E F G H 9 A B C D 5
5 4 3 2 1 0 8 7 6 D I F G H 9 A B C E
6 5 4 3 2 1 0 8 7 C D E I G H 9 A B F
7 6 5 4 3 2 1 0 8 B C D E F I H 9 A G
8 7 6 5 4 3 2 1 0 A B C D E F G I 9 H
9 H G F E D C B A I 1 2 3 4 5 6 7 8 0
A 9 H G F E D C B 8 0 I 2 3 4 5 6 7 1
B A 9 H G F E D C 7 8 0 1 I 3 4 5 6 2
C B A 9 H G F E D 6 7 8 0 1 2 I 4 5 3
D C B A 9 H G F E 5 6 7 8 0 1 2 3 I 4
E I C B A 9 H G F 4 5 6 7 8 0 1 2 3 D
F E D I B A 9 H G 3 4 5 6 7 8 0 1 2 C
G F E D C I A 9 H 2 3 4 5 6 7 8 0 1 B
H G F E D C B I 9 1 2 3 4 5 6 7 8 0 A
I D 8 C 7 B 6 A 5 0 E 1 F 2 G 3 H 4 9
```

Partial square

```
0 8 7 6 5 4 . . . 9 A B . . . . G . .
. . . 7 6 5 4 3 2 H 9 A B C D E F G .
. . . 8 . 6 5 4 3 G H 9 A B C D E F .
. . . . 8 7 I 5 4 F G H 9 A B . D E .
. . . . . 8 7 6 . E F G H 9 A B C D .
. . . . . 8 7 6 D . F G H 9 A . C E
. . . . . 8 7 C D E . G H 9 A B F
. . . . . 8 B C D E F . H 9 . .
. . . . . A B C D E F G . 9 H
9 . G F E . C . . . . 2 3 4 5 6 . . .
. . H G F E D C B . . . 2 3 4 5 6 7 1
. . . H G F E D C . . . 1 . 3 4 5 6 2
. . . . H G F E D . . . . 1 2 . 4 5 3
. . . . . H G F E . . . . . 1 2 3 I .
. I . . . . H G F . . . . . 1 2 3 D
. . . I . . . . G . . . . . . . 1 2 C
. . . . . I . . H . . . . . . . . 1 .
. . . . . I . . . . . . . . . . A
I . 8 . 7 B . A . 0 . . . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E095: order 19, 181 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m9_seed4.json`

SHA256: `b98086a01633794a5bc28b15ed863a984a0bf441fc9e9b49c7dc9bba1c3a6207`

Completion

```
9 8 7 6 5 I 3 2 1 0 A B C D E F G H 4
I 0 8 7 6 5 4 3 2 H 9 A B C D E F G 1
2 1 0 8 7 6 5 4 3 G H 9 A B I D E F C
3 2 1 0 I 7 6 5 4 F G H 9 A B C D E 8
4 3 2 1 0 8 7 6 5 E F I H 9 A B C D G
5 4 3 2 1 0 8 7 6 D E F G H 9 I B C A
6 5 4 I 2 1 0 8 7 C D E F G H 9 A B 3
7 6 5 4 3 2 1 I 8 B C D E F G H 9 A 0
8 7 6 5 4 3 2 1 0 A B C D I F G H 9 E
0 H G F E D C B A I 1 2 3 4 5 6 7 8 9
A 9 H G F E I C B 8 0 1 2 3 4 5 6 7 D
B A 9 H G F E D C 7 8 0 1 2 3 4 I 6 5
C I A 9 H G F E D 6 7 8 0 1 2 3 4 5 B
D C B A 9 H G F E 5 I 7 8 0 1 2 3 4 6
E D C B A 9 H G F 4 5 6 I 8 0 1 2 3 7
F E D C B A 9 H G 3 4 5 6 7 8 0 1 I 2
G F E D C B A 9 I 2 3 4 5 6 7 8 0 1 H
H G I E D C B A 9 1 2 3 4 5 6 7 8 0 F
1 B F 3 8 4 D 0 H 9 6 G 7 E C A 5 2 I
```

Partial square

```
. 8 . . . I . . . 0 A B C D E . . H .
I . 8 7 6 5 4 . 2 . 9 A . C D E F G 1
. . . 8 7 6 5 4 3 G H 9 A B . D E . .
. . . . . 7 6 5 4 F G H 9 A B C D E .
. . . . . 8 7 6 5 E F . H 9 A B C D .
. . . . . . 8 7 6 D E F G H 9 . B C A
. . . I . . . 8 7 C D E F G H 9 A B .
. . . . . I . B C D E F G H 9 . 0
. . . . . A B C D . F G H 9 .
0 . G F . . . B . . . . . . . 8 9
. . . G F E I C B . . 1 2 3 4 5 6 7 .
. . . H G F E D C . . . 1 2 3 4 I 6 .
. I . . H G F E D . . . . 1 2 3 4 5 .
. . . . . H G F E . I . . . 1 2 3 4 .
. . . . . H G . . . . I . . 1 2 3 7
. . . . . H G . . . . . . . 1 . 2
. . . . . 1 H
. . I . . . . . F
1 . F 3 8 4 D 0 . 9 6 . . . . . 5 . I
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E096: order 19, 180 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_four_m_plus_three/near_examples/n19_seed2.json`

SHA256: `1be0f6e69081ed5dd6b478fd13f289d77983cd834e11c5e268eb536655099484`

Completion

```
0 8 7 6 5 4 3 2 1 9 A B C D E F G H I
1 0 I 7 6 5 4 3 2 H 9 A B C D E F G 8
2 1 0 8 I 6 5 4 3 G H 9 A B C D E F 7
3 2 1 0 8 7 I 5 4 F G H 9 A B C D E 6
4 3 2 1 0 8 7 6 I E F G H 9 A B C D 5
5 4 3 2 1 0 8 7 6 D I F G H 9 A B C E
6 5 4 3 2 1 0 8 7 C D E I G H 9 A B F
7 6 5 4 3 2 1 0 8 B C D E F I H 9 A G
8 7 6 5 4 3 2 1 0 A B C D E F G I 9 H
9 H G F E D C B A I 1 2 3 4 5 6 7 8 0
A 9 H G F E D C B 8 0 I 2 3 4 5 6 7 1
B A 9 H G F E D C 7 8 0 1 I 3 4 5 6 2
C B A 9 H G F E D 6 7 8 0 1 2 I 4 5 3
D C B A 9 H G F E 5 6 7 8 0 1 2 3 I 4
E I C B A 9 H G F 4 5 6 7 8 0 1 2 3 D
F E D I B A 9 H G 3 4 5 6 7 8 0 1 2 C
G F E D C I A 9 H 2 3 4 5 6 7 8 0 1 B
H G F E D C B I 9 1 2 3 4 5 6 7 8 0 A
I D 8 C 7 B 6 A 5 0 E 1 F 2 G 3 H 4 9
```

Partial square

```
0 . 7 . . . . 2 1 9 A B C D E F G . .
. . I 7 6 5 4 3 2 H 9 A B C D E F G .
. . . 8 . 6 5 4 3 G H 9 A B C D E F .
. . . . 8 7 I 5 4 F G H 9 A B C D E .
. . . . . 8 7 6 . E F G H 9 A B C D .
. . . . . 8 7 6 D . F G H 9 A B C E
. . . . . 8 7 C D E . G H 9 A B F
. . . . . 8 B C D E F . H 9 . G
. . . . . A B C D E F G I 9 .
9 H . . . . . I . . . . . 7 . 0
. . H G F E D C B . . . 2 3 4 5 6 7 .
. . . H G F E D C . . . 1 . 3 4 5 6 .
. . . . H G F E D . . . . 1 2 . 4 5 .
. . . . . H G F E . . . . . 1 2 3 I 4
. . . . . H G F . . . . . 1 2 3 D
. . . I . . . H G . . . . . 1 2 C
. . . . . I . . . . . 1 .
. . . . . A
I D 8 C 7 . 6 A . 0 . . . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E097: order 19, 179 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/examples/m9_seed1.json`

SHA256: `358fd6e27ddf92b17f52269fed82094c12557c3dddcd7503cb84aa9fd0918fa6`

Completion

```
9 8 7 6 5 I 3 2 1 0 A B C D E F G H 4
I 0 8 7 6 5 4 3 2 H 9 A B C D E F G 1
2 1 0 8 7 6 5 4 3 G H 9 A B I D E F C
3 2 1 0 I 7 6 5 4 F G H 9 A B C D E 8
4 3 2 1 0 8 7 6 5 E F I H 9 A B C D G
5 4 3 2 1 0 8 7 6 D E F G H 9 I B C A
6 5 4 I 2 1 0 8 7 C D E F G H 9 A B 3
7 6 5 4 3 2 1 I 8 B C D E F G H 9 A 0
8 7 6 5 4 3 2 1 0 A B C D I F G H 9 E
0 H G F E D C B A I 1 2 3 4 5 6 7 8 9
A 9 H G F E I C B 8 0 1 2 3 4 5 6 7 D
B A 9 H G F E D C 7 8 0 1 2 3 4 I 6 5
C I A 9 H G F E D 6 7 8 0 1 2 3 4 5 B
D C B A 9 H G F E 5 I 7 8 0 1 2 3 4 6
E D C B A 9 H G F 4 5 6 I 8 0 1 2 3 7
F E D C B A 9 H G 3 4 5 6 7 8 0 1 I 2
G F E D C B A 9 I 2 3 4 5 6 7 8 0 1 H
H G I E D C B A 9 1 2 3 4 5 6 7 8 0 F
1 B F 3 8 4 D 0 H 9 6 G 7 E C A 5 2 I
```

Partial square

```
9 8 7 6 5 I 3 2 . . . . . . . . . . .
. . . 7 6 5 4 . 2 H 9 A B C D E F G 1
. . . 8 7 6 5 4 3 G H 9 A B I D E . C
. . . . . 7 6 5 4 F G H 9 A B C D E .
. . . . . 8 7 6 5 E F . H 9 A B C D G
. . . . . 8 7 6 D E F G H 9 . B C A
. . . I . . . 8 7 C D E F G H 9 A B .
. . . . . I . B C D E F G H 9 . 0
. . . . . A B C D I F G H 9 E
. . G . . . . . A I . . . . . . . 9
. . . G F E . C B . . 1 2 3 4 5 6 7 .
. . . H G F E D C . . . 1 2 3 4 . 6 .
. I . . H G F E D . . . . 1 2 3 4 5 .
. . . . . H G F E . . . . . 1 2 3 4 .
. . . . . H G F . . . . . 1 2 3 7
. . . . . H G . . . . . 1 . .
. . . . . 1 H
. . I . . . . . . . . . . . . . . . F
1 . F 3 8 4 D 0 . . . G 7 . . . . 2 I
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E098: order 19, 174 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_four_m_plus_three/near_examples/n19_seed1.json`

SHA256: `9123323f315fc7fe3622293be59ec21e54512e9671d594ada371ad4c784977d0`

Completion

```
0 8 7 6 5 4 3 2 1 9 A B C D E F G H I
1 0 I 7 6 5 4 3 2 H 9 A B C D E F G 8
2 1 0 8 I 6 5 4 3 G H 9 A B C D E F 7
3 2 1 0 8 7 I 5 4 F G H 9 A B C D E 6
4 3 2 1 0 8 7 6 I E F G H 9 A B C D 5
5 4 3 2 1 0 8 7 6 D I F G H 9 A B C E
6 5 4 3 2 1 0 8 7 C D E I G H 9 A B F
7 6 5 4 3 2 1 0 8 B C D E F I H 9 A G
8 7 6 5 4 3 2 1 0 A B C D E F G I 9 H
9 H G F E D C B A I 1 2 3 4 5 6 7 8 0
A 9 H G F E D C B 8 0 I 2 3 4 5 6 7 1
B A 9 H G F E D C 7 8 0 1 I 3 4 5 6 2
C B A 9 H G F E D 6 7 8 0 1 2 I 4 5 3
D C B A 9 H G F E 5 6 7 8 0 1 2 3 I 4
E I C B A 9 H G F 4 5 6 7 8 0 1 2 3 D
F E D I B A 9 H G 3 4 5 6 7 8 0 1 2 C
G F E D C I A 9 H 2 3 4 5 6 7 8 0 1 B
H G F E D C B I 9 1 2 3 4 5 6 7 8 0 A
I D 8 C 7 B 6 A 5 0 E 1 F 2 G 3 H 4 9
```

Partial square

```
0 8 7 6 5 4 3 2 1 . . . . . . . G . .
. . . 7 6 5 4 3 2 H 9 A B C . E F G .
. . . 8 . 6 5 4 3 G H 9 A B C D E F .
. . . . 8 7 . 5 4 F G H 9 A B . D E .
. . . . . 8 7 6 . E F G H 9 A B C D .
. . . . . 8 7 6 D I F G H 9 A . C E
. . . . . 8 7 C D E I G H 9 A B .
. . . . . 8 B C D E F I H 9 . .
. . . . . A B C D E F G I 9 H
9 H . . . . . I . . . . . . . . 0
. . H G F E D C B . . . 2 3 4 5 6 7 .
. . . H G F E D C . . . 1 . 3 4 5 6 2
. . . . H G F E D . . . . 1 2 I 4 5 3
. . . . . H G F E . . . . . 1 2 3 . .
. I . . . . H G F . . . . . 1 2 3 D
. . . . . H G . . . . . 1 2 .
. . . . . 1 B
. . . . . I . . . . . A
I . 8 . . . . . 0 E . F . G . . . 9
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E099: order 21, 267 clues

Status I. Previously independently checked uniqueness and all clue-deletion witnesses; uniqueness is solver-dependent.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m5_seed0.json`

SHA256: `45c71aee8dec1d7575e8f4a82b677cda8412401fcc8a48f6175b0248985d821f`

Completion

```
K 4 3 2 1 5 6 7 8 9 A B C D E F G H I J 0
1 0 4 3 K 9 5 6 7 8 E A B C D G H I J F 2
2 1 0 K 3 8 9 5 6 7 D E A B C H I J F G 4
3 2 K 0 4 7 8 9 5 6 C D E A B I J F G H 1
4 K 2 1 0 6 7 8 9 5 B C D E A J F G H I 3
5 9 8 7 6 0 1 2 3 4 K G H I J A B C D E F
6 5 9 8 7 4 0 1 2 3 G K I J F E A B C D H
7 6 5 9 8 3 4 0 1 2 H I K F G D E A B C J
8 7 6 5 9 2 3 4 0 1 I J F K H C D E A B G
9 8 7 6 5 1 2 3 4 0 J F G H K B C D E A I
A E D C B F G H I J 0 1 2 3 4 K 6 7 8 9 5
B A E D C G H I J F 4 0 1 2 3 9 5 6 7 K 8
C B A E D H I J F G 3 4 0 1 2 8 9 5 K 7 6
D C B A E I J F G H 2 3 4 0 1 7 8 K 5 6 9
E D C B A J F G H I 1 2 3 4 0 6 K 8 9 5 7
F G H I J K E D C B 5 9 8 7 6 0 4 3 2 1 A
G H I J F B A E D K 6 5 9 8 7 1 0 4 3 2 C
H I J F G C B A K D 7 6 5 9 8 2 1 0 4 3 E
I J F G H D C K A E 8 7 6 5 9 3 2 1 0 4 B
J F G H I E K C B A 9 8 7 6 5 4 3 2 1 0 D
0 3 1 4 2 A D B E C F H J G I 5 7 9 6 8 K
```

Partial square

```
K . . . . . . . . . . . . . . . . . . . .
. 0 . . . . 5 . . . . A . . . . . . . F 2
. 1 0 K . . 9 5 . . . E A . . . . . F G .
. 2 K 0 . . 8 9 5 . . D E A . . . F G H .
. . 2 . 0 . 7 8 9 5 . C D E A . F G H I 3
. . . . . . . 1 2 3 4 . . . . . A B C D E .
. 5 9 8 7 4 . 1 2 3 . K . . F E A B C D .
. . 5 9 8 3 4 . 1 2 . . K . G D E A B . J
. . . 5 9 2 3 4 . 1 . . F K H C D E A B .
. . . . 5 1 2 3 4 0 . F G H K B C D E A .
. . . . . . . . . . . 0 1 2 3 4 . 6 7 8 9 5
. A E D C . . . . F 4 0 1 2 3 9 5 6 7 K .
. . A E D . . . F G 3 4 0 1 2 8 9 5 K 7 .
. . . A E . . F G H 2 3 4 0 1 7 8 K 5 6 .
. . . . A . F G H I 1 2 3 4 0 6 . 8 9 5 7
. . . . . K E D C B 5 9 8 7 6 0 4 3 2 1 .
. . . . F B A E D K 6 5 9 8 7 1 0 4 3 2 .
. . . F G C B A K D 7 6 5 9 8 2 1 0 4 3 .
. . F G H D C K A E 8 7 6 5 9 3 2 1 0 4 .
. F G H I E K C B A 9 8 . 6 5 4 3 2 1 0 .
. . . 4 . . . . . C F . J . . . . . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E100: order 21, 265 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_shifted_examples/thesis_m5_seed0.json`

SHA256: `df1c62cb01b8663e8855c0eebd94538e90bff7537a0ee463c8f54dfdba4c3f8d`

Completion

```
0 4 3 2 K 5 6 7 8 9 A B C D E F G H I J 1
1 0 4 K 2 9 5 6 7 8 E A B C D G H I J F 3
2 1 K 4 3 8 9 5 6 7 D E A B C H I J F G 0
3 K 1 0 4 7 8 9 5 6 C D E A B I J F G H 2
K 3 2 1 0 6 7 8 9 5 B C D E A J F G H I 4
5 9 8 7 6 0 1 2 3 4 K G H I J A B C D E F
6 5 9 8 7 4 0 1 2 3 G K I J F E A B C D H
7 6 5 9 8 3 4 0 1 2 H I K F G D E A B C J
8 7 6 5 9 2 3 4 0 1 I J F K H C D E A B G
9 8 7 6 5 1 2 3 4 0 J F G H K B C D E A I
A E D C B F G H I J 0 1 2 3 4 K 6 7 8 9 5
B A E D C G H I J F 4 0 1 2 3 9 5 6 7 K 8
C B A E D H I J F G 3 4 0 1 2 8 9 5 K 7 6
D C B A E I J F G H 2 3 4 0 1 7 8 K 5 6 9
E D C B A J F G H I 1 2 3 4 0 6 K 8 9 5 7
F G H I J K E D C B 5 9 8 7 6 0 4 3 2 1 A
G H I J F B A E D K 6 5 9 8 7 1 0 4 3 2 C
H I J F G C B A K D 7 6 5 9 8 2 1 0 4 3 E
I J F G H D C K A E 8 7 6 5 9 3 2 1 0 4 B
J F G H I E K C B A 9 8 7 6 5 4 3 2 1 0 D
4 2 0 3 1 A D B E C F H J G I 5 7 9 6 8 K
```

Partial square

```
. . . . K . . . . . . . . . . . . . . .
. 0 . . . . 5 . . . . A . . . . . . . F 3
. 1 K . . . 9 5 . . . E A . . . . . F G .
. K 1 0 . . 8 9 5 . . D E A . . . F G H .
. 3 2 1 0 . 7 8 9 5 . C D E A . F G H I .
. . . . . 0 . 2 3 4 . . . . . A B C D E .
. 5 9 8 7 4 0 1 2 3 . K . . F E A B C D .
. . 5 9 8 3 4 0 1 2 . . K . G D E A B . J
. . . 5 9 2 3 4 0 1 . . F K H C D E A B .
. . . . 5 1 2 3 4 0 . F G H K B C D E A .
. . . . . 0 1 2 3 4 . 6 7 8 9 5
. A E D C . . . . F . 0 1 2 3 9 5 6 7 K .
. . A E D . . . F G 3 4 0 1 2 8 9 5 K 7 .
. . . A E . . F G H 2 3 4 0 1 7 8 K 5 6 .
. . . . A . F G H I 1 2 3 4 0 6 . 8 9 5 7
. . . . . K E D C B . 9 8 7 6 0 4 3 2 . .
. . . . F B A E D . 6 5 9 8 7 1 0 4 3 2 .
. . . F G C B A K D 7 6 5 9 8 2 1 0 4 3 .
. . F G H D C K A E 8 7 6 5 9 3 2 1 0 4 .
. F G H I E K C B A 9 8 . 6 5 . 3 2 1 0 .
4 . . . . . C F . J . . . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E101: order 21, 265 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m5_seed3.json`

SHA256: `f8f6b1ec21b38fd53dcf6d9d7eb8736d67561f8b7de74fc20b102b581e1ecaa4`

Completion

```
K 4 3 2 1 5 6 7 8 9 A B C D E F G H I J 0
1 0 4 3 K 9 5 6 7 8 E A B C D G H I J F 2
2 1 0 K 3 8 9 5 6 7 D E A B C H I J F G 4
3 2 K 0 4 7 8 9 5 6 C D E A B I J F G H 1
4 K 2 1 0 6 7 8 9 5 B C D E A J F G H I 3
5 9 8 7 6 0 1 2 3 4 K G H I J A B C D E F
6 5 9 8 7 4 0 1 2 3 G K I J F E A B C D H
7 6 5 9 8 3 4 0 1 2 H I K F G D E A B C J
8 7 6 5 9 2 3 4 0 1 I J F K H C D E A B G
9 8 7 6 5 1 2 3 4 0 J F G H K B C D E A I
A E D C B F G H I J 0 1 2 3 4 K 6 7 8 9 5
B A E D C G H I J F 4 0 1 2 3 9 5 6 7 K 8
C B A E D H I J F G 3 4 0 1 2 8 9 5 K 7 6
D C B A E I J F G H 2 3 4 0 1 7 8 K 5 6 9
E D C B A J F G H I 1 2 3 4 0 6 K 8 9 5 7
F G H I J K E D C B 5 9 8 7 6 0 4 3 2 1 A
G H I J F B A E D K 6 5 9 8 7 1 0 4 3 2 C
H I J F G C B A K D 7 6 5 9 8 2 1 0 4 3 E
I J F G H D C K A E 8 7 6 5 9 3 2 1 0 4 B
J F G H I E K C B A 9 8 7 6 5 4 3 2 1 0 D
0 3 1 4 2 A D B E C F H J G I 5 7 9 6 8 K
```

Partial square

```
K . . . . . . . . . . . . . . . . . . . .
. 0 . . . . 5 . . . . A . . . . . . . F .
. . 0 K . . 9 5 . . . E A . . . . . F G 4
. 2 K 0 . . 8 9 5 . . D E A . . . F G H .
. K 2 1 0 . 7 8 9 5 . C D E A . F G H I .
. . . . . 0 1 2 3 4 . . . . . A B C D E .
. 5 9 8 7 4 0 1 2 3 . K . . F E A B C D .
. . 5 9 8 3 4 0 1 2 . . K . G D E A B . J
. . . 5 9 2 3 4 . 1 . . F K H C D E A B .
. . . . 5 1 2 3 4 . . . . H K B C D E A I
. . . . . 0 1 2 3 4 K 6 7 8 9 .
. A E D C . . . . F 4 0 1 2 3 9 5 6 7 K .
. . A E D . . . F G 3 4 . 1 2 8 9 5 K 7 .
. . . A E . . F G H 2 3 4 0 1 7 8 K 5 6 .
. . . . A . F G H I 1 2 3 4 . 6 K 8 9 5 .
. . . . . E D C B . 9 8 7 6 0 4 3 2 1 A
. . . . F B A E D . 6 5 9 8 7 1 0 4 3 2 .
. . . F G C B A K D 7 6 5 9 8 2 1 0 4 3 .
. . F G H D C K A E 8 7 6 5 9 3 2 1 0 4 .
. F G H I E K C B A 9 8 . 6 5 4 3 2 1 0 .
. 3 . . 2 . . . . C F . J . . . . . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E102: order 21, 262 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m5_seed2.json`

SHA256: `046b53d50fb7ba3734ee67cddfc053e4a8e7f78ce12ee23fca4e91b93e871597`

Completion

```
K 4 3 2 1 5 6 7 8 9 A B C D E F G H I J 0
1 0 4 3 K 9 5 6 7 8 E A B C D G H I J F 2
2 1 0 K 3 8 9 5 6 7 D E A B C H I J F G 4
3 2 K 0 4 7 8 9 5 6 C D E A B I J F G H 1
4 K 2 1 0 6 7 8 9 5 B C D E A J F G H I 3
5 9 8 7 6 0 1 2 3 4 K G H I J A B C D E F
6 5 9 8 7 4 0 1 2 3 G K I J F E A B C D H
7 6 5 9 8 3 4 0 1 2 H I K F G D E A B C J
8 7 6 5 9 2 3 4 0 1 I J F K H C D E A B G
9 8 7 6 5 1 2 3 4 0 J F G H K B C D E A I
A E D C B F G H I J 0 1 2 3 4 K 6 7 8 9 5
B A E D C G H I J F 4 0 1 2 3 9 5 6 7 K 8
C B A E D H I J F G 3 4 0 1 2 8 9 5 K 7 6
D C B A E I J F G H 2 3 4 0 1 7 8 K 5 6 9
E D C B A J F G H I 1 2 3 4 0 6 K 8 9 5 7
F G H I J K E D C B 5 9 8 7 6 0 4 3 2 1 A
G H I J F B A E D K 6 5 9 8 7 1 0 4 3 2 C
H I J F G C B A K D 7 6 5 9 8 2 1 0 4 3 E
I J F G H D C K A E 8 7 6 5 9 3 2 1 0 4 B
J F G H I E K C B A 9 8 7 6 5 4 3 2 1 0 D
0 3 1 4 2 A D B E C F H J G I 5 7 9 6 8 K
```

Partial square

```
K . . . . . . . . . . . . . . . . . . . .
. 0 . . . . 5 . . . . A . . . . . . . F .
. 1 0 K . . 9 5 . . . E A . . . . . F G 4
. . K 0 . . 8 9 5 . . D E A . . . F G H .
. . 2 . 0 . 7 8 9 5 . C D E A . F G H I 3
. . . . . 0 1 2 3 4 K . . . . . B C D E F
. 5 9 8 7 4 0 1 2 3 . K . . F E A B C D .
. . 5 9 8 3 4 0 1 2 . . . F G D E A B . J
. . . 5 9 2 3 4 . 1 . . . . . C D E A B .
. . . . 5 1 2 3 4 . . F . H . B C D E A I
. . . . . 0 1 2 3 4 K 6 7 8 9 .
. A E D C . . . . F 4 0 1 2 3 9 5 6 7 K .
. . A E D . . . F G 3 4 . 1 2 8 9 5 K 7 .
. . . A E . . F G H 2 3 4 . 1 7 8 K 5 6 .
. . . . A . F G H I 1 2 3 4 . 6 . 8 9 5 .
. . . . . K E D C B . 9 8 7 6 . 4 3 2 1 A
. . . . F B A E D K 6 5 9 8 7 1 . 4 3 2 .
. . . F G C B A . D 7 6 5 9 8 2 1 . 4 3 E
. . F G H D C K A E 8 7 6 5 9 3 2 1 0 4 .
. F G H I E . C B A 9 8 . 6 5 4 3 2 1 . D
. . . 4 2 . D . . . . H J G . 5 7 9 . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E103: order 21, 260 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m5_seed6.json`

SHA256: `933c104d25deccf6c88be47dc3314eee93f6d8072065e38368040651f5b9717c`

Completion

```
K 4 3 2 1 5 6 7 8 9 A B C D E F G H I J 0
1 0 4 3 K 9 5 6 7 8 E A B C D G H I J F 2
2 1 0 K 3 8 9 5 6 7 D E A B C H I J F G 4
3 2 K 0 4 7 8 9 5 6 C D E A B I J F G H 1
4 K 2 1 0 6 7 8 9 5 B C D E A J F G H I 3
5 9 8 7 6 0 1 2 3 4 K G H I J A B C D E F
6 5 9 8 7 4 0 1 2 3 G K I J F E A B C D H
7 6 5 9 8 3 4 0 1 2 H I K F G D E A B C J
8 7 6 5 9 2 3 4 0 1 I J F K H C D E A B G
9 8 7 6 5 1 2 3 4 0 J F G H K B C D E A I
A E D C B F G H I J 0 1 2 3 4 K 6 7 8 9 5
B A E D C G H I J F 4 0 1 2 3 9 5 6 7 K 8
C B A E D H I J F G 3 4 0 1 2 8 9 5 K 7 6
D C B A E I J F G H 2 3 4 0 1 7 8 K 5 6 9
E D C B A J F G H I 1 2 3 4 0 6 K 8 9 5 7
F G H I J K E D C B 5 9 8 7 6 0 4 3 2 1 A
G H I J F B A E D K 6 5 9 8 7 1 0 4 3 2 C
H I J F G C B A K D 7 6 5 9 8 2 1 0 4 3 E
I J F G H D C K A E 8 7 6 5 9 3 2 1 0 4 B
J F G H I E K C B A 9 8 7 6 5 4 3 2 1 0 D
0 3 1 4 2 A D B E C F H J G I 5 7 9 6 8 K
```

Partial square

```
K . . . . . . . . . . . . . . . . . . . .
. 0 . . K . 5 . . . . A . . . . . . . F 2
. 1 0 . . . 9 5 . . . E A . . . . . F G .
. 2 . . . . 8 9 5 . . D E A . . . F G H .
. . 2 . 0 . 7 8 9 5 . C D E A . F G H I 3
. . . . . 0 1 2 3 4 K . . . . . B C D E F
. 5 9 8 7 4 0 1 2 3 . . . . F E A B C D .
. . 5 9 8 3 4 0 1 2 . . K F G D E A B C .
. . . 5 9 2 3 4 0 1 . . . K H C D E A B G
. . . . 5 1 2 3 4 0 . F . H . B C D E A I
. . . . . 0 1 2 3 4 K 6 7 8 9 .
. A E D C . . . . F 4 . 1 2 3 9 5 6 7 . .
. . A E D . . . F G 3 4 . 1 2 8 9 5 . 7 6
. . . A E . . F G H 2 3 4 . 1 7 8 . 5 6 .
. . . . A . F G H I 1 2 3 4 0 6 K 8 9 5 .
. . . . . E D C B 5 9 8 7 6 0 4 3 2 1 .
. . . . F B A E D . 6 5 9 8 7 1 0 4 3 2 .
. . . F G C B A . D 7 6 5 9 8 2 1 0 4 3 .
. . F G H D C . A E 8 7 6 5 9 3 2 1 0 4 .
. F G H I E . C B A 9 8 . 6 5 4 3 2 1 0 .
. . . 4 . A . . . C . . J . . . . . . . K
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E104: order 21, 259 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_shifted_examples/thesis_m5_seed2.json`

SHA256: `6cb167cbb62ed1a0992f9e3e34078ea16db9ce2e65e0ea6dadd6f915c3ef3ade`

Completion

```
0 4 3 2 K 5 6 7 8 9 A B C D E F G H I J 1
1 0 4 K 2 9 5 6 7 8 E A B C D G H I J F 3
2 1 K 4 3 8 9 5 6 7 D E A B C H I J F G 0
3 K 1 0 4 7 8 9 5 6 C D E A B I J F G H 2
K 3 2 1 0 6 7 8 9 5 B C D E A J F G H I 4
5 9 8 7 6 0 1 2 3 4 K G H I J A B C D E F
6 5 9 8 7 4 0 1 2 3 G K I J F E A B C D H
7 6 5 9 8 3 4 0 1 2 H I K F G D E A B C J
8 7 6 5 9 2 3 4 0 1 I J F K H C D E A B G
9 8 7 6 5 1 2 3 4 0 J F G H K B C D E A I
A E D C B F G H I J 0 1 2 3 4 K 6 7 8 9 5
B A E D C G H I J F 4 0 1 2 3 9 5 6 7 K 8
C B A E D H I J F G 3 4 0 1 2 8 9 5 K 7 6
D C B A E I J F G H 2 3 4 0 1 7 8 K 5 6 9
E D C B A J F G H I 1 2 3 4 0 6 K 8 9 5 7
F G H I J K E D C B 5 9 8 7 6 0 4 3 2 1 A
G H I J F B A E D K 6 5 9 8 7 1 0 4 3 2 C
H I J F G C B A K D 7 6 5 9 8 2 1 0 4 3 E
I J F G H D C K A E 8 7 6 5 9 3 2 1 0 4 B
J F G H I E K C B A 9 8 7 6 5 4 3 2 1 0 D
4 2 0 3 1 A D B E C F H J G I 5 7 9 6 8 K
```

Partial square

```
. . . . K . . . . . . . . . . . . . . 1
. 0 . . . . 5 . . . . A . . . . . . . F .
. . K . . . 9 5 . . . E A . . . . . F G .
. K 1 0 . . 8 9 5 . . D E A . . . F G H 2
. . 2 1 . . 7 8 9 5 . C D E A . F G H I 4
. . . . . 0 . 2 3 4 K . . . . . B C D E F
. 5 9 8 7 . 0 1 2 3 . K . . F E A B C D .
. . 5 9 8 3 4 0 1 2 . . . F G D E A B . J
. . . 5 9 2 3 4 0 1 . . . . . C D E A B .
. . . . 5 1 2 3 4 0 . F . H . B C D E A I
. . . . . 0 1 2 3 4 . 6 7 8 9 .
. A E D C . . . . F 4 0 1 2 3 9 5 6 7 . .
. . A E D . . . F G 3 4 0 1 2 8 9 5 . 7 .
. . . A E . . F G H 2 3 4 0 1 7 8 . 5 6 .
. . . . A . F G H I 1 2 3 4 0 6 . 8 9 5 .
. . . . . K E D C B . 9 8 7 6 0 4 3 2 . A
. . . . F B A E D . 6 5 9 8 7 1 0 4 3 2 .
. . . F G C B A . D 7 6 5 9 8 2 1 0 4 3 .
. . F G H D C . A E 8 7 6 5 9 3 2 1 0 4 .
. F G H I E . C B A 9 8 . 6 5 . 3 2 1 0 D
4 . . . 1 . D . . . . H J . . 5 7 . . . K
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E105: order 21, 257 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m5_seed5.json`

SHA256: `5aa93f1b060044df8913abf1ecbcfed76c39f4b981516aa8d51c2f9c6a4a9fc5`

Completion

```
K 4 3 2 1 5 6 7 8 9 A B C D E F G H I J 0
1 0 4 3 K 9 5 6 7 8 E A B C D G H I J F 2
2 1 0 K 3 8 9 5 6 7 D E A B C H I J F G 4
3 2 K 0 4 7 8 9 5 6 C D E A B I J F G H 1
4 K 2 1 0 6 7 8 9 5 B C D E A J F G H I 3
5 9 8 7 6 0 1 2 3 4 K G H I J A B C D E F
6 5 9 8 7 4 0 1 2 3 G K I J F E A B C D H
7 6 5 9 8 3 4 0 1 2 H I K F G D E A B C J
8 7 6 5 9 2 3 4 0 1 I J F K H C D E A B G
9 8 7 6 5 1 2 3 4 0 J F G H K B C D E A I
A E D C B F G H I J 0 1 2 3 4 K 6 7 8 9 5
B A E D C G H I J F 4 0 1 2 3 9 5 6 7 K 8
C B A E D H I J F G 3 4 0 1 2 8 9 5 K 7 6
D C B A E I J F G H 2 3 4 0 1 7 8 K 5 6 9
E D C B A J F G H I 1 2 3 4 0 6 K 8 9 5 7
F G H I J K E D C B 5 9 8 7 6 0 4 3 2 1 A
G H I J F B A E D K 6 5 9 8 7 1 0 4 3 2 C
H I J F G C B A K D 7 6 5 9 8 2 1 0 4 3 E
I J F G H D C K A E 8 7 6 5 9 3 2 1 0 4 B
J F G H I E K C B A 9 8 7 6 5 4 3 2 1 0 D
0 3 1 4 2 A D B E C F H J G I 5 7 9 6 8 K
```

Partial square

```
K . . . . . . . . . . . . . . . . . . . 0
. 0 . . . . 5 . . . . A . . . . . . . F .
. . 0 K . . 9 5 . . . E A . . . . . F G .
. 2 K 0 . . 8 9 5 . . D E A . . . F G H 1
. . 2 1 0 . 7 8 9 5 . C D E A . F G H I 3
. . . . . . . 1 2 3 4 K . . . . . B C D E .
. 5 9 8 7 4 . 1 2 3 . . . . F E A B C D .
. . 5 9 8 3 4 . 1 2 . . K . . D E A B . J
. . . 5 9 2 3 4 . 1 . . . K H C D E A B G
. . . . 5 1 2 3 4 . . F . H . B C D E A .
. . . . . . . . . . . 0 1 2 3 4 . 6 7 8 9 .
. A E D C . . . . F 4 0 1 2 3 9 5 6 7 . .
. . A E D . . . F G 3 4 0 1 2 8 9 5 . 7 .
. . . A E . . F G H 2 3 4 0 1 7 8 . 5 6 .
. . . . A . F G H I 1 2 3 4 0 6 . 8 9 5 7
. . . . . K E D C B . 9 8 7 6 . 4 3 2 1 A
. . . . F B A E D . 6 5 9 8 7 1 . 4 3 2 .
. . . F G C B A . D 7 6 5 9 8 2 1 . 4 3 .
. . F G H D C . A E 8 7 6 5 9 3 2 1 0 4 .
. F G H I E . C B A 9 8 . 6 5 4 3 2 1 0 .
. 3 1 . . A . B . . . . J G . 5 7 . . . K
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E106: order 21, 256 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_shifted_examples/thesis_m5_seed1.json`

SHA256: `4736a1c61ba17fa32c6cf9805b0144497bd22f1382bda192bebe85aa56dcc9b1`

Completion

```
0 4 3 2 K 5 6 7 8 9 A B C D E F G H I J 1
1 0 4 K 2 9 5 6 7 8 E A B C D G H I J F 3
2 1 K 4 3 8 9 5 6 7 D E A B C H I J F G 0
3 K 1 0 4 7 8 9 5 6 C D E A B I J F G H 2
K 3 2 1 0 6 7 8 9 5 B C D E A J F G H I 4
5 9 8 7 6 0 1 2 3 4 K G H I J A B C D E F
6 5 9 8 7 4 0 1 2 3 G K I J F E A B C D H
7 6 5 9 8 3 4 0 1 2 H I K F G D E A B C J
8 7 6 5 9 2 3 4 0 1 I J F K H C D E A B G
9 8 7 6 5 1 2 3 4 0 J F G H K B C D E A I
A E D C B F G H I J 0 1 2 3 4 K 6 7 8 9 5
B A E D C G H I J F 4 0 1 2 3 9 5 6 7 K 8
C B A E D H I J F G 3 4 0 1 2 8 9 5 K 7 6
D C B A E I J F G H 2 3 4 0 1 7 8 K 5 6 9
E D C B A J F G H I 1 2 3 4 0 6 K 8 9 5 7
F G H I J K E D C B 5 9 8 7 6 0 4 3 2 1 A
G H I J F B A E D K 6 5 9 8 7 1 0 4 3 2 C
H I J F G C B A K D 7 6 5 9 8 2 1 0 4 3 E
I J F G H D C K A E 8 7 6 5 9 3 2 1 0 4 B
J F G H I E K C B A 9 8 7 6 5 4 3 2 1 0 D
4 2 0 3 1 A D B E C F H J G I 5 7 9 6 8 K
```

Partial square

```
. . . . K . . . . . . . . . . . . . . 1
. . . K . . 5 . . . . A . . . . . . . F 3
. 1 . . . . 9 5 . . . E A . . . . . F G .
. K 1 0 . . 8 9 5 . . D E A . . . F G H .
. . 2 . . . 7 8 9 5 . C D E A . F G H I 4
. . . . . 0 . 2 3 4 K . . . . . B C D E F
. 5 9 8 7 4 0 1 2 3 . K . . . E A B C D .
. . 5 9 8 3 4 0 1 2 . . K F . D E A B . J
. . . 5 9 2 3 4 0 1 . . . . H C D E A B .
. . . . 5 1 2 3 4 0 . . G . . B C D E A I
. . . . . 0 1 2 3 4 . 6 7 8 9 5
. A E D C . . . . F . 0 1 2 3 9 5 6 7 . .
. . A E D . . . F G 3 4 0 1 2 8 9 5 . 7 6
. . . A E . . F G H 2 3 4 0 1 7 8 . 5 6 .
. . . . A . F G H I 1 2 3 4 0 6 . 8 9 5 .
. . . . . E D C B . 9 8 7 6 0 4 3 2 . .
. . . . F B A E D . 6 5 9 8 7 1 0 4 3 2 .
. . . F G C B A . D 7 6 5 9 8 2 1 0 4 3 .
. . F G H D C . A E 8 7 6 5 9 3 2 1 0 4 .
. F G H I E . C B A 9 8 . 6 5 . 3 2 1 0 .
4 2 0 . . . . . . . . H J . I 5 . . . . K
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E107: order 21, 255 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m5_seed4.json`

SHA256: `2900616342d2862439a0f75f72c580031e5037cc885bafa033e2e683574bf821`

Completion

```
K 4 3 2 1 5 6 7 8 9 A B C D E F G H I J 0
1 0 4 3 K 9 5 6 7 8 E A B C D G H I J F 2
2 1 0 K 3 8 9 5 6 7 D E A B C H I J F G 4
3 2 K 0 4 7 8 9 5 6 C D E A B I J F G H 1
4 K 2 1 0 6 7 8 9 5 B C D E A J F G H I 3
5 9 8 7 6 0 1 2 3 4 K G H I J A B C D E F
6 5 9 8 7 4 0 1 2 3 G K I J F E A B C D H
7 6 5 9 8 3 4 0 1 2 H I K F G D E A B C J
8 7 6 5 9 2 3 4 0 1 I J F K H C D E A B G
9 8 7 6 5 1 2 3 4 0 J F G H K B C D E A I
A E D C B F G H I J 0 1 2 3 4 K 6 7 8 9 5
B A E D C G H I J F 4 0 1 2 3 9 5 6 7 K 8
C B A E D H I J F G 3 4 0 1 2 8 9 5 K 7 6
D C B A E I J F G H 2 3 4 0 1 7 8 K 5 6 9
E D C B A J F G H I 1 2 3 4 0 6 K 8 9 5 7
F G H I J K E D C B 5 9 8 7 6 0 4 3 2 1 A
G H I J F B A E D K 6 5 9 8 7 1 0 4 3 2 C
H I J F G C B A K D 7 6 5 9 8 2 1 0 4 3 E
I J F G H D C K A E 8 7 6 5 9 3 2 1 0 4 B
J F G H I E K C B A 9 8 7 6 5 4 3 2 1 0 D
0 3 1 4 2 A D B E C F H J G I 5 7 9 6 8 K
```

Partial square

```
K . . . . . . . . . . . . . . . . . . . 0
. 0 . . K . 5 . . . . A . . . . . . . F .
. 1 0 K . . 9 5 . . . E A . . . . . F G 4
. . . 0 . . 8 9 5 . . D E A . . . F G H 1
. . 2 . 0 . 7 8 9 5 . C D E A . F G H I .
. . . . . . . 1 2 3 4 K . . . . . B C D E .
. 5 9 8 7 4 . 1 2 3 . K . . F E A B C D .
. . 5 9 8 3 4 . 1 2 . . K . G D E A B . J
. . . 5 9 2 3 4 . 1 . . . . . C D E A B G
. . . . 5 1 2 3 4 . . . G H . B C D E A I
. . . . . 0 1 2 3 4 . 6 7 8 9 5
. A E D C . . . . F 4 0 1 2 3 9 5 6 7 . .
. . A E D . . . F G 3 4 0 1 2 8 9 5 . 7 .
. . . A E . . F G H 2 3 4 0 1 7 8 . 5 6 .
. . . . A . F G H I 1 2 3 4 0 6 . 8 9 5 .
. . . . . E D C B . 9 8 7 6 . 4 3 2 1 A
. . . . F B A E D . 6 5 9 8 7 1 . 4 3 2 .
. . . F G C B A . D 7 6 5 9 8 2 1 . 4 3 .
. . F G H D C . A E 8 7 6 5 9 3 2 1 . 4 .
. F G H I E . C B A 9 8 . 6 5 4 3 2 1 . .
0 3 1 . 2 A . . . C . H J . . 5 . . . . K
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E108: order 21, 253 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m5_seed1.json`

SHA256: `138a199877d6a1ab4ded44a6b2acc14c55d770c5095f6e17b7be1a414ed239f5`

Completion

```
K 4 3 2 1 5 6 7 8 9 A B C D E F G H I J 0
1 0 4 3 K 9 5 6 7 8 E A B C D G H I J F 2
2 1 0 K 3 8 9 5 6 7 D E A B C H I J F G 4
3 2 K 0 4 7 8 9 5 6 C D E A B I J F G H 1
4 K 2 1 0 6 7 8 9 5 B C D E A J F G H I 3
5 9 8 7 6 0 1 2 3 4 K G H I J A B C D E F
6 5 9 8 7 4 0 1 2 3 G K I J F E A B C D H
7 6 5 9 8 3 4 0 1 2 H I K F G D E A B C J
8 7 6 5 9 2 3 4 0 1 I J F K H C D E A B G
9 8 7 6 5 1 2 3 4 0 J F G H K B C D E A I
A E D C B F G H I J 0 1 2 3 4 K 6 7 8 9 5
B A E D C G H I J F 4 0 1 2 3 9 5 6 7 K 8
C B A E D H I J F G 3 4 0 1 2 8 9 5 K 7 6
D C B A E I J F G H 2 3 4 0 1 7 8 K 5 6 9
E D C B A J F G H I 1 2 3 4 0 6 K 8 9 5 7
F G H I J K E D C B 5 9 8 7 6 0 4 3 2 1 A
G H I J F B A E D K 6 5 9 8 7 1 0 4 3 2 C
H I J F G C B A K D 7 6 5 9 8 2 1 0 4 3 E
I J F G H D C K A E 8 7 6 5 9 3 2 1 0 4 B
J F G H I E K C B A 9 8 7 6 5 4 3 2 1 0 D
0 3 1 4 2 A D B E C F H J G I 5 7 9 6 8 K
```

Partial square

```
K . . . . . . . . . . . . . . . . . . . . 0
. 0 . . K . 5 . . . . A . . . . . . . F . .
. 1 0 K . . 9 5 . . . E A . . . . . F G 4
. . . 0 . . 8 9 5 . . D E A . . . F G H .
. . 2 . . . 7 8 9 5 . C D E A . F G H I 3
. . . . . 0 1 2 3 4 K . . . . . B C D E F
. 5 9 8 7 4 0 1 2 3 . K . . . E A B C D .
. . 5 9 8 3 4 . 1 2 . . K F . D E A B . J
. . . 5 9 2 3 4 . 1 . . . . H C D E A B .
. . . . 5 1 2 3 4 . . . G . . B C D E A I
. . . . . 0 1 2 3 4 . 6 7 8 9 5
. A E D C . . . . F 4 0 1 2 3 9 5 6 7 . .
. . A E D . . . F G 3 4 . 1 2 8 9 5 . 7 .
. . . A E . . F G H 2 3 4 . 1 7 8 K 5 6 .
. . . . A . F G H I 1 2 3 4 0 6 . 8 9 5 .
. . . . . E D C B . 9 8 7 6 0 4 3 2 1 .
. . . . F B A E D . 6 5 9 8 7 1 . 4 3 2 .
. . . F G C B A . D 7 6 5 9 8 2 1 . 4 3 .
. . F G H D C . A E 8 7 6 5 9 3 2 1 . 4 .
. F G H I E . C B A 9 8 . 6 5 4 3 2 1 0 .
0 3 . 4 . . . . . . . H J . I 5 . . . . K
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E109: order 22, 303 clues

Status I. Previously independently verified example; see the archived certificate directory.

Source: `outputs/lcs_examples.json:9`

SHA256: `bc9f5d45dea73255335e3d71ed2bc6d1257cd3337591c7c79c54e06897b8b675`

Completion

```
8 9 1 2 3 4 5 6 7 A 0 J K C D E F G H I L B
0 8 9 3 4 2 6 7 5 1 A B J K E F D H I G C L
9 A 8 4 2 3 7 5 6 0 1 K L J F D E I G H B C
2 3 4 8 9 7 A 0 1 5 6 D E F J K I L B C G H
3 4 2 6 8 9 0 1 A 7 5 E F D H J K B C L I G
4 2 3 9 5 8 1 A 0 6 7 F D E K G J C L B H I
5 6 7 A 0 1 8 9 4 2 3 G H I L B C J K F D E
6 7 5 0 1 A 3 8 9 4 2 H I G B C L E J K F D
7 5 6 1 A 0 9 2 8 3 4 I G H C L B K D J E F
A 1 0 5 7 6 2 4 3 9 8 L C B G I H D F E K J
1 0 A 7 6 5 4 3 2 8 9 C B L I H G F E D J K
J K C D E F G H I L B 8 9 1 2 3 4 5 6 7 A 0
B J K E F D H I G C L 0 8 9 3 4 2 6 7 5 1 A
K L J F D E I G H B C 9 A 8 4 2 3 7 5 6 0 1
D E F J K I L B C G H 2 3 4 8 9 7 A 0 1 5 6
E F D H J K B C L I G 3 4 2 6 8 9 0 1 A 7 5
F D E K G J C L B H I 4 2 3 9 5 8 1 A 0 6 7
G H I L B C J K F D E 5 6 7 A 0 1 8 9 4 2 3
H I G B C L E J K F D 6 7 5 0 1 A 3 8 9 4 2
I G H C L B K D J E F 7 5 6 1 A 0 9 2 8 3 4
L C B G I H D F E K J A 1 0 5 7 6 2 4 3 9 8
C B L I H G F E D J K 1 0 A 7 6 5 4 3 2 8 9
```

Partial square

```
8 . . 2 3 4 5 . 7 A . . K C D E F . H . . B
. 8 9 3 4 2 . 7 . 1 . B . . E F D H . G . L
. . 8 4 . . 7 5 6 0 1 K L . . D E I G H . C
2 . 4 8 . . A . 1 5 6 . E F . K I L B C . H
. . 2 . 8 9 0 1 A 7 . E F D H . K . C L . G
. 2 3 . . 8 1 A . 6 . F . E K G . C L B . I
5 6 7 . 0 . 8 9 4 2 3 G H I L B C . . F . .
6 7 . 0 1 A . 8 . 4 . H I G B . . E . K . D
7 5 6 1 . 0 . . 8 3 4 I G . . L B K D . . .
A 1 0 5 7 6 2 4 3 9 8 . . . . . . . . . . .
1 . . . 6 . . 3 . 8 9 . B L I . G F E D . K
. K C D E F . H . . B . 9 1 2 3 4 . 6 . . 0
B . . E F D H . G . L 0 . . 3 4 2 6 . 5 . A
K L . . D E I G H . C 9 A . . 2 3 7 5 6 . 1
. E F . K I L B C . H . 3 4 . 9 7 A 0 1 . 6
E F D H . K . C L . G 3 4 2 6 . 9 . 1 A . 5
F . E K G . C L B . I 4 . 3 9 5 . 1 A 0 . 7
G H I L B C . . F . . 5 6 7 A 0 1 . . 4 . .
H I G B . . E . K . D 6 7 5 0 . . 3 . 9 . 2
I G . . L B K D . . . 7 5 . . A 0 9 2 . . .
. . . . . . . . . . . . . . . . . . . . . .
. B L I . G F E D . K . 0 A 7 . 5 4 3 2 . 9
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E110: order 23, 366 clues

Status I. Previously independently verified example; see the archived certificate directory.

Source: `outputs/lcs_bunya_verification_round2/n23/bunya_record.json`

SHA256: `ee95cc084a51310f8a2fab0301740a3a7e368afd08306ad256b415e37180f83e`

Completion

```
8 9 1 M 3 4 5 6 7 A 0 J K C D E F G H I L B 2
0 8 9 3 4 2 6 7 5 1 A B J K E F D H M G C L I
9 A 8 4 2 3 7 5 6 0 1 K L M F D E I G H B C J
2 3 4 8 M 7 A E 1 5 6 D 0 F J K I L B C G H 9
3 4 2 6 8 9 0 1 A 7 5 E F D H J K B C L I M G
B 2 3 9 F 8 1 A G 6 7 0 D E K 4 J C L M H I 5
5 6 7 A E 1 8 9 4 2 M G H I L B C J K F D 0 3
6 7 5 0 1 A 3 8 9 4 2 H I G B C M E J K F D L
7 5 6 1 A 0 9 2 8 3 4 I G H C L B M D J E F K
A 1 0 5 7 6 2 M 3 9 8 L C B G I H D F E K J 4
M 0 A 7 6 5 4 3 2 8 9 C B L I H G F E D J K 1
J K C D 0 F G H I L B M 9 1 2 3 4 5 6 7 A E 8
4 J K E 5 M H I 0 C L F 8 9 3 G 2 A 7 B 1 6 D
K L J F D E I G H B C 9 A 8 4 2 3 7 5 6 M 1 0
D E F J K I L B M G H 2 3 4 8 9 7 6 0 1 5 A C
E M D H J L B C K I G 3 4 2 6 8 9 0 1 A 7 5 F
F D E K G J C L B M I 4 2 3 9 5 8 1 A 0 6 7 H
G H I L B C J K F D E 5 M 7 A 0 1 8 9 4 2 3 6
H I G B C K M J L F D 6 7 5 0 1 A 3 8 9 4 2 E
I G H C L B K D J E F 7 5 6 1 M 0 9 2 8 3 4 A
L C M G I H D F E K J A 1 0 5 7 6 2 4 3 9 8 B
C B L I H G F 0 D J K 1 E A M 6 5 4 3 2 8 9 7
1 F B 2 9 D E 4 C H 3 8 6 J 7 A L K I 5 0 G M
```

Partial square

```
8 9 . . 3 . 5 6 7 A 0 J K C D E F G H I L B .
. 8 9 3 4 2 6 7 5 1 . . J K E F D H . G C L .
9 . 8 4 2 3 7 5 6 0 1 K . . F D E I G . B C .
2 3 4 8 . . A . 1 5 6 D . F J K I L B C G H .
3 4 2 . 8 9 0 1 . 7 5 E F . H J K B C . I . .
. 2 . 9 . . 1 A . 6 7 0 D E K . . C L . H I .
5 6 . A . 1 8 9 4 2 . G H I L B C J K . D . .
. 7 5 0 1 A . . 9 4 2 H . G B . . . J . F D .
7 5 6 1 A 0 9 . 8 3 4 I G H C L B . . J E F .
A 1 0 . 7 6 2 . 3 . . L . B G I H D F E . . .
. 0 A 7 6 5 4 3 2 . 9 . B L I H G F E D . K .
J K . D . F . H I L B . 9 . 2 3 . 5 6 . A E .
4 J . E . . . I 0 C L . . 9 3 . . . . . 1 . .
K . J F D E I G H B C 9 A 8 4 2 . 7 5 6 . 1 .
D E F J . . L . . G H 2 3 4 8 9 . . 0 1 5 A .
E . D . J . B C . I G 3 4 2 . 8 9 0 1 A 7 5 .
F D E K . J C L B . I 4 2 3 9 5 8 . A . 6 7 .
G H I L B . J K . D E 5 . 7 A 0 1 8 9 . 2 3 .
. I G B C . . J L . D . 7 5 . 1 . . . 9 4 2 .
I G H C L B K . J . F 7 5 6 1 . 0 9 2 8 3 4 .
L C . G . H . . E K J A 1 0 5 . 6 . 4 3 . . .
C . L I H G F 0 D . K 1 E A . 6 5 4 3 2 8 9 .
. . . . . M
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E111: order 23, 365 clues

Status I. Previously independently verified example; see the archived certificate directory.

Source: `outputs/lcs_bunya_verification_2026_10_15/n23/bunya_record.json`

SHA256: `60a922b62d228803948533ea38b0de55778535b88f69971c7ede9a3f4c4c8781`

Completion

```
8 0 9 2 M 4 5 6 7 A 1 J B K D E F G H I L 3 C
9 8 M 3 4 2 6 7 5 1 0 K J L E F D H I G C B A
1 9 8 4 2 3 7 5 6 0 A C K J F M E I G H B L D
2 3 4 8 6 9 A 0 1 5 7 D E F J H K L B C G M I
3 4 2 9 8 5 0 1 A 7 6 E F D K J G B C M I H L
4 2 3 7 9 8 1 A 0 6 5 F D E I K J C L B M G H
5 6 7 A 0 1 8 3 M 2 4 G H I L B C J E K D F 9
6 7 5 M 1 A 9 8 2 4 3 H I G B C L K J D F E 0
7 M 6 1 A 0 4 9 8 3 2 I G H C L B F K J E D 5
A 1 0 5 7 6 2 4 3 9 8 L C B G I H D M E K J F
0 A 1 6 5 M 3 2 4 8 9 B L C H G I E D F J K 7
J B K D E F G H I L C 8 0 9 M 3 4 5 6 7 A 1 2
K I L E F D H J G C M 9 8 A 3 4 2 6 7 5 1 0 B
C K J F D E I M H B L 1 9 8 4 2 3 7 5 6 0 A G
D E F J H K L B C G I 2 M 4 8 6 9 A 0 1 5 7 3
M F D K J G B C L I H 3 4 2 9 8 5 0 1 A 7 6 E
F D E I K J C L B H G 4 2 3 7 9 M 1 A 0 6 5 8
G H I L B C M E K D F 5 6 7 A 0 1 8 3 9 2 4 J
H J G B 3 L K I D F E M 7 5 0 1 A 9 8 2 4 C 6
I G H C L B F K J E D 7 5 6 1 A 0 M 9 8 3 2 4
L C B G I H D F E M J A 1 0 5 7 6 2 4 3 9 8 K
B L C H G I E D F J K 0 A M 6 5 7 3 2 4 8 9 1
E 5 A 0 C 7 J G 9 K B 6 3 1 2 D 8 4 F L H I M
```

Partial square

```
8 0 9 2 . 4 5 6 . A 1 J . K D . F G H I L . .
. . . . . 2 6 7 5 1 0 K . . E F D H . G C B .
. 9 8 4 2 3 7 . 6 0 A . K J F . E I G H B L .
2 . 4 8 . 9 . . . 5 7 D E F J . K L B C G . .
. 4 2 9 . . 0 1 . 7 6 E F D K . . . C . I H .
4 2 3 . 9 8 1 A 0 6 5 F D E . K . C L B . G .
5 6 7 A 0 1 8 . . 2 4 G H . L B C J . K D F .
6 7 5 . 1 A 9 . . 4 3 H . G B C L K J . F E .
7 . 6 1 A 0 . 9 8 3 2 I . H C L B . K J E D .
A 1 0 . 7 6 2 4 3 . 8 L C B . I H D . E . J .
0 A 1 6 5 . 3 2 4 . 9 B L C H G I E D F J K .
. . K D E . G H I L C 8 . 9 . 3 . 5 6 7 A 1 .
K . . . . . . . G C . 9 8 . . 4 2 6 . 5 1 0 .
C K J . D E . . H B L . 9 8 4 2 3 7 5 6 0 A .
D E F . . K L B . G I 2 . 4 8 6 9 A 0 . 5 7 .
. F D K . G B C L I H 3 4 2 9 8 . 0 1 A . 6 .
F D E I K J . L B H G 4 2 . . 9 . 1 A 0 6 5 .
G H I L B C . E K D F 5 6 7 A 0 1 8 . 9 2 4 .
. . G . . . K I D F . . . . . . . 9 . . . . .
. G H C L B . K J E D 7 5 6 1 A 0 . 9 8 3 2 .
L C B G I H D . E . . A 1 0 5 . 6 2 4 3 . . .
B L C H G I E D F . K 0 A . 6 5 7 3 2 4 . 9 .
. . . . . . . . . . . . . . . . . . . . . . M
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E112: order 27, 446 clues

Status I. Previously independently verified example; see the archived certificate directory.

Source: `outputs/lcs_bunya_verification_round3/n27/bunya_record.json`

SHA256: `132d18b9403eb7a8dc8b3f1aafe4c737e73a6b4ac63149e122be4750af72e8b2`

Completion

```
0 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q
1 0 7 6 5 2 3 8 4 A 9 G F E B C H D J I P O N K L Q M
2 5 0 4 8 1 7 3 6 B E 9 D H A G C F K N I M Q J P L O
8 6 4 0 2 3 1 5 7 H F D 9 B C A E G Q O M I K L J N P
5 2 1 8 0 7 4 6 3 E B A H 9 G D F C N K J Q I P M O L
6 3 5 1 7 0 8 4 2 F C E A G 9 H D B O L N J P I Q M K
3 4 6 7 1 8 0 2 5 C D F G A H 9 B E L M O P J Q I K N
7 8 3 5 6 4 2 0 1 G H C E F D B 9 A P Q L N O M K I J
4 7 8 2 3 6 5 1 0 D G H B C F E A 9 M P Q K L O N J I
I J K L M N O P Q 0 1 2 3 4 5 6 7 8 9 A B C D E F G H
J I P O N K L Q M 1 0 7 6 5 2 3 8 4 A 9 G F E B C H D
K N I M Q J P L O 2 5 0 4 8 1 7 3 6 B E 9 D H A G C F
Q O M I K L J N P 8 6 4 0 2 3 1 5 7 H F D 9 B C A E G
N K J Q I P M O L 5 2 1 8 0 7 4 6 3 E B A H 9 G D F C
O L N J P I Q M K 6 3 5 1 7 0 8 4 2 F C E A G 9 H D B
L M O P J Q I K N 3 4 6 7 1 8 0 2 5 C D F G A H 9 B E
P Q L N O M K I J 7 8 3 5 6 4 2 0 1 G H C E F D B 9 A
M P Q K L O N J I 4 7 8 2 3 6 5 1 0 D G H B C F E A 9
9 A B C D E F G H I J K L M N O P Q 0 1 2 3 4 5 6 7 8
A 9 G F E B C H D J I P O N K L Q M 1 0 7 6 5 2 3 8 4
B E 9 D H A G C F K N I M Q J P L O 2 5 0 4 8 1 7 3 6
H F D 9 B C A E G Q O M I K L J N P 8 6 4 0 2 3 1 5 7
E B A H 9 G D F C N K J Q I P M O L 5 2 1 8 0 7 4 6 3
F C E A G 9 H D B O L N J P I Q M K 6 3 5 1 7 0 8 4 2
C D F G A H 9 B E L M O P J Q I K N 3 4 6 7 1 8 0 2 5
G H C E F D B 9 A P Q L N O M K I J 7 8 3 5 6 4 2 0 1
D G H B C F E A 9 M P Q K L O N J I 4 7 8 2 3 6 5 1 0
```

Partial square

```
. . . . . . . . . . . . . . . . . . . . . . . . . . .
1 . 7 . . . 3 8 4 A . G F E . C . . . I . O . K L Q M
. 5 0 4 8 1 7 3 6 . . 9 D . A . . F K N I . . . P L O
8 6 4 0 . . . . 7 . . D 9 B C A E G Q O . . K L . N P
. 2 . 8 0 7 4 6 3 E . A H 9 G . F C N K . Q I . M . .
6 3 . 1 7 0 8 4 2 F . E A . 9 H D B O L . . P . . . K
3 . 6 7 1 8 . . . . . . . A H 9 B . . M O P . Q . K N
7 8 3 5 6 4 2 0 1 . . C E F . B 9 A P Q L N O . K I .
4 7 8 . 3 6 5 1 0 D . H B C . E A 9 . P . K L . N . I
I . K L M N O . Q O . 2 3 4 . 6 . 8 9 A B C D E F G .
. I P O N K L Q . . . . . 5 2 3 8 . . . G F E B . . D
K N . . Q . . L . 2 . 0 4 . 1 . . 6 B E 9 . H A G C .
. . M I . L . N P 8 . . 0 2 3 1 5 7 H . D 9 . C A . G
N . . Q I P . O L 5 . 1 8 . . 4 6 3 . B A H 9 G D . C
. L N . P I Q M K 6 . . 1 7 . 8 4 2 . C . A . . H . B
L . . . . Q I . N 3 . 6 7 . 8 0 2 5 C D F . A H 9 B .
P Q L N O . . I . 7 . 3 5 . . . 0 1 G . C E F D B 9 .
. P . K L . N . I 4 . 8 2 3 6 5 1 0 D . H . C . E A 9
. A B C D . F . . I . K L M N O P Q 0 1 2 3 . . 6 7 8
. 9 G . E . C H D . . P O . K L . . 1 . 7 . 5 2 3 . 4
. . . . H A . . . . . . . . . . . . 2 . . . . . . . .
H F D . . C . E G Q . M I K L . . P . 6 . 0 2 3 1 5 .
. B . . 9 . D F C N . . Q I P . O . 5 2 1 8 0 . . 6 .
. C E A G . . D B O . N . P . . . K 6 3 . . 7 0 . 4 .
C D F G . H 9 . . L . O . . Q . . N 3 4 6 . 1 . . 2 5
G H C . F . B 9 . . . L . O M K I . . . . 5 6 4 2 0 1
. G . B . F . A 9 . . . K . O N . I . 7 8 2 3 6 5 1 .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E113: order 27, 442 clues

Status I. Previously independently verified example; see the archived certificate directory.

Source: `outputs/lcs_bunya_verification_round2/n27/bunya_record.json`

SHA256: `fe35dcad40ad9b345a1b0c2de99278b2038ec4caf812ab59b124ee4fc0332c91`

Completion

```
0 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q
1 0 7 6 5 2 3 8 4 A 9 G F E B C H D J I P O N K L Q M
2 5 0 4 8 1 7 3 6 B E 9 D H A G C F K N I M Q J P L O
8 6 4 0 2 3 1 5 7 H F D 9 B C A E G Q O M I K L J N P
5 2 1 8 0 7 4 6 3 E B A H 9 G D F C N K J Q I P M O L
6 3 5 1 7 0 8 4 2 F C E A G 9 H D B O L N J P I Q M K
3 4 6 7 1 8 0 2 5 C D F G A H 9 B E L M O P J Q I K N
7 8 3 5 6 4 2 0 1 G H C E F D B 9 A P Q L N O M K I J
4 7 8 2 3 6 5 1 0 D G H B C F E A 9 M P Q K L O N J I
I J K L M N O P Q 0 1 2 3 4 5 6 7 8 9 A B C D E F G H
J I P O N K L Q M 1 0 7 6 5 2 3 8 4 A 9 G F E B C H D
K N I M Q J P L O 2 5 0 4 8 1 7 3 6 B E 9 D H A G C F
Q O M I K L J N P 8 6 4 0 2 3 1 5 7 H F D 9 B C A E G
N K J Q I P M O L 5 2 1 8 0 7 4 6 3 E B A H 9 G D F C
O L N J P I Q M K 6 3 5 1 7 0 8 4 2 F C E A G 9 H D B
L M O P J Q I K N 3 4 6 7 1 8 0 2 5 C D F G A H 9 B E
P Q L N O M K I J 7 8 3 5 6 4 2 0 1 G H C E F D B 9 A
M P Q K L O N J I 4 7 8 2 3 6 5 1 0 D G H B C F E A 9
9 A B C D E F G H I J K L M N O P Q 0 1 2 3 4 5 6 7 8
A 9 G F E B C H D J I P O N K L Q M 1 0 7 6 5 2 3 8 4
B E 9 D H A G C F K N I M Q J P L O 2 5 0 4 8 1 7 3 6
H F D 9 B C A E G Q O M I K L J N P 8 6 4 0 2 3 1 5 7
E B A H 9 G D F C N K J Q I P M O L 5 2 1 8 0 7 4 6 3
F C E A G 9 H D B O L N J P I Q M K 6 3 5 1 7 0 8 4 2
C D F G A H 9 B E L M O P J Q I K N 3 4 6 7 1 8 0 2 5
G H C E F D B 9 A P Q L N O M K I J 7 8 3 5 6 4 2 0 1
D G H B C F E A 9 M P Q K L O N J I 4 7 8 2 3 6 5 1 0
```

Partial square

```
. . . . . . . . . . . . . . . . . . . . . . . . . . .
1 . 7 . . . 3 8 4 A . G F E . C . . . I . O . K L Q M
2 5 . 4 8 1 7 3 6 B . 9 D . A . . F . N I . Q . P . O
8 6 4 0 2 . . . 7 . . D 9 B C A E G Q O . . K L . N P
5 2 . 8 0 7 . 6 3 E . A H 9 G D . . N K . Q I . M . .
6 3 . 1 7 0 8 4 2 F . E A . . H D B O L . . P . . . K
3 4 6 7 1 8 . 2 . C . F . A H . B . . M O P . Q . K .
7 8 3 5 6 4 2 0 1 G . C . F . . 9 A P Q L N O . . I .
4 7 8 . 3 6 5 1 0 D . . . C . E A 9 . P . K L . N . I
I . K L M N O . Q O . 2 3 4 5 6 . 8 9 A B C D E . G .
. . P O . K L Q . . . 7 . 5 2 3 8 . A . G F E B . . D
K N . . Q . . . . 2 . 0 4 . 1 . . 6 B . 9 D . A G C .
. . M I . L . N P 8 . . 0 2 3 1 5 . H . . . B C A E G
N . . Q I P . O L 5 . . 8 . . 4 6 3 . B A H 9 G D . C
. L N . P I Q M K . . . 1 7 . 8 4 2 . C . A G . H . B
. . . . . Q I . N 3 . 6 . . 8 0 2 5 C D F . A H 9 B E
P Q L N O . . I . 7 . . 5 . . . . 1 G H C E F D B 9 .
. P . K L . N . I 4 . 8 2 3 6 5 1 0 D . . . C F E A 9
. A B C D . F . . I . K L M N O P Q 0 1 2 3 . . 6 7 8
A 9 G . . . C H D . . P O . K L . . 1 0 7 6 5 2 3 . 4
. . . . . A . . . . . . . . . L . 2 . . . . . . . . .
H F D . . C . E G Q . M I K . . N P . 6 . 0 2 3 1 5 7
. B A H 9 G D F . N . . Q I P . . . 5 2 1 8 0 . . 6 .
. C E . G . . D B O . N . P . . . K 6 3 . . 7 0 . 4 .
C D F G . H 9 . . . . O . . Q . K N 3 4 6 . 1 . . 2 .
. H C . F . B 9 . . . L . O M K I . 7 8 . 5 6 4 2 0 1
. G H B . F . A 9 . . . K L O N . I . 7 8 . 3 6 5 1 0
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E114: order 29, 621 clues

Status I. Previously independently verified example; see the archived certificate directory.

Source: `outputs/lcs_bunya_verification_2026_10_15/n29/bunya_record.json`

SHA256: `910b48b6018605b143f149715bb7e902363b39e205eb2d2f8a7e926a283cb7f6`

Completion

```
S 6 5 4 3 2 1 7 8 9 A B C D E F G H I J K L M N O P Q R 0
1 0 6 5 4 3 S D 7 8 9 A B C K E F G H I J M N O P Q R L 2
2 1 0 6 5 S 3 C D 7 8 9 A B J K E F G H I N O P Q R L M 4
3 2 1 0 S 5 4 B C D 7 8 9 A I J K E F G H O P Q R L M N 6
4 3 2 S 0 6 5 A B C D 7 8 9 H I J K E F G P Q R L M N O 1
5 4 S 2 1 0 6 9 A B C D 7 8 G H I J K E F Q R L M N O P 3
6 S 4 3 2 1 0 8 9 A B C D 7 F G H I J K E R L M N O P Q 5
7 D C B A 9 8 0 1 2 3 4 5 6 S M N O P Q R E F G H I J K L
8 7 D C B A 9 6 0 1 2 3 4 5 M S O P Q R L K E F G H I J N
9 8 7 D C B A 5 6 0 1 2 3 4 N O S Q R L M J K E F G H I P
A 9 8 7 D C B 4 5 6 0 1 2 3 O P Q S L M N I J K E F G H R
B A 9 8 7 D C 3 4 5 6 0 1 2 P Q R L S N O H I J K E F G M
C B A 9 8 7 D 2 3 4 5 6 0 1 Q R L M N S P G H I J K E F O
D C B A 9 8 7 1 2 3 4 5 6 0 R L M N O P S F G H I J K E Q
E K J I H G F L M N O P Q R 0 1 2 3 4 5 6 S 8 9 A B C D 7
F E K J I H G M N O P Q R L 6 0 1 2 3 4 5 D 7 8 9 A B S C
G F E K J I H N O P Q R L M 5 6 0 1 2 3 4 C D 7 8 9 S B A
H G F E K J I O P Q R L M N 4 5 6 0 1 2 3 B C D 7 S 9 A 8
I H G F E K J P Q R L M N O 3 4 5 6 0 1 2 A B C S 7 8 9 D
J I H G F E K Q R L M N O P 2 3 4 5 6 0 1 9 A S C D 7 8 B
K J I H G F E R L M N O P Q 1 2 3 4 5 6 0 8 S A B C D 7 9
L M N O P Q R S K J I H G F 7 D C B A 9 8 0 6 5 4 3 2 1 E
M N O P Q R L F E K J I H S 8 7 D C B A 9 1 0 6 5 4 3 2 G
N O P Q R L M G F E K J S H 9 8 7 D C B A 2 1 0 6 5 4 3 I
O P Q R L M N H G F E S J I A 9 8 7 D C B 3 2 1 0 6 5 4 K
P Q R L M N O I H G S E K J B A 9 8 7 D C 4 3 2 1 0 6 5 F
Q R L M N O P J I S G F E K C B A 9 8 7 D 5 4 3 2 1 0 6 H
R L M N O P Q K S I H G F E D C B A 9 8 7 6 5 4 3 2 1 0 J
0 5 3 1 6 4 2 E J H F K I G L N P R M O Q 7 9 B D 8 A C S
```

Partial square

```
. . 5 . 3 2 1 7 . . A B C D E F . H I . K . M N O P Q . .
1 0 6 5 4 3 . . 7 8 9 A B . K . . G H I J M N O . Q R L .
2 1 0 6 5 . . C D . 8 9 A B J K . . G H I N O P Q R L M .
3 2 1 0 . 5 4 B C D 7 8 9 A . J K E F G H O P Q . L M N .
4 . 2 . . 6 5 A B . D 7 8 9 H I J K E F G P Q R L M . O .
5 4 . 2 . 0 6 9 A . . . . 8 G H I J K E F Q R L M N O P .
6 . 4 3 2 1 0 8 9 . B . D 7 F G H I . . . R L . N O P Q .
7 D . . . 9 8 0 1 2 3 4 5 6 . M N O P Q R E . G H I J K .
8 7 D . B A 9 6 . 1 2 3 4 5 M . O P Q R L K . . . H I J .
9 8 7 . C B A 5 6 0 . 2 . 4 N O . Q R L M . K E F G H . .
A . 8 7 D C . 4 5 6 . 1 2 3 O . Q . L M N I J K E F G H .
B A . 8 7 D C . . 5 6 0 1 2 P Q R L . N O . I J K E F . .
C B A 9 8 7 . 2 3 4 5 6 0 . Q R L M N . P G H . . K . F .
D C . A 9 8 . 1 2 3 4 5 6 . R L M N O P . F G . I J K E .
. K . I . G . L M N O P Q R 0 1 2 3 4 5 6 . 8 9 A B C D .
F . . J I H G . N O P Q R . . 0 1 2 3 4 5 D 7 8 9 A B . .
G . . K J I H N . . Q R L M 5 6 0 . 2 3 4 . D 7 . 9 . . .
. G F E K J I O P Q R L . N 4 5 6 . 1 2 . B C D . . . A .
. H G . . K J P Q R L M N O 3 . 5 6 . 1 2 A B C . 7 8 9 .
. . H G F E K . R L M N . . 2 3 . 5 6 0 1 9 A . C D 7 8 .
. J I H G F E R L M N . P Q 1 2 3 4 5 6 0 8 . A B C D 7 .
L M N O P Q . . K J . H G F 7 D . . A 9 . . 6 5 4 . 2 1 .
M N . P Q R L F E K J . H . 8 7 D C B A 9 1 0 6 . . 3 . .
N O P Q R . M G F E K J . H 9 8 . D C B A . 1 0 6 5 . 3 .
O . Q R L M . H G F E . J I A 9 8 7 . . B 3 2 1 . . . 4 .
. Q R L M N O I H G . E K J B A . 8 7 D C 4 3 2 1 0 6 5 .
Q R . M N O P J . . G F E K C B . 9 . 7 D 5 4 3 2 1 . 6 .
R L M . . P Q K . I H G F E D C B A . 8 7 6 5 4 . 2 1 0 .
. . . . . . . . . . . . . . . . . . . . . . . . . . . . S
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E115: order 29, 539 clues

Status I. Previously independently checked uniqueness and all clue-deletion witnesses; uniqueness is solver-dependent.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m7_seed2.json`

SHA256: `86eb3b9134eea3075da628fde283eb022ce2d2f86f67735be3f98ee91342561e`

Completion

```
S 6 5 4 3 2 1 7 8 9 A B C D E F G H I J K L M N O P Q R 0
1 0 6 5 4 3 S D 7 8 9 A B C K E F G H I J M N O P Q R L 2
2 1 0 6 5 S 3 C D 7 8 9 A B J K E F G H I N O P Q R L M 4
3 2 1 0 S 5 4 B C D 7 8 9 A I J K E F G H O P Q R L M N 6
4 3 2 S 0 6 5 A B C D 7 8 9 H I J K E F G P Q R L M N O 1
5 4 S 2 1 0 6 9 A B C D 7 8 G H I J K E F Q R L M N O P 3
6 S 4 3 2 1 0 8 9 A B C D 7 F G H I J K E R L M N O P Q 5
7 D C B A 9 8 0 1 2 3 4 5 6 S M N O P Q R E F G H I J K L
8 7 D C B A 9 6 0 1 2 3 4 5 M S O P Q R L K E F G H I J N
9 8 7 D C B A 5 6 0 1 2 3 4 N O S Q R L M J K E F G H I P
A 9 8 7 D C B 4 5 6 0 1 2 3 O P Q S L M N I J K E F G H R
B A 9 8 7 D C 3 4 5 6 0 1 2 P Q R L S N O H I J K E F G M
C B A 9 8 7 D 2 3 4 5 6 0 1 Q R L M N S P G H I J K E F O
D C B A 9 8 7 1 2 3 4 5 6 0 R L M N O P S F G H I J K E Q
E K J I H G F L M N O P Q R 0 1 2 3 4 5 6 S 8 9 A B C D 7
F E K J I H G M N O P Q R L 6 0 1 2 3 4 5 D 7 8 9 A B S C
G F E K J I H N O P Q R L M 5 6 0 1 2 3 4 C D 7 8 9 S B A
H G F E K J I O P Q R L M N 4 5 6 0 1 2 3 B C D 7 S 9 A 8
I H G F E K J P Q R L M N O 3 4 5 6 0 1 2 A B C S 7 8 9 D
J I H G F E K Q R L M N O P 2 3 4 5 6 0 1 9 A S C D 7 8 B
K J I H G F E R L M N O P Q 1 2 3 4 5 6 0 8 S A B C D 7 9
L M N O P Q R S K J I H G F 7 D C B A 9 8 0 6 5 4 3 2 1 E
M N O P Q R L F E K J I H S 8 7 D C B A 9 1 0 6 5 4 3 2 G
N O P Q R L M G F E K J S H 9 8 7 D C B A 2 1 0 6 5 4 3 I
O P Q R L M N H G F E S J I A 9 8 7 D C B 3 2 1 0 6 5 4 K
P Q R L M N O I H G S E K J B A 9 8 7 D C 4 3 2 1 0 6 5 F
Q R L M N O P J I S G F E K C B A 9 8 7 D 5 4 3 2 1 0 6 H
R L M N O P Q K S I H G F E D C B A 9 8 7 6 5 4 3 2 1 0 J
0 5 3 1 6 4 2 E J H F K I G L N P R M O Q 7 9 B D 8 A C S
```

Partial square

```
S . . . . . . . . . . . . . . . . . . . . . . . . . . . .
. 0 . . . . S . 7 . . . . . . E . . . . . . . . . . L . .
. 1 0 . . S . . D 7 . . . . . K E . . . . . . . . . L M 4
. 2 . 0 . . . . C D 7 . . . . J K E . . . . . . . L M N .
. 3 2 S 0 . . . B C D 7 . . . I J K E . . . . . L M N O 1
. 4 S 2 1 0 . . A B C D 7 . . H I J K E . . . L M N O P 3
. . 4 3 2 1 0 . 9 A B C D 7 . G H I J K E . L M N O P Q 5
. . . . . . . . 1 2 3 4 5 6 . . . . . . . . F G H I J K .
. 7 D C B A 9 6 0 1 2 3 4 5 . S . . . . L K E F G H I J .
. . 7 D C B A 5 6 . 1 2 3 4 . . S . . . M J K E F G H I P
. . . 7 D C B 4 5 6 . 1 2 3 . . . S L M N I J K E F G . .
. . . . 7 D C 3 4 5 6 . 1 2 . . . L S N O H I J K E F G M
. . . . . 7 D 2 3 4 5 6 0 1 . . L M N S . G H I J K E F O
. . . . . 7 1 2 3 4 5 6 . . L M . O P S F G H I J K E Q
. . . . . 0 1 2 3 4 5 6 . 8 9 A B C D 7
. E K J I H G . . . . . L 6 0 1 2 3 4 5 D 7 8 9 A B . .
. . E K J I H . . . . . L M 5 6 0 1 2 3 4 C D 7 8 9 . B .
. . . E K J I . . . . L M N 4 5 6 0 1 2 3 B C D 7 . 9 A 8
. . . . E K J . . . L M N O 3 4 5 6 0 1 2 A B C S 7 8 9 .
. . . . . E K . . L M N O P 2 3 4 5 6 0 1 9 A . C D 7 8 B
. . . . . E . L M N O P Q 1 2 3 4 5 6 0 8 . A B C D 7 9
. . . . . S K J I H G F . D C B A 9 8 . 6 5 4 3 2 1 E
. . . . . L F E K J I H . 8 7 D C B A 9 1 . 6 5 4 3 2 G
. . . . . L M G F E K J . H 9 8 7 D C B A 2 1 0 6 5 4 3 .
. . . . L M N H G F E . J I A 9 8 7 D C B 3 2 1 . 6 5 4 .
. . . L M N O I H G S E K J B A 9 8 7 D C 4 3 2 1 . 6 5 F
. . L M N O P J I . G F E K C B A 9 8 7 D 5 4 3 2 1 0 6 H
. L M N O P Q K . I H G F E D C B . 9 8 7 6 5 4 3 2 1 . .
. 5 3 . . . 2 E J H F K . . . N . R . . Q 7 . B . . . C S
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E116: order 29, 537 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m7_seed3.json`

SHA256: `f5a1ea5bcac9165ae5c8e923b6c599ede3823b1dbfc37cf330b1594571da411d`

Completion

```
S 6 5 4 3 2 1 7 8 9 A B C D E F G H I J K L M N O P Q R 0
1 0 6 5 4 3 S D 7 8 9 A B C K E F G H I J M N O P Q R L 2
2 1 0 6 5 S 3 C D 7 8 9 A B J K E F G H I N O P Q R L M 4
3 2 1 0 S 5 4 B C D 7 8 9 A I J K E F G H O P Q R L M N 6
4 3 2 S 0 6 5 A B C D 7 8 9 H I J K E F G P Q R L M N O 1
5 4 S 2 1 0 6 9 A B C D 7 8 G H I J K E F Q R L M N O P 3
6 S 4 3 2 1 0 8 9 A B C D 7 F G H I J K E R L M N O P Q 5
7 D C B A 9 8 0 1 2 3 4 5 6 S M N O P Q R E F G H I J K L
8 7 D C B A 9 6 0 1 2 3 4 5 M S O P Q R L K E F G H I J N
9 8 7 D C B A 5 6 0 1 2 3 4 N O S Q R L M J K E F G H I P
A 9 8 7 D C B 4 5 6 0 1 2 3 O P Q S L M N I J K E F G H R
B A 9 8 7 D C 3 4 5 6 0 1 2 P Q R L S N O H I J K E F G M
C B A 9 8 7 D 2 3 4 5 6 0 1 Q R L M N S P G H I J K E F O
D C B A 9 8 7 1 2 3 4 5 6 0 R L M N O P S F G H I J K E Q
E K J I H G F L M N O P Q R 0 1 2 3 4 5 6 S 8 9 A B C D 7
F E K J I H G M N O P Q R L 6 0 1 2 3 4 5 D 7 8 9 A B S C
G F E K J I H N O P Q R L M 5 6 0 1 2 3 4 C D 7 8 9 S B A
H G F E K J I O P Q R L M N 4 5 6 0 1 2 3 B C D 7 S 9 A 8
I H G F E K J P Q R L M N O 3 4 5 6 0 1 2 A B C S 7 8 9 D
J I H G F E K Q R L M N O P 2 3 4 5 6 0 1 9 A S C D 7 8 B
K J I H G F E R L M N O P Q 1 2 3 4 5 6 0 8 S A B C D 7 9
L M N O P Q R S K J I H G F 7 D C B A 9 8 0 6 5 4 3 2 1 E
M N O P Q R L F E K J I H S 8 7 D C B A 9 1 0 6 5 4 3 2 G
N O P Q R L M G F E K J S H 9 8 7 D C B A 2 1 0 6 5 4 3 I
O P Q R L M N H G F E S J I A 9 8 7 D C B 3 2 1 0 6 5 4 K
P Q R L M N O I H G S E K J B A 9 8 7 D C 4 3 2 1 0 6 5 F
Q R L M N O P J I S G F E K C B A 9 8 7 D 5 4 3 2 1 0 6 H
R L M N O P Q K S I H G F E D C B A 9 8 7 6 5 4 3 2 1 0 J
0 5 3 1 6 4 2 E J H F K I G L N P R M O Q 7 9 B D 8 A C S
```

Partial square

```
S . . . . . . . . . . . . . . . . . . . . . . . . . . . .
. 0 . . . . . 7 . . . . . E . . . . . . . . . . L 2
. 1 0 . . S . . D 7 . . . . . K E . . . . . . . L M .
. . 1 0 S . . . C D 7 . . . . J K E . . . . . . . L M N 6
. 3 2 S 0 . . . B C D 7 . . . I J K E . . . . . L M N O .
. 4 S 2 1 0 . . A B C D 7 . . H I J K E . . . L M N O P .
. . 4 3 2 1 0 . 9 A B C D 7 . G H I J K E . L M N O P Q .
. . . . . 0 1 2 3 4 5 6 S . . . . . F G H I J K L
. 7 D C B A 9 6 0 1 2 3 4 5 . . . . . L K E F G H I J N
. . 7 D C B A 5 6 . 1 2 3 4 . . S . . . M J K E F G H I .
. . . 7 D C B 4 5 6 0 1 2 3 . . . S . M . I J K E F G . .
. . . . 7 D C 3 4 5 6 0 1 2 . . . . S N O H I J K E F G .
. . . . . 7 D 2 3 4 5 6 0 1 . . . M N S P G H I J K E F O
. . . . . 7 1 2 3 4 5 6 . . . M N O P . F G H I J K E Q
. . . . . 0 1 2 3 4 5 6 S 8 9 A B C D .
. E K J I H G . . . . . L 6 0 1 2 3 4 5 D 7 8 9 A B S .
. . E K J I H . . . . . L M 5 6 . 1 2 3 4 C D 7 8 9 . B A
. . . E K J I . . . . L M N 4 5 6 0 1 2 3 B C D 7 S 9 A 8
. . . . E K J . . . L M N O 3 4 5 6 0 1 2 A B C S 7 8 9 .
. . . . . E K . . L M N O P 2 3 4 5 6 . 1 9 A S C D 7 8 .
. . . . . E . L M N O P Q 1 2 3 4 5 6 0 8 S A B C D 7 .
. . . . . S K J I H G F 7 D C B A 9 8 0 6 5 4 3 2 1 .
. . . . . L F E K J I H S 8 7 D C B A 9 1 0 6 5 4 3 2 .
. . . . . L M G F E K J S H 9 8 7 D C B A 2 1 0 6 5 4 3 .
. . . . L M N H G F E S J I A 9 8 7 D C B 3 2 1 0 6 5 4 .
. . . L M N O I H G S E K J B A 9 8 7 D C 4 3 2 1 0 6 5 .
. . L M N O P J I . G F E K C B A 9 8 7 D 5 4 3 2 1 0 6 .
. L M N O P Q K S I H G F E D C B . 9 8 7 6 5 4 3 2 1 0 .
. 5 3 1 . . . . . H F . . . L . P . . O Q . . . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E117: order 29, 536 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m7_seed6.json`

SHA256: `08b53dd7c8313c56a16168b087f767074a1c6cfc387ca6da46f0a58f900dd10f`

Completion

```
S 6 5 4 3 2 1 7 8 9 A B C D E F G H I J K L M N O P Q R 0
1 0 6 5 4 3 S D 7 8 9 A B C K E F G H I J M N O P Q R L 2
2 1 0 6 5 S 3 C D 7 8 9 A B J K E F G H I N O P Q R L M 4
3 2 1 0 S 5 4 B C D 7 8 9 A I J K E F G H O P Q R L M N 6
4 3 2 S 0 6 5 A B C D 7 8 9 H I J K E F G P Q R L M N O 1
5 4 S 2 1 0 6 9 A B C D 7 8 G H I J K E F Q R L M N O P 3
6 S 4 3 2 1 0 8 9 A B C D 7 F G H I J K E R L M N O P Q 5
7 D C B A 9 8 0 1 2 3 4 5 6 S M N O P Q R E F G H I J K L
8 7 D C B A 9 6 0 1 2 3 4 5 M S O P Q R L K E F G H I J N
9 8 7 D C B A 5 6 0 1 2 3 4 N O S Q R L M J K E F G H I P
A 9 8 7 D C B 4 5 6 0 1 2 3 O P Q S L M N I J K E F G H R
B A 9 8 7 D C 3 4 5 6 0 1 2 P Q R L S N O H I J K E F G M
C B A 9 8 7 D 2 3 4 5 6 0 1 Q R L M N S P G H I J K E F O
D C B A 9 8 7 1 2 3 4 5 6 0 R L M N O P S F G H I J K E Q
E K J I H G F L M N O P Q R 0 1 2 3 4 5 6 S 8 9 A B C D 7
F E K J I H G M N O P Q R L 6 0 1 2 3 4 5 D 7 8 9 A B S C
G F E K J I H N O P Q R L M 5 6 0 1 2 3 4 C D 7 8 9 S B A
H G F E K J I O P Q R L M N 4 5 6 0 1 2 3 B C D 7 S 9 A 8
I H G F E K J P Q R L M N O 3 4 5 6 0 1 2 A B C S 7 8 9 D
J I H G F E K Q R L M N O P 2 3 4 5 6 0 1 9 A S C D 7 8 B
K J I H G F E R L M N O P Q 1 2 3 4 5 6 0 8 S A B C D 7 9
L M N O P Q R S K J I H G F 7 D C B A 9 8 0 6 5 4 3 2 1 E
M N O P Q R L F E K J I H S 8 7 D C B A 9 1 0 6 5 4 3 2 G
N O P Q R L M G F E K J S H 9 8 7 D C B A 2 1 0 6 5 4 3 I
O P Q R L M N H G F E S J I A 9 8 7 D C B 3 2 1 0 6 5 4 K
P Q R L M N O I H G S E K J B A 9 8 7 D C 4 3 2 1 0 6 5 F
Q R L M N O P J I S G F E K C B A 9 8 7 D 5 4 3 2 1 0 6 H
R L M N O P Q K S I H G F E D C B A 9 8 7 6 5 4 3 2 1 0 J
0 5 3 1 6 4 2 E J H F K I G L N P R M O Q 7 9 B D 8 A C S
```

Partial square

```
S . . . . . . . . . . . . . . . . . . . . . . . . . . . .
. 0 . . . . S . 7 . . . . . . E . . . . . . . . . . L . .
. 1 0 . . S . . D 7 . . . . . K E . . . . . . . . . L M 4
. 2 1 0 . . . . C D 7 . . . . J K E . . . . . . . L M N 6
. 3 2 . 0 . . . B C D 7 . . . I J K E . . . . . L M N O .
. 4 S 2 . 0 . . A B C D 7 . . H I J K E . . . L M N O P .
. S 4 3 2 1 0 . 9 A B C D 7 . G H I J K E . L M N O P Q .
. . . . . . . 0 1 2 3 4 5 6 . . . . . . . E F G H I J K .
. 7 D C B A 9 6 0 1 2 3 4 5 . S . . . . L K E F G H I J .
. . 7 D C B A 5 6 0 1 2 3 4 . . S . . . M J K E F G H I P
. . . 7 D C B 4 5 6 0 1 2 3 . . . S L . . I J K E F G . R
. . . . 7 D C 3 4 5 6 0 1 2 . . . L S N O H I J K E F G .
. . . . . 7 D 2 3 4 5 6 0 1 . . L M N S P G H I J K E F .
. . . . . 7 1 2 3 4 5 6 0 . . M N O P S F G H I J K E Q
. . . . . 1 2 3 4 5 6 S 8 9 A B C D .
. E K J I H G . . . . . L 6 . 1 2 3 4 5 D 7 8 9 A B S .
. . E K J I H . . . . . L M 5 6 . 1 2 3 4 C D 7 8 9 S B .
. . . E K J I . . . . L M N 4 5 6 0 1 2 3 B C D 7 S 9 A .
. . . . E K J . . . L M N O 3 4 5 6 0 1 2 A B C S 7 8 9 .
. . . . . E K . . L M N O P 2 3 4 5 6 0 1 9 A . C D 7 8 B
. . . . . E . L M N O P Q 1 2 3 4 5 6 . 8 S A B C D 7 .
. . . . . S K J I H G F . D C B A 9 8 0 6 5 4 3 2 1 E
. . . . . L F E K J I H S 8 7 D C B A 9 1 . 6 5 4 3 2 .
. . . . . L M G F E K J S H 9 8 7 D C B A 2 1 0 6 5 4 3 .
. . . . L M N H G F E S J I A 9 8 7 D C B 3 2 1 0 6 5 4 .
. . . L M N O I H G . E K J B A 9 8 7 D C 4 3 2 1 0 6 5 .
. . L M N O P J I . G F E K C B A 9 8 7 D 5 4 3 2 1 0 6 .
. L M N O P Q K S I H G F E D C B . 9 8 7 6 5 4 3 2 1 0 .
. . . 1 6 . . . . H F . . . L . P . . O Q . . . . . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E118: order 29, 534 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_shifted_examples/thesis_m7_seed2.json`

SHA256: `1f74cc17db80715de857b89b426b70a600b6d69d85d8bd4a481505679f5dbb2a`

Completion

```
0 6 5 4 3 2 S 7 8 9 A B C D E F G H I J K L M N O P Q R 1
1 0 6 5 4 S 2 D 7 8 9 A B C K E F G H I J M N O P Q R L 3
2 1 0 6 S 4 3 C D 7 8 9 A B J K E F G H I N O P Q R L M 5
3 2 1 S 6 5 4 B C D 7 8 9 A I J K E F G H O P Q R L M N 0
4 3 S 1 0 6 5 A B C D 7 8 9 H I J K E F G P Q R L M N O 2
5 S 3 2 1 0 6 9 A B C D 7 8 G H I J K E F Q R L M N O P 4
S 5 4 3 2 1 0 8 9 A B C D 7 F G H I J K E R L M N O P Q 6
7 D C B A 9 8 0 1 2 3 4 5 6 S M N O P Q R E F G H I J K L
8 7 D C B A 9 6 0 1 2 3 4 5 M S O P Q R L K E F G H I J N
9 8 7 D C B A 5 6 0 1 2 3 4 N O S Q R L M J K E F G H I P
A 9 8 7 D C B 4 5 6 0 1 2 3 O P Q S L M N I J K E F G H R
B A 9 8 7 D C 3 4 5 6 0 1 2 P Q R L S N O H I J K E F G M
C B A 9 8 7 D 2 3 4 5 6 0 1 Q R L M N S P G H I J K E F O
D C B A 9 8 7 1 2 3 4 5 6 0 R L M N O P S F G H I J K E Q
E K J I H G F L M N O P Q R 0 1 2 3 4 5 6 S 8 9 A B C D 7
F E K J I H G M N O P Q R L 6 0 1 2 3 4 5 D 7 8 9 A B S C
G F E K J I H N O P Q R L M 5 6 0 1 2 3 4 C D 7 8 9 S B A
H G F E K J I O P Q R L M N 4 5 6 0 1 2 3 B C D 7 S 9 A 8
I H G F E K J P Q R L M N O 3 4 5 6 0 1 2 A B C S 7 8 9 D
J I H G F E K Q R L M N O P 2 3 4 5 6 0 1 9 A S C D 7 8 B
K J I H G F E R L M N O P Q 1 2 3 4 5 6 0 8 S A B C D 7 9
L M N O P Q R S K J I H G F 7 D C B A 9 8 0 6 5 4 3 2 1 E
M N O P Q R L F E K J I H S 8 7 D C B A 9 1 0 6 5 4 3 2 G
N O P Q R L M G F E K J S H 9 8 7 D C B A 2 1 0 6 5 4 3 I
O P Q R L M N H G F E S J I A 9 8 7 D C B 3 2 1 0 6 5 4 K
P Q R L M N O I H G S E K J B A 9 8 7 D C 4 3 2 1 0 6 5 F
Q R L M N O P J I S G F E K C B A 9 8 7 D 5 4 3 2 1 0 6 H
R L M N O P Q K S I H G F E D C B A 9 8 7 6 5 4 3 2 1 0 J
6 4 2 0 5 3 1 E J H F K I G L N P R M O Q 7 9 B D 8 A C S
```

Partial square

```
. . . . . S . . . . . . . . . . . . . . . . . . . . . .
. 0 . . . . . 7 . . . . . E . . . . . . . . . . . L . .
. 1 0 . S . . . D 7 . . . . . K E . . . . . . . . . L M 5
. 2 1 S . . . . C D 7 . . . . J K E . . . . . . . L M N 0
. 3 S 1 0 . . . B C D 7 . . . I J K E . . . . . L M N O .
. . 3 2 1 0 . . A B C D 7 . . H I J K E . . . L M N O P .
S . . . . . . . 9 A B C D 7 . G H I J K E . L M N O P Q 6
. . . . . . . . 0 . 2 3 4 5 6 . . . . . . . . F G H I J K .
. 7 D C B A 9 . 0 1 2 3 4 5 . S . . . . L K E F G H I J .
. . 7 D C B A 5 6 0 1 2 3 4 . . S . . . M J K E F G H I P
. . . 7 D C B 4 5 6 0 1 2 3 . . . S L M N I J K E F G . .
. . . . 7 D C 3 4 5 6 0 1 2 . . . L S N O H I J K E F G .
. . . . . 7 D 2 3 4 5 6 0 1 . . L M N S . G H I J K E F O
. . . . . 7 1 2 3 4 5 6 0 . L M . O P S F G H I J K E Q
. . . . . 0 1 2 3 4 5 6 . 8 9 A B C D 7
. E K J I H G . . . . . L 6 0 1 2 3 4 5 D 7 8 9 A B . .
. . E K J I H . . . . . L M 5 6 0 1 2 3 4 C D 7 8 9 . B .
. . . E K J I . . . . L M N 4 5 6 0 1 2 3 B C D 7 . 9 A 8
. . . . E K J . . . L M N O 3 4 5 6 0 1 2 A B C . 7 8 9 .
. . . . . E K . . L M N O P 2 3 4 5 6 0 1 9 A . C D 7 8 .
. . . . . E . L M N O P Q 1 2 3 4 5 6 0 8 . A B C D 7 9
. . . . . S K J I H G F 7 D C B A 9 8 0 6 5 4 3 2 . .
. . . . . L F E K J I H S 8 7 D C B A 9 1 0 6 5 4 3 2 G
. . . . . L M G F E K J . H 9 8 7 D C B A 2 1 0 6 5 4 3 .
. . . . L M N H G F E . J I A 9 8 7 D C B 3 2 1 0 6 5 4 .
. . . L M N O I H G . E K J B A 9 8 7 D C 4 3 2 1 0 6 5 F
. . L M N O P J I . G F E K C B A 9 8 7 D 5 4 3 2 1 0 6 H
. L M N O P Q K . I H G F E D C B . 9 8 7 . 5 4 3 2 1 0 .
. 4 2 0 5 3 1 E J . . . . . N . R . . Q . . B . . . . S
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E119: order 29, 533 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_shifted_examples/thesis_m7_seed0.json`

SHA256: `89062932222964b06b24ae1e42c734c4d0aac1a57d33d0f7f8cafcea09b2dcbb`

Completion

```
0 6 5 4 3 2 S 7 8 9 A B C D E F G H I J K L M N O P Q R 1
1 0 6 5 4 S 2 D 7 8 9 A B C K E F G H I J M N O P Q R L 3
2 1 0 6 S 4 3 C D 7 8 9 A B J K E F G H I N O P Q R L M 5
3 2 1 S 6 5 4 B C D 7 8 9 A I J K E F G H O P Q R L M N 0
4 3 S 1 0 6 5 A B C D 7 8 9 H I J K E F G P Q R L M N O 2
5 S 3 2 1 0 6 9 A B C D 7 8 G H I J K E F Q R L M N O P 4
S 5 4 3 2 1 0 8 9 A B C D 7 F G H I J K E R L M N O P Q 6
7 D C B A 9 8 0 1 2 3 4 5 6 S M N O P Q R E F G H I J K L
8 7 D C B A 9 6 0 1 2 3 4 5 M S O P Q R L K E F G H I J N
9 8 7 D C B A 5 6 0 1 2 3 4 N O S Q R L M J K E F G H I P
A 9 8 7 D C B 4 5 6 0 1 2 3 O P Q S L M N I J K E F G H R
B A 9 8 7 D C 3 4 5 6 0 1 2 P Q R L S N O H I J K E F G M
C B A 9 8 7 D 2 3 4 5 6 0 1 Q R L M N S P G H I J K E F O
D C B A 9 8 7 1 2 3 4 5 6 0 R L M N O P S F G H I J K E Q
E K J I H G F L M N O P Q R 0 1 2 3 4 5 6 S 8 9 A B C D 7
F E K J I H G M N O P Q R L 6 0 1 2 3 4 5 D 7 8 9 A B S C
G F E K J I H N O P Q R L M 5 6 0 1 2 3 4 C D 7 8 9 S B A
H G F E K J I O P Q R L M N 4 5 6 0 1 2 3 B C D 7 S 9 A 8
I H G F E K J P Q R L M N O 3 4 5 6 0 1 2 A B C S 7 8 9 D
J I H G F E K Q R L M N O P 2 3 4 5 6 0 1 9 A S C D 7 8 B
K J I H G F E R L M N O P Q 1 2 3 4 5 6 0 8 S A B C D 7 9
L M N O P Q R S K J I H G F 7 D C B A 9 8 0 6 5 4 3 2 1 E
M N O P Q R L F E K J I H S 8 7 D C B A 9 1 0 6 5 4 3 2 G
N O P Q R L M G F E K J S H 9 8 7 D C B A 2 1 0 6 5 4 3 I
O P Q R L M N H G F E S J I A 9 8 7 D C B 3 2 1 0 6 5 4 K
P Q R L M N O I H G S E K J B A 9 8 7 D C 4 3 2 1 0 6 5 F
Q R L M N O P J I S G F E K C B A 9 8 7 D 5 4 3 2 1 0 6 H
R L M N O P Q K S I H G F E D C B A 9 8 7 6 5 4 3 2 1 0 J
6 4 2 0 5 3 1 E J H F K I G L N P R M O Q 7 9 B D 8 A C S
```

Partial square

```
. . . . . S . . . . . . . . . . . . . . . . . . . . . 1
. 0 . . . . . 7 . . . . . E . . . . . . . . . . . L . .
. 1 0 . S . . . D 7 . . . . . K E . . . . . . . . . L M .
. 2 1 S . . . . C D 7 . . . . J K E . . . . . . . L M N .
. 3 S 1 0 . . . B C D 7 . . . I J K E . . . . . L M N O 2
. . 3 2 1 0 . . A B C D 7 . . H I J K E . . . L M N O P 4
. 5 . 3 . 1 0 . 9 A B C D 7 . G H I J K E . L M N O P Q .
. . . . . . . . 0 . 2 3 4 5 6 S . . . . . . E F G H I J K .
. 7 D C B A 9 . 0 1 2 3 4 5 . S . . . . . K E F G H I J N
. . 7 D C B A 5 6 0 1 2 3 4 . . S . . L M J K E F G H I .
. . . 7 D C B 4 5 6 0 1 2 3 . . . . . M N I J K E F G . R
. . . . 7 D C 3 4 5 6 0 1 2 . . . L . N O H I J K E F G .
. . . . . 7 D 2 3 4 5 6 0 1 . . L M N S . G H I J K E F O
. . . . . 7 1 2 3 4 5 6 0 . L M . O P . F G H I J K E .
. . . . . . . . . . . . . . . . 0 1 2 3 4 5 6 S 8 9 A B C D .
. E K J I H G . . . . . L 6 0 1 2 3 4 5 D 7 8 9 A B S C
. . E K J I H . . . . . L M 5 6 0 1 2 3 4 C D 7 8 9 S B .
. . . E K J I . . . . L M N 4 5 6 0 1 2 3 B C D 7 S 9 A .
. . . . E K J . . . L M N O 3 4 5 6 0 1 2 A B C . 7 8 9 .
. . . . . E K . . L M N O P 2 3 4 5 6 0 1 9 A . C D 7 8 .
. . . . . E . L M N O P Q 1 2 3 4 5 6 0 8 S A B C D 7 .
. . . . . K J I H G F 7 D C B A 9 8 0 6 5 4 3 2 . E
. . . . . L F E K J I H . 8 7 D C B A 9 1 0 6 5 4 3 2 G
. . . . . L M G F E K J . H 9 8 7 D C B A 2 1 0 6 5 4 3 .
. . . . L M N H G F E . J I A 9 8 7 D C B 3 2 1 0 6 5 4 .
. . . L M N O I H G . E K J B A 9 8 7 D C 4 3 2 1 0 6 5 .
. . L M N O P J I S G F E K C B A 9 8 7 D 5 4 3 2 1 0 6 .
. L M N O P Q K S I H G F E D C B . 9 8 7 . 5 4 3 2 1 0 .
6 . 2 . . . . . G . . . R M . Q 7 . B . . . . S
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E120: order 29, 532 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m7_seed0.json`

SHA256: `13336b440f4d05f6d816f86d713c8ecaf2d32fd96bb5bf351243f5819aa21651`

Completion

```
S 6 5 4 3 2 1 7 8 9 A B C D E F G H I J K L M N O P Q R 0
1 0 6 5 4 3 S D 7 8 9 A B C K E F G H I J M N O P Q R L 2
2 1 0 6 5 S 3 C D 7 8 9 A B J K E F G H I N O P Q R L M 4
3 2 1 0 S 5 4 B C D 7 8 9 A I J K E F G H O P Q R L M N 6
4 3 2 S 0 6 5 A B C D 7 8 9 H I J K E F G P Q R L M N O 1
5 4 S 2 1 0 6 9 A B C D 7 8 G H I J K E F Q R L M N O P 3
6 S 4 3 2 1 0 8 9 A B C D 7 F G H I J K E R L M N O P Q 5
7 D C B A 9 8 0 1 2 3 4 5 6 S M N O P Q R E F G H I J K L
8 7 D C B A 9 6 0 1 2 3 4 5 M S O P Q R L K E F G H I J N
9 8 7 D C B A 5 6 0 1 2 3 4 N O S Q R L M J K E F G H I P
A 9 8 7 D C B 4 5 6 0 1 2 3 O P Q S L M N I J K E F G H R
B A 9 8 7 D C 3 4 5 6 0 1 2 P Q R L S N O H I J K E F G M
C B A 9 8 7 D 2 3 4 5 6 0 1 Q R L M N S P G H I J K E F O
D C B A 9 8 7 1 2 3 4 5 6 0 R L M N O P S F G H I J K E Q
E K J I H G F L M N O P Q R 0 1 2 3 4 5 6 S 8 9 A B C D 7
F E K J I H G M N O P Q R L 6 0 1 2 3 4 5 D 7 8 9 A B S C
G F E K J I H N O P Q R L M 5 6 0 1 2 3 4 C D 7 8 9 S B A
H G F E K J I O P Q R L M N 4 5 6 0 1 2 3 B C D 7 S 9 A 8
I H G F E K J P Q R L M N O 3 4 5 6 0 1 2 A B C S 7 8 9 D
J I H G F E K Q R L M N O P 2 3 4 5 6 0 1 9 A S C D 7 8 B
K J I H G F E R L M N O P Q 1 2 3 4 5 6 0 8 S A B C D 7 9
L M N O P Q R S K J I H G F 7 D C B A 9 8 0 6 5 4 3 2 1 E
M N O P Q R L F E K J I H S 8 7 D C B A 9 1 0 6 5 4 3 2 G
N O P Q R L M G F E K J S H 9 8 7 D C B A 2 1 0 6 5 4 3 I
O P Q R L M N H G F E S J I A 9 8 7 D C B 3 2 1 0 6 5 4 K
P Q R L M N O I H G S E K J B A 9 8 7 D C 4 3 2 1 0 6 5 F
Q R L M N O P J I S G F E K C B A 9 8 7 D 5 4 3 2 1 0 6 H
R L M N O P Q K S I H G F E D C B A 9 8 7 6 5 4 3 2 1 0 J
0 5 3 1 6 4 2 E J H F K I G L N P R M O Q 7 9 B D 8 A C S
```

Partial square

```
S . . . . . . . . . . . . . . . . . . . . . . . . . . . .
. 0 . . . . . 7 . . . . . E . . . . . . . . . . . L 2
. 1 0 . . S . . D 7 . . . . . K E . . . . . . . L M .
. . 1 0 S . . . C D 7 . . . . J K E . . . . . . . L M N 6
. 3 2 S 0 . . . B C D 7 . . . I J K E . . . . . L M N O .
. 4 S 2 . 0 . . A B C D 7 . . H I J K E . . . L M N O P .
. S 4 3 . 1 0 . 9 A B C D 7 . G H I J K E . L M N O P Q .
. . . . . 0 1 2 3 4 5 6 S . . . . . E F G H I J K .
. 7 D C B A 9 6 . 1 2 3 4 5 . S . . . . . K E F G H I J N
. . 7 D C B A 5 6 . 1 2 3 4 . . S . . L M J K E F G H I .
. . . 7 D C B 4 5 6 . 1 2 3 . . . . . M N I J K E F G . R
. . . . 7 D C 3 4 5 6 . 1 2 . . . L . N O H I J K E F G .
. . . . . 7 D 2 3 4 5 6 . 1 . . L M N S . G H I J K E F O
. . . . . 7 1 2 3 4 5 6 0 . L M . O P . F G H I J K E .
. . . . . 0 1 2 3 4 5 6 S 8 9 A B C D .
. E K J I H G . . . . . L 6 0 1 2 3 4 5 D 7 8 9 A B . C
. . E K J I H . . . . . L M 5 6 0 1 2 3 4 C D 7 8 9 S B .
. . . E K J I . . . . L M N 4 5 6 0 1 2 3 B C D 7 S 9 A .
. . . . E K J . . . L M N O 3 4 5 6 0 1 2 A B C S 7 8 9 .
. . . . . E K . . L M N O P 2 3 4 5 6 0 1 9 A . C D 7 8 .
. . . . . E . L M N O P Q 1 2 3 4 5 6 0 8 S A B C D 7 .
. . . . . K J I H G F 7 D C B A 9 8 . 6 5 4 3 2 1 E
. . . . . L F E K J I H . 8 7 D C B A 9 1 . 6 5 4 3 2 G
. . . . . L M G F E K J S H 9 8 7 D C B A 2 1 0 6 5 4 3 .
. . . . L M N H G F E S J I A 9 8 7 D C B 3 2 1 0 6 5 4 .
. . . L M N O I H G S E K J B A 9 8 7 D C 4 3 2 1 0 6 5 .
. . L M N O P J I S G F E K C B A 9 8 7 D 5 4 3 2 1 0 6 .
. L M N O P Q K S I H G F E D C B . 9 8 7 6 5 4 3 2 1 0 .
. . . . 6 . 2 . . . . . G . . . R M . Q 7 . B . . . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E121: order 29, 527 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m7_seed5.json`

SHA256: `63d79870be47fedd6f0f2341d6f3f573a3fc3038beb5c38695b102254257fb95`

Completion

```
S 6 5 4 3 2 1 7 8 9 A B C D E F G H I J K L M N O P Q R 0
1 0 6 5 4 3 S D 7 8 9 A B C K E F G H I J M N O P Q R L 2
2 1 0 6 5 S 3 C D 7 8 9 A B J K E F G H I N O P Q R L M 4
3 2 1 0 S 5 4 B C D 7 8 9 A I J K E F G H O P Q R L M N 6
4 3 2 S 0 6 5 A B C D 7 8 9 H I J K E F G P Q R L M N O 1
5 4 S 2 1 0 6 9 A B C D 7 8 G H I J K E F Q R L M N O P 3
6 S 4 3 2 1 0 8 9 A B C D 7 F G H I J K E R L M N O P Q 5
7 D C B A 9 8 0 1 2 3 4 5 6 S M N O P Q R E F G H I J K L
8 7 D C B A 9 6 0 1 2 3 4 5 M S O P Q R L K E F G H I J N
9 8 7 D C B A 5 6 0 1 2 3 4 N O S Q R L M J K E F G H I P
A 9 8 7 D C B 4 5 6 0 1 2 3 O P Q S L M N I J K E F G H R
B A 9 8 7 D C 3 4 5 6 0 1 2 P Q R L S N O H I J K E F G M
C B A 9 8 7 D 2 3 4 5 6 0 1 Q R L M N S P G H I J K E F O
D C B A 9 8 7 1 2 3 4 5 6 0 R L M N O P S F G H I J K E Q
E K J I H G F L M N O P Q R 0 1 2 3 4 5 6 S 8 9 A B C D 7
F E K J I H G M N O P Q R L 6 0 1 2 3 4 5 D 7 8 9 A B S C
G F E K J I H N O P Q R L M 5 6 0 1 2 3 4 C D 7 8 9 S B A
H G F E K J I O P Q R L M N 4 5 6 0 1 2 3 B C D 7 S 9 A 8
I H G F E K J P Q R L M N O 3 4 5 6 0 1 2 A B C S 7 8 9 D
J I H G F E K Q R L M N O P 2 3 4 5 6 0 1 9 A S C D 7 8 B
K J I H G F E R L M N O P Q 1 2 3 4 5 6 0 8 S A B C D 7 9
L M N O P Q R S K J I H G F 7 D C B A 9 8 0 6 5 4 3 2 1 E
M N O P Q R L F E K J I H S 8 7 D C B A 9 1 0 6 5 4 3 2 G
N O P Q R L M G F E K J S H 9 8 7 D C B A 2 1 0 6 5 4 3 I
O P Q R L M N H G F E S J I A 9 8 7 D C B 3 2 1 0 6 5 4 K
P Q R L M N O I H G S E K J B A 9 8 7 D C 4 3 2 1 0 6 5 F
Q R L M N O P J I S G F E K C B A 9 8 7 D 5 4 3 2 1 0 6 H
R L M N O P Q K S I H G F E D C B A 9 8 7 6 5 4 3 2 1 0 J
0 5 3 1 6 4 2 E J H F K I G L N P R M O Q 7 9 B D 8 A C S
```

Partial square

```
S . . . . . . . . . . . . . . . . . . . . . . . . . . . 0
. 0 . . . . S . 7 . . . . . . . E . . . . . . . . . . L .
. 1 0 . . S . . D 7 . . . . . K E . . . . . . . . . L M 4
. 2 1 0 S . . . C D 7 . . . . J K E . . . . . . . L M N 6
. 3 2 . . . . . B C D 7 . . . I J K E . . . . . L M N O .
. . . 2 1 . . . A B C D 7 . . H I J K E . . . L M N O P .
. S . 3 2 1 0 . 9 A B C D 7 . G H I J K E . L M N O P Q .
. . . . . . . . 0 1 2 3 4 5 6 . . . . . . . . F G H I J K L
. 7 D C B A 9 6 0 1 2 3 4 5 . S . . . . . K E F G H I J .
. . 7 D C B A 5 6 0 1 2 3 4 . . S . . . M J K E F G H I P
. . . 7 D C B 4 5 6 0 1 2 3 . . . S L M . I J K E F G . R
. . . . 7 D C 3 4 5 6 0 1 2 . . . L S N O H I J K E F G .
. . . . . 7 D 2 3 4 5 6 0 1 . . L . N S P G H I J K E F .
. . . . . 7 1 2 3 4 5 6 0 . . M N O . S F G H I J K E .
. . . . . 0 1 2 3 4 5 6 . 8 9 A B C D .
. E K J I H G . . . . . L 6 . 1 2 3 4 5 D 7 8 9 A B . .
. . E K J I H . . . . . L M 5 6 . 1 2 3 4 C D 7 8 9 . B A
. . . E K J I . . . . L M N 4 5 6 . 1 2 3 B C D 7 . 9 A 8
. . . . E K J . . . L M N O 3 4 5 6 . 1 2 A B C . 7 8 9 .
. . . . . E K . . L M N O P 2 3 4 5 6 0 1 9 A . C D 7 8 .
. . . . . E . L M N O P Q 1 2 3 4 5 6 0 8 . A B C D 7 9
. . . . . K J I H G F 7 D C B A 9 8 0 6 5 4 3 2 1 .
. . . . . L F E K J I H . 8 7 D C B A 9 1 0 6 5 4 3 2 .
. . . . . L M G F E K J . H 9 8 7 D C B A 2 1 0 6 5 4 3 I
. . . . L M N H G F E . J I A 9 8 7 D C B 3 2 1 0 6 5 4 .
. . . L M N O I H G . E K J B A 9 8 7 D C 4 3 2 1 0 6 5 .
. . L M N O P J I . G F E K C B A 9 8 7 D 5 4 3 2 1 . 6 .
. L M N O P Q K . I H G F E D C B . 9 8 7 6 5 4 3 2 1 0 .
. . . . 6 4 . . J H . . . G L N P R . . . . . B . . A . S
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E122: order 29, 525 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_shifted_examples/thesis_m7_seed1.json`

SHA256: `622f08202b3a07ad8d019edf0cbd1d05dba7c06d7a1f403554b94e7747a44117`

Completion

```
0 6 5 4 3 2 S 7 8 9 A B C D E F G H I J K L M N O P Q R 1
1 0 6 5 4 S 2 D 7 8 9 A B C K E F G H I J M N O P Q R L 3
2 1 0 6 S 4 3 C D 7 8 9 A B J K E F G H I N O P Q R L M 5
3 2 1 S 6 5 4 B C D 7 8 9 A I J K E F G H O P Q R L M N 0
4 3 S 1 0 6 5 A B C D 7 8 9 H I J K E F G P Q R L M N O 2
5 S 3 2 1 0 6 9 A B C D 7 8 G H I J K E F Q R L M N O P 4
S 5 4 3 2 1 0 8 9 A B C D 7 F G H I J K E R L M N O P Q 6
7 D C B A 9 8 0 1 2 3 4 5 6 S M N O P Q R E F G H I J K L
8 7 D C B A 9 6 0 1 2 3 4 5 M S O P Q R L K E F G H I J N
9 8 7 D C B A 5 6 0 1 2 3 4 N O S Q R L M J K E F G H I P
A 9 8 7 D C B 4 5 6 0 1 2 3 O P Q S L M N I J K E F G H R
B A 9 8 7 D C 3 4 5 6 0 1 2 P Q R L S N O H I J K E F G M
C B A 9 8 7 D 2 3 4 5 6 0 1 Q R L M N S P G H I J K E F O
D C B A 9 8 7 1 2 3 4 5 6 0 R L M N O P S F G H I J K E Q
E K J I H G F L M N O P Q R 0 1 2 3 4 5 6 S 8 9 A B C D 7
F E K J I H G M N O P Q R L 6 0 1 2 3 4 5 D 7 8 9 A B S C
G F E K J I H N O P Q R L M 5 6 0 1 2 3 4 C D 7 8 9 S B A
H G F E K J I O P Q R L M N 4 5 6 0 1 2 3 B C D 7 S 9 A 8
I H G F E K J P Q R L M N O 3 4 5 6 0 1 2 A B C S 7 8 9 D
J I H G F E K Q R L M N O P 2 3 4 5 6 0 1 9 A S C D 7 8 B
K J I H G F E R L M N O P Q 1 2 3 4 5 6 0 8 S A B C D 7 9
L M N O P Q R S K J I H G F 7 D C B A 9 8 0 6 5 4 3 2 1 E
M N O P Q R L F E K J I H S 8 7 D C B A 9 1 0 6 5 4 3 2 G
N O P Q R L M G F E K J S H 9 8 7 D C B A 2 1 0 6 5 4 3 I
O P Q R L M N H G F E S J I A 9 8 7 D C B 3 2 1 0 6 5 4 K
P Q R L M N O I H G S E K J B A 9 8 7 D C 4 3 2 1 0 6 5 F
Q R L M N O P J I S G F E K C B A 9 8 7 D 5 4 3 2 1 0 6 H
R L M N O P Q K S I H G F E D C B A 9 8 7 6 5 4 3 2 1 0 J
6 4 2 0 5 3 1 E J H F K I G L N P R M O Q 7 9 B D 8 A C S
```

Partial square

```
. . . . . S . . . . . . . . . . . . . . . . . . . . . .
. 0 . . . S . . 7 . . . . . E . . . . . . . . . . L . .
. 1 0 . S . . . D 7 . . . . . K E . . . . . . . . L M .
. 2 1 . . . . . C D 7 . . . . J K E . . . . . . . L M N 0
. 3 . 1 0 . . . B C D 7 . . . I J K E . . . . . L M N O 2
. . 3 2 1 0 . . A B C D 7 . . H I J K E . . . L M N O P 4
S . . . . . 9 A B C D 7 . G H I J K E . L M N O P Q 6
. . . . . 0 1 2 3 4 5 6 S . . . . . F G H I J K L
. 7 D C B A 9 6 0 1 2 3 4 5 . S . . . . . K E F G H I J N
. . 7 D C B A 5 6 0 1 2 3 4 . . S . . . M J K E F G H I P
. . . 7 D C B 4 5 6 0 1 2 3 . . . . L . . I J K E F G . .
. . . . 7 D C 3 4 5 6 0 1 2 . . . . . N O H I J K E F G M
. . . . . 7 D 2 3 4 5 6 0 1 . . . M N . P G H I J K E F O
. . . . . 7 1 2 3 4 5 6 0 . . M N O P . F G H I J K E Q
. . . . . 0 . 2 3 4 5 6 . 8 9 A B C D .
. E K J I H G . . . . . L . 0 1 2 3 4 5 D 7 8 9 A B . .
. . E K J I H . . . . . L M 5 6 0 1 2 3 4 C D 7 8 9 . B .
. . . E K J I . . . . L M N 4 5 6 0 1 2 3 B C D 7 . 9 A .
. . . . E K J . . . L M N O 3 4 5 6 0 1 2 A B C . 7 8 9 .
. . . . . E K . . L M N O P 2 3 4 5 6 0 1 9 A . C D 7 8 .
. . . . . E . L M N O P Q 1 2 3 4 5 6 0 8 . A B C D 7 9
. . . . . S K J I H G F . D C B A 9 8 0 6 5 4 3 2 . E
. . . . . L F E K J I H . 8 7 D C B A 9 1 0 6 5 4 3 2 .
. . . . . L M G F E K J . H 9 8 7 D C B A 2 1 0 6 5 4 3 .
. . . . L M N H G F E . J I A 9 8 7 D C B 3 2 1 0 6 5 4 .
. . . L M N O I H G . E K J B A 9 8 7 D C 4 3 2 1 0 6 5 F
. . L M N O P J I S G F E K C B A 9 8 7 D 5 4 3 2 1 0 6 .
. L M N O P Q K . I H G F E D C B . 9 8 7 . 5 4 3 2 1 0 .
6 . . . 5 . 1 . J . F . . G L N . R . . . . . B . . . C S
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E123: order 29, 518 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m7_seed4.json`

SHA256: `8bf06cdded6750ebf1363a9d696e80f5c8ea358000122de83bacb1f4b8f708a3`

Completion

```
S 6 5 4 3 2 1 7 8 9 A B C D E F G H I J K L M N O P Q R 0
1 0 6 5 4 3 S D 7 8 9 A B C K E F G H I J M N O P Q R L 2
2 1 0 6 5 S 3 C D 7 8 9 A B J K E F G H I N O P Q R L M 4
3 2 1 0 S 5 4 B C D 7 8 9 A I J K E F G H O P Q R L M N 6
4 3 2 S 0 6 5 A B C D 7 8 9 H I J K E F G P Q R L M N O 1
5 4 S 2 1 0 6 9 A B C D 7 8 G H I J K E F Q R L M N O P 3
6 S 4 3 2 1 0 8 9 A B C D 7 F G H I J K E R L M N O P Q 5
7 D C B A 9 8 0 1 2 3 4 5 6 S M N O P Q R E F G H I J K L
8 7 D C B A 9 6 0 1 2 3 4 5 M S O P Q R L K E F G H I J N
9 8 7 D C B A 5 6 0 1 2 3 4 N O S Q R L M J K E F G H I P
A 9 8 7 D C B 4 5 6 0 1 2 3 O P Q S L M N I J K E F G H R
B A 9 8 7 D C 3 4 5 6 0 1 2 P Q R L S N O H I J K E F G M
C B A 9 8 7 D 2 3 4 5 6 0 1 Q R L M N S P G H I J K E F O
D C B A 9 8 7 1 2 3 4 5 6 0 R L M N O P S F G H I J K E Q
E K J I H G F L M N O P Q R 0 1 2 3 4 5 6 S 8 9 A B C D 7
F E K J I H G M N O P Q R L 6 0 1 2 3 4 5 D 7 8 9 A B S C
G F E K J I H N O P Q R L M 5 6 0 1 2 3 4 C D 7 8 9 S B A
H G F E K J I O P Q R L M N 4 5 6 0 1 2 3 B C D 7 S 9 A 8
I H G F E K J P Q R L M N O 3 4 5 6 0 1 2 A B C S 7 8 9 D
J I H G F E K Q R L M N O P 2 3 4 5 6 0 1 9 A S C D 7 8 B
K J I H G F E R L M N O P Q 1 2 3 4 5 6 0 8 S A B C D 7 9
L M N O P Q R S K J I H G F 7 D C B A 9 8 0 6 5 4 3 2 1 E
M N O P Q R L F E K J I H S 8 7 D C B A 9 1 0 6 5 4 3 2 G
N O P Q R L M G F E K J S H 9 8 7 D C B A 2 1 0 6 5 4 3 I
O P Q R L M N H G F E S J I A 9 8 7 D C B 3 2 1 0 6 5 4 K
P Q R L M N O I H G S E K J B A 9 8 7 D C 4 3 2 1 0 6 5 F
Q R L M N O P J I S G F E K C B A 9 8 7 D 5 4 3 2 1 0 6 H
R L M N O P Q K S I H G F E D C B A 9 8 7 6 5 4 3 2 1 0 J
0 5 3 1 6 4 2 E J H F K I G L N P R M O Q 7 9 B D 8 A C S
```

Partial square

```
S . . . . . . . . . . . . . . . . . . . . . . . . . . . 0
. 0 . . . . S . 7 . . . . . . E . . . . . . . . . . . L 2
. 1 0 . . S . . D 7 . . . . . K E . . . . . . . . . L M .
. . 1 0 S . . . C D 7 . . . . J K E . . . . . . . L M N 6
. 3 2 . 0 . . . B C D 7 . . . I J K E . . . . . L M N O .
. 4 . 2 . 0 . . A B C D 7 . . H I J K E . . . L M N O P .
. . 4 3 . 1 . . 9 A B C D 7 . G H I J K E . L M N O P Q .
. . . . . . . 0 1 2 3 4 5 6 S . . . . . . . F G H I J K L
. 7 D C B A 9 6 0 1 2 3 4 5 . S . . . . L K E F G H I J N
. . 7 D C B A 5 6 . 1 2 3 4 . . S . . L M J K E F G H I .
. . . 7 D C B 4 5 6 . 1 2 3 . . . S . M . I J K E F G . R
. . . . 7 D C 3 4 5 6 . 1 2 . . . . . N O H I J K E F G .
. . . . . 7 D 2 3 4 5 6 . 1 . . . M N . P G H I J K E F .
. . . . . 7 1 2 3 4 5 6 . . . M N O . . F G H I J K E Q
. . . . . 0 1 2 3 4 5 6 . 8 9 A B C D .
. E K J I H G . . . . . L 6 0 1 2 3 4 5 D 7 8 9 A B . C
. . E K J I H . . . . . L M 5 6 0 1 2 3 4 C D 7 8 9 . B .
. . . E K J I . . . . L M N 4 5 6 0 1 2 3 B C D 7 . 9 A .
. . . . E K J . . . L M N O 3 4 5 6 0 1 2 A B C . 7 8 9 .
. . . . . E K . . L M N O P 2 3 4 5 6 . 1 9 A . C D 7 8 B
. . . . . E . L M N O P Q 1 2 3 4 5 6 . 8 . A B C D 7 .
. . . . . K J I H G F . D C B A 9 8 0 6 5 4 3 2 1 .
. . . . . L F E K J I H S 8 7 D C B A 9 1 . 6 5 4 3 2 G
. . . . . L M G F E K J S H 9 8 7 D C B A 2 1 . 6 5 4 3 .
. . . . L M N H G F E . J I A 9 8 7 D C B 3 2 1 . 6 5 4 .
. . . L M N O I H G . E K J B A 9 8 7 D C 4 3 2 1 . 6 5 .
. . L M N O P J I . G F E K C B A 9 8 7 D 5 4 3 2 1 . 6 .
. L M N O P Q K . I H G F E D C B . 9 8 7 6 5 4 3 2 1 . .
. . . . 6 . 2 E . . . . . L N P R . O Q . . . . 8 A C S
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E124: order 29, 515 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_odd_prolongation/thesis_examples/thesis_m7_seed1.json`

SHA256: `8e4ce69543066ceeb83db042f10f054f0e130753fec6947704e9b3cb8a8e5dc4`

Completion

```
S 6 5 4 3 2 1 7 8 9 A B C D E F G H I J K L M N O P Q R 0
1 0 6 5 4 3 S D 7 8 9 A B C K E F G H I J M N O P Q R L 2
2 1 0 6 5 S 3 C D 7 8 9 A B J K E F G H I N O P Q R L M 4
3 2 1 0 S 5 4 B C D 7 8 9 A I J K E F G H O P Q R L M N 6
4 3 2 S 0 6 5 A B C D 7 8 9 H I J K E F G P Q R L M N O 1
5 4 S 2 1 0 6 9 A B C D 7 8 G H I J K E F Q R L M N O P 3
6 S 4 3 2 1 0 8 9 A B C D 7 F G H I J K E R L M N O P Q 5
7 D C B A 9 8 0 1 2 3 4 5 6 S M N O P Q R E F G H I J K L
8 7 D C B A 9 6 0 1 2 3 4 5 M S O P Q R L K E F G H I J N
9 8 7 D C B A 5 6 0 1 2 3 4 N O S Q R L M J K E F G H I P
A 9 8 7 D C B 4 5 6 0 1 2 3 O P Q S L M N I J K E F G H R
B A 9 8 7 D C 3 4 5 6 0 1 2 P Q R L S N O H I J K E F G M
C B A 9 8 7 D 2 3 4 5 6 0 1 Q R L M N S P G H I J K E F O
D C B A 9 8 7 1 2 3 4 5 6 0 R L M N O P S F G H I J K E Q
E K J I H G F L M N O P Q R 0 1 2 3 4 5 6 S 8 9 A B C D 7
F E K J I H G M N O P Q R L 6 0 1 2 3 4 5 D 7 8 9 A B S C
G F E K J I H N O P Q R L M 5 6 0 1 2 3 4 C D 7 8 9 S B A
H G F E K J I O P Q R L M N 4 5 6 0 1 2 3 B C D 7 S 9 A 8
I H G F E K J P Q R L M N O 3 4 5 6 0 1 2 A B C S 7 8 9 D
J I H G F E K Q R L M N O P 2 3 4 5 6 0 1 9 A S C D 7 8 B
K J I H G F E R L M N O P Q 1 2 3 4 5 6 0 8 S A B C D 7 9
L M N O P Q R S K J I H G F 7 D C B A 9 8 0 6 5 4 3 2 1 E
M N O P Q R L F E K J I H S 8 7 D C B A 9 1 0 6 5 4 3 2 G
N O P Q R L M G F E K J S H 9 8 7 D C B A 2 1 0 6 5 4 3 I
O P Q R L M N H G F E S J I A 9 8 7 D C B 3 2 1 0 6 5 4 K
P Q R L M N O I H G S E K J B A 9 8 7 D C 4 3 2 1 0 6 5 F
Q R L M N O P J I S G F E K C B A 9 8 7 D 5 4 3 2 1 0 6 H
R L M N O P Q K S I H G F E D C B A 9 8 7 6 5 4 3 2 1 0 J
0 5 3 1 6 4 2 E J H F K I G L N P R M O Q 7 9 B D 8 A C S
```

Partial square

```
S . . . . . . . . . . . . . . . . . . . . . . . . . . . 0
. 0 . . . . S . 7 . . . . . . E . . . . . . . . . . . L 2
. 1 0 . . S . . D 7 . . . . . K E . . . . . . . . . L M .
. . 1 0 S . . . C D 7 . . . . J K E . . . . . . . L M N 6
. 3 2 . 0 . . . B C D 7 . . . I J K E . . . . . L M N O .
. 4 . 2 . 0 . . A B C D 7 . . H I J K E . . . L M N O P .
. . 4 3 . 1 . . 9 A B C D 7 . G H I J K E . L M N O P Q .
. . . . . . . 0 1 2 3 4 5 6 S . . . . . . . F G H I J K L
. 7 D C B A 9 6 . 1 2 3 4 5 . S . . . . . K E F G H I J N
. . 7 D C B A 5 6 . 1 2 3 4 . . S . . . M J K E F G H I P
. . . 7 D C B 4 5 6 . 1 2 3 . . . . L . . I J K E F G . .
. . . . 7 D C 3 4 5 6 . 1 2 . . . . . N O H I J K E F G M
. . . . . 7 D 2 3 4 5 6 . 1 . . . M N . P G H I J K E F O
. . . . . 7 1 2 3 4 5 6 . . . M N O P S F G H I J K E Q
. . . . . 0 1 2 3 4 5 6 . 8 9 A B C D .
. E K J I H G . . . . . L 6 0 1 2 3 4 5 D 7 8 9 A B . .
. . E K J I H . . . . . L M 5 6 0 1 2 3 4 C D 7 8 9 . B .
. . . E K J I . . . . L M N 4 5 6 0 1 2 3 B C D 7 . 9 A .
. . . . E K J . . . L M N O 3 4 5 6 0 1 2 A B C . 7 8 9 .
. . . . . E K . . L M N O P 2 3 4 5 6 0 1 9 A . C D 7 8 .
. . . . . E . L M N O P Q 1 2 3 4 5 6 . 8 . A B C D 7 .
. . . . . K J I H G F . D C B A 9 8 0 6 5 4 3 2 1 E
. . . . . L F E K J I H . 8 7 D C B A 9 1 . 6 5 4 3 2 .
. . . . . L M G F E K J . H 9 8 7 D C B A 2 1 . 6 5 4 3 .
. . . . L M N H G F E . J I A 9 8 7 D C B 3 2 1 . 6 5 4 .
. . . L M N O I H G . E K J B A 9 8 7 D C 4 3 2 1 . 6 5 F
. . L M N O P J I . G F E K C B A 9 8 7 D 5 4 3 2 1 . 6 .
. L M N O P Q K . I H G F E D C B . 9 8 7 6 5 4 3 2 1 . .
0 . . . 6 . 2 . J . F . . G L N . R . . . . . B . . . C S
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E125: order 33, 841 clues

Status I. Previously independently checked uniqueness and all clue-deletion witnesses; uniqueness is solver-dependent.

Source: `outputs/lcs33_844_bunya_29909858/independent_audit/previous_record.json`

SHA256: `0eb7cb87d2223e4d3347ffcee631cdac549f3163b98c957020acf9c89a67807c`

Completion

```
W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
2 3 0 1 W 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 6
3 2 1 0 7 6 W 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 5
4 5 6 7 0 1 2 3 W D E F 8 9 A B K L M N G H I J S T U V O P Q R C
5 4 7 6 1 0 3 2 D C W E 9 8 B A L K N M H G J I T S V U P O R Q F
6 7 4 5 2 3 0 1 E F C D W B 8 9 M N K L I J G H U V S T Q R O P A
7 6 5 4 3 2 1 0 F E D C B A W 8 N M L K J I H G V U T S R Q P O 9
8 9 A B C D E F 0 1 2 3 4 5 6 7 W P Q R S T U V G H I J K L M N O
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O W Q T S V U H G J I L K N M R
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P W V S T I J G H M N K L U
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U W S J I H G N M L K T
C D E F 8 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R W L M N G H I J K
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K W M H G J I N
E F C D A B 8 9 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L W J G H I
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I W G H
G H I J K W M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 9 A B C D E F L
H G J I L K N W P O R Q T S V U 1 0 3 2 5 4 7 6 9 8 B A D C F E M
I W G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D J
J I H W N M L K R Q P O V U T S 3 2 1 0 7 6 5 4 B A 9 8 F E D C G
K L M N G H I J S T U V O W Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B P
L K N M H G J I T S V U P O R W 5 4 7 6 1 0 3 2 D C F E 9 8 B A Q
M N K L I J G H U W S T Q R O P 6 7 4 5 2 3 0 1 E F C D A B 8 9 V
N M L K J I H G V U T W R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 S
O P Q R S T U V G H I J K L M N 8 9 A B C W E F 0 1 2 3 4 5 6 7 D
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
Q R O P U V S T I J G H M N K L A W 8 9 E F C D 2 3 0 1 6 7 4 5 B
R Q P O V U T S J I H G N M L K B A 9 W F E D C 3 2 1 0 7 6 5 4 8
S T U V O P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 W 2 3 1
T S V U P O R Q L K N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 W 2
U V S T Q R O P M N K L I J G H E F C D A B 8 9 6 W 4 5 2 3 0 1 7
V U T S R Q P O N M L K J I H G F E D C B A 9 8 7 6 5 W 3 2 1 0 4
0 J 3 G 6 L 5 M C V F S A P 9 Q O B R 8 U D T E K 7 N 4 I 1 H 2 W
```

Partial square

```
. . . 3 . 5 6 . . . . . C D . F . . . . K L M N . . . . S T U V 0
. . W . 5 4 7 6 . 8 B A . C . E . G J I L K N M . O . Q T . V U 3
. 3 0 1 . 7 4 5 A B . . . F . D I . G H M N . . . R O P U V . T 6
3 2 1 0 . 6 W 4 B A . 8 F E . C J I H G N M L K R Q . . . U T S 5
4 5 6 7 0 1 . . W D E . 8 . . B . L M N G H I . S T U V O P Q R .
. 4 7 6 . . 3 2 D C . E . 8 . A L K . . H G J I T . V U P . R Q F
6 7 4 5 2 3 . 1 E . C D . . . . M N . L I . G H U V S T . R O P A
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
8 . A B C D E F 0 1 2 3 . 5 . 7 W P Q R S T U V G H I J . L . . O
. . B A D C F E 1 0 3 2 5 4 . 6 P O W Q T . V U . G J I L . N M R
A B 8 . E F C D 2 . 0 1 6 7 . 5 Q R O . W V S T I J G H M N K L U
B A . 8 . E D C 3 2 . 0 7 . . . R Q P O V . W S J I . G . M L K T
C D E F 8 . A B . 5 . 7 0 1 . 3 S T U V . P Q R W L M N G H I J K
. C F E . 8 . A 5 4 7 6 1 0 . 2 T S V . P O R Q L K . M H . J I N
E F . . A B 8 . 6 7 4 5 2 3 . 1 U V S T . R O P M N K L W J G H I
F E D C B A . 8 7 6 5 4 3 2 . 0 V U T . R Q P O N M L K J I W G H
G H I J K W M N O P Q R S T . V 0 1 2 3 4 5 6 7 8 . . B C D E F .
H G J I L . N W P O R Q T . . U 1 0 3 2 5 . 7 6 . 8 B A D C F E M
I W G H M N K L . R O P U V . T . 3 0 1 6 7 4 5 A B 8 . E F C D J
J I H W N M L K R Q P . V U . S 3 2 1 0 7 6 5 4 B A . 8 F E . C G
K L M N G H . J S . U V O W . R . 5 6 7 0 1 2 3 C . E F 8 . . B P
. K N M H G . I T S V U P O . . 5 4 7 6 1 0 . 2 D C F E . 8 B A Q
M N K L . J G H U W S T Q . . P 6 7 4 5 2 3 0 1 E F . D A B 8 . V
N M L . . I H G V U T W . Q . O 7 6 5 4 3 2 1 0 F E D C B A . 8 .
O P Q R S T . V G H I J . L . . 8 . A B C W E . 0 1 . 3 4 . 6 7 .
P O . . T S V U H G . . L K . M . 8 B A D C . W 1 0 . 2 5 4 7 6 E
Q R O P U V S T I J G H M N . L A W 8 . E F C . 2 3 0 1 6 7 4 5 B
R Q P O V U T S J I . G N M . K . A . W . E D C 3 . 1 0 7 6 5 4 8
S T U V O P Q R K L M N G . . J . . E . . . A B 4 5 6 . 0 W 2 3 .
T S V U P O R Q L K N M H . . I D C F E . 8 B A 5 4 7 6 1 0 3 W 2
U V . T Q R O . . N K L I J . H E F C D A B 8 . 6 W 4 5 . 3 0 . 7
V U T S R Q . O N . L K J I . G F E D C . A . . 7 6 5 W 3 . 1 0 .
0 . 3 G 6 L 5 M C V F S A P . . O B . 8 U D T E K 7 N 4 I 1 H 2 W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E126: order 33, 837 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs_dyadic_plus_sharp_progress/forced_after_delete_n33.json`

SHA256: `6dc672522b75468d9442e042e1f692276e517d61ba32b2ba07cabf37efe21730`

Completion

```
W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
2 3 0 W 6 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 1
3 W 1 0 7 6 5 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 2
4 5 6 7 0 1 2 3 C D E F 8 9 A B K L M N G H I J W T U V O P Q R S
5 4 7 6 1 0 3 2 D C F E 9 8 B A L K N M H G J I T S W U P O R Q V
6 7 4 5 2 3 0 1 E F C D A B 8 9 M N K L I J G H U V S W Q R O P T
7 6 5 4 3 2 1 0 F E D C B A 9 8 N M L K J I H G V W T S R Q P O U
8 9 A B C D E F 0 1 2 3 4 5 6 7 O P Q R S T U V G H I J W L M N K
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O R Q T S V U H G J I L K W M N
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P U V S T I J G H M N K W L
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U T S J I H G N W L K M
C D E F W 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R K L M N G H I J 8
D C F E 9 8 W A 5 4 7 6 1 0 3 2 T S V U P O R Q L K N M H G J I B
E F C D A B 8 W 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L I J G H 9
F E D C B W 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I H G A
G H I J K L M N O P Q R S T U V 0 1 2 3 W 5 6 7 8 9 A B C D E F 4
H G J I L K N M P O R Q T S V U 1 0 3 2 5 4 W 6 9 8 B A D C F E 7
I J G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 W A B 8 9 E F C D 5
J I H G N M L K R Q P O V U T S 3 2 1 0 7 W 5 4 B A 9 8 F E D C 6
K L M N G H I J S T U V W P Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B O
L K N M H G J I T S V U P O W Q 5 4 7 6 1 0 3 2 D C F E 9 8 B A R
M N K L I J G H U V S T Q R O W 6 7 4 5 2 3 0 1 E F C D A B 8 9 P
N M L K J I H G V U T S R W P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 Q
O P Q R S T U V W H I J K L M N 8 9 A B C D E F 0 1 2 3 4 5 6 7 G
P O R Q T S V U H G W I L K N M 9 8 B A D C F E 1 0 3 2 5 4 7 6 J
Q R O P U V S T I J G W M N K L A B 8 9 E F C D 2 3 0 1 6 7 4 5 H
R Q P O V U T S J W H G N M L K B A 9 8 F E D C 3 2 1 0 7 6 5 4 I
S T U V O P Q R K L M N G H I J W D E F 8 9 A B 4 5 6 7 0 1 2 3 C
T S V U P O R Q L K N M H G J I D C W E 9 8 B A 5 4 7 6 1 0 3 2 F
U V S T Q R O P M N K L I J G H E F C W A B 8 9 6 7 4 5 2 3 0 1 D
V U T S R Q P O N M L K J I H G F W D C B A 9 8 7 6 5 4 3 2 1 0 E
0 2 3 1 8 A B 9 G I J H O Q R P C E F D 4 6 7 5 S U V T K M N L W
```

Partial square

```
. 1 2 3 . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
1 0 . 2 . 4 7 6 . 8 B A . C F E . G J I . K N M . O R Q . S V U 3
2 3 0 W . 7 4 . . B 8 . . F C . . J G . . N K . . R O . . V S . 1
3 W 1 0 . 6 . 4 . A . 8 . E . C . I . G . M . K . Q . O . U . S 2
. . . . . . . . . C D E F 8 9 A B K L M N G H I J W T U V O P Q R .
. 4 7 6 . 0 3 2 D C F E 9 8 B A L K N M H G J I T S W U P O R Q V
. 7 4 . . 3 0 . E F C D A B 8 9 M N K L I J G H U V S W Q R O P T
. 6 . 4 . 2 . 0 F E D C B A 9 8 N M L K J I H G V W T S R Q P O U
. . . . C D E F . . . . 4 5 6 7 O P Q R S T U V G H I J W L M N .
. 8 B A D C F E . 0 3 2 5 4 7 6 P O R Q T S V U H G J I L K W M N
. B 8 . E F C D . 3 0 . 6 7 4 5 Q R O P U V S T I J G H M N K W L
. A . 8 F E D C . 2 . 0 7 6 5 4 R Q P O V U T S J I H G N W L K M
. . . . . . . . . 4 5 6 7 . . . . S T U V . . . . K L M N G H I J 8
. C F E . 8 W A 5 4 7 6 . 0 3 2 T S V U . O R Q L K N M H G J I B
. F C . . B 8 . 6 7 4 5 . 3 0 . U V S T . R O . M N K L I J G H 9
. E . C . W . 8 7 6 5 4 . 2 . 0 V U T S . Q . O N M L K J I H G A
. . . . K L M N O P Q R S T U V . . . . W 5 6 7 8 9 A B . . . . 4
. G J I L K N M P O R Q T S V U . 0 3 2 5 4 W 6 9 8 B A . C F E 7
. J G . M N K L Q R O P U V S T . 3 0 . 6 7 4 W A B 8 9 . F C . 5
. I . G N M L K R Q P O V U T S . 2 . 0 7 W 5 4 B A 9 8 . E . C 6
. . . . G H I J S T U V W P Q R 4 5 6 7 . . . . . . . . . 8 9 A B O
. K N M H G J I T S V U P O W Q 5 4 7 6 . 0 3 2 . C F E 9 8 B A R
. N K . I J G H U V S T Q R O W 6 7 4 5 . 3 0 . . F C . A B 8 9 P
. M . K J I H G V U T S R W P O 7 6 5 4 . 2 . 0 . E . C B A 9 8 Q
. . . . S T U V W H I J K L M N . . . . C D E F . . . . 4 5 6 7 .
. O R Q T S V U H G W I L K N M . 8 B A D C F E . 0 3 2 5 4 7 6 J
. R O . U V S T I J G W M N K L . B 8 . E F C D . 3 0 . 6 7 4 5 H
. Q . O V U T S J W H G N M L K . A . 8 F E D C . 2 . 0 7 6 5 4 I
. . . . O P Q R . . . . G H I J W D E F 8 9 A B 4 5 6 7 . . . . C
. S V U P O R Q . K N M H G J I D C W E 9 8 B A 5 4 7 6 . 0 3 2 F
. V S . Q R O P . N K . I J G H E F C W A B 8 9 6 7 4 5 . 3 0 . D
. U . S R Q P O . M . K J I H G F W D C B A 9 8 7 6 5 4 . 2 . 0 E
. 2 3 1 8 A B 9 G I J H . Q R P C E F D . 6 7 5 S U V T K M N L W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E127: order 33, 836 clues

Status I. Previously independently checked uniqueness and all clue-deletion witnesses; uniqueness is solver-dependent.

Source: `outputs/lcs33_review_29814974/n33_836.json`

SHA256: `15bd01ca03373dcad3e69b80788fe56721efdbe97c1fcb7e5c14f1d91550f5ff`

Completion

```
W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
2 3 0 1 W 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 6
3 2 1 0 7 6 W 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 5
4 5 6 7 0 1 2 3 W D E F 8 9 A B K L M N G H I J S T U V O P Q R C
5 4 7 6 1 0 3 2 D C W E 9 8 B A L K N M H G J I T S V U P O R Q F
6 7 4 5 2 3 0 1 E F C D W B 8 9 M N K L I J G H U V S T Q R O P A
7 6 5 4 3 2 1 0 F E D C B A W 8 N M L K J I H G V U T S R Q P O 9
8 9 A B C D E F 0 1 2 3 4 5 6 7 W P Q R S T U V G H I J K L M N O
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O W Q T S V U H G J I L K N M R
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P W V S T I J G H M N K L U
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U W S J I H G N M L K T
C D E F 8 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R W L M N G H I J K
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K W M H G J I N
E F C D A B 8 9 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L W J G H I
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I W G H
G H I J K W M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 9 A B C D E F L
H G J I L K N W P O R Q T S V U 1 0 3 2 5 4 7 6 9 8 B A D C F E M
I W G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D J
J I H W N M L K R Q P O V U T S 3 2 1 0 7 6 5 4 B A 9 8 F E D C G
K L M N G H I J S T U V O W Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B P
L K N M H G J I T S V U P O R W 5 4 7 6 1 0 3 2 D C F E 9 8 B A Q
M N K L I J G H U W S T Q R O P 6 7 4 5 2 3 0 1 E F C D A B 8 9 V
N M L K J I H G V U T W R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 S
O P Q R S T U V G H I J K L M N 8 9 A B C W E F 0 1 2 3 4 5 6 7 D
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
Q R O P U V S T I J G H M N K L A W 8 9 E F C D 2 3 0 1 6 7 4 5 B
R Q P O V U T S J I H G N M L K B A 9 W F E D C 3 2 1 0 7 6 5 4 8
S T U V O P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 W 2 3 1
T S V U P O R Q L K N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 W 2
U V S T Q R O P M N K L I J G H E F C D A B 8 9 6 W 4 5 2 3 0 1 7
V U T S R Q P O N M L K J I H G F E D C B A 9 8 7 6 5 W 3 2 1 0 4
0 J 3 G 6 L 5 M C V F S A P 9 Q O B R 8 U D T E K 7 N 4 I 1 H 2 W
```

Partial square

```
. . . 3 . 5 6 . . . . . C D . F . . . . K L M N . . . . S T U V 0
. . W . 5 4 7 6 . 8 B A . C . E . G J I L K N M . O . Q T . V U 3
. . 0 1 . 7 4 . A B . . . F . D I . G H M N . . Q R O P U V . T 6
3 2 1 0 7 6 W 4 B A . 8 F E . C J I H G N M L K R . P . . U T S 5
4 5 6 . 0 1 2 3 W D E . 8 . . B . L M N G H I J S T U V O P Q R .
. 4 7 6 . . 3 2 D C . E . 8 . A L K . M H G . I T . . U P . R Q F
6 7 4 5 2 3 0 1 E . C D . . . . M N . L I . G H U V . T . R O P A
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
8 . A B C D E F 0 1 2 3 4 5 . 7 W P Q R S T U V G H I J K L M . O
. . B A D C F E 1 0 3 2 5 4 . 6 P O W Q T . V U . G J I L K N . R
A B 8 . E F C D 2 . 0 1 6 7 . 5 Q R O . W V S T I J G H M N K L U
B A . 8 . E D C 3 2 . 0 7 . . . R Q P O V . W S J I . G N M L . T
C D E F 8 . A B . 5 . 7 0 1 . 3 S T U V O P Q R W L . N G H I J K
. C F E . 8 B A 5 4 7 6 1 0 . . T S V . P O R Q L K . M H . J . N
E F . . A B . . 6 7 4 5 2 3 . 1 U . S T . R . P M N . L W J G H I
F E D C B A . 8 7 . 5 4 3 . . 0 V U T . R Q P O N M L K . I W G H
G H I J K W M N O P Q R S T . V 0 1 2 3 4 5 6 7 8 . . B C D E F .
H G J I L . N W P O R Q T . . U 1 0 . 2 5 . 7 6 . 8 B A D C F E M
I W G H M N K L . R O P . V . T . 3 0 1 6 7 4 5 A B 8 . E F C D J
J I H W N M L K R Q P . V U . S 3 2 1 0 7 6 5 4 B A . 8 F E . C G
K L M N G H I J S . U . O W . R . 5 6 . 0 1 2 3 C D E F 8 . . B P
L K N M H G J I T S V U P O . . 5 4 7 6 1 0 . 2 D C F E . 8 B A Q
M N K L I J G H U W S T Q . . P 6 7 4 5 2 3 0 1 E F . D A B 8 . V
N M L . . I H G V U T W . Q . O 7 6 5 4 3 2 1 0 F E D C B A . 8 .
O P Q R S T . V G H I J K L . . 8 . A . C W E F 0 1 2 3 4 . 6 7 .
. O . . T S V U H G . . L K . M . 8 B A D C . W 1 0 . 2 5 4 7 6 E
Q R O P U V S T I J G H M N . L A W 8 . E F C . 2 3 0 1 6 7 4 5 B
R . P O V U T S J I H G N M . K . A . W . E D C 3 . 1 0 7 6 5 4 8
S T U V O P Q R K L M N . . . J . . E . . . A B 4 5 6 . 0 W 2 3 .
T S V U P O R Q L K N M H . . I D C F E . 8 B A 5 4 . 6 1 0 3 W 2
U V . T Q R O . . N K L I J . H E F C D A B 8 . 6 W 4 5 . 3 0 . 7
V U T S R Q P O N . L K J I . G F E D C . A . . 7 6 5 W 3 . 1 0 .
0 . 3 . 6 L 5 M C V F S A P . . O B . 8 U D T E K 7 N 4 I 1 H . W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E128: order 33, 831 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs33_review_29814974/n33_831.json`

SHA256: `658e93e9819e7fa54511cf6dd35a76098f1170337ac01c20ded13a9241aa4ed9`

Completion

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W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
2 3 0 1 W 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 6
3 2 1 0 7 6 W 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 5
4 5 6 7 0 1 2 3 W D E F 8 9 A B K L M N G H I J S T U V O P Q R C
5 4 7 6 1 0 3 2 D C W E 9 8 B A L K N M H G J I T S V U P O R Q F
6 7 4 5 2 3 0 1 E F C D W B 8 9 M N K L I J G H U V S T Q R O P A
7 6 5 4 3 2 1 0 F E D C B A W 8 N M L K J I H G V U T S R Q P O 9
8 9 A B C D E F 0 1 2 3 4 5 6 7 W P Q R S T U V G H I J K L M N O
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O W Q T S V U H G J I L K N M R
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P W V S T I J G H M N K L U
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U W S J I H G N M L K T
C D E F 8 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R W L M N G H I J K
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K W M H G J I N
E F C D A B 8 9 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L W J G H I
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I W G H
G H I J K W M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 9 A B C D E F L
H G J I L K N W P O R Q T S V U 1 0 3 2 5 4 7 6 9 8 B A D C F E M
I W G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D J
J I H W N M L K R Q P O V U T S 3 2 1 0 7 6 5 4 B A 9 8 F E D C G
K L M N G H I J S T U V O W Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B P
L K N M H G J I T S V U P O R W 5 4 7 6 1 0 3 2 D C F E 9 8 B A Q
M N K L I J G H U W S T Q R O P 6 7 4 5 2 3 0 1 E F C D A B 8 9 V
N M L K J I H G V U T W R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 S
O P Q R S T U V G H I J K L M N 8 9 A B C W E F 0 1 2 3 4 5 6 7 D
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
Q R O P U V S T I J G H M N K L A W 8 9 E F C D 2 3 0 1 6 7 4 5 B
R Q P O V U T S J I H G N M L K B A 9 W F E D C 3 2 1 0 7 6 5 4 8
S T U V O P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 W 2 3 1
T S V U P O R Q L K N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 W 2
U V S T Q R O P M N K L I J G H E F C D A B 8 9 6 W 4 5 2 3 0 1 7
V U T S R Q P O N M L K J I H G F E D C B A 9 8 7 6 5 W 3 2 1 0 4
0 J 3 G 6 L 5 M C V F S A P 9 Q O B R 8 U D T E K 7 N 4 I 1 H 2 W
```

Partial square

```
. . . 3 . 5 6 . . . . . C D . F . . . . K L M N . . . . S T U V 0
. . W . 5 4 7 6 . 8 B A . C . E . G J I L K N M . O . Q T . V U 3
2 3 0 1 . 7 4 . A B . . . F . D I . G H M N . . Q R O . U V S T .
3 2 1 0 7 6 W 4 B A . 8 F E . C J I H G N M L K R Q P . . U T S 5
4 5 6 7 0 . 2 3 W D E . 8 . . B . L M N G H I J S T U V O P Q R .
. 4 7 6 . . 3 2 D C . E . 8 . A L K . M H G J I T . V U P . R Q F
6 7 4 5 2 3 0 . E . C D . . . . M N K L I . G H U V . T . R O P A
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
8 . A B C D E F 0 1 2 3 . 5 . 7 W P Q R S T U V G . I J K L M . O
. . B A D C F E 1 0 3 . 5 4 . 6 P O W Q T . V U . G J I L K N . .
A B 8 . E F C D 2 . 0 . 6 7 . 5 Q R O . W V S T I J G H M N K L U
B A . 8 . E D C 3 2 . 0 7 . . . R Q P O V . W S J I . G N M L . T
C D E F 8 . A B . 5 . 7 0 1 . 3 S T U V . P Q R W L . N G H I J K
. C F E . 8 B A 5 4 7 6 1 0 . 2 T S V . P O R Q L K . M H . J I N
E F . . A B 8 . 6 7 4 5 2 3 . 1 U V S T . R O P M N K L W J G H I
F E D C B A . 8 7 . 5 4 . 2 . 0 V U T . R Q P O N M L K J I W . H
G H I . K W M N O P Q R S T . V 0 . 2 3 4 5 6 7 8 . . B . . E F .
H G J I L . N W P O R Q T . . U 1 0 3 2 5 . 7 6 . 8 B A D C F E M
I W G H M N K L . R O P U V . T . 3 0 1 6 7 4 5 A B 8 . E F C D J
. I H W N M L . R Q P . . U . S 3 2 1 0 7 6 5 4 B A . 8 F E . C G
K L M N G H I J S . U . O W . R 4 5 6 7 0 1 2 3 C D E F 8 . . B P
L K N M H G J I T S V U P O . . 5 4 7 6 . 0 . 2 D C F E . 8 B A Q
M N K L I J G H U W S T Q . . P 6 7 4 5 2 3 0 1 E F . D A B 8 . V
N M L . . I H G V U T W . Q . O 7 6 5 4 3 2 1 0 F E D C B A . 8 .
O P Q R S T U V G H I J K L . . 8 . A B C W E . 0 1 2 . . . 6 7 .
P O . . T S V U H G . . L K . M . 8 B A D C . W . 0 . 2 5 4 7 6 E
Q R O P U V S T I J G H M N . L A W 8 . E F C . 2 3 0 . 6 7 4 5 B
R Q P O . U T S J I H G N M . K . A . W . E . C 3 . . 0 7 6 5 4 8
S T U . O P Q R K L M N . H . J . . E . . . A B 4 5 6 . 0 W 2 3 1
T S V U P O R Q L K N M H . . I D C F E . 8 B A 5 4 7 6 1 0 3 W 2
U V . T Q R O . . N K L I J . . E F C D A B 8 . 6 W 4 5 . 3 0 1 7
. U T S R Q P . N . L K J I . G F E D C . A . . 7 6 5 W 3 . . 0 .
0 . . . 6 L 5 M C V F S A P . . O B . 8 U D T E K 7 N 4 I 1 H 2 W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E129: order 33, 830 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs33_review_29814974/n33_830.json`

SHA256: `763bfd358d56a1f03175e1d06abc43725d8ee56c7139edfd4c84bbb62b2b1944`

Completion

```
W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
2 3 0 1 W 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 6
3 2 1 0 7 6 W 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 5
4 5 6 7 0 1 2 3 W D E F 8 9 A B K L M N G H I J S T U V O P Q R C
5 4 7 6 1 0 3 2 D C W E 9 8 B A L K N M H G J I T S V U P O R Q F
6 7 4 5 2 3 0 1 E F C D W B 8 9 M N K L I J G H U V S T Q R O P A
7 6 5 4 3 2 1 0 F E D C B A W 8 N M L K J I H G V U T S R Q P O 9
8 9 A B C D E F 0 1 2 3 4 5 6 7 W P Q R S T U V G H I J K L M N O
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O W Q T S V U H G J I L K N M R
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P W V S T I J G H M N K L U
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U W S J I H G N M L K T
C D E F 8 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R W L M N G H I J K
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K W M H G J I N
E F C D A B 8 9 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L W J G H I
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I W G H
G H I J K W M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 9 A B C D E F L
H G J I L K N W P O R Q T S V U 1 0 3 2 5 4 7 6 9 8 B A D C F E M
I W G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D J
J I H W N M L K R Q P O V U T S 3 2 1 0 7 6 5 4 B A 9 8 F E D C G
K L M N G H I J S T U V O W Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B P
L K N M H G J I T S V U P O R W 5 4 7 6 1 0 3 2 D C F E 9 8 B A Q
M N K L I J G H U W S T Q R O P 6 7 4 5 2 3 0 1 E F C D A B 8 9 V
N M L K J I H G V U T W R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 S
O P Q R S T U V G H I J K L M N 8 9 A B C W E F 0 1 2 3 4 5 6 7 D
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
Q R O P U V S T I J G H M N K L A W 8 9 E F C D 2 3 0 1 6 7 4 5 B
R Q P O V U T S J I H G N M L K B A 9 W F E D C 3 2 1 0 7 6 5 4 8
S T U V O P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 W 2 3 1
T S V U P O R Q L K N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 W 2
U V S T Q R O P M N K L I J G H E F C D A B 8 9 6 W 4 5 2 3 0 1 7
V U T S R Q P O N M L K J I H G F E D C B A 9 8 7 6 5 W 3 2 1 0 4
0 J 3 G 6 L 5 M C V F S A P 9 Q O B R 8 U D T E K 7 N 4 I 1 H 2 W
```

Partial square

```
. . . 3 . 5 6 . . . . . C D . F . . . . K L M N . . . . S T U V 0
. . W . 5 4 7 6 . 8 B A . C . E . G J I L K N M . O . Q T . V U 3
. 3 0 1 . 7 4 5 A B . . . F . D I . G H M N . . . R O . U V S T .
3 2 1 0 7 6 W 4 B A . 8 F E . C J I H G N M L K R Q P . . U T S 5
4 5 6 7 0 1 2 3 W D E . 8 . . B . L M N G H I J S T U V O P Q R .
. 4 7 6 . . 3 2 D C . E . 8 . A L . . M H G J . T . . U P . R Q F
6 7 4 5 2 3 0 . E . C D . . . . M N K . I . G H U V S T . R O P A
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
8 . A B C D E F 0 1 2 3 . 5 . 7 W P Q . S T U V G H . J K L M . O
. . B A D C F E 1 . 3 2 5 4 . 6 P O W Q T . V U . . . I L . N . R
A B 8 . E F C D 2 . 0 1 6 7 . 5 Q R O . W V S T I J G H M N K L U
B A . 8 . E D C 3 2 . 0 7 . . . R Q . O V . W S J I H G N M L K T
C D E F 8 . A B . 5 . 7 0 1 . 3 S T U V O P Q R W L M N G H I J K
. C F E . 8 B A 5 4 7 6 1 0 . 2 T S V . P O R Q L K . M H . J I N
E F . . A B 8 . 6 7 4 5 2 3 . 1 U V S T . R O P M N K L W J G . I
F E D C B A . 8 7 6 5 . 3 2 . 0 V U T . R Q P O N M L K J I W G H
G H I . K W M N O P Q . S T . V 0 1 2 3 4 5 6 7 8 . . B C D E . .
H G J I L . N W P O R Q T . . U 1 0 . 2 5 . 7 6 . 8 B A D C F E M
I W G H M N K L . R O P U V . T . 3 0 1 6 7 4 5 A B 8 . E F C D J
J I H W N M L K R Q P . V U . . 3 2 1 0 7 6 5 4 B A . 8 F E . C G
K L M N G H I J S . U V O W . R . 5 6 7 0 . 2 3 C D E F 8 . . B P
L K N M H G . I T S V U P O . . 5 4 7 6 1 0 . 2 D C F E . 8 B A Q
M N K L I J G H U W S T Q . . P 6 7 4 5 2 3 0 1 E F . D A B 8 . V
N M L . . I H G V U T W . Q . O 7 6 5 4 3 2 1 0 F E D C B A . 8 .
O P Q . S T U V G H I J K L . . 8 . A B C W E F 0 . 2 . 4 . 6 7 .
P O . . T S V U H G . . L K . M . 8 B A D C . W . 0 . 2 5 4 7 6 E
Q R O P U V S T I J G H M N . L A W 8 . E F C . 2 . 0 1 6 7 4 5 B
R Q P O V U T S J I H G N M . . . A . W . E D C 3 . . 0 7 6 5 4 8
S . U . O . . R K L M N G H . J . . E . . . . B 4 5 6 . 0 . 2 . 1
T S V U P O R Q L K N M H . . I D C F E . 8 B A 5 4 7 6 1 0 3 W 2
U V . T Q R O . . N K L I J . H E F C D A B 8 . 6 W . 5 . 3 0 . 7
V U T S R Q P O N . L K J I . G F E D C . A . . 7 6 5 W 3 . . 0 .
0 . 3 G 6 L 5 M C V F S A P . . O B . 8 U D T E K 7 N 4 I 1 H 2 W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E130: order 33, 829 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs33_review_29814974/n33_829.json`

SHA256: `4fa47219d7f667c85f402e57984e258634ae5d19da550364e7e396273d76eae2`

Completion

```
W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
2 3 0 1 W 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 6
3 2 1 0 7 6 W 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 5
4 5 6 7 0 1 2 3 W D E F 8 9 A B K L M N G H I J S T U V O P Q R C
5 4 7 6 1 0 3 2 D C W E 9 8 B A L K N M H G J I T S V U P O R Q F
6 7 4 5 2 3 0 1 E F C D W B 8 9 M N K L I J G H U V S T Q R O P A
7 6 5 4 3 2 1 0 F E D C B A W 8 N M L K J I H G V U T S R Q P O 9
8 9 A B C D E F 0 1 2 3 4 5 6 7 W P Q R S T U V G H I J K L M N O
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O W Q T S V U H G J I L K N M R
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P W V S T I J G H M N K L U
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U W S J I H G N M L K T
C D E F 8 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R W L M N G H I J K
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K W M H G J I N
E F C D A B 8 9 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L W J G H I
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I W G H
G H I J K W M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 9 A B C D E F L
H G J I L K N W P O R Q T S V U 1 0 3 2 5 4 7 6 9 8 B A D C F E M
I W G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D J
J I H W N M L K R Q P O V U T S 3 2 1 0 7 6 5 4 B A 9 8 F E D C G
K L M N G H I J S T U V O W Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B P
L K N M H G J I T S V U P O R W 5 4 7 6 1 0 3 2 D C F E 9 8 B A Q
M N K L I J G H U W S T Q R O P 6 7 4 5 2 3 0 1 E F C D A B 8 9 V
N M L K J I H G V U T W R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 S
O P Q R S T U V G H I J K L M N 8 9 A B C W E F 0 1 2 3 4 5 6 7 D
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
Q R O P U V S T I J G H M N K L A W 8 9 E F C D 2 3 0 1 6 7 4 5 B
R Q P O V U T S J I H G N M L K B A 9 W F E D C 3 2 1 0 7 6 5 4 8
S T U V O P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 W 2 3 1
T S V U P O R Q L K N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 W 2
U V S T Q R O P M N K L I J G H E F C D A B 8 9 6 W 4 5 2 3 0 1 7
V U T S R Q P O N M L K J I H G F E D C B A 9 8 7 6 5 W 3 2 1 0 4
0 J 3 G 6 L 5 M C V F S A P 9 Q O B R 8 U D T E K 7 N 4 I 1 H 2 W
```

Partial square

```
. . . 3 . 5 6 . . . . . C D . F . . . . K L M N . . . . S T U V 0
. . W . 5 4 7 6 . 8 B A . C . E . G J I L K N M . O . Q T . V U 3
. . 0 1 . 7 . . A B . . . F . D I . G H M N . . . . O P U V . T 6
3 2 1 0 . 6 W 4 B A . 8 F E . C J I . G N M L K R . P . . U T S 5
4 5 6 . 0 . 2 . W D E . 8 . . B . L M N G H I J S T U V O P Q R .
. 4 7 6 . . 3 2 D C . E . 8 . A L K . . H . J I T . . U P . R Q F
6 7 4 5 2 3 0 . E . C D . . . . M N . L I . . H U V . T . R . P A
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
8 . A B C D E F 0 1 2 3 4 5 . 7 W P Q R S T U V G H . J K L . . O
. . B A D C F E 1 0 3 2 5 4 . 6 P O W Q T . V U . G J I L K N . R
A B 8 . E F C D 2 . 0 1 6 7 . 5 Q R O . W V S T I J G H M N K L U
B A . 8 . E D C 3 2 . 0 7 . . . R Q . O V . W S J I . G N M L . T
C D E F 8 . A B . 5 . 7 0 1 . 3 S T U V . P Q . W L . N G H I J K
. C F E . 8 B A 5 4 7 6 1 0 . 2 T S V . P O R Q L K . M H . J I N
E F . . A B 8 . 6 7 4 5 2 3 . 1 U V S T . R O P M N K L W J G H I
F E . C B A . 8 7 6 5 4 3 2 . 0 . U T . R Q P . N M L K J I W G H
G H I . K W M N O P Q R S T . . 0 1 2 3 4 5 6 . 8 . . B C D E . .
H G J I L . N W P O R Q T . . U 1 0 . 2 5 . 7 6 . 8 B A D C F E M
I W G H M N K L . R O P U V . T . 3 0 1 6 7 4 5 A B 8 . E F C D J
J I H W N M L K R Q P . V U . S 3 2 1 0 7 6 5 4 B A . 8 F E . C G
K L M N G H I J S . U . O W . R . 5 6 7 0 . 2 3 C D E F 8 . . B P
L K N M H G J I T S V U P O . . 5 4 7 6 1 0 . 2 D C F E . 8 B A Q
M N K L I J G H U W S T Q . . P 6 7 4 5 2 3 0 1 E F . D A B 8 . V
N M L . . I H G V U T W . Q . O 7 6 5 4 3 2 1 0 F E D C B A . 8 .
O P Q R S T U V G H I J . L . . 8 . A B C W E F 0 1 2 3 4 . 6 7 .
P O . . T S V U H G . . L K . M . 8 B A D C . W 1 0 . 2 5 4 7 6 E
Q R O P U V S T I J G H M N . L A W 8 . E F C . 2 3 0 1 6 7 4 5 B
R Q P O V U T S J I H G N M . K . A . . . E D C 3 . . 0 7 6 . 4 8
S T U V O P Q R K L M N G . . J . . E . . . A B 4 5 6 . 0 W 2 3 1
T S V U P O R Q L K N M H . . I D C F E . 8 B A 5 4 7 6 1 0 3 W 2
U V . T Q R O . . N K L I J . H E F C D A B 8 . 6 W 4 5 . 3 0 1 7
V U T S R Q P O N . L K J I . G F E D C . A . . 7 6 5 W 3 . 1 0 .
0 . 3 G 6 L 5 M C V F S A P . . O B . 8 U D T E K 7 N 4 I 1 H 2 W
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Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E131: order 33, 824 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs33_review_29814974/random_repairs/task_2_n33_rank_1.pending_restart3/repair.json`

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Completion

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W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
2 3 0 1 W 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 6
3 2 1 0 7 6 W 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 5
4 5 6 7 0 1 2 3 W D E F 8 9 A B K L M N G H I J S T U V O P Q R C
5 4 7 6 1 0 3 2 D C W E 9 8 B A L K N M H G J I T S V U P O R Q F
6 7 4 5 2 3 0 1 E F C D W B 8 9 M N K L I J G H U V S T Q R O P A
7 6 5 4 3 2 1 0 F E D C B A W 8 N M L K J I H G V U T S R Q P O 9
8 9 A B C D E F 0 1 2 3 4 5 6 7 W P Q R S T U V G H I J K L M N O
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O W Q T S V U H G J I L K N M R
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P W V S T I J G H M N K L U
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U W S J I H G N M L K T
C D E F 8 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R W L M N G H I J K
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K W M H G J I N
E F C D A B 8 9 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L W J G H I
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I W G H
G H I J K W M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 9 A B C D E F L
H G J I L K N W P O R Q T S V U 1 0 3 2 5 4 7 6 9 8 B A D C F E M
I W G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D J
J I H W N M L K R Q P O V U T S 3 2 1 0 7 6 5 4 B A 9 8 F E D C G
K L M N G H I J S T U V O W Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B P
L K N M H G J I T S V U P O R W 5 4 7 6 1 0 3 2 D C F E 9 8 B A Q
M N K L I J G H U W S T Q R O P 6 7 4 5 2 3 0 1 E F C D A B 8 9 V
N M L K J I H G V U T W R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 S
O P Q R S T U V G H I J K L M N 8 9 A B C W E F 0 1 2 3 4 5 6 7 D
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
Q R O P U V S T I J G H M N K L A W 8 9 E F C D 2 3 0 1 6 7 4 5 B
R Q P O V U T S J I H G N M L K B A 9 W F E D C 3 2 1 0 7 6 5 4 8
S T U V O P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 W 2 3 1
T S V U P O R Q L K N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 W 2
U V S T Q R O P M N K L I J G H E F C D A B 8 9 6 W 4 5 2 3 0 1 7
V U T S R Q P O N M L K J I H G F E D C B A 9 8 7 6 5 W 3 2 1 0 4
0 J 3 G 6 L 5 M C V F S A P 9 Q O B R 8 U D T E K 7 N 4 I 1 H 2 W
```

Partial square

```
. . . 3 . 5 6 . . . . . C D . F . . . . K L M N . . . . S T U V 0
. . W . 5 4 7 6 . 8 B A . C . E . G J I L K N M . O . Q T . V U 3
. 3 0 1 . 7 4 5 A B . . . F . D I . G H M N . . Q R O . U V S T .
3 2 1 0 7 6 W 4 B A . 8 F E . C J I H G N M L K R Q . . . U T S 5
4 5 6 7 0 1 2 3 W D E . 8 . . B . L M N G H I J S T U V O P Q R .
. 4 7 6 . . 3 2 D C . E . 8 . A L K . M H G J I T . . U P . R Q F
6 7 4 5 2 3 0 1 E . C D . . . . M N . L I . G H U V S T . R . P A
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
8 . A B C D E F 0 1 2 3 . 5 . 7 W P Q R S T U V G H . J K L M . O
. . B . D C F E 1 0 3 2 5 4 . 6 P O W Q T . V . . G J I L . N . .
A B 8 . E F C D 2 . 0 1 6 7 . 5 Q R O . W V S T I J G H M N K L U
B A . 8 . E D C 3 2 . 0 7 . . . R Q P O . . W S J I H G N M . . T
C D E F 8 . A B . . . 7 0 1 . 3 S T U V O P Q R W L M N G H I J K
. C F E . 8 B A 5 4 7 6 1 0 . 2 T S V . P O R Q L K . M H . J . N
E F . . A B 8 . 6 7 4 5 2 3 . 1 . V S T . R O P M N . L W J G H .
F E D C B A . 8 7 . 5 . . . . . . . T . . Q P . N M L K J I W G .
G H I J K W M N O P Q . S T . V 0 1 2 3 4 5 6 7 8 . . B C D E . .
H G J I L . N W P O R Q T . . U 1 0 3 2 5 . 7 6 . 8 B A D C F E M
I W G H M N K L . R O P . V . T . 3 0 1 6 7 4 5 A B 8 . E F C D J
. I H W N M L K R Q P . V U . S 3 2 1 0 7 6 5 4 B A . 8 F E . C G
K L M N G H I J S . U . O W . R 4 5 6 7 0 1 2 3 C . E F 8 . . B P
L K N M H G J I T S V U P O . . 5 4 7 6 1 0 . 2 D C F E . 8 B A Q
M N K L I J G H U W S T Q . . P 6 7 4 5 2 3 0 1 E F . D A B 8 . V
N M L . . I H G V U T W . Q . O 7 6 5 4 3 2 . 0 F E D C B A . 8 .
O P Q R S T U V G H I J . L . . 8 . A B C W E F 0 1 . . . . 6 7 .
P O . . T S V U H G . . L K . M . 8 B A D C . W 1 0 . 2 5 4 7 6 E
Q R O P U V S T I J G H M N . L A W 8 . E F C . 2 3 0 1 6 7 4 5 B
R Q P O V U T S J I H G N M . K . A . . . E . C 3 . . 0 7 6 5 4 8
S T U V O P Q R K L M N . H . J . . E . . . A B 4 5 6 . 0 . 2 3 1
T S V U P O R Q L K N M H . . I D C F E . 8 B A 5 4 7 6 1 0 3 W 2
U V . T Q R O . . N K L I J . H E F C D A B 8 . 6 W 4 5 . 3 0 1 7
V U T S . Q P O N . L K J I . G F E D C . A . . 7 6 5 W 3 . 1 . .
0 . 3 G 6 L 5 M C V F S A P . . O B . 8 U D T E K 7 N 4 I 1 H . W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E132: order 33, 823 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs33_review_29814974/random_repairs/task_2_n33_rank_1.pending_restart0/repair.json`

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Completion

```
W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
2 3 0 1 W 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 6
3 2 1 0 7 6 W 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 5
4 5 6 7 0 1 2 3 W D E F 8 9 A B K L M N G H I J S T U V O P Q R C
5 4 7 6 1 0 3 2 D C W E 9 8 B A L K N M H G J I T S V U P O R Q F
6 7 4 5 2 3 0 1 E F C D W B 8 9 M N K L I J G H U V S T Q R O P A
7 6 5 4 3 2 1 0 F E D C B A W 8 N M L K J I H G V U T S R Q P O 9
8 9 A B C D E F 0 1 2 3 4 5 6 7 W P Q R S T U V G H I J K L M N O
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O W Q T S V U H G J I L K N M R
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P W V S T I J G H M N K L U
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U W S J I H G N M L K T
C D E F 8 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R W L M N G H I J K
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K W M H G J I N
E F C D A B 8 9 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L W J G H I
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I W G H
G H I J K W M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 9 A B C D E F L
H G J I L K N W P O R Q T S V U 1 0 3 2 5 4 7 6 9 8 B A D C F E M
I W G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D J
J I H W N M L K R Q P O V U T S 3 2 1 0 7 6 5 4 B A 9 8 F E D C G
K L M N G H I J S T U V O W Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B P
L K N M H G J I T S V U P O R W 5 4 7 6 1 0 3 2 D C F E 9 8 B A Q
M N K L I J G H U W S T Q R O P 6 7 4 5 2 3 0 1 E F C D A B 8 9 V
N M L K J I H G V U T W R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 S
O P Q R S T U V G H I J K L M N 8 9 A B C W E F 0 1 2 3 4 5 6 7 D
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
Q R O P U V S T I J G H M N K L A W 8 9 E F C D 2 3 0 1 6 7 4 5 B
R Q P O V U T S J I H G N M L K B A 9 W F E D C 3 2 1 0 7 6 5 4 8
S T U V O P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 W 2 3 1
T S V U P O R Q L K N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 W 2
U V S T Q R O P M N K L I J G H E F C D A B 8 9 6 W 4 5 2 3 0 1 7
V U T S R Q P O N M L K J I H G F E D C B A 9 8 7 6 5 W 3 2 1 0 4
0 J 3 G 6 L 5 M C V F S A P 9 Q O B R 8 U D T E K 7 N 4 I 1 H 2 W
```

Partial square

```
. . . 3 . 5 6 . . . . . C D . F . . . . K L M N . . . . S T U V 0
. . W . 5 4 7 6 . 8 B A . C . E . G J I L K N M . O . Q T . V U 3
2 3 0 1 . 7 4 5 A B . . . F . D I . G H M N . . Q R O . U . S T 6
3 2 1 0 7 6 W 4 B A . 8 F E . C J I H G N M L K R Q P . . U T S 5
4 5 6 7 0 1 2 3 W D E . 8 . . B . L M N G H I J S T U V O P Q R .
. 4 7 6 . . 3 2 D C . E . 8 . A L K . . H G J I T . . U P . R Q F
6 7 4 5 2 3 0 1 E . C D . . . . M N . L I . G H U V S . . R O P A
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
8 . A B C D E F 0 1 2 3 . 5 . 7 W P Q R S T U . G H I J K L M . O
. . B A D C F E 1 0 . . 5 4 . 6 P O W Q T . V U . G J I L . N . .
A B 8 . E F C D 2 . 0 1 6 7 . 5 Q R O . W V S T I J G H M N K L U
B A . 8 . E D C 3 2 . 0 7 . . . R Q . O V . W S J I . G N M L . T
C D E F 8 . A B . 5 . 7 0 1 . 3 S T U V O P Q R W L M N G H I J K
. C F E . 8 B A 5 4 7 6 1 0 . 2 T S V . P O R Q L K . M H . J I N
E F . . A B 8 . 6 7 4 5 2 3 . . U V S T . R O P M N K L W J G H I
F E . C B A . 8 7 6 5 . . . . 0 . U T . . Q P O N M . K J I . . H
G H I J K W M N O P Q R S T . . . 1 2 3 4 5 6 7 8 . . . . . E . .
H G . I L . N W P O R Q T . . U 1 0 3 2 5 . 7 6 . 8 B A D C F E M
I W G H M N K L . R O P U V . T . 3 0 1 6 7 4 5 A B 8 . E F C D .
J I H W N M L K R Q P . V U . S 3 2 1 0 7 6 5 4 B A . 8 F E . C G
K L M N G H I . S . U V O W . R 4 5 6 7 0 1 2 3 C D E F 8 . . B P
L K N M H G J I T S V U P O . . 5 4 7 6 1 0 . 2 D C F E . 8 B A Q
M N K L I J G H U W S T Q . . P 6 7 4 5 2 3 0 1 E F . D A B 8 . V
N M L . . I H G V U . W . Q . O 7 6 5 4 3 2 . 0 F E D C B A . 8 .
O P Q . S T U V G H I J K L . . 8 . A B C W E F 0 1 2 . 4 . 6 7 .
. O . . T S V U H G . . L K . M . 8 B A D C . W 1 0 . 2 5 4 7 6 E
Q R O P U V S T I J G H M N . L A W 8 . E F C . 2 3 0 1 6 7 4 5 B
R Q P O V U T . J I H G N M . K . A . . . E . C . . 1 0 7 6 . . 8
S T U . . P Q R K L M N G H . J . . E . . . A B 4 5 6 . 0 W 2 3 .
T S V U P O R Q L K N M H . . I D C F E . 8 B A 5 4 7 6 1 0 3 W 2
U V . T Q R O . . N K L I J . H E F C D A B 8 . 6 W . 5 . 3 0 1 7
V U T S . Q P O N . L K J I . G F E D C . A . . 7 6 5 W 3 . 1 0 .
0 . 3 G 6 L 5 M C V F S A P . . O B . 8 U D T E K 7 N 4 I 1 H . W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E133: order 33, 823 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs33_review_29814974/random_repairs/task_2_n33_rank_1.pending_restart4/repair.json`

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Completion

```
W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
2 3 0 1 W 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 6
3 2 1 0 7 6 W 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 5
4 5 6 7 0 1 2 3 W D E F 8 9 A B K L M N G H I J S T U V O P Q R C
5 4 7 6 1 0 3 2 D C W E 9 8 B A L K N M H G J I T S V U P O R Q F
6 7 4 5 2 3 0 1 E F C D W B 8 9 M N K L I J G H U V S T Q R O P A
7 6 5 4 3 2 1 0 F E D C B A W 8 N M L K J I H G V U T S R Q P O 9
8 9 A B C D E F 0 1 2 3 4 5 6 7 W P Q R S T U V G H I J K L M N O
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O W Q T S V U H G J I L K N M R
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P W V S T I J G H M N K L U
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U W S J I H G N M L K T
C D E F 8 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R W L M N G H I J K
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K W M H G J I N
E F C D A B 8 9 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L W J G H I
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I W G H
G H I J K W M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 9 A B C D E F L
H G J I L K N W P O R Q T S V U 1 0 3 2 5 4 7 6 9 8 B A D C F E M
I W G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D J
J I H W N M L K R Q P O V U T S 3 2 1 0 7 6 5 4 B A 9 8 F E D C G
K L M N G H I J S T U V O W Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B P
L K N M H G J I T S V U P O R W 5 4 7 6 1 0 3 2 D C F E 9 8 B A Q
M N K L I J G H U W S T Q R O P 6 7 4 5 2 3 0 1 E F C D A B 8 9 V
N M L K J I H G V U T W R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 S
O P Q R S T U V G H I J K L M N 8 9 A B C W E F 0 1 2 3 4 5 6 7 D
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
Q R O P U V S T I J G H M N K L A W 8 9 E F C D 2 3 0 1 6 7 4 5 B
R Q P O V U T S J I H G N M L K B A 9 W F E D C 3 2 1 0 7 6 5 4 8
S T U V O P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 W 2 3 1
T S V U P O R Q L K N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 W 2
U V S T Q R O P M N K L I J G H E F C D A B 8 9 6 W 4 5 2 3 0 1 7
V U T S R Q P O N M L K J I H G F E D C B A 9 8 7 6 5 W 3 2 1 0 4
0 J 3 G 6 L 5 M C V F S A P 9 Q O B R 8 U D T E K 7 N 4 I 1 H 2 W
```

Partial square

```
. . . 3 . 5 6 . . . . . C D . F . . . . K L M N . . . . S T U V 0
. . W . 5 4 7 6 . 8 B A . C . E . G J I L K N M . O . Q T . V U 3
2 3 0 . . 7 4 5 A B . . . F . D I . G H M N . . . R O P U V . T 6
3 2 1 0 7 6 W 4 B A . 8 F E . C J I H G N M L K R Q . . . U T S 5
4 5 6 . 0 1 2 . W D E . 8 . . B . L M N G H I . S T U V O P Q R .
. 4 7 6 . . 3 2 D C . E . 8 . A L K . . H G . I T . V U P . R Q F
6 7 4 5 2 3 0 . E . C D . . . . M N . L I . G H U V S T . R O P A
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
8 . A B C . E F 0 1 2 3 4 5 . 7 W P Q . S T U V G H I J K L M . O
. . B A D C F E 1 0 3 . 5 4 . 6 P O W Q T . V U . . . I L . N . R
A B 8 . E F C D 2 . 0 1 6 7 . 5 Q R O . W V S T I J G H M N K L U
B A . 8 . E D C 3 2 . 0 7 . . . R Q P O V . W S J I H G N M L . T
C D E F 8 . A B . 5 . 7 0 1 . 3 S T U . O P Q . W L M N G H I J K
. C F E . 8 B A 5 4 7 6 1 0 . 2 T S V . P O R Q L K . M H . J I N
E F . . A B 8 . 6 7 4 5 2 3 . 1 U V S T . R O P M N K L W J G . I
F E D C B A . 8 7 6 5 4 . . . 0 V U T . R Q P . N M L K J I W G H
G H I . K W M N O P Q . S T . V . 1 2 3 4 5 6 7 . . . B C D E F .
H G J I L . N W P O R Q T . . U 1 0 . 2 5 . 7 6 . 8 B A D C F E M
I W G H M N . L . R O P U V . T . 3 0 1 6 7 4 5 A B 8 . E F C D J
J I H W N M L K R Q P . V U . S 3 2 1 0 7 6 5 4 B A . 8 F E . C G
K L M N G H I . S . U . O W . R . 5 . 7 0 1 2 3 C . E F 8 . . B P
L K N M H G J I T S V U P O . . 5 4 7 6 1 0 . 2 D C F E . 8 B A Q
M N K L I J G H U W S T Q . . P 6 7 4 5 2 3 0 1 E F . D A B 8 . V
N M L . . I H G V U T W . Q . O 7 6 5 4 3 2 1 0 F E D C B A . 8 .
O P Q R . T U V G H I J . L . . 8 . A B C W E F 0 1 2 3 . . 6 7 .
. O . . T S . U H G . . L K . M . 8 B A D C . W 1 0 . 2 5 4 7 6 E
Q R O P U V S T I J G H M N . L A W 8 . E F C . 2 3 0 1 6 7 4 5 B
R Q P . . U T S J I H G N M . K . A . W . E D C 3 . . 0 7 6 . 4 8
S T U V O P . R K L M N G H . J . . E . . . A B 4 5 . . 0 W 2 . 1
T S V U P O R Q L K N M H . . I D C F E . 8 B A 5 4 7 6 1 0 3 W 2
U V . T Q R O . . N K L I J . . E F C D A B 8 . 6 W . 5 . 3 0 1 7
. U T S R Q P O N . L K J I . G F E D C . A . . 7 6 . W 3 . 1 0 .
0 . 3 G 6 L 5 M C V F S A P . . O B . 8 U D T E K 7 N 4 I 1 H 2 W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E134: order 33, 819 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs33_review_29814974/random_repairs/task_2_n33_rank_1.pending_restart7/repair.json`

SHA256: `3048506c331eca68627a5515bf137989088a44f2b4176e0442adb1a1959f38d3`

Completion

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W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
2 3 0 1 W 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 6
3 2 1 0 7 6 W 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 5
4 5 6 7 0 1 2 3 W D E F 8 9 A B K L M N G H I J S T U V O P Q R C
5 4 7 6 1 0 3 2 D C W E 9 8 B A L K N M H G J I T S V U P O R Q F
6 7 4 5 2 3 0 1 E F C D W B 8 9 M N K L I J G H U V S T Q R O P A
7 6 5 4 3 2 1 0 F E D C B A W 8 N M L K J I H G V U T S R Q P O 9
8 9 A B C D E F 0 1 2 3 4 5 6 7 W P Q R S T U V G H I J K L M N O
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O W Q T S V U H G J I L K N M R
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P W V S T I J G H M N K L U
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U W S J I H G N M L K T
C D E F 8 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R W L M N G H I J K
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K W M H G J I N
E F C D A B 8 9 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L W J G H I
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I W G H
G H I J K W M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 9 A B C D E F L
H G J I L K N W P O R Q T S V U 1 0 3 2 5 4 7 6 9 8 B A D C F E M
I W G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D J
J I H W N M L K R Q P O V U T S 3 2 1 0 7 6 5 4 B A 9 8 F E D C G
K L M N G H I J S T U V O W Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B P
L K N M H G J I T S V U P O R W 5 4 7 6 1 0 3 2 D C F E 9 8 B A Q
M N K L I J G H U W S T Q R O P 6 7 4 5 2 3 0 1 E F C D A B 8 9 V
N M L K J I H G V U T W R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 S
O P Q R S T U V G H I J K L M N 8 9 A B C W E F 0 1 2 3 4 5 6 7 D
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
Q R O P U V S T I J G H M N K L A W 8 9 E F C D 2 3 0 1 6 7 4 5 B
R Q P O V U T S J I H G N M L K B A 9 W F E D C 3 2 1 0 7 6 5 4 8
S T U V O P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 W 2 3 1
T S V U P O R Q L K N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 W 2
U V S T Q R O P M N K L I J G H E F C D A B 8 9 6 W 4 5 2 3 0 1 7
V U T S R Q P O N M L K J I H G F E D C B A 9 8 7 6 5 W 3 2 1 0 4
0 J 3 G 6 L 5 M C V F S A P 9 Q O B R 8 U D T E K 7 N 4 I 1 H 2 W
```

Partial square

```
. . . 3 . 5 6 . . . . . C D . F . . . . K L M N . . . . S T U V 0
. . W . 5 4 7 6 . 8 B A . C . E . G J I L K N M . O . Q T . V U 3
. 3 0 1 . 7 4 5 A B . . . F . D I . G H M N . . . R O . U V S T 6
3 2 1 0 7 6 W 4 B A . 8 F E . C J I H G N M L K R Q P . . U T S 5
4 5 6 7 0 1 2 3 W D E . 8 . . B . L M N G H I J S T U V O P Q R .
. 4 7 6 . . 3 2 D C . E . 8 . A L . . M H G J I T . . U P . R Q F
6 7 4 5 2 3 0 1 E . C D . . . . M N . L I . G H U V . T . R . P A
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
8 . A B C D E F 0 1 2 3 . 5 . 7 W P Q R S T U . G H . J K L M . O
. . B . D C F E 1 . 3 2 5 4 . 6 P O . Q T . V U . G J I L . N . R
A B 8 . E F C D 2 . 0 . 6 7 . 5 Q R O . W V S T I J G H M N K L U
B A . 8 . E D C 3 2 . 0 7 . . . R Q . O V . W S J I . G N M L K T
C D E F 8 . A B . 5 . 7 0 1 . 3 S T U V O P Q R W L M N G H I J .
. C F E . 8 B A 5 4 7 6 1 0 . . T S V . P O R Q L K . M H . J I N
E F . . A B 8 . 6 7 4 5 2 3 . . . V S T . R O P M N . L W J G . .
F E . C B A . 8 7 . 5 4 3 . . 0 . . . . R . . . N M L K J I W G H
G H I J K W M N O P Q R S T . V 0 1 2 3 4 5 6 7 8 . . B C D E F .
H G J I L . N W P O R Q T . . U 1 0 3 2 5 . 7 6 . 8 B A D C F E M
I W G H M N K L . R O P U V . T . 3 0 1 6 7 4 5 A B 8 . E F C D J
J I H W N M L K R Q P . V U . S 3 2 1 0 7 6 5 4 B A . 8 F E . C G
K L M N G H I . S . U . O W . R . 5 6 7 0 1 2 3 C . E F 8 . . B P
L K N M H G J I T . V . P O . . 5 4 7 6 1 0 . 2 D C F E . 8 B A Q
M N K L I J G H U W S T Q . . P 6 7 4 5 2 3 0 1 E F . D A B 8 . V
N M L . . I H G V U T W . Q . O 7 6 5 4 3 2 1 0 F E D C B A . 8 .
O P Q . S T U V G H . J K L . . 8 . A B C W E . 0 1 2 . 4 . 6 7 .
P O . . T S V U H G . . L K . M . 8 B A D C . W 1 0 . 2 5 4 7 6 E
Q R O P U V S T I J G H M N . L A W 8 . E F C . 2 3 0 1 6 7 4 5 B
R Q P O . U T S J I H G N M . K . A . W . E D C 3 . 1 0 7 6 5 . 8
S T U V O P Q R K L M N . H . J . . E . . . A B 4 5 6 . 0 W 2 3 .
T S . U P O R Q L K N M H . . I D C F E . 8 B A 5 4 7 6 1 0 3 W 2
. V . T Q R O . . N K L I J . H E F C D A B 8 . 6 W . 5 . 3 0 . 7
V U T S R Q P O N . L K J . . G F E D C . A . . 7 6 5 W 3 . . 0 .
0 . 3 G 6 L 5 M C V F S A P . . O B . 8 U D T E K 7 N 4 I 1 H . W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E135: order 33, 819 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs33_review_29814974/random_repairs/task_2_n33_rank_1.pending_restart6/repair.json`

SHA256: `62b6d99fccdfacdc24c370ce8ca54db9298a41eaf205486de9603a41fa94e049`

Completion

```
W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
2 3 0 1 W 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 6
3 2 1 0 7 6 W 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 5
4 5 6 7 0 1 2 3 W D E F 8 9 A B K L M N G H I J S T U V O P Q R C
5 4 7 6 1 0 3 2 D C W E 9 8 B A L K N M H G J I T S V U P O R Q F
6 7 4 5 2 3 0 1 E F C D W B 8 9 M N K L I J G H U V S T Q R O P A
7 6 5 4 3 2 1 0 F E D C B A W 8 N M L K J I H G V U T S R Q P O 9
8 9 A B C D E F 0 1 2 3 4 5 6 7 W P Q R S T U V G H I J K L M N O
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O W Q T S V U H G J I L K N M R
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P W V S T I J G H M N K L U
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U W S J I H G N M L K T
C D E F 8 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R W L M N G H I J K
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K W M H G J I N
E F C D A B 8 9 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L W J G H I
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I W G H
G H I J K W M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 9 A B C D E F L
H G J I L K N W P O R Q T S V U 1 0 3 2 5 4 7 6 9 8 B A D C F E M
I W G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D J
J I H W N M L K R Q P O V U T S 3 2 1 0 7 6 5 4 B A 9 8 F E D C G
K L M N G H I J S T U V O W Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B P
L K N M H G J I T S V U P O R W 5 4 7 6 1 0 3 2 D C F E 9 8 B A Q
M N K L I J G H U W S T Q R O P 6 7 4 5 2 3 0 1 E F C D A B 8 9 V
N M L K J I H G V U T W R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 S
O P Q R S T U V G H I J K L M N 8 9 A B C W E F 0 1 2 3 4 5 6 7 D
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
Q R O P U V S T I J G H M N K L A W 8 9 E F C D 2 3 0 1 6 7 4 5 B
R Q P O V U T S J I H G N M L K B A 9 W F E D C 3 2 1 0 7 6 5 4 8
S T U V O P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 W 2 3 1
T S V U P O R Q L K N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 W 2
U V S T Q R O P M N K L I J G H E F C D A B 8 9 6 W 4 5 2 3 0 1 7
V U T S R Q P O N M L K J I H G F E D C B A 9 8 7 6 5 W 3 2 1 0 4
0 J 3 G 6 L 5 M C V F S A P 9 Q O B R 8 U D T E K 7 N 4 I 1 H 2 W
```

Partial square

```
. . . 3 . 5 6 . . . . . C D . F . . . . K L M N . . . . S T U V 0
. . W . 5 4 7 6 . 8 B A . C . E . G J I L K N M . O . Q T . V U 3
2 3 0 . . 7 4 5 A B . . . F . D I . G H M N . . Q R O P U V . T 6
3 2 1 0 7 6 W 4 B A . 8 F E . C J I H G N M L K R Q . . . U T S 5
4 5 6 7 0 1 2 3 W D E . 8 . . B . L M N G H I J S T U V O P Q R .
. 4 7 6 . . 3 2 D C . E . 8 . A L K . M H G . I T . V U P . R Q F
6 7 4 5 2 3 0 1 E . C . . . . . M N . L I . G H U V S . . R O P A
. . . . . .
8 . A B C D E F 0 1 2 3 . 5 . 7 W P Q R S T U . G H I J K L M . O
. . B A D C F E 1 0 3 . 5 4 . 6 P O . Q T . V U . . J I L . N . R
A B 8 . E F C D 2 . 0 1 6 7 . 5 Q R O . W V S T I J G H M N K L U
B A . 8 . E D C 3 2 . 0 7 . . . R Q P O V . W S J I . G N M L K T
C D E F 8 . A B . 5 . 7 0 1 . 3 S T U V O P Q R W L M N G H I J .
. C F E . 8 B A 5 4 7 6 1 0 . 2 T S V . P O R Q L K . M H . J I N
E F . . A B 8 . 6 7 4 5 2 3 . 1 . V S T . R O P M N . L W J G H I
F E D C B A . 8 7 6 5 4 . . . . . U T . . Q . . N M . K J I W G H
G H . J . W M N O P Q R S T . V 0 1 2 3 4 5 6 7 8 . . B C D E . .
H G . I L . N W P O R Q T . . U 1 0 3 2 5 . 7 6 . 8 . A D C F E M
I W G H M N K L . R O P U V . T . 3 0 1 6 7 4 5 A B 8 . E F C D .
J I H W N M L K R Q P . . U . S 3 2 . 0 7 6 5 4 B A . 8 F E . C G
K L M N G H I . S . U . O W . R . 5 . 7 0 . 2 3 C . E F 8 . . B P
L K N M H G J I T S V U P O . . 5 4 7 6 1 0 . 2 D C F E . 8 B A Q
M N K L I J G H U W S T Q . . P 6 7 4 5 2 3 0 1 E F . D A B 8 . V
N M L . . I H G V U T W . Q . O 7 6 5 4 3 2 1 0 F E D C B A . 8 .
O P Q R S T U V G H . J K L . . 8 . A . C W E . 0 1 . . . . 6 7 .
P O . . T S V U H G . . L K . M . 8 B A D C . W 1 0 . 2 5 4 7 6 E
Q R O P U V S T I J G H M N . L A W 8 . E F C . 2 3 0 . 6 7 4 5 B
R Q P O . U T S J I H G N M . K . A . W . E . C 3 . 1 0 7 6 5 4 8
S T U V O . Q R K L M N . H . J . . E . . . A B 4 5 6 . 0 W 2 3 1
T S V U P O R Q L K N M H . . I D C F E . 8 B A 5 4 7 6 1 0 3 W 2
U V . T Q R O . . N . L I J . H E F C D A B 8 . 6 W 4 5 . 3 0 . 7
. U T S R Q P O N . L K J I . G F E D C . A . . 7 6 5 W 3 . . 0 .
0 . 3 G 6 L 5 M C V F S A P . . O B . 8 U D T E K 7 N 4 I 1 H . W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E136: order 33, 818 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs33_review_29814974/random_repairs/task_2_n33_rank_1.pending_restart1/repair.json`

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Completion

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W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
2 3 0 1 W 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 6
3 2 1 0 7 6 W 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 5
4 5 6 7 0 1 2 3 W D E F 8 9 A B K L M N G H I J S T U V O P Q R C
5 4 7 6 1 0 3 2 D C W E 9 8 B A L K N M H G J I T S V U P O R Q F
6 7 4 5 2 3 0 1 E F C D W B 8 9 M N K L I J G H U V S T Q R O P A
7 6 5 4 3 2 1 0 F E D C B A W 8 N M L K J I H G V U T S R Q P O 9
8 9 A B C D E F 0 1 2 3 4 5 6 7 W P Q R S T U V G H I J K L M N O
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O W Q T S V U H G J I L K N M R
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P W V S T I J G H M N K L U
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U W S J I H G N M L K T
C D E F 8 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R W L M N G H I J K
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K W M H G J I N
E F C D A B 8 9 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L W J G H I
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I W G H
G H I J K W M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 9 A B C D E F L
H G J I L K N W P O R Q T S V U 1 0 3 2 5 4 7 6 9 8 B A D C F E M
I W G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D J
J I H W N M L K R Q P O V U T S 3 2 1 0 7 6 5 4 B A 9 8 F E D C G
K L M N G H I J S T U V O W Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B P
L K N M H G J I T S V U P O R W 5 4 7 6 1 0 3 2 D C F E 9 8 B A Q
M N K L I J G H U W S T Q R O P 6 7 4 5 2 3 0 1 E F C D A B 8 9 V
N M L K J I H G V U T W R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 S
O P Q R S T U V G H I J K L M N 8 9 A B C W E F 0 1 2 3 4 5 6 7 D
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
Q R O P U V S T I J G H M N K L A W 8 9 E F C D 2 3 0 1 6 7 4 5 B
R Q P O V U T S J I H G N M L K B A 9 W F E D C 3 2 1 0 7 6 5 4 8
S T U V O P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 W 2 3 1
T S V U P O R Q L K N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 W 2
U V S T Q R O P M N K L I J G H E F C D A B 8 9 6 W 4 5 2 3 0 1 7
V U T S R Q P O N M L K J I H G F E D C B A 9 8 7 6 5 W 3 2 1 0 4
0 J 3 G 6 L 5 M C V F S A P 9 Q O B R 8 U D T E K 7 N 4 I 1 H 2 W
```

Partial square

```
. . . 3 . 5 6 . . . . . C D . F . . . . K L M N . . . . S T U V 0
. . W . 5 4 7 6 . 8 B A . C . E . G J I L K N M . O . Q T . V U 3
2 3 0 . . 7 4 . A B . . . F . D I . G H M N . . Q R O . U V S T 6
3 2 1 0 7 6 W 4 B A . 8 F E . C J I H G N M L K R Q P . . U T S 5
4 5 6 7 0 1 2 3 W D E . 8 . . B . L M N G H I J S T U V O P Q R .
. 4 7 6 . . 3 2 D C . E . 8 . A L K . . H . J I T . . U P . R Q F
6 7 4 5 2 3 0 . E . C . . . . . M N K L I . . H U V . T . R O P A
. . . . . .
8 . A B C D E F 0 1 2 3 . 5 . 7 . P Q R S T . . G H I J K L . . O
. . B A D C F E 1 0 . . 5 4 . 6 P O W Q T . V U . G J I L . N M .
A B 8 . E F C D 2 . 0 1 6 7 . 5 Q R O . W V . T I J G H M N K L U
B A . 8 . E D C 3 2 . 0 7 . . . R Q P O V . W S J I . G N M L . T
C D E F 8 . A B . 5 . 7 0 1 . . S T U V . P Q . W L M N G H I J .
. C F E . 8 B A 5 4 7 6 1 0 . 2 T S V . P O R Q L K . M H . J I N
E F . . A B 8 . 6 7 4 5 2 3 . 1 . V S T . R . P M N K L W J G H I
F E . C B A . 8 7 6 5 4 . . . 0 . U T . R . P O N M . K J I W . .
G H I . K W M N O P Q R S T . V 0 1 2 3 4 5 6 7 8 . . B C D E F .
H G J I L . N W P O R Q T . . U 1 0 3 2 5 . 7 6 . 8 B A D C F E M
I W G H M N K L . R O P . V . T . 3 0 1 6 7 4 5 A B 8 . E F C D J
J I H W N M L K R Q P . V U . S 3 2 1 0 7 6 5 4 B A . 8 F E . C G
K L M N G H I J S . U . O W . R . 5 6 7 0 1 2 3 C D E F 8 . . B P
L K N M H G J I T S V . P O . . 5 4 7 6 . 0 . 2 D C F E . 8 B A Q
M N K L I J G H U W S T Q . . P 6 7 4 5 2 3 0 1 E F . D A B 8 . V
N M L . . I H G V . T W . Q . O 7 6 5 4 3 2 1 0 F E D C B A . 8 .
O . Q . . T U V G H I J K L . . 8 . A B C W E . 0 1 2 3 4 . 6 7 .
P O . . T S V U H G . . L K . M . 8 B A D C . W . 0 . 2 5 4 7 6 E
Q R O P U V S T I J G H M N . L A W 8 . E F C . 2 3 0 1 6 7 4 5 B
R Q P O V U T . J I H G N M . . . A . W . E . C 3 . 1 0 7 6 . 4 8
S T U V O P Q R K L M N G H . J . . E . . . A B 4 5 6 . 0 W 2 . 1
T S V U P O R Q L K N M H . . I D C F E . 8 B A 5 4 7 6 1 0 3 W 2
. V . T Q R O . . N K L I J . H E F C D A B 8 . 6 W . 5 . 3 0 1 7
. U T S R Q P O N . L K J I . G F E D C . A . . 7 6 5 W 3 . . 0 .
0 . 3 G 6 L 5 M C V F S A P . . O B . 8 U D T E K 7 N 4 I 1 H 2 W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E137: order 33, 807 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs33_review_29814974/repairs/task_1_n33_rank_6.pending/repair.json`

SHA256: `81d7f569af893bd895828a13e089ea8886629be1c49216f04a11b4a25b012cfb`

Completion

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W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
2 3 0 W 6 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 1
3 W 1 0 7 6 5 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 2
4 5 6 7 0 1 2 3 C D E F 8 9 A B K L M N G H I J W T U V O P Q R S
5 4 7 6 1 0 3 2 D C F E 9 8 B A L K N M H G J I T S W U P O R Q V
6 7 4 5 2 3 0 1 E F C D A B 8 9 M N K L I J G H U V S W Q R O P T
7 6 5 4 3 2 1 0 F E D C B A 9 8 N M L K J I H G V W T S R Q P O U
8 9 A B C D E F 0 1 2 3 4 5 6 7 O P Q R S T U V G H I J W L M N K
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O R Q T S V U H G J I L K W M N
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P U V S T I J G H M N K W L
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U T S J I H G N W L K M
C D E F W 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R K L M N G H I J 8
D C F E 9 8 W A 5 4 7 6 1 0 3 2 T S V U P O R Q L K N M H G J I B
E F C D A B 8 W 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L I J G H 9
F E D C B W 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I H G A
G H I J K L M N O P Q R S T U V 0 1 2 3 W 5 6 7 8 9 A B C D E F 4
H G J I L K N M P O R Q T S V U 1 0 3 2 5 4 W 6 9 8 B A D C F E 7
I J G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 W A B 8 9 E F C D 5
J I H G N M L K R Q P O V U T S 3 2 1 0 7 W 5 4 B A 9 8 F E D C 6
K L M N G H I J S T U V W P Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B O
L K N M H G J I T S V U P O W Q 5 4 7 6 1 0 3 2 D C F E 9 8 B A R
M N K L I J G H U V S T Q R O W 6 7 4 5 2 3 0 1 E F C D A B 8 9 P
N M L K J I H G V U T S R W P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 Q
O P Q R S T U V W H I J K L M N 8 9 A B C D E F 0 1 2 3 4 5 6 7 G
P O R Q T S V U H G W I L K N M 9 8 B A D C F E 1 0 3 2 5 4 7 6 J
Q R O P U V S T I J G W M N K L A B 8 9 E F C D 2 3 0 1 6 7 4 5 H
R Q P O V U T S J W H G N M L K B A 9 8 F E D C 3 2 1 0 7 6 5 4 I
S T U V O P Q R K L M N G H I J W D E F 8 9 A B 4 5 6 7 0 1 2 3 C
T S V U P O R Q L K N M H G J I D C W E 9 8 B A 5 4 7 6 1 0 3 2 F
U V S T Q R O P M N K L I J G H E F C W A B 8 9 6 7 4 5 2 3 0 1 D
V U T S R Q P O N M L K J I H G F W D C B A 9 8 7 6 5 4 3 2 1 0 E
0 2 3 1 8 A B 9 G I J H O Q R P C E F D 4 6 7 5 S U V T K M N L W
```

Partial square

```
W . 2 . 4 5 . . . . . . . . . . G H . J K L M N O . Q R S T . V .
1 . . 2 5 . . 6 9 . B A D . F . H . . . . K N M P O R Q T S V U 3
2 3 0 . . 7 4 . . . 8 9 E F . . . J . H M . K L Q . O . U V S T 1
3 W . 0 . 6 5 4 B . 9 8 F E D . J . . G N M L K R Q P O V U T S 2
4 5 . . 0 1 2 3 . . E F 8 9 A B . L . N . . . J . T U . O P Q R S
5 4 7 6 . . 3 2 D . . . 9 8 B A L K . M H G J . T S W U . . R . V
6 . . 5 2 . 0 1 . . C D A . 8 9 M N K L . J G H U V S . Q R O P T
. 6 5 4 . 2 1 0 F . D C B A 9 8 N . L K J . H G V W T S R Q P O U
8 9 A B C . E . 0 . 2 3 4 5 6 7 O . Q . S T U V G . . J W L M N K
9 8 B A D C F E . . 3 2 5 . . 6 P O R Q T S V U H . J . L K W M N
A B 8 9 E F C D 2 . 0 . 6 7 4 5 Q R O P U V S T . . G H M N K . L
B A . 8 . E D . . . 1 0 7 . 5 4 R . P . V U T S J . H G N W L . M
. D E . . . A B 4 . . 7 . . . 3 S T U V O . . R K L M N G H . . 8
D C F E 9 8 W . 5 . 7 6 1 0 3 2 T S . U P O R Q . K N M H . J . B
E F . D A B 8 W 6 . 4 5 2 3 0 1 U . S T Q R O P . N K L . J . . 9
. E D C B W 9 8 . . 5 . 3 2 1 0 V U T . R Q P O N . L . J . H G A
G H . J K L M . O . Q R S T U V 0 1 2 3 W 5 6 7 . 9 A B C D E F 4
H G J . L K N M . . R . T . . . . . 3 2 5 . W 6 . 8 B A D C F E .
. J G H M N K L Q . O P U V S . 2 . 0 1 6 7 . W A B 8 9 E F C D .
J . H G . M L K . . P O V U . . . 2 1 0 7 W 5 4 B A 9 8 F E D C 6
K . M N . H . . S . U V W . Q R 4 . 6 7 0 1 . 3 C D E F 8 9 A B .
L K N M H G J . T . V U P O . Q 5 . 7 6 . 0 3 2 D C F . 9 8 B A R
M N K L . J G H U . S T Q R O W 6 7 4 5 2 3 0 1 E F C D A B 8 9 .
N M L K J . . G V . T S . . P O . 6 5 4 3 2 1 0 F E D . B . 9 . .
O P . R S T U V W . . . . . M N 8 9 A B C D E F . 1 2 3 4 5 6 7 G
P O R Q T . V U H . W . L K N M 9 8 B A D C F E 1 0 3 2 5 4 7 6 J
Q R O P U . S T . . G W M N K L A . 8 9 E F C D 2 3 0 1 6 7 4 5 .
. . . . . .
S T U V O . Q R K . M N G H . J W D E F 8 9 A B 4 5 6 7 0 . 2 3 C
T S V U . . R Q . . N M H G J . D C . E 9 . B A 5 . 7 6 1 0 3 2 .
. V S T Q R O P M . K L . J . . E F C W A B 8 9 6 7 . 5 2 . 0 1 .
V U . S R Q P O N . L K J . H G F . D C B A 9 8 7 6 5 4 3 2 1 0 E
0 2 . . 8 A B . G . J . . Q R P . E F . 4 6 7 5 S . V . K M N . W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E138: order 33, 805 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs33_review_29814974/repairs/task_1_n33_rank_1.pending/repair.json`

SHA256: `753272b546c86494b92c069490d9181ba937d692fe5e43cf186ec9fddc7b4926`

Completion

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W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
2 3 0 1 W 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 6
3 2 1 0 7 6 W 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 5
4 5 6 7 0 1 2 3 W D E F 8 9 A B K L M N G H I J S T U V O P Q R C
5 4 7 6 1 0 3 2 D C W E 9 8 B A L K N M H G J I T S V U P O R Q F
6 7 4 5 2 3 0 1 E F C D W B 8 9 M N K L I J G H U V S T Q R O P A
7 6 5 4 3 2 1 0 F E D C B A W 8 N M L K J I H G V U T S R Q P O 9
8 9 A B C D E F 0 1 2 3 4 5 6 7 W P Q R S T U V G H I J K L M N O
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O W Q T S V U H G J I L K N M R
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P W V S T I J G H M N K L U
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U W S J I H G N M L K T
C D E F 8 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R W L M N G H I J K
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K W M H G J I N
E F C D A B 8 9 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L W J G H I
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I W G H
G H I J K W M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 9 A B C D E F L
H G J I L K N W P O R Q T S V U 1 0 3 2 5 4 7 6 9 8 B A D C F E M
I W G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D J
J I H W N M L K R Q P O V U T S 3 2 1 0 7 6 5 4 B A 9 8 F E D C G
K L M N G H I J S T U V O W Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B P
L K N M H G J I T S V U P O R W 5 4 7 6 1 0 3 2 D C F E 9 8 B A Q
M N K L I J G H U W S T Q R O P 6 7 4 5 2 3 0 1 E F C D A B 8 9 V
N M L K J I H G V U T W R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 S
O P Q R S T U V G H I J K L M N 8 9 A B C W E F 0 1 2 3 4 5 6 7 D
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
Q R O P U V S T I J G H M N K L A W 8 9 E F C D 2 3 0 1 6 7 4 5 B
R Q P O V U T S J I H G N M L K B A 9 W F E D C 3 2 1 0 7 6 5 4 8
S T U V O P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 W 2 3 1
T S V U P O R Q L K N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 W 2
U V S T Q R O P M N K L I J G H E F C D A B 8 9 6 W 4 5 2 3 0 1 7
V U T S R Q P O N M L K J I H G F E D C B A 9 8 7 6 5 W 3 2 1 0 4
0 J 3 G 6 L 5 M C V F S A P 9 Q O B R 8 U D T E K 7 N 4 I 1 H 2 W
```

Partial square

```
W . . . 4 . 6 . . . A . . D . F . H . J K L M . O P Q . S . U . .
. . . . . 4 . . . 8 . A . C . . H G J I L K N . P . R Q T S . . 3
. . . 1 . . 4 5 . . 8 9 . F C D . J G H . N K L Q R O P . V S . .
3 . 1 0 7 . W 4 . A 9 8 F E . C J I H G . M L K R Q P O . U T . .
. 5 . 7 . 1 . . . D . . . 9 . B . . M N . H . J S T U V O P Q . C
5 4 7 6 . . 3 . D C W E . 8 B A L K N M H G J I T S V U P O R . F
6 7 4 5 . . 0 1 . F C D W . 8 9 . N . L I . G H . V . T . R O . A
7 6 5 4 3 . 1 0 F E D C . A W 8 N M L K J . H G V U T . R Q P . 9
8 . A B . . E F 0 1 . 3 4 5 6 7 W P Q R S T U V G H . . . L M . O
. 8 B . . C F E 1 . . . 5 4 7 6 P O W Q T . . U H G J I . K N . R
A B 8 9 E . C . . 3 0 1 6 7 4 5 Q R O P W . S T I J . H M N . . U
B A 9 8 F . . C 3 . 1 0 7 6 5 4 R Q P O V . W S J . H G N M L . T
. D E F 8 . A B 4 5 6 7 0 1 . 3 S T U . O P Q R W . M N G H I . K
D C . E . 8 . A 5 4 7 6 1 . 3 . T S V U P O R . L K W . . G J . N
E F C D A B 8 9 6 7 4 5 . 3 0 1 U V S T . R O P M . K L W . . . .
F . . C B A 9 8 7 6 5 4 3 . 1 0 V . T S R Q P . N M L K . I W . H
G H I . K . . N O P . R . T U V 0 1 . . 4 5 6 7 8 9 A B C . E . L
H G J I . K N . P O R Q T S V U 1 0 3 . 5 4 7 6 . 8 B A D . F . .
I W G H M N K L Q R O P U V S T . 3 0 1 6 . 4 5 A B 8 . . F C . J
J . H W N M L K R Q P O V U T S 3 . 1 . 7 6 5 4 . A 9 8 F E . . .
. L M N . H I . S T U V O W Q R 4 5 . 7 0 1 . 3 C . . F 8 9 A . P
. . N M H G J I T S V U . O R W . 4 7 6 1 0 . . D C F . 9 . B . Q
M N K L I J G H U W S T Q R O P 6 7 4 5 . 3 0 1 E F C D A B . . V
. M L K J I H G V U T W R Q P O 7 6 5 4 3 . 1 0 . E D C B A 9 . S
O P Q R S T U V G . I J K L M N 8 9 . B C W E F 0 1 . 3 4 . 6 . D
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 . 5 4 . . E
. R O P U V . T . J G H M . . L A . 8 9 E F C D . 3 0 1 . 7 4 . B
R . P O . . T S . I H . N M L . B . . W F E D C 3 . . . 7 6 5 . 8
. T U V O P Q R K L M N G H I J C D E F 8 9 A . 4 5 6 7 0 . . . 1
. . . . . .
U V S T Q R O P M N K L I J G H E F . D A . 8 9 6 W 4 5 . 3 0 . 7
V U T . R Q P . . M L K J I H G F E D C B A 9 8 7 6 5 . . . 1 . 4
0 J 3 G 6 L . . C V F S A P 9 Q O B R 8 U D T . K 7 N 4 I 1 H . W
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Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E139: order 33, 801 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs33_review_29814974/repairs/task_0_n33_rank_0.pending/repair.json`

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Completion

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W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 3 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V W U
2 3 0 W 6 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 1
3 2 1 0 7 6 5 4 B A 9 8 F E D C J I H G N M L K R Q P O W U T S V
4 5 6 7 0 1 2 3 C D E F 8 9 A B K L M N G H I J S T U V O P W R Q
5 W 7 6 1 0 3 2 D C F E 9 8 B A L K N M H G J I T S V U P O R Q 4
6 7 4 5 2 3 0 1 E F C D A B 8 9 M N K L I J G H U V S T Q W O P R
7 6 W 4 3 2 1 0 F E D C B A 9 8 N M L K J I H G V U T S R Q P O 5
8 9 A B C D E F 0 1 2 3 4 5 6 7 O P Q R S T U V G H W J K L M N I
9 8 B A D W F E 1 0 3 2 5 4 7 6 P O R Q T S V U H G J I L K N M C
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P U V S T I W G H M N K L J
B A 9 8 F E W C 3 2 1 0 7 6 5 4 R Q P O V U T S J I H G N M L K D
C D E F W 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R K L M N G H I J 8
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K N W H G J I M
E F C D A B 8 W 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L I J G H 9
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O W M L K J I H G N
G H I J K L M N O P Q R S T U V 0 1 W 3 4 5 6 7 8 9 A B C D E F 2
H G J I L K N M P O R Q T W V U 1 0 3 2 5 4 7 6 9 8 B A D C F E S
I J G H M N K L Q R O P U V S T 2 W 0 1 6 7 4 5 A B 8 9 E F C D 3
J I H G N M L K R Q P O V U W S 3 2 1 0 7 6 5 4 B A 9 8 F E D C T
K L M N G H I J S T U V W P Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B O
L K N M H G J I T S V U P O R Q 5 4 7 W 1 0 3 2 D C F E 9 8 B A 6
M N K L I J G H U V S T Q R O W 6 7 4 5 2 3 0 1 E F C D A B 8 9 P
N M L K J I H G V U T S R Q P O W 6 5 4 3 2 1 0 F E D C B A 9 8 7
O P Q R S T U V W H I J K L M N 8 9 A B C D E F 0 1 2 3 4 5 6 7 G
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
Q R O P U V S T I J G W M N K L A B 8 9 E F C D 2 3 0 1 6 7 4 5 H
R Q P O V U T S J I H G N M L K B A 9 8 W E D C 3 2 1 0 7 6 5 4 F
S T U V O P Q R K L M N G H I J C D E F 8 9 W B 4 5 6 7 0 1 2 3 A
T S V U P O R Q L W N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 2 K
U V S T Q R O P M N K L I J G H E F C D A W 8 9 6 7 4 5 2 3 0 1 B
V U T S R Q P O N M W K J I H G F E D C B A 9 8 7 6 5 4 3 2 1 0 L
0 4 5 1 8 C D 9 G K L H O S T P 7 3 2 6 F B A E N J I M V R Q U W
```

Partial square

```
W . . 3 . . 6 7 . 9 . B . D E F . . I J K . . N . P Q R S T . . .
. . 3 . . 4 . 6 . . B A D C F E . G . . . K . M P O R Q . S V . U
. . 0 . . . 4 . A B . 9 . F C D I . G . M N K . . R . . U V S T .
. 2 1 . . 6 5 4 B A . . F E D C J . . G . M L K . . P O . U T . .
4 5 6 7 . . 2 3 C D E F . 9 A B K L . . . . I J S . U V O P W R Q
5 W . . . 0 3 . D . F E 9 . B . . . . M H G J I T S V U P O R Q 4
6 7 4 5 . 3 0 1 E . C D A B . 9 M N K L I J G . U V . T Q W O . .
7 6 . 4 . 2 1 0 F E D . B A 9 . N M . . J . . . V U T S R Q P O .
. 9 A B . D E F . 1 2 3 4 5 6 7 O P . . . T U V G H W J K L M N I
9 . . A . W F E 1 0 3 2 5 4 7 6 P O R Q . S . U . G J I L K N M C
. B . 9 . . C D 2 . . 1 6 7 4 5 Q R . . U V . T . . G H M N K L J
B . . . . E W . 3 2 1 0 7 6 5 . R Q P . V U . . J . H G N M L . D
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
D C F E . . B A 5 4 7 6 1 0 3 . T S V U P O R Q L K N W H G J I M
E F C . . B . . 6 7 4 5 2 3 . 1 U . S T Q R O P M N K L I J G H .
F E D C . A 9 . 7 . 5 . 3 2 . . V U T . R Q P O . M . K J I . G .
G H I J . L M N O . Q R S T U V 0 1 . 3 4 5 6 7 . . . B C D E F 2
H G J I . K N M P O R Q T W V U 1 . 3 2 5 4 7 6 9 . B A D C F E S
I J G . . . K L Q R O P U V S . 2 W 0 1 . 7 4 5 A B . 9 . F C . 3
J I H G . M L K R . P O V U W S 3 . 1 0 7 6 5 . B A 9 . F E D . .
K L M N . . I J S T U V W P Q R 4 5 6 7 0 1 . 3 C . E F . 9 A B .
L . . M . G . I T . . U P O R Q 5 4 7 W . 0 3 2 D . F E . . B A 6
M . K L . J G H U V S T Q R O W 6 7 . 5 2 3 0 1 E F . D A B . 9 P
N M L K . I H . V U . S R Q P O W 6 5 . 3 2 1 0 F E D C B A 9 . .
O P . R . T U . . H I J K L . N . 9 . B C D E F 0 . 2 3 4 5 . 7 G
P O R Q . S V U H G J I L K N M 9 . . A D C F W . 0 3 2 5 4 . 6 E
Q R O . . V S T I J . W M N K L A B . 9 E F C D 2 3 0 1 6 7 4 5 H
R Q P O . U T S J I H G N M L K B A . . W E D C 3 2 . . 7 6 5 4 F
S T . V . . Q . K L M N G . I J C D E F . 9 W B 4 5 6 7 0 1 2 . .
T . V U . O . Q L W . M H G J I D C F E 9 . B A 5 4 . 6 . 0 3 2 K
U V S T . R O P M N . L I J G H E F C D A W . . 6 7 . 5 2 3 . . B
V U T S . Q P O . . . K J . H G F E D C B A 9 . 7 6 5 4 3 2 1 . L
0 4 5 1 . C D . G K L H O S . P 7 3 2 6 F B A E . J . M V R Q U W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E140: order 33, 799 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs33_review_29814974/repairs/task_1_n33_rank_0.pending/repair.json`

SHA256: `410e19f4ba97d07ec986abf4c163281b34ed018be157e9b8f483734417eeca35`

Completion

```
W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 3 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V W U
2 3 0 W 6 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 1
3 2 1 0 7 6 5 4 B A 9 8 F E D C J I H G N M L K R Q P O W U T S V
4 5 6 7 0 1 2 3 C D E F 8 9 A B K L M N G H I J S T U V O P W R Q
5 W 7 6 1 0 3 2 D C F E 9 8 B A L K N M H G J I T S V U P O R Q 4
6 7 4 5 2 3 0 1 E F C D A B 8 9 M N K L I J G H U V S T Q W O P R
7 6 W 4 3 2 1 0 F E D C B A 9 8 N M L K J I H G V U T S R Q P O 5
8 9 A B C D E F 0 1 2 3 4 5 6 7 O P Q R S T U V G H W J K L M N I
9 8 B A D W F E 1 0 3 2 5 4 7 6 P O R Q T S V U H G J I L K N M C
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P U V S T I W G H M N K L J
B A 9 8 F E W C 3 2 1 0 7 6 5 4 R Q P O V U T S J I H G N M L K D
C D E F W 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R K L M N G H I J 8
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K N W H G J I M
E F C D A B 8 W 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L I J G H 9
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O W M L K J I H G N
G H I J K L M N O P Q R S T U V 0 1 W 3 4 5 6 7 8 9 A B C D E F 2
H G J I L K N M P O R Q T W V U 1 0 3 2 5 4 7 6 9 8 B A D C F E S
I J G H M N K L Q R O P U V S T 2 W 0 1 6 7 4 5 A B 8 9 E F C D 3
J I H G N M L K R Q P O V U W S 3 2 1 0 7 6 5 4 B A 9 8 F E D C T
K L M N G H I J S T U V W P Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B O
L K N M H G J I T S V U P O R Q 5 4 7 W 1 0 3 2 D C F E 9 8 B A 6
M N K L I J G H U V S T Q R O W 6 7 4 5 2 3 0 1 E F C D A B 8 9 P
N M L K J I H G V U T S R Q P O W 6 5 4 3 2 1 0 F E D C B A 9 8 7
O P Q R S T U V W H I J K L M N 8 9 A B C D E F 0 1 2 3 4 5 6 7 G
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
Q R O P U V S T I J G W M N K L A B 8 9 E F C D 2 3 0 1 6 7 4 5 H
R Q P O V U T S J I H G N M L K B A 9 8 W E D C 3 2 1 0 7 6 5 4 F
S T U V O P Q R K L M N G H I J C D E F 8 9 W B 4 5 6 7 0 1 2 3 A
T S V U P O R Q L W N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 2 K
U V S T Q R O P M N K L I J G H E F C D A W 8 9 6 7 4 5 2 3 0 1 B
V U T S R Q P O N M W K J I H G F E D C B A 9 8 7 6 5 4 3 2 1 0 L
0 4 5 1 8 C D 9 G K L H O S T P 7 3 2 6 F B A E N J I M V R Q U W
```

Partial square

```
W . . . 4 . . . . . A . . . . F G H I . K L M . O . Q R S . U V 0
. . 3 . 5 . . 6 9 . B A D . . . . G . . . K N . P O . Q . S V . U
. 3 0 . 6 7 . . . B . 9 . F C . I J G H M N K . . . . P U V . T 1
3 . 1 0 7 6 5 4 B A . . . . D C J I . . N . L . . Q P O . U T S V
. . 6 7 0 1 2 . . D . F 8 9 . B . L M N . H I . . . U V O P . R Q
. W 7 6 1 0 . 2 D . F . 9 8 B . L K N M H G J . T S V U P . R Q 4
6 7 4 . 2 3 0 1 . . C D A B 8 . M N K L . J G . . V S . Q W O P R
7 6 W 4 3 2 . 0 F . D C B A 9 8 N M L K J . H . V . T S . Q P O 5
8 9 A B C D . . 0 1 . 3 4 . 6 7 O P . R S . U . G H W J K . M . I
9 8 . A D . F . . . 3 2 5 . 7 . . O R Q T S V . H G J . L K N M C
A B 8 9 . F C D . . 0 1 . 7 . . Q . . . . V S . I W G H M N . L .
. . 9 . F . W C 3 2 1 . 7 . 5 4 R Q P O V U T . J I H G N . L K .
C D . F W 9 A B 4 . 6 7 0 . 2 3 S T U . . P Q . K L M N G . . J 8
. C F . 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R . L K N . . . J I M
. F C D A B 8 W 6 7 4 5 2 . 0 1 U V S T Q R O . M N K L I J G H 9
. . D C B A . 8 7 6 5 4 . 2 1 0 V U T S R . P . W M L K . I H . N
G H I J K L M N . . Q . S T U V 0 1 W 3 . 5 6 . 8 9 A B C D . F 2
H G J I L K N M . O R Q T W V U 1 0 . 2 . 4 7 . 9 8 B A D C . . S
I J G H M N K L Q R O P U V S T 2 W 0 1 6 7 4 . A B 8 9 . F C D 3
J I H G N M L K R . P O V U . S 3 2 1 0 7 6 . . . . 9 8 F . . C T
K L M N G H I J S T U V W P . R 4 5 6 7 0 1 2 . C D . F 8 . A B O
L K N M H G J I T S V U . O R Q . 4 7 W 1 0 3 . . C . . 9 8 B A 6
M N K L I J G H U V S . Q R O W 6 7 4 5 2 3 0 . . F C D . B 8 9 P
. M L K J I . G V U T S R . P O W 6 5 4 . 2 1 . . . D C . A 9 8 7
O . Q R S . U V W H . J K . M . . 9 A B C D . . 0 1 2 3 4 5 6 7 G
. . . . . .
Q R O P U V S T I . G W M N K L A B 8 9 . F C . 2 3 0 1 6 7 4 5 H
. Q P O V U T . J I H G N M . K B A . 8 W . D . 3 2 1 0 7 6 5 . F
S T U V O P Q R . . . N G H I J C D . . 8 9 W . 4 . 6 . 0 1 2 3 A
. S V U P . R . L W N M H G J I D C F . 9 . . . 5 4 7 6 1 . . 2 K
. V S T Q . O P M N K L . J G H . F C D A W 8 . 6 7 4 5 2 3 0 1 B
V . T S R . P O N M W K J . . . F . . C B A 9 . 7 . 5 4 . . 1 0 L
. 4 5 1 8 . D 9 G K L H O S T P 7 3 2 . F B A . N J . M V R . U W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E141: order 33, 794 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs33_review_29814974/repairs/task_3_n33_rank_0.pending/repair.json`

SHA256: `2f798bee918315e0eb864db0755cc89cc7d0ff7470ddc9654d04e57a07584fa6`

Completion

```
W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 3 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V W U
2 3 0 W 6 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 1
3 2 1 0 7 6 5 4 B A 9 8 F E D C J I H G N M L K R Q P O W U T S V
4 5 6 7 0 1 2 3 C D E F 8 9 A B K L M N G H I J S T U V O P W R Q
5 W 7 6 1 0 3 2 D C F E 9 8 B A L K N M H G J I T S V U P O R Q 4
6 7 4 5 2 3 0 1 E F C D A B 8 9 M N K L I J G H U V S T Q W O P R
7 6 W 4 3 2 1 0 F E D C B A 9 8 N M L K J I H G V U T S R Q P O 5
8 9 A B C D E F 0 1 2 3 4 5 6 7 O P Q R S T U V G H W J K L M N I
9 8 B A D W F E 1 0 3 2 5 4 7 6 P O R Q T S V U H G J I L K N M C
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P U V S T I W G H M N K L J
B A 9 8 F E W C 3 2 1 0 7 6 5 4 R Q P O V U T S J I H G N M L K D
C D E F W 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R K L M N G H I J 8
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K N W H G J I M
E F C D A B 8 W 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L I J G H 9
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O W M L K J I H G N
G H I J K L M N O P Q R S T U V 0 1 W 3 4 5 6 7 8 9 A B C D E F 2
H G J I L K N M P O R Q T W V U 1 0 3 2 5 4 7 6 9 8 B A D C F E S
I J G H M N K L Q R O P U V S T 2 W 0 1 6 7 4 5 A B 8 9 E F C D 3
J I H G N M L K R Q P O V U W S 3 2 1 0 7 6 5 4 B A 9 8 F E D C T
K L M N G H I J S T U V W P Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B O
L K N M H G J I T S V U P O R Q 5 4 7 W 1 0 3 2 D C F E 9 8 B A 6
M N K L I J G H U V S T Q R O W 6 7 4 5 2 3 0 1 E F C D A B 8 9 P
N M L K J I H G V U T S R Q P O W 6 5 4 3 2 1 0 F E D C B A 9 8 7
O P Q R S T U V W H I J K L M N 8 9 A B C D E F 0 1 2 3 4 5 6 7 G
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
Q R O P U V S T I J G W M N K L A B 8 9 E F C D 2 3 0 1 6 7 4 5 H
R Q P O V U T S J I H G N M L K B A 9 8 W E D C 3 2 1 0 7 6 5 4 F
S T U V O P Q R K L M N G H I J C D E F 8 9 W B 4 5 6 7 0 1 2 3 A
T S V U P O R Q L W N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 2 K
U V S T Q R O P M N K L I J G H E F C D A W 8 9 6 7 4 5 2 3 0 1 B
V U T S R Q P O N M W K J I H G F E D C B A 9 8 7 6 5 4 3 2 1 0 L
0 4 5 1 8 C D 9 G K L H O S T P 7 3 2 6 F B A E N J I M V R Q U W
```

Partial square

```
W . 2 . . . . . . . A B . D . F . H . J K . M . O P . R S T U V 0
1 0 . 2 5 4 . 6 . 8 . A D C F . . . . I . . N . . . . Q . . V . U
2 . . . . . 4 5 . B 8 9 . F C . I J . . M N K L Q . O P U . S T 1
. 2 1 . 7 . 5 4 . . 9 . . E . . J . H . N M L K R . P O . . T . .
. . 6 7 0 1 2 . . . E F 8 . A B . L M N . H I . . . . V O P . . Q
. W . 6 1 0 3 2 . C F E 9 8 B A L K N . H . . I . S V U P O R Q 4
6 7 4 . 2 3 . 1 . F C . A B 8 9 . N K . . J . H U V S . Q W O . R
. 6 . 4 3 2 1 0 . E D C B A 9 8 N M L K J I . . . . T S . Q P O 5
8 9 A B C D E F . 1 2 . . . 6 . . P Q . S T U V . H W J K L M N I
. 8 . . . W F E . 0 3 2 . 4 7 6 P O R Q T S . U H . . I L K N . C
. B . 9 E F C . . 3 0 1 6 . 4 5 Q R O . U V S T . W . . . . . L J
B A 9 8 F E W C . 2 1 0 7 6 5 . . Q P O . . T S J I H . N M L K D
C D E F W 9 A B . . 6 . 0 1 2 3 S T U . . P Q R K L M N . H I J 8
D C F E 9 8 B A . 4 . 6 1 0 . . . S V U P O . Q L K N W H . J I .
E . C D A B 8 W . 7 4 5 2 . 0 . . . . . Q R . P M N K L I J . H 9
F E D C B A 9 8 . 6 5 4 3 2 1 0 V U T S R Q P O W M . K J I H . N
. H I J K L . N . . . . S T U V 0 1 W 3 4 5 6 7 . 9 A B C D E . 2
. . J I . K . M . O R . . W V U 1 0 3 2 5 4 7 6 9 8 B A D C F E S
. J . H . N K L . . O P . V S T 2 W 0 1 . 7 . . A B 8 9 E F C D 3
J I H . N M L K . Q P O V U W S 3 2 1 0 7 6 5 4 B A 9 8 F E D C T
K L M . . H I J . . . V W P Q R 4 5 6 . 0 1 . 3 C D E F 8 9 A B .
L K N M H . J I . S . U P . . . 5 4 7 W . 0 3 2 D C F E 9 8 B . 6
M N K L . J . H . V S . Q R O W 6 7 4 5 2 3 0 . E F C D A B 8 9 P
N . L K . I H . . U T S . Q P . . 6 5 4 3 . 1 . F E D C B A 9 8 7
. . . . . .
P O R Q T S . U . . . I . K N M 9 8 B A D C F W 1 0 . 2 5 . . 6 E
Q R O . U V S T . J . W M . K L . B 8 9 E F C D 2 3 . 1 6 7 . 5 H
R Q . O V U T S . I H . N M L K . . 9 8 W E D C 3 2 1 0 7 6 5 . F
S T U . O P Q R . L M N . H I J C D E F 8 9 W B 4 5 6 . 0 1 2 3 .
T S . U P O . Q . W N M H . J I D C F E 9 8 B . 5 4 7 6 1 0 3 2 K
U V . T Q R O P . N . L I J . . E F C D A W 8 9 6 7 . 5 2 3 0 1 B
V U T S R Q P O . M . K J I H . F E D C B . 9 8 7 6 5 4 3 2 1 0 L
0 4 5 1 8 C D 9 . K . H O S T . 7 3 2 6 F B A E N J I M V R Q U W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E142: order 33, 791 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs33_review_29814974/repairs/task_2_n33_rank_6.pending/repair.json`

SHA256: `68593b57ba5b6e71b368330fa50ea37b48d7efeb8b6a720731d0253e96dd738a`

Completion

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W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
2 3 0 W 6 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 1
3 W 1 0 7 6 5 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 2
4 5 6 7 0 1 2 3 C D E F 8 9 A B K L M N G H I J W T U V O P Q R S
5 4 7 6 1 0 3 2 D C F E 9 8 B A L K N M H G J I T S W U P O R Q V
6 7 4 5 2 3 0 1 E F C D A B 8 9 M N K L I J G H U V S W Q R O P T
7 6 5 4 3 2 1 0 F E D C B A 9 8 N M L K J I H G V W T S R Q P O U
8 9 A B C D E F 0 1 2 3 4 5 6 7 O P Q R S T U V G H I J W L M N K
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O R Q T S V U H G J I L K W M N
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P U V S T I J G H M N K W L
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U T S J I H G N W L K M
C D E F W 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R K L M N G H I J 8
D C F E 9 8 W A 5 4 7 6 1 0 3 2 T S V U P O R Q L K N M H G J I B
E F C D A B 8 W 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L I J G H 9
F E D C B W 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I H G A
G H I J K L M N O P Q R S T U V 0 1 2 3 W 5 6 7 8 9 A B C D E F 4
H G J I L K N M P O R Q T S V U 1 0 3 2 5 4 W 6 9 8 B A D C F E 7
I J G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 W A B 8 9 E F C D 5
J I H G N M L K R Q P O V U T S 3 2 1 0 7 W 5 4 B A 9 8 F E D C 6
K L M N G H I J S T U V W P Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B O
L K N M H G J I T S V U P O W Q 5 4 7 6 1 0 3 2 D C F E 9 8 B A R
M N K L I J G H U V S T Q R O W 6 7 4 5 2 3 0 1 E F C D A B 8 9 P
N M L K J I H G V U T S R W P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 Q
O P Q R S T U V W H I J K L M N 8 9 A B C D E F 0 1 2 3 4 5 6 7 G
P O R Q T S V U H G W I L K N M 9 8 B A D C F E 1 0 3 2 5 4 7 6 J
Q R O P U V S T I J G W M N K L A B 8 9 E F C D 2 3 0 1 6 7 4 5 H
R Q P O V U T S J W H G N M L K B A 9 8 F E D C 3 2 1 0 7 6 5 4 I
S T U V O P Q R K L M N G H I J W D E F 8 9 A B 4 5 6 7 0 1 2 3 C
T S V U P O R Q L K N M H G J I D C W E 9 8 B A 5 4 7 6 1 0 3 2 F
U V S T Q R O P M N K L I J G H E F C W A B 8 9 6 7 4 5 2 3 0 1 D
V U T S R Q P O N M L K J I H G F W D C B A 9 8 7 6 5 4 3 2 1 0 E
0 2 3 1 8 A B 9 G I J H O Q R P C E F D 4 6 7 5 S U V T K M N L W
```

Partial square

```
. . . 3 . . . . 8 . . . . . . . . . . . . . . . . . . Q . S . U . .
1 . . . 5 . 7 . 9 . B . D C . E . G . I L . . M . O . . T S . U .
2 3 0 . 6 . . 5 A B 8 . E F . . I J . . M N . L . . O P U . . T .
3 . 1 0 7 6 . 4 B A 9 . F E D . J I . G N M L K R Q P O V U T S .
4 . 6 7 0 1 2 3 . . E . . 9 A B K L M N . . I J . T U . O . Q R .
5 4 . 6 1 0 . 2 D . F E 9 . . . L K . M H . J I T S . U P . R . .
6 7 4 . . 3 0 1 E F C D . B 8 9 . N K L . J G . . V S . . R O P .
7 . . 4 3 2 1 0 F E . . B A 9 8 . M . K J I H G V . . . . Q P . .
8 9 A B C D E F 0 1 2 . 4 5 6 7 O P Q R S T U V G H I . . L M . .
9 8 B A D . F E 1 0 3 . 5 4 7 6 P O R Q T S V U H G J I L K . M .
A B 8 9 . F C D 2 3 0 . . 7 4 . . R O . U V S . I J G H M N . . .
B A 9 8 F E . C 3 2 . 0 7 6 5 4 R Q P O V U T S . . . G N . L . .
C D E F . 9 A B 4 5 6 7 0 1 . 3 . . U V O P Q R K L M N G H . J .
D C F E 9 8 . A 5 4 7 6 1 0 3 2 T S . U P O . Q L K N M H G J I .
E F . D A B . . . 7 4 5 2 3 0 . U . . T . R . P M N K L I . G H .
F E D C B . 9 8 7 6 5 4 3 2 1 0 . U T S R Q P O . M L . J . H G .
G H I . K L M N O P Q R S T U V 0 . 2 . . 5 6 7 8 9 . B C D . F .
. . J . L K N M P O R . T S V U 1 0 3 2 5 4 . . 9 . B A D . . E .
I J G H M N K L Q R O P U V S T 2 3 0 . 6 . 4 . A B 8 9 E F . . .
J I H . . M L K . Q P O V U T S 3 2 1 0 7 . 5 4 B . 9 8 F E D C .
K . M N G H I J . T U V . P Q R . 5 6 7 0 1 2 . C D E F . 9 A B .
L . N M . G . . T S . U P . . . 5 4 7 6 1 0 3 2 D C F . 9 8 B A .
M N . . I J G H U V S T Q R O . 6 7 . 5 2 3 0 1 E F C D A B 8 9 .
N . L K J . H G V U T S R . P O 7 6 5 4 3 2 1 0 F E . . B A . 8 .
O P Q R S T U V . H I J K L M N . 9 A B C D . . . 1 2 3 4 5 6 7 .
P O R Q . S V U H G . . L K N M . 8 B A D C F E 1 . 3 2 5 4 7 . .
Q R O P U V S T . J G . M N . L A B 8 . E F C D 2 3 0 1 6 7 4 5 .
R Q P O V U T S . . H G N M L K B A 9 8 F E . C 3 2 1 0 7 6 5 4 .
S T U V O P Q R K L M N G H I . . D E F 8 9 A B 4 5 6 7 0 1 2 3 .
T . V U P O R Q L K N M H G J I D . . E 9 8 . . 5 4 7 6 1 . 3 2 .
U . S T . R O P M N K L . J G H E . C . A B 8 9 6 7 . 5 2 3 0 1 .
V U T . R Q P O . M L K J I H G F . D . B A 9 8 7 6 5 4 3 2 1 0 .
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . W
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Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E143: order 33, 785 clues

Status P. Producer-reported critical; not promoted here to independently verified status.

Source: `outputs/lcs33_review_29814974/repairs/task_0_n33_rank_6.pending/repair.json`

SHA256: `25b2ab6f9447cdfd53e51d7afd740c8a0296e0d9ba9b76de9df90f2fbffb351f`

Completion

```
W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
2 3 0 W 6 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 1
3 W 1 0 7 6 5 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 2
4 5 6 7 0 1 2 3 C D E F 8 9 A B K L M N G H I J W T U V O P Q R S
5 4 7 6 1 0 3 2 D C F E 9 8 B A L K N M H G J I T S W U P O R Q V
6 7 4 5 2 3 0 1 E F C D A B 8 9 M N K L I J G H U V S W Q R O P T
7 6 5 4 3 2 1 0 F E D C B A 9 8 N M L K J I H G V W T S R Q P O U
8 9 A B C D E F 0 1 2 3 4 5 6 7 O P Q R S T U V G H I J W L M N K
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O R Q T S V U H G J I L K W M N
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P U V S T I J G H M N K W L
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U T S J I H G N W L K M
C D E F W 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R K L M N G H I J 8
D C F E 9 8 W A 5 4 7 6 1 0 3 2 T S V U P O R Q L K N M H G J I B
E F C D A B 8 W 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L I J G H 9
F E D C B W 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I H G A
G H I J K L M N O P Q R S T U V 0 1 2 3 W 5 6 7 8 9 A B C D E F 4
H G J I L K N M P O R Q T S V U 1 0 3 2 5 4 W 6 9 8 B A D C F E 7
I J G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 W A B 8 9 E F C D 5
J I H G N M L K R Q P O V U T S 3 2 1 0 7 W 5 4 B A 9 8 F E D C 6
K L M N G H I J S T U V W P Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B O
L K N M H G J I T S V U P O W Q 5 4 7 6 1 0 3 2 D C F E 9 8 B A R
M N K L I J G H U V S T Q R O W 6 7 4 5 2 3 0 1 E F C D A B 8 9 P
N M L K J I H G V U T S R W P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 Q
O P Q R S T U V W H I J K L M N 8 9 A B C D E F 0 1 2 3 4 5 6 7 G
P O R Q T S V U H G W I L K N M 9 8 B A D C F E 1 0 3 2 5 4 7 6 J
Q R O P U V S T I J G W M N K L A B 8 9 E F C D 2 3 0 1 6 7 4 5 H
R Q P O V U T S J W H G N M L K B A 9 8 F E D C 3 2 1 0 7 6 5 4 I
S T U V O P Q R K L M N G H I J W D E F 8 9 A B 4 5 6 7 0 1 2 3 C
T S V U P O R Q L K N M H G J I D C W E 9 8 B A 5 4 7 6 1 0 3 2 F
U V S T Q R O P M N K L I J G H E F C W A B 8 9 6 7 4 5 2 3 0 1 D
V U T S R Q P O N M L K J I H G F W D C B A 9 8 7 6 5 4 3 2 1 0 E
0 2 3 1 8 A B 9 G I J H O Q R P C E F D 4 6 7 5 S U V T K M N L W
```

Partial square

```
. 1 . 3 4 . . . . . A B C . . . G H I . . L . N O P . R . T U V 0
1 0 . 2 5 4 7 6 . 8 . . . . . E . . . . . . . M P . R . T S V . 3
. . . W . 7 . 5 . B . 9 . . C D I J . H M N . L . R . P U V S T .
3 W 1 0 . 6 5 4 . . 9 8 F . D C J I H G N M L K . . P O V U . . 2
4 5 6 . 0 . . 3 C D . F 8 . . . K L . . . H I J W T U . O P . . S
5 4 7 6 1 0 3 2 D C F E 9 . B . L K N M H G J I T S W U P O R . V
6 7 4 5 2 3 0 1 E F C D A . 8 9 M N K L I J G H U V S W . R . . T
. . . 4 3 2 1 0 F E . C B . 9 8 N M L . J I . G . W T . . . . O U
8 . A B . D E F . . . . 4 . . 7 . P . R S T U V G H I J . L M . K
9 . . A D C F E . 0 3 . 5 . . . P O R . T S V U H . J I L K W M N
A . 8 9 E F C D 2 3 . 1 . . 4 . . R O P U . S T I . G H M N . W L
B A . 8 F . D C . 2 1 0 7 . 5 4 R . P . V U . S J I . G N W L K M
C . E F . 9 A B 4 5 6 . . . 2 3 S T U V . P . R K L M N G H I J 8
D C F E . 8 W A 5 4 7 6 1 . 3 . . S V . . O R . . . . M . G J I B
. F C D . B 8 W 6 7 4 5 2 . 0 1 U V S T . R O P . N K L I J G H 9
F . D C B W 9 8 7 6 5 4 3 . 1 0 V U T S R . P O N M L K J I H G A
. H I J . L M N O P . R S . U . 0 1 2 3 W 5 6 7 8 9 A B C D E . 4
H G J I L . N M P O R . . . V . 1 0 3 2 5 4 W 6 9 8 . A D C F E 7
I J G H . N K L . . O P U . S T . 3 0 1 6 7 4 W A B 8 9 E F C D 5
J . H G . M L K R . P O V . T S 3 2 1 0 7 W 5 4 . A 9 8 F E D C .
K L M N G . I J S . U . W . . . 4 5 6 7 0 1 . 3 C . E F . 9 A B O
L K N M H G J I T S V U P . W . 5 4 7 6 1 0 3 2 D C F E 9 8 B A R
M N K L I J G H U V S T . . O W 6 7 4 5 2 . 0 1 E F . D A B . 9 P
. . . . . .
. P . R . T U V W H I J . . M N . 9 . B C D E F 0 . 2 3 . 5 6 7 G
P O . . T S V U H G W I L . N M 9 8 . A D C F . 1 0 . . 5 4 7 . J
. . O P U V S . . J G . M . K . A . . 9 E . C D 2 3 0 1 6 7 4 5 H
R . . O V U T S J . . . N . L . B A 9 8 F E D C 3 2 1 . . 6 5 4 I
S . U V . P . R . L M N . . I J W D E F 8 9 A B 4 5 6 7 0 1 2 . .
T . V U . . . . L . N M H . J I D C W E 9 . B . . 4 7 6 . 0 3 2 F
U V S T . R O P M N K L I . . H E F C W A . 8 9 6 7 4 5 2 3 0 . D
V U T . R . . O N M . K J . H . F W D C B A . . . 6 5 4 . . 1 0 .
. . 3 1 8 A B 9 . . J . O . R . C E F D . 6 7 5 . U V T K M . L W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E144: order 33, 768 clues

Status I. Previously independently verified example; see the archived certificate directory.

Source: `outputs/lcs_dyadic_plus_lower/order33_size768.in`

SHA256: `58c1ff81c01e403340544f166111d190c5580daa93e3de56d6d45be7364a00e8`

Completion

```
W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 3 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V W U
2 3 0 W 6 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 1
3 2 1 0 7 6 5 4 B A 9 8 F E D C J I H G N M L K R Q P O W U T S V
4 5 6 7 0 1 2 3 C D E F 8 9 A B K L M N G H I J S T U V O P W R Q
5 W 7 6 1 0 3 2 D C F E 9 8 B A L K N M H G J I T S V U P O R Q 4
6 7 4 5 2 3 0 1 E F C D A B 8 9 M N K L I J G H U V S T Q W O P R
7 6 W 4 3 2 1 0 F E D C B A 9 8 N M L K J I H G V U T S R Q P O 5
8 9 A B C D E F 0 1 2 3 4 5 6 7 O P Q R S T U V G H W J K L M N I
9 8 B A D W F E 1 0 3 2 5 4 7 6 P O R Q T S V U H G J I L K N M C
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P U V S T I W G H M N K L J
B A 9 8 F E W C 3 2 1 0 7 6 5 4 R Q P O V U T S J I H G N M L K D
C D E F W 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R K L M N G H I J 8
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K N W H G J I M
E F C D A B 8 W 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L I J G H 9
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O W M L K J I H G N
G H I J K L M N O P Q R S T U V 0 1 W 3 4 5 6 7 8 9 A B C D E F 2
H G J I L K N M P O R Q T W V U 1 0 3 2 5 4 7 6 9 8 B A D C F E S
I J G H M N K L Q R O P U V S T 2 W 0 1 6 7 4 5 A B 8 9 E F C D 3
J I H G N M L K R Q P O V U W S 3 2 1 0 7 6 5 4 B A 9 8 F E D C T
K L M N G H I J S T U V W P Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B O
L K N M H G J I T S V U P O R Q 5 4 7 W 1 0 3 2 D C F E 9 8 B A 6
M N K L I J G H U V S T Q R O W 6 7 4 5 2 3 0 1 E F C D A B 8 9 P
N M L K J I H G V U T S R Q P O W 6 5 4 3 2 1 0 F E D C B A 9 8 7
O P Q R S T U V W H I J K L M N 8 9 A B C D E F 0 1 2 3 4 5 6 7 G
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
Q R O P U V S T I J G W M N K L A B 8 9 E F C D 2 3 0 1 6 7 4 5 H
R Q P O V U T S J I H G N M L K B A 9 8 W E D C 3 2 1 0 7 6 5 4 F
S T U V O P Q R K L M N G H I J C D E F 8 9 W B 4 5 6 7 0 1 2 3 A
T S V U P O R Q L W N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 2 K
U V S T Q R O P M N K L I J G H E F C D A W 8 9 6 7 4 5 2 3 0 1 B
V U T S R Q P O N M W K J I H G F E D C B A 9 8 7 6 5 4 3 2 1 0 L
0 4 5 1 8 C D 9 G K L H O S T P 7 3 2 6 F B A E N J I M V R Q U W
```

Partial square

```
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
. 0 . 2 . 4 . 6 . 8 . A . C . E . G . I . K . M . O . Q . S . W .
. . 0 . . . 4 5 . . 8 9 . . C D . . G H . . K L . . O P . . S T 1
. 2 1 0 . 6 5 4 . A 9 8 . E D C . I H G . M L K . Q P O W . T . V
. . . . . 1 2 3 . . . . 8 9 A B . . . . G H I J . . . . O P . R Q
. . . 6 1 . 3 2 . C . E 9 8 B A . K . M H G J I . S . U P O R Q .
. . 4 5 2 3 0 1 . . C D A B 8 9 . . K L I J G H . . S T Q . O P .
. 6 W 4 3 2 1 . . E D C B A 9 8 . M L K J I H G . U T S R Q P O .
. . . . . . . . 0 1 2 3 4 5 6 7 . . . . . . . . G H W J K L M N .
. 8 . A . . . E 1 0 3 2 5 4 7 6 . O . Q . S . U H G J I L K N M C
. . 8 9 . . C D 2 3 0 1 6 7 4 5 . . O P . . S T I W G H M N K L .
. A 9 8 . E . C 3 2 1 0 7 6 5 4 . Q P O . U T S J I H G . M L K .
. . . . . 9 A B 4 5 6 7 0 1 2 3 . . . . O P Q R K L M N G H I J .
. C . E 9 8 B A 5 4 7 6 1 . 3 2 . S . U P O R Q L K N W H G J I .
. . C D A B 8 W 6 7 4 5 2 3 . 1 . . S T Q R O P M N K L I J G H .
. E D C B A 9 8 7 6 5 4 3 2 1 . . U T S R Q P O W M L K . I H G .
. . . . . . . . . . . . . . . . . . 0 1 . 3 4 5 6 7 8 9 A B C D E F .
. G . I . K . M . O . Q . . . U 1 0 3 2 5 4 7 6 9 8 B A D C F E S
. . G H . . K L . . O P . . S T 2 . 0 1 6 7 4 5 A B 8 9 E F C D .
. I H G . M L K . Q P O . U . S 3 2 1 0 7 6 5 4 B A 9 8 F E D C T
. . . . G H I J . . . . . P Q R 4 5 6 7 . 1 2 3 C D E F 8 9 A B O
. K . M H G J I . S . U P O R Q 5 4 7 W 1 0 3 2 D C F E 9 8 B A .
. . K L I J G H . . S T Q R O W 6 7 4 5 2 3 . 1 E F C D A B 8 9 .
. M L K J I H G . U T S R Q P O W 6 5 4 3 2 1 . F E D C B A 9 8 7
. . . . . W H I J K L M N 8 9 A B C D E F 0 1 2 3 4 5 6 7 .
. O . Q . S . U H G J I L K N M 9 8 B A D C F . 1 0 3 2 5 4 7 6 .
. . O P . . S T I J G . M N K L A B 8 9 E F C D 2 3 0 1 6 7 4 5 .
. Q P O . U T S J I H G N M L K B A 9 8 . E D C 3 2 1 0 . 6 5 4 F
. . . . O P Q R K L M N G H I J C D E F 8 9 W B 4 5 6 7 . 1 2 3 .
. S . U P O R Q L W N M H G J I D C F E 9 8 B A 5 4 7 6 1 . 3 2 .
. . S T Q R O P M N K L I J G H E F C D A . 8 9 6 7 4 5 2 3 . 1 B
. U T S R Q P O N M . K J I H G F E D C B A 9 8 7 6 5 4 . 2 1 0 .
0 4 . . 8 . D . . K L H . . . . . 3 2 . . . . E . . . . . R . . .
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E145: order 32, 752 clues

Status U. Unresolved producer output; criticality is NOT asserted and its size is not a lower bound.

Source: `outputs/lcs_full_review/stripping/pruned_direct_n32_seed0.json`

SHA256: `afae8cfab5025fd7e885d0877e418aa6ded28accd7e4239924501f0764cc59da`

Completion

```
0 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V
1 0 3 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U
2 3 0 1 6 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T
3 2 1 0 7 6 5 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S
4 5 6 7 0 1 2 3 C D E F 8 9 A B K L M N G H I J S T U V O P Q R
5 4 7 6 1 0 3 2 D C F E 9 8 B A L K N M H G J I T S V U P O R Q
6 7 4 5 2 3 0 1 E F C D A B 8 9 M N K L I J G H U V S T Q R O P
7 6 5 4 3 2 1 0 F E D C B A 9 8 N M L K J I H G V U T S R Q P O
8 9 A B C D E F 0 1 2 3 4 5 6 7 O P Q R S T U V G H I J K L M N
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O R Q T S V U H G J I L K N M
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P U V S T I J G H M N K L
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U T S J I H G N M L K
C D E F 8 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R K L M N G H I J
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K N M H G J I
E F C D A B 8 9 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L I J G H
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I H G
G H I J K L M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 9 A B C D E F
H G J I L K N M P O R Q T S V U 1 0 3 2 5 4 7 6 9 8 B A D C F E
I J G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D
J I H G N M L K R Q P O V U T S 3 2 1 0 7 6 5 4 B A 9 8 F E D C
K L M N G H I J S T U V O P Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B
L K N M H G J I T S V U P O R Q 5 4 7 6 1 0 3 2 D C F E 9 8 B A
M N K L I J G H U V S T Q R O P 6 7 4 5 2 3 0 1 E F C D A B 8 9
N M L K J I H G V U T S R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8
O P Q R S T U V G H I J K L M N 8 9 A B C D E F 0 1 2 3 4 5 6 7
P O R Q T S V U H G J I L K N M 9 8 B A D C F E 1 0 3 2 5 4 7 6
Q R O P U V S T I J G H M N K L A B 8 9 E F C D 2 3 0 1 6 7 4 5
R Q P O V U T S J I H G N M L K B A 9 8 F E D C 3 2 1 0 7 6 5 4
S T U V O P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 1 2 3
T S V U P O R Q L K N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 2
U V S T Q R O P M N K L I J G H E F C D A B 8 9 6 7 4 5 2 3 0 1
V U T S R Q P O N M L K J I H G F E D C B A 9 8 7 6 5 4 3 2 1 0
```

Partial square

```
0 . . 3 . 5 6 . . . . . C D . F . . . . K L M N . . . . S T U V
. . 3 . 5 4 7 6 . 8 B A . C . E . G . I L K N M . O . Q T . V U
2 3 0 1 6 . 4 5 A B . . . F . D I . G H M N . . . R O . U V S T
3 2 1 0 . 6 5 4 B A . 8 F E . C J I H G N M L K R Q . . . U T S
4 5 6 7 0 1 2 3 C . E . . . . B . L M N G H I . S T U V . P Q R
. 4 7 6 . . 3 2 D C F E . 8 . A L K . M H G . I . . V U P . R Q
. 7 4 5 2 . 0 1 E . C D A . . . M N K . I . G H U V S T . R O P
. . . . . . . . . . . . . . . 9 . . . . . . . . . . . . . . . .
8 . A B C . E F 0 1 2 3 4 5 . 7 O P Q R . T U V G H I J K L M .
. . B A D C F E 1 0 3 2 5 4 . 6 P O R Q . . V . . G J I . K N M
A B 8 . E F C D 2 . 0 1 6 7 . 5 Q . O . U V S T I J G H M N K L
B A . 8 . E D C 3 2 . 0 7 . . . R Q P O . . T S J I H G N M L .
C D E F 8 . A B . 5 . 7 0 1 . 3 S . U V . P Q R K L M N G H I J
. . F E . 8 B A 5 4 7 6 1 0 . 2 T S V . P . R Q L K N M H . J I
E F . . . B 8 . 6 7 4 5 2 3 . 1 U . S T . R O P M N K . . J G .
F E D . B A . . 7 . 5 4 . . . 0 V U T . . Q P O N M L K J I . G
G H . J K L M N O P Q R S T . V . 1 2 3 . 5 6 7 . . . B C D E F
H G . I L . N M P O R Q T . . U 1 0 3 2 5 . 7 . . 8 B A D C F E
I J G H M N K L . R O P U V . T . . 0 1 6 7 . 5 A B 8 . E F C D
J I H . N M L K R Q P . V U . S 3 2 1 . 7 6 5 4 B A . 8 F E . C
. L M N G H I J S . U V O P . R 4 5 6 7 0 . 2 3 C D E F 8 . . B
L K N M H . . I T . V . P O . Q 5 4 7 6 1 0 . 2 D C F E . 8 B A
M N K L I J G H U V S T Q . . P 6 7 4 5 2 3 0 1 E . . D A B 8 .
. M L . . I H G V U T S . . . . 7 6 5 4 3 2 1 0 F E D C B A . 8
O P Q R S T U V G H I J K . . . 8 . A B . D E . 0 1 2 3 4 . . .
P O . . T S V U H G . . L K . M . 8 B A D C . E 1 0 . 2 5 4 7 6
Q R O P U V S T I J G H M N . L A B 8 . E F C . 2 3 0 1 6 7 4 5
R . P O V U T S J I H G N M . . . A . 8 . E D C . . 1 0 7 6 . 4
S T U V O P Q R K L M N . H . J . . E . . . A B 4 5 6 . 0 1 2 .
T S V U P O R Q L K N M H . . I D C F E . 8 B A 5 4 7 6 1 0 3 2
U V . T Q R O . . N K L I J . H E F C D A B 8 . 6 7 . 5 . 3 0 1
V U T S R Q . . N . L K J . . G F E D . . A . . 7 6 5 4 3 . 1 0
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E146: order 33, 810 clues

Status U. Unresolved producer output; criticality is NOT asserted and its size is not a lower bound.

Source: `outputs/lcs33_review_29814974/repairs/task_3_n33_rank_1.pending/repair.json`

SHA256: `6d7c34f83c91906890e1fd7f4345a2ae421669351a2e373dd335dd667cac149f`

Completion

```
W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
2 3 0 1 W 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 6
3 2 1 0 7 6 W 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 5
4 5 6 7 0 1 2 3 W D E F 8 9 A B K L M N G H I J S T U V O P Q R C
5 4 7 6 1 0 3 2 D C W E 9 8 B A L K N M H G J I T S V U P O R Q F
6 7 4 5 2 3 0 1 E F C D W B 8 9 M N K L I J G H U V S T Q R O P A
7 6 5 4 3 2 1 0 F E D C B A W 8 N M L K J I H G V U T S R Q P O 9
8 9 A B C D E F 0 1 2 3 4 5 6 7 W P Q R S T U V G H I J K L M N O
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O W Q T S V U H G J I L K N M R
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P W V S T I J G H M N K L U
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U W S J I H G N M L K T
C D E F 8 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R W L M N G H I J K
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K W M H G J I N
E F C D A B 8 9 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L W J G H I
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I W G H
G H I J K W M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 9 A B C D E F L
H G J I L K N W P O R Q T S V U 1 0 3 2 5 4 7 6 9 8 B A D C F E M
I W G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D J
J I H W N M L K R Q P O V U T S 3 2 1 0 7 6 5 4 B A 9 8 F E D C G
K L M N G H I J S T U V O W Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B P
L K N M H G J I T S V U P O R W 5 4 7 6 1 0 3 2 D C F E 9 8 B A Q
M N K L I J G H U W S T Q R O P 6 7 4 5 2 3 0 1 E F C D A B 8 9 V
N M L K J I H G V U T W R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 S
O P Q R S T U V G H I J K L M N 8 9 A B C W E F 0 1 2 3 4 5 6 7 D
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
Q R O P U V S T I J G H M N K L A W 8 9 E F C D 2 3 0 1 6 7 4 5 B
R Q P O V U T S J I H G N M L K B A 9 W F E D C 3 2 1 0 7 6 5 4 8
S T U V O P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 W 2 3 1
T S V U P O R Q L K N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 W 2
U V S T Q R O P M N K L I J G H E F C D A B 8 9 6 W 4 5 2 3 0 1 7
V U T S R Q P O N M L K J I H G F E D C B A 9 8 7 6 5 W 3 2 1 0 4
0 J 3 G 6 L 5 M C V F S A P 9 Q O B R 8 U D T E K 7 N 4 I 1 H 2 W
```

Partial square

```
. 1 . 3 . . 6 7 . . A B C . E . . . I J K . . . O P . . . T . V 0
. 0 . . 5 . 7 . . 8 . . D . . . H G J I . . N . . O R Q T . V U .
2 . . 1 . . . 5 . B 8 9 E F . . I J G H M N K . . R O P U V S T 6
. . 1 0 . . W 4 . . 9 8 F E D C J I H G N M . K . Q P O V . . S 5
4 . . . 0 . . 3 W . E F . 9 . B K . M N G H I J S . U V . P Q R .
. . 7 6 1 . 3 . . C . E . 8 B A . . N M H G J I . S V . P O R Q F
6 7 4 5 2 . . 1 E F . . W B 8 . M N . . I J . . . V . T Q R O P A
7 6 5 4 . . 1 0 F . D C B A W 8 N M . K J I H G V U T . R Q P O 9
8 9 . B . . E F 0 1 2 3 . 5 6 7 W P . . S T U V G H I J K . M N O
9 8 . A D . . . 1 0 3 2 5 4 7 6 . O W . . S . U . G . I . K N M R
A . . . E . C . 2 3 0 1 6 7 4 5 . R O P W . S . I J G H M N K . U
B A 9 . F . D C 3 2 1 0 7 6 5 4 R Q P O V U . S J I H G N M . K T
. D E F . . A . 4 5 6 7 0 1 2 3 S T U V O . . . W . M N G H I J K
D C F E 9 . B A 5 4 7 6 1 0 3 . T S V U P O R Q . K W M H G J I N
E F C D . . . 9 6 7 4 5 2 3 0 . U V S T . R O P M N K . W J G H I
F . D C B . . . 7 6 5 4 3 . 1 0 V U T S . Q P O N M . K J I W . H
. . . . . .
H G J I . . N W . O R Q T S V U 1 0 3 2 5 4 7 6 9 8 B A D C F E M
I W G H M . K . . R O P . V S T 2 3 0 . 6 7 4 5 A B 8 9 E F C D J
. I H W N . . K R Q . O V . T S 3 2 . 0 7 . 5 4 B A 9 8 F E D C G
K . M . G . I J . T U V O W Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B P
. K N M H . J I T S V U P O R W 5 4 7 6 1 0 3 . . C F E . 8 B A Q
M N K . I . . H U W S T Q R O . 6 7 . 5 2 . 0 1 E F C . A B 8 9 V
N M . K . . H G V U . W R Q P O 7 6 5 . 3 2 . 0 . E D . B A . 8 S
. P Q R S . U . G H I J K . M N . . A B C W E F 0 1 2 3 4 5 6 7 .
P . . Q T . V U H G J I . K . M 9 8 B A D . F W . 0 3 2 5 4 7 . E
. R O P . . . T I J G H M N K . A W 8 9 E F C D 2 3 0 1 . 7 4 5 B
R Q P O V . T S J I H G N M . K B A 9 W F . . C 3 2 1 0 7 6 5 4 8
S . . V . . Q R K . M N G H I J C D E . 8 9 A B 4 5 6 . 0 W 2 3 .
T S V U P . . Q . K N M H G J I D . F E 9 8 B A 5 4 7 6 1 0 3 W .
U . S . Q . O P M . K . . J G H E . C D A B 8 9 6 W 4 5 . 3 . 1 7
V U T S . . P . N M . K J I H G F E D C B . 9 8 7 6 5 W 3 2 1 0 4
0 . 3 G 6 . 5 M C V F S A P . Q . B R 8 U D T . K 7 N 4 I 1 H 2 W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E147: order 33, 801 clues

Status U. Unresolved producer output; criticality is NOT asserted and its size is not a lower bound.

Source: `outputs/lcs33_review_29814974/repairs/task_3_n33_rank_6.pending/repair.json`

SHA256: `48631d4de311baa40df1849b3b4b8f8875dda5010f277b7a5f32e67f5fc00b74`

Completion

```
W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
2 3 0 W 6 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 1
3 W 1 0 7 6 5 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 2
4 5 6 7 0 1 2 3 C D E F 8 9 A B K L M N G H I J W T U V O P Q R S
5 4 7 6 1 0 3 2 D C F E 9 8 B A L K N M H G J I T S W U P O R Q V
6 7 4 5 2 3 0 1 E F C D A B 8 9 M N K L I J G H U V S W Q R O P T
7 6 5 4 3 2 1 0 F E D C B A 9 8 N M L K J I H G V W T S R Q P O U
8 9 A B C D E F 0 1 2 3 4 5 6 7 O P Q R S T U V G H I J W L M N K
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O R Q T S V U H G J I L K W M N
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P U V S T I J G H M N K W L
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U T S J I H G N W L K M
C D E F W 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R K L M N G H I J 8
D C F E 9 8 W A 5 4 7 6 1 0 3 2 T S V U P O R Q L K N M H G J I B
E F C D A B 8 W 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L I J G H 9
F E D C B W 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I H G A
G H I J K L M N O P Q R S T U V 0 1 2 3 W 5 6 7 8 9 A B C D E F 4
H G J I L K N M P O R Q T S V U 1 0 3 2 5 4 W 6 9 8 B A D C F E 7
I J G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 W A B 8 9 E F C D 5
J I H G N M L K R Q P O V U T S 3 2 1 0 7 W 5 4 B A 9 8 F E D C 6
K L M N G H I J S T U V W P Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B O
L K N M H G J I T S V U P O W Q 5 4 7 6 1 0 3 2 D C F E 9 8 B A R
M N K L I J G H U V S T Q R O W 6 7 4 5 2 3 0 1 E F C D A B 8 9 P
N M L K J I H G V U T S R W P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 Q
O P Q R S T U V W H I J K L M N 8 9 A B C D E F 0 1 2 3 4 5 6 7 G
P O R Q T S V U H G W I L K N M 9 8 B A D C F E 1 0 3 2 5 4 7 6 J
Q R O P U V S T I J G W M N K L A B 8 9 E F C D 2 3 0 1 6 7 4 5 H
R Q P O V U T S J W H G N M L K B A 9 8 F E D C 3 2 1 0 7 6 5 4 I
S T U V O P Q R K L M N G H I J W D E F 8 9 A B 4 5 6 7 0 1 2 3 C
T S V U P O R Q L K N M H G J I D C W E 9 8 B A 5 4 7 6 1 0 3 2 F
U V S T Q R O P M N K L I J G H E F C W A B 8 9 6 7 4 5 2 3 0 1 D
V U T S R Q P O N M L K J I H G F W D C B A 9 8 7 6 5 4 3 2 1 0 E
0 2 3 1 8 A B 9 G I J H O Q R P C E F D 4 6 7 5 S U V T K M N L W
```

Partial square

```
. . . 3 . . . . . . . . . . . F G . . J . . . N . P Q R . T . . .
1 0 . . 5 . . 6 . 8 B . D . F . H G . . L K N M . O . Q . S V U .
2 . 0 . 6 . 4 5 A B 8 . E . C D I . G H M N K L . R . P U V S T .
3 . 1 0 7 6 . 4 B A . . F E D C J . H G . M L K R . P O V U T S .
4 5 6 7 0 . 2 3 C D . F . 9 A . K . . . G H I . . . . V O P Q R .
5 4 7 6 1 . 3 2 D . . E 9 . . A L K N M H G . I T S . U P O R Q .
6 . . 5 2 3 0 1 E F C D A B 8 9 M . K L I J G H U . S . Q . . P .
. 6 5 . 3 2 1 0 F . D . B A 9 . N M . K . I . . . . T S R . P O .
8 9 . B C . . . 0 . 2 3 . 5 6 7 O . Q R S T . V G H I J . L M N .
9 . B A D C F E 1 . 3 2 5 4 . 6 P O R Q T S V U H G J . L K . M .
. B 8 9 E F . D 2 3 . 1 6 7 4 5 . R O P . V S . . J G H M N K . .
B A 9 8 . E . C 3 2 1 0 7 6 5 . R Q P O V . T S . . H G . . L K .
C . E F . 9 A B 4 . 6 7 . 1 2 3 S . U V O P . R K L M N G H I J .
D . F E 9 8 . A 5 4 . 6 1 0 3 2 T S V U P O R Q L K . M H G . I .
E F . . A B 8 . 6 7 4 5 2 3 . 1 U V S T Q R O P M N K L I J G H .
F E D C . . 9 . 7 6 5 4 . . 1 . . U T S R Q . O . M L K J . H G .
G . I J . . M N O P . R . . U V 0 1 2 3 . 5 6 7 8 9 A B C D E F .
H G J I L K N M P O R Q T S V U . 0 3 2 5 4 . 6 9 8 B A D C F E .
I . G H M N K L . R O P . V S T 2 3 0 1 6 . 4 . A B 8 9 E F C . .
. . H G N . L K R Q P O V U . S . 2 1 0 7 . 5 4 B A 9 8 . . D C .
K L M . G . I . S T U . . P Q R 4 5 6 . 0 1 2 3 C D E F 8 9 A B .
. . N M H G J I T S V U P . . . 5 4 7 6 1 0 . 2 D C F E 9 8 B A .
M N K L I J G . U . S T Q R O . 6 7 . 5 2 3 0 1 E F C D A B 8 9 .
N M L K J . H G V U T S R . P O 7 6 5 4 . 2 1 0 F E . C B A 9 8 .
. P . R S T U . . H I J K L M . 8 9 . B C D . F 0 1 2 3 4 5 . 7 .
P O R Q T S V U H G . . L K N M 9 8 B A D . F E 1 0 3 2 5 4 7 6 .
Q R O P U V . T I . G . M N K L A B 8 9 E F C D 2 3 0 . 6 7 4 5 .
R Q P O V U . S J . H G N M L . B A 9 . F E D C 3 2 1 0 7 6 5 4 .
S T U V O P Q R K L M . . H I J . D E . 8 9 A B 4 5 6 7 0 1 2 3 .
T S V U P O . . L K N M H G . I D . . E 9 8 B A 5 4 7 6 1 . 3 2 .
U . S T Q R O P M . K L I . G H E F C . A B 8 . 6 7 4 5 2 3 . 1 .
V U T S R Q P O N M L K J I H . . . D C B A 9 8 7 6 5 4 3 2 1 0 .
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E148: order 33, 797 clues

Status U. Unresolved producer output; criticality is NOT asserted and its size is not a lower bound.

Source: `outputs/lcs33_review_29814974/repairs/task_2_n33_rank_0.pending/repair.json`

SHA256: `738fb44d28c53d385677830d243594a00f0ace9a779169bcac4e001b03460667`

Completion

```
W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 3 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V W U
2 3 0 W 6 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 1
3 2 1 0 7 6 5 4 B A 9 8 F E D C J I H G N M L K R Q P O W U T S V
4 5 6 7 0 1 2 3 C D E F 8 9 A B K L M N G H I J S T U V O P W R Q
5 W 7 6 1 0 3 2 D C F E 9 8 B A L K N M H G J I T S V U P O R Q 4
6 7 4 5 2 3 0 1 E F C D A B 8 9 M N K L I J G H U V S T Q W O P R
7 6 W 4 3 2 1 0 F E D C B A 9 8 N M L K J I H G V U T S R Q P O 5
8 9 A B C D E F 0 1 2 3 4 5 6 7 O P Q R S T U V G H W J K L M N I
9 8 B A D W F E 1 0 3 2 5 4 7 6 P O R Q T S V U H G J I L K N M C
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P U V S T I W G H M N K L J
B A 9 8 F E W C 3 2 1 0 7 6 5 4 R Q P O V U T S J I H G N M L K D
C D E F W 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R K L M N G H I J 8
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K N W H G J I M
E F C D A B 8 W 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L I J G H 9
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O W M L K J I H G N
G H I J K L M N O P Q R S T U V 0 1 W 3 4 5 6 7 8 9 A B C D E F 2
H G J I L K N M P O R Q T W V U 1 0 3 2 5 4 7 6 9 8 B A D C F E S
I J G H M N K L Q R O P U V S T 2 W 0 1 6 7 4 5 A B 8 9 E F C D 3
J I H G N M L K R Q P O V U W S 3 2 1 0 7 6 5 4 B A 9 8 F E D C T
K L M N G H I J S T U V W P Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B O
L K N M H G J I T S V U P O R Q 5 4 7 W 1 0 3 2 D C F E 9 8 B A 6
M N K L I J G H U V S T Q R O W 6 7 4 5 2 3 0 1 E F C D A B 8 9 P
N M L K J I H G V U T S R Q P O W 6 5 4 3 2 1 0 F E D C B A 9 8 7
O P Q R S T U V W H I J K L M N 8 9 A B C D E F 0 1 2 3 4 5 6 7 G
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
Q R O P U V S T I J G W M N K L A B 8 9 E F C D 2 3 0 1 6 7 4 5 H
R Q P O V U T S J I H G N M L K B A 9 8 W E D C 3 2 1 0 7 6 5 4 F
S T U V O P Q R K L M N G H I J C D E F 8 9 W B 4 5 6 7 0 1 2 3 A
T S V U P O R Q L W N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 2 K
U V S T Q R O P M N K L I J G H E F C D A W 8 9 6 7 4 5 2 3 0 1 B
V U T S R Q P O N M W K J I H G F E D C B A 9 8 7 6 5 4 3 2 1 0 L
0 4 5 1 8 C D 9 G K L H O S T P 7 3 2 6 F B A E N J I M V R Q U W
```

Partial square

```
. . . 3 . . 6 . . . . . . D . . . H . . . L . . O . Q R . T . . .
. 0 . 2 5 . 7 6 . 8 . A . . . E . G J . . K . M . . . Q T S V . .
. 3 0 . 6 7 4 . A . 8 9 E F . D I J . H M N . L Q R O P U . . T .
3 2 1 0 7 6 5 4 B . 9 8 F E . C J . H . . M L K R Q P O . U T S .
. . 6 7 0 1 2 . C . E F . 9 A . K . M N G H . J S T U . O P . R .
5 . 7 . . 0 3 2 D C F . 9 8 B A L . N M H . . I T . V . P O R Q .
. 7 4 . 2 3 0 . E F C D A . 8 9 M N K L I J G H U V S T Q . O P .
7 . . 4 3 2 1 0 F . D C . A 9 8 . . L K J I H G . U T S R Q . O .
. . . . C . E F 0 1 2 3 4 5 6 7 . P . R S T U . G H . J . L M N .
9 8 B . D . F E 1 . 3 2 5 4 7 . P . R Q T S V U H . J I L K N M .
. B 8 . E F . . 2 3 0 1 6 7 4 5 Q R O P U . S . I . G H M . . L .
B A 9 . F E . C 3 2 1 0 7 6 5 . R . P O V U . S J I H G N M L K .
C D E F . 9 A B 4 . 6 . 0 1 . . S T U V . P . R K L M N . H I . .
D . F E 9 8 B A 5 . 7 6 1 0 3 2 T S V U . O R Q L K N . H G J . .
E F C D . B 8 . 6 7 4 5 2 3 0 1 U V S T Q R . P . N K L . J G H .
F . . C B A 9 . 7 6 5 4 3 2 1 0 V U T S R Q P O . . L K J I . G .
G H I . K L M N O P Q . . T U V 0 1 . . 4 5 6 7 8 9 A B C . E F .
H G J I L K N M P O R Q T . V U . 0 3 2 5 4 7 6 9 8 B A . C F E .
I J G H M . K L Q R O P U . . T . . 0 1 6 7 4 5 A B 8 9 E F C . .
J I H G N . L K R . P O . U . S 3 2 1 0 7 6 5 4 B A 9 8 F E D C .
K L . N G H I J S T U V . . Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A . .
L K N M H G J I T . V U . O R Q 5 4 7 . 1 0 3 2 D C F E 9 . B A .
M N K L I J G . . V S . . R O . 6 . 4 5 . 3 0 1 E F C D A . 8 9 .
N M . K J I H G V . T S R . P . . 6 . . 3 . 1 0 F E D C B . 9 8 .
. P Q R S T U V . H I J K L . N 8 . A B C D E F 0 1 2 3 4 5 6 7 .
P . R . . S . . H . J I L K N M 9 8 B A D C F . . 0 . 2 5 4 7 . .
. R O P U V S T . J G . . N K L A B 8 9 E F C D 2 3 0 1 . 7 4 5 .
. Q . . V . T S J I H . N M . . B A 9 8 . E . . 3 2 1 . 7 6 5 4 .
S T U V O P Q R K L M N . H I J C . E F 8 9 . B 4 5 6 7 0 1 2 3 .
T S V U P O R Q L . N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 . .
U V S T . R O P M N K L I J G H E F C D A . 8 9 6 7 4 5 2 3 0 1 .
V U T S . Q P O N M . K J I H . F E D C . A . . 7 6 5 4 3 2 1 0 .
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E149: order 33, 790 clues

Status U. Unresolved producer output; criticality is NOT asserted and its size is not a lower bound.

Source: `outputs/lcs33_review_29814974/repairs/task_3_n33_rank_4.pending/repair.json`

SHA256: `5efa5d71a26b3daab2ade8a830f1be6a85ca0571a8e1e36d415fd0a5b7adf7c0`

Completion

```
0 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K W M N O P Q R S T U V L
1 W 3 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 0
2 3 0 1 6 7 4 5 A B 8 9 E W C D I J G H M N K L Q R O P U V S T F
3 2 1 0 7 6 5 4 B A 9 8 F E D W J I H G N M L K R Q P O V U T S C
4 5 6 7 0 1 2 3 C D E F 8 9 A B K L M N G H I J S T U V O P Q W R
5 4 7 6 1 0 3 2 D C F E 9 8 B A L K W M H G J I T S V U P O R Q N
6 7 W 5 2 3 0 1 E F C D A B 8 9 M N K L I J G H U V S T Q R O P 4
7 6 5 4 3 2 1 0 F E D C B A 9 8 N M L K J I H G V U T S R W P O Q
8 9 A B C D E F 0 1 2 3 4 5 6 7 O P Q R S T U V W H I J K L M N G
9 8 B A D C F E 1 0 3 2 5 4 W 6 P O R Q T S V U H G J I L K N M 7
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P U V S T I J G W M N K L H
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U T W J I H G N M L K S
C D E F 8 9 A B 4 W 6 7 0 1 2 3 S T U V O P Q R K L M N G H I J 5
D C F E 9 8 B W 5 4 7 6 1 0 3 2 T S V U P O R Q L K N M H G J I A
E F C D A B 8 9 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L W J G H I
F E D C B A 9 8 7 6 5 4 W 2 1 0 V U T S R Q P O N M L K J I H G 3
G H I J K L M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 W A B C D E F 9
H G J I L K N M P O R Q T S V U W 0 3 2 5 4 7 6 9 8 B A D C F E 1
I J G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 5 A B W 9 E F C D 8
J I H G N W L K R Q P O V U T S 3 2 1 0 7 6 5 4 B A 9 8 F E D C M
W L M N G H I J S T U V O P Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B K
L K N M H G J I T S V W P O R Q 5 4 7 6 1 0 3 2 D C F E 9 8 B A U
M N K L I J G H U V S T Q R O P 6 7 4 5 W 3 0 1 E F C D A B 8 9 2
N M L K J I H G W U T S R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 V
O P Q R S T U V G H I J K L M N 8 9 A B C D E F 0 1 2 3 4 5 W 7 6
P O R Q T S V U H G W I L K N M 9 8 B A D C F E 1 0 3 2 5 4 7 6 J
Q R O W U V S T I J G H M N K L A B 8 9 E F C D 2 3 0 1 6 7 4 5 P
R Q P O V U W S J I H G N M L K B A 9 8 F E D C 3 2 1 0 7 6 5 4 T
S T U V W P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 1 2 3 O
T S V U P O R Q L K N M H G J I D C F E 9 8 W A 5 4 7 6 1 0 3 2 B
U V S T Q R O P M N K L I J G H E F C W A B 8 9 6 7 4 5 2 3 0 1 D
V U T S R Q P O N M L K J I H G F W D C B A 9 8 7 6 5 4 3 2 1 0 E
K 0 4 P O M T A V 5 J U 3 F 7 C 1 E N D 2 L B S G 9 8 H I Q 6 R W
```

Partial square

```
. . 2 . . . . . . . . . . . E F . H . J K . M N . . . . S T U V .
. . 3 2 . 4 7 . 9 . . A D . F E H G J . . K N M P . R Q T . V . .
2 3 . . . . 4 5 . B 8 9 E . . D . . G . M . . . Q . O P U V S . .
3 2 1 . 7 6 . . B . 9 8 F E D . J . . . N M . K R . P O V . T S .
. . 6 7 0 1 2 3 . D . . 8 9 A B . L . . G H I J . T U V . . . . .
. 4 7 6 1 0 3 2 D C F E 9 . B A L K . M H G J I T S V . P O R Q .
6 7 . 5 2 3 0 1 E F C D A B 8 9 M . K L I J G H U V S T Q R O P .
7 6 . 4 3 2 1 . F E D . B A 9 8 . M L K . I H G V U T S R . P O .
. 9 . . C . E F 0 1 . 3 4 5 . 7 O P Q R S T U V . . . J K . M . .
9 . B A . C F . 1 0 3 . . 4 . 6 P O R Q . . V U H G J I L . N M .
A B 8 9 E F C . . 3 0 . . 7 . 5 Q R O P U V S T I . G . M N K L .
B A 9 8 F E . C 3 2 . . 7 6 . . R Q P O V U T . J I H . N M L K .
C D E F 8 9 A . 4 . 6 7 0 1 2 3 S T U V O P Q R K L M N G H I J .
. C F . . 8 B . . 4 7 6 1 0 3 2 . S V U P O R Q L K . M H . J I .
. F . D . . . 9 6 7 4 5 2 . 0 1 U V . T Q R O . M N K L . J G . .
F E D C B A 9 8 7 6 5 . . 2 1 0 V . T S . Q . O . M L . J I . G .
G H I J K L M . O P Q R S T . V 0 1 2 3 4 5 6 7 . . A B C D E F .
H G J I L K N M P O R Q . . V U . 0 3 2 5 4 7 6 9 8 B A D C F E .
I J G H . N . L Q R O P U V S T 2 3 0 . 6 7 4 5 A B . 9 E F C D .
J I H G N . L . R Q P O . U T S 3 2 1 0 7 6 5 4 B A 9 8 F E . . .
. L M N G H I J S . U V O P Q R 4 . 6 7 0 1 2 3 C D E F 8 9 A B .
L K N . H G J I T S V . P O . Q 5 4 7 6 1 . 3 2 D C F E 9 . B A .
M N K . I J G H . V S T Q R O P . 7 . 5 . 3 0 1 E . C D A B . . .
. . L K J I H G . U T S . . . O 7 6 5 4 3 2 1 0 F E D C B A 9 8 .
. . . R S . U V . . . . K L M N 8 9 A B C D E F 0 1 2 3 4 5 . 7 .
P O R Q T S V U H G . I . K N M 9 . B A D . F E 1 0 3 2 5 4 7 6 .
Q R O . . V . T I . G H M N K L A . 8 9 E F C . 2 3 0 1 6 7 4 5 .
R Q P O V U . S . I H G N M L K B A 9 8 F E D C 3 2 1 0 7 . 5 4 .
S T . V . . Q R K L . N G . I J . . E . 8 9 A B . 5 6 7 0 . 2 3 .
T S V U P O . Q L K . M H G J I D C F E . 8 . A . 4 7 6 . 0 3 2 .
U V S T Q R O P M N K L . . G H E F C . A B 8 9 6 7 4 5 2 3 0 1 .
V U . . . Q P O N . . K . I H . F . . C B A 9 . 7 6 5 4 3 2 . 0 .
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E150: order 33, 790 clues

Status U. Unresolved producer output; criticality is NOT asserted and its size is not a lower bound.

Source: `outputs/lcs33_review_29814974/repairs/task_2_n33_rank_4.pending/repair.json`

SHA256: `c564ce672ab31b7578197e09984c7dd987119a3f4c1ead4adb8c9625d0013e59`

Completion

```
0 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S W U V T
1 0 3 2 5 W 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 4
2 3 0 1 6 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V W T S
3 2 1 0 7 6 5 4 B A 9 8 F E D C J I H G N M L W R Q P O V U T S K
4 5 6 7 0 1 2 3 C D E F 8 9 A B K L M W G H I J S T U V O P Q R N
5 4 7 6 W 0 3 2 D C F E 9 8 B A L K N M H G J I T S V U P O R Q 1
6 7 4 5 2 3 0 1 E F C D A B 8 9 M N K L W J G H U V S T Q R O P I
7 6 5 4 3 2 1 0 F E D C B A 9 8 N M W K J I H G V U T S R Q P O L
8 W A B C D E F 0 1 2 3 4 5 6 7 O P Q R S T U V G H I J K L M N 9
9 8 B A D C W E 1 0 3 2 5 4 7 6 P O R Q T S V U H G J I L K N M F
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P U V S T I J G W M N K L H
B A 9 W F E D C 3 2 1 0 7 6 5 4 R Q P O V U T S J I H G N M L K 8
C D E F 8 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R K L M N W H I J G
D C F E 9 8 B A 5 4 7 6 1 W 3 2 T S V U P O R Q L K N M H G J I 0
E F C D A B 8 9 6 7 4 W 2 3 0 1 U V S T Q R O P M N K L I J G H 5
F E W C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I H G D
G H I J K L M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 9 W B C D E F A
H G J I L K N M P O R Q T S V U 1 0 3 2 5 4 W 6 9 8 B A D C F E 7
I J G H M N K L W R O P U V S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D Q
J I H G N M L K R Q P O V U T S 3 W 1 0 7 6 5 4 B A 9 8 F E D C 2
K L M N G H I J S T U V O P Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A W B
L K N M H G J I T S W U P O R Q 5 4 7 6 1 0 3 2 D C F E 9 8 B A V
M N K L I J G H U V S T Q R O W 6 7 4 5 2 3 0 1 E F C D A B 8 9 P
N M L K J I H G V W T S R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 U
W P Q R S T U V G H I J K L M N 8 9 A B C D E F 0 1 2 3 4 5 6 7 O
P O R Q T S V U H G J I L K N M 9 8 B A D W F E 1 0 3 2 5 4 7 6 C
Q R O P U V S T I J G H W N K L A B 8 9 E F C D 2 3 0 1 6 7 4 5 M
R Q P O V U T S J I H G N M L K B A 9 8 F E D C W 2 1 0 7 6 5 4 3
S T U V O P Q W K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 1 2 3 R
T S V U P O R Q L K N M H G W I D C F E 9 8 B A 5 4 7 6 1 0 3 2 J
U V S T Q R O P M N K L I J G H W F C D A B 8 9 6 7 4 5 2 3 0 1 E
V U T S R Q P O N M L K J I H G F E D C B A 9 8 7 W 5 4 3 2 1 0 6
0 9 D 8 1 4 F R Q U V 5 M 0 J P E 2 L N I C 7 K 3 6 A H G T S B W
```

Partial square

```
. . . . 4 5 . . 8 9 . . . . . F . H . J . . M N . P . R . . U . .
1 0 . 2 . . 7 . 9 8 B . . C F . H G . I L . . M . . R . T . V . .
2 . 0 1 6 . 4 5 A B . . E F C D I J G H M . K L Q R . P U V . T .
3 2 . 0 7 . . 4 B A 9 8 . E . . . I H . N M L . . Q P O . U T S .
4 5 6 . . 1 2 3 . D E F 8 9 . . . L . . G H . J S T . . . P Q . .
. 4 7 . . 0 3 . . C F E 9 8 B . L K N . . G J I T . V U P O R Q .
. 7 4 5 2 . 0 . . F C D . B 8 . M N K L . . G H U . S T Q . O . .
. 6 5 4 3 2 . 0 F E D C . A . 8 N M . K J I . G . U T S R Q . O .
. . . B C D E . . . 2 3 . 5 6 . O P Q R S T . V G H I . K . M N .
9 8 B A D C . E . 0 3 2 5 4 . 6 P O R Q T S V . H G J I . K N M .
A B 8 9 E F C D . 3 0 . 6 7 4 5 Q R . . U V S T I J G . M N . L .
B A 9 . F E D C . 2 1 0 7 6 5 4 R Q P O V . T . J I H G N . L K .
. D E F 8 9 A B 4 5 6 7 0 1 . . S T U V O P Q R K L . N . H I . .
. C F E 9 8 B A 5 4 7 . 1 . 3 . T S V U P O . Q . K . . . G J I .
E F C D A B . 9 . . 4 . 2 . 0 1 . V S . Q R O P M N K . . J . H .
F E . . B . 9 . 7 6 5 4 3 2 1 . . U T . R Q P O N M L K J I H G .
G H I J K L M N O P Q R S T U V 0 1 2 . 4 5 6 7 8 9 . B C D E F .
H G J I L K N M P O R Q T S V U 1 0 3 2 5 4 . 6 9 8 B . D C F E .
I . G . . N K L . . O P . V S T 2 . 0 1 6 . 4 5 A . 8 . E F C D .
J I . G N M L K R Q P O V . T S 3 . 1 0 . 6 . . B A . 8 . . . C .
. L . . G H I . S T U V O P Q R 4 . 6 7 0 1 2 3 C . E F . . A . .
L K N M H G J I T S . . P O R . 5 4 7 6 . . 3 2 D C . E 9 8 . A .
M N K L I J G H U V S T Q R O . 6 7 4 5 2 3 0 1 E . C D . B 8 9 .
N M L K . I . G V . . S R Q P O 7 6 5 4 . . 1 0 F E D . . A . 8 .
. P Q R S T U V G H I . K L M N . 9 A . C D E F 0 1 2 3 4 5 6 7 .
P O R Q T S V U H G J I L K N M 9 8 B . . . F E 1 0 . 2 5 4 7 . .
Q R O . U V . . I J G H . N K L A B . 9 . F C D 2 . 0 1 6 7 4 . .
R Q P O V U T S J I H G N M L K B A 9 8 F . D C . 2 . 0 7 6 5 4 .
S T U V O P Q . K L M N G H I J C D E F 8 9 A B 4 5 6 7 . 1 2 3 .
T . V U P O R Q L K N M H G . I D C F E 9 8 B A 5 4 . 6 1 0 3 2 .
U V S T Q R . P M N K L I J G H . F C D A B 8 9 6 7 4 5 2 3 0 1 .
. U T S R Q P O . M L K J I H G F E D C B A 9 8 7 . 5 4 3 2 1 0 .
. . . . . W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E151: order 33, 784 clues

Status U. Unresolved producer output; criticality is NOT asserted and its size is not a lower bound.

Source: `outputs/lcs33_review_29814974/repairs/task_0_n33_rank_5.pending/repair.json`

SHA256: `eba93bc363f10c7e3f3326c1a6fdff8452054bbbc14a64fa0d5097d046c7b1bb`

Completion

```
W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 3 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S W U V
2 3 0 1 6 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S W T
3 2 1 0 7 6 5 4 W A 9 8 F E D C J I H G N M L K R Q P O V U T S B
4 5 6 7 0 1 2 3 C D E F 8 9 A B K L W N G H I J S T U V O P Q R M
5 4 7 6 1 0 3 2 D C W E 9 8 B A L K N M H G J I T S V U P O R Q F
6 7 W 5 2 3 0 1 E F C D A B 8 9 M N K L I J G H U V S T Q R O P 4
7 6 5 4 3 2 1 0 F E D C B A 9 8 N M L K J I H W V U T S R Q P O G
8 9 A B C D E F 0 W 2 3 4 5 6 7 O P Q R S T U V G H I J K L M N 1
9 8 B A D C F E 1 0 3 2 5 4 7 W P O R Q T S V U H G J I L K N M 6
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P U V W T I J G H M N K L S
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U T S W I H G N M L K J
C D E F 8 9 A B 4 5 6 7 0 1 2 3 S T U V O W Q R K L M N G H I J P
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K N M W G J I H
E F C D A B 8 9 6 7 4 W 2 3 0 1 U V S T Q R O P M N K L I J G H 5
F E D C B A 9 8 7 6 5 4 3 W 1 0 V U T S R Q P O N M L K J I H G 2
G H I J K L M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 9 W B C D E F A
H G J I W K N M P O R Q T S V U 1 0 3 2 5 4 7 6 9 8 B A D C F E L
I J G H M N W L Q R O P U V S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D K
J I H G N M L K R Q P O V U T S W 2 1 0 7 6 5 4 B A 9 8 F E D C 3
K L M W G H I J S T U V O P Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B N
L K N M H G J I T S V U P O R Q 5 4 7 6 1 0 3 2 D C F E 9 W B A 8
M N K T I J G H U V S L Q R W P 6 7 4 5 2 3 0 1 E F C D A B 8 9 O
N M L K J I H G V U T S R Q P O 7 6 5 4 3 2 1 0 F W D C B A 9 8 E
O P Q R S T U V G H I J K L M N 8 W A B C D E F 0 1 2 3 4 5 6 7 9
P O R Q T S V U H G J I L K N M 9 8 B A W C F E 1 0 3 2 5 4 7 6 D
Q W O P U V S T I J G H M N K L A B 8 9 E F C D 2 3 0 1 6 7 4 5 R
R Q P O V W T S J I H G N M L K B A 9 8 F E D C 3 2 1 0 7 6 5 4 U
S T U V O P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 W 0 1 2 3 7
T S V U P O R W L K N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 2 Q
U V S L Q R O P M N K T W J G H E F C D A B 8 9 6 7 4 5 2 3 0 1 I
V U T S R Q P O N M L K J I H G F E D W B A 9 8 7 6 5 4 3 2 1 0 C
0 R 4 N L U K Q B 1 F 5 I 2 0 6 3 9 M C D P S G J E A 7 H 8 V T W
```

Partial square

```
. . . 3 . 5 . . 8 . A . C . . . G H . . . . . N . P . R S . U . .
. 0 3 . 5 . 7 . 9 8 . . . C . E . G J I . K N . P O . Q . . . U .
2 . 0 . 6 7 . 5 . B 8 9 E F C D I J G . M N K . . . O . U V S . .
3 . 1 . 7 6 5 4 . A . . F E D . J I H G N M . K R . P O . U T S .
4 . 6 . 0 1 2 3 C D E . 8 9 A B K . . N G . I J S . . V O . Q R .
5 4 7 6 1 0 3 . D C . . 9 8 B A . K N M H G J I . S . U P O R Q .
. 7 . 5 2 3 0 1 . F C . A B 8 9 M N K . I J . H U V S . . . O P .
7 . 5 4 3 2 1 0 F . D . . A 9 8 N M . K J I H . V U . S R Q P O .
. 9 A . C D E F 0 . 2 . . 5 6 7 O . Q . S . U V G H I J K L M N .
9 8 B . D C F E 1 0 . 2 5 4 7 . P O R Q T S V U H G J I . K N M .
A B 8 . E F C D . 3 0 . . 7 4 5 Q R O P U V . . I . G H M N K . .
B . 9 . F E D C 3 2 1 0 . 6 5 4 R Q . O V U . S . I H G N M L K .
C . E . 8 9 A B . 5 . 7 0 1 2 3 S T U V O . Q R K . M N . H I J .
D C F . 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q . K N M . G J I .
E F C . A B 8 9 . 7 4 . 2 3 0 1 U V S T Q R O P M N K L I J G H .
F . D . B A 9 8 7 6 5 . 3 . 1 0 V U T S R Q P O N M . K J I H G .
G H I . K L M N O P Q . S . U V 0 1 2 3 4 5 6 7 . 9 . B C D E F .
H G J I . K N M P O R Q . S V U 1 0 3 2 5 4 7 6 . 8 B A D C F E .
I J G H M N . . Q R O . U V S . 2 3 0 1 6 7 4 5 A B 8 . E F . D .
J I H . N M . K . Q P O . U T . . 2 . 0 7 6 5 4 B A . 8 F E D C .
K . M . G H I J S T U . O P Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B .
. K N M H G J I . S V . P O R Q 5 4 7 6 1 0 3 2 D . F E . . B A .
M N . . I . G H . . S . . . . . . . . 5 2 3 0 . E F . . . B . 9 .
N M . . J I H G V U T . R Q P O 7 6 5 4 3 2 1 0 F . D C B A 9 8 .
O P Q R S T U V G H I . K . M N 8 . A B C D E F 0 1 2 3 4 5 6 7 .
. O R . . S V U H G J I L K N M 9 8 B A . C F E 1 0 3 2 5 4 7 6 .
Q . O P U V S . I J G . M N K L A B 8 9 E F C D 2 3 0 1 6 7 4 5 .
. Q P . V . . S . I H . . M . K B A 9 8 F E D . . 2 1 0 7 6 5 4 .
S . U . O P Q R . . M . G H I . C D E F 8 9 A B 4 5 6 . 0 1 2 3 .
T . V . P O R . L K N . H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 2 .
. V . L Q . O . . N . . . J . . . . . . . . . . . 7 4 . . . . . .
V U T . R Q P O N M . K . I H G F E D . B A 9 8 7 6 5 4 3 2 1 0 .
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.

E152: order 33, 775 clues

Status U. Unresolved producer output; criticality is NOT asserted and its size is not a lower bound.

Source: `outputs/lcs33_review_29814974/repairs/task_0_n33_rank_1.pending/repair.json`

SHA256: `9db3d70d6cb1664ea715e0a19232a85df57d2a85a1a316d860a6f00cf501139c`

Completion

```
W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
2 3 0 1 W 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 6
3 2 1 0 7 6 W 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 5
4 5 6 7 0 1 2 3 W D E F 8 9 A B K L M N G H I J S T U V O P Q R C
5 4 7 6 1 0 3 2 D C W E 9 8 B A L K N M H G J I T S V U P O R Q F
6 7 4 5 2 3 0 1 E F C D W B 8 9 M N K L I J G H U V S T Q R O P A
7 6 5 4 3 2 1 0 F E D C B A W 8 N M L K J I H G V U T S R Q P O 9
8 9 A B C D E F 0 1 2 3 4 5 6 7 W P Q R S T U V G H I J K L M N O
9 8 B A D C F E 1 0 3 2 5 4 7 6 P O W Q T S V U H G J I L K N M R
A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P W V S T I J G H M N K L U
B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U W S J I H G N M L K T
C D E F 8 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R W L M N G H I J K
D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K W M H G J I N
E F C D A B 8 9 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L W J G H I
F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I W G H
G H I J K W M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 9 A B C D E F L
H G J I L K N W P O R Q T S V U 1 0 3 2 5 4 7 6 9 8 B A D C F E M
I W G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D J
J I H W N M L K R Q P O V U T S 3 2 1 0 7 6 5 4 B A 9 8 F E D C G
K L M N G H I J S T U V O W Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B P
L K N M H G J I T S V U P O R W 5 4 7 6 1 0 3 2 D C F E 9 8 B A Q
M N K L I J G H U W S T Q R O P 6 7 4 5 2 3 0 1 E F C D A B 8 9 V
N M L K J I H G V U T W R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 S
O P Q R S T U V G H I J K L M N 8 9 A B C W E F 0 1 2 3 4 5 6 7 D
P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
Q R O P U V S T I J G H M N K L A W 8 9 E F C D 2 3 0 1 6 7 4 5 B
R Q P O V U T S J I H G N M L K B A 9 W F E D C 3 2 1 0 7 6 5 4 8
S T U V O P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 W 2 3 1
T S V U P O R Q L K N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 W 2
U V S T Q R O P M N K L I J G H E F C D A B 8 9 6 W 4 5 2 3 0 1 7
V U T S R Q P O N M L K J I H G F E D C B A 9 8 7 6 5 W 3 2 1 0 4
0 J 3 G 6 L 5 M C V F S A P 9 Q O B R 8 U D T E K 7 N 4 I 1 H 2 W
```

Partial square

```
. . . . 4 . 6 7 . 9 . B . D E . G . . . K . M N . . . R S T . V 0
. 0 . 2 5 . 7 6 9 . . A D C . E . . J . L K . M . O R Q T . . . .
. . 0 1 . . 4 5 A B . 9 . . C . I . . H . N K L . . O . U . S T .
. 2 1 0 . . . . B . . . . E . C . I H . . M L K . Q . O V U T . 5
4 5 . 7 0 . 2 . W D . . 8 9 A . K . . N G H . J S . U V . P Q . .
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
6 7 4 5 2 . 0 1 E . . D W B 8 9 M N K . I J G H . V . T . R O P A
7 6 5 . . . 1 0 . . . C B A . 8 . M L . J . H G . . T S . Q P O .
8 9 A B . D E . 0 1 . 3 4 5 . 7 . . Q R . T U V G H . J K L . N .
9 8 B A D C . E 1 0 . . 5 4 7 . P O W Q T S V U H G . . . K N . .
. . . . . . . . . 2 3 . 1 6 7 4 5 . . . . . . . . . I J G . M N K L U
B A . 8 . E D C 3 2 . 0 7 . 5 4 R Q P O V U W S J . H G . . . K T
C D E . 8 9 A B 4 5 . 7 0 1 2 3 S T U V O P Q R W L M N G H I J K
D . . . 9 8 B A 5 4 . 6 1 0 . 2 T S V U . . R Q L K W M H G . . N
E . C D A B 8 9 . . . 5 . 3 0 1 U V S . Q . . P M N . L . J G H I
. E . C B A . 8 7 6 . . 3 2 1 0 V U T . R . P O . M L K . I W G .
G H I J K . M N O . . R S T U V 0 1 . . 4 5 . 7 8 9 A . C D E . L
H G J I L . N W P O . Q T S V U 1 0 . 2 5 4 7 . 9 8 . A D C . E M
I W G H M . K L Q R . . U V S T 2 . 0 . 6 7 4 5 A B 8 . E . C D J
J I H W N . L K R Q . O V U T S . 2 1 0 7 . 5 4 B A 9 8 . E . C G
K L . . G H I J S . . . O W Q R 4 5 6 7 0 1 . . C D E . 8 9 A B P
L K N . H . J I T S . U P . R W 5 4 7 6 1 0 . 2 D C . E 9 8 B A Q
. N K L I . G H U W . T Q R O P 6 7 . 5 2 . 0 1 . . C . A B 8 9 V
N M L K J . H G V U . W R Q P O 7 6 5 4 . 2 1 0 . E D C B A 9 8 S
O . Q . . T U V G . . J K . M . 8 9 A B C W E . 0 1 2 3 4 . 6 7 D
P O . . . S V U H G . I L K N M 9 8 B A D C . W 1 0 3 2 5 4 7 6 E
Q R O . U V S . I J . H . . K L A . 8 9 E . . D 2 3 0 . 6 7 4 5 B
R Q P O V . T S J I . G N M L . B A 9 . . . . C 3 2 . 0 7 6 5 4 .
S T U V O P Q R K L . N G H I J C D E . 8 9 A B 4 5 6 7 0 W 2 3 1
T S V U P O R Q L K . M H G J I D C . E 9 8 B A 5 4 7 6 1 0 3 W .
U V S T . R O P . N . L . J G H E . C D A B . 9 6 . 4 5 2 3 0 1 7
V U T S R Q P O N . . . J I H G . E D C B A . 8 . . 5 W 3 2 1 0 4
0 J 3 G 6 . 5 M C V . S A P 9 Q O B R 8 U D T E K 7 . 4 I 1 H 2 W
```

Symbols 0–9 and A–W represent 0–32; dot = hole. Rows and columns start at 0.